



HYDRAULIC & PNEUMATIC



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NAK HYDRAULIC & PNEUMATIC SEALS

NAK was established in 1976, and became the first sealing manufacturing company in Taiwan to take its stock public in January of 2002. NAK's value-added products have been in high demand across both global and domestic markets due to its brand recognition, created by the quality of its products. Through the years, NAK has evolved into the No.1 leading manufacturer of sealing products in Taiwan, and secured its position and reputation in five continents of the world.

NAK has a strong and dedicated R&D team, where we not only design, develop and do material research but we take the initiative following up with our customers. Our R&D team has constant communication with our customers to obtain their specifications, requirements and demands, taking customer satisfaction to a higher level.

In a desire to reduce overhead costs, we have developed complex seal designs by a full implementation of FEA software (Finite Element Analysis). Our process initiates a theoretical analysis for material composition, pressure, temperature rise, and oil film, combining the designed prototype product and tooling with an in-house self-developed software known as SCAD to decrease failure rates during trial runs.



As for the material research at NAK we have a research lab entirely aligned as per regulations for complex testing procedures such as, rubber compounding, material characteristic testing, processibility of un-vulcanized rubber, hardness of plastic and rubber, material tensile strength and adhesive strength, heat aging / oil immersion / fuel immersion / low temp. properties / compression set, test of rebound properties /microscopy of compound dispersion, FT-IR analysis, down to the quality inspections which are all executed systematically.

At NAK all our CNC machines are internally connected to the SMART CAM computer and the mainframe network called the Machine center that provides shared processing resources to all the machines on demand. Thus, making the manufacture of our tooling molds much more precise and efficiently connecting to our mold manufacturing network to further secure our product quality.

For NAK to become an international enterprise and operate under a global strategic management model, our future orientations will focus on product differentiations and global positioning. NAK will primarily focus in the aggressive development of high performance and high value added seals, expansion of company size and boost technical competence so as to strengthen the functions and capabilities of our technical service team to meet market demands and also raise our global competing leverages. NAK will also use our overseas investment and strategic joint ventures to conceive global distributing channels and products varieties to meet the customer's never ending changes and demands!



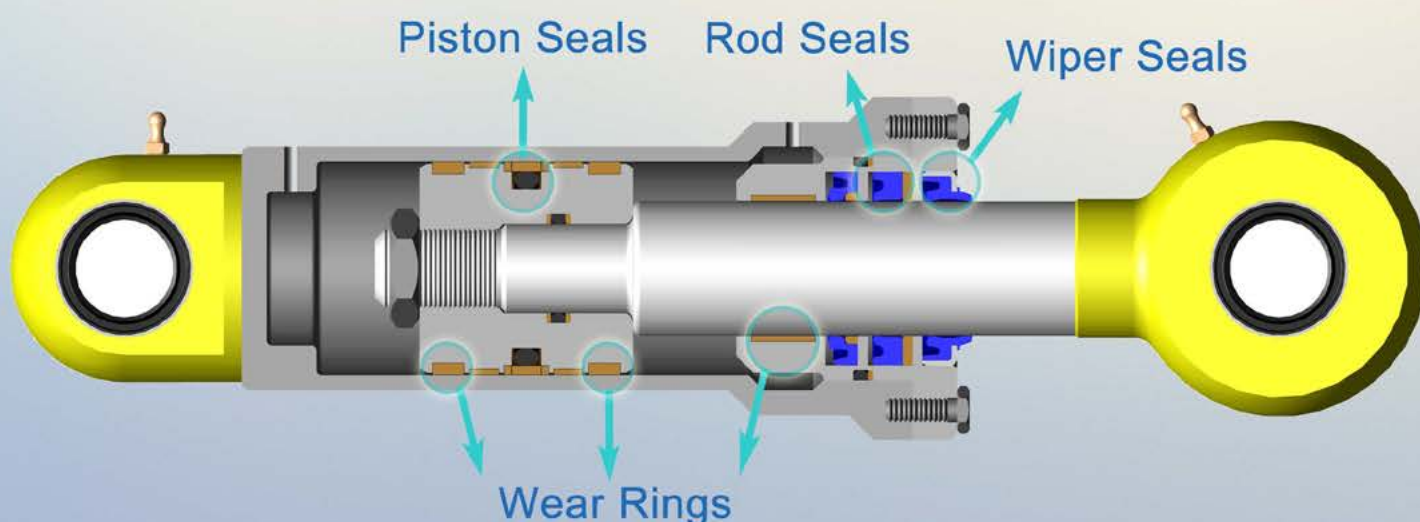
INTRODUCTION

NAK Hydraulic & Pneumatic Seals include Piston Seals, Rod Seals, Wiper Seals, Wear Rings, and etc. Rubber, Polyurethane, PTFE, Nylon and other materials are used to create a variety of seals to work under different hydraulic and pneumatic conditions. Most of these seals are designed for the reciprocating motion used commonly in hydraulic and pneumatic applications such as cylinders. To meet the critical requirements of the hydraulic and pneumatic equipment industries, NAK offers a wide range of designs.

For the hydraulic and pneumatic equipment, the cylinder design is getting more and more complicated. As a result, for the cylinder application, a variety of different seals are required, and the seals need to offer long service life and reliability. Sealing elements are critical components for the hydraulic and pneumatic cylinder application. Their main function is to prevent the cylinder fluid or gas from escaping, so as to ensure the normal functioning of the cylinder whether in a pressurized or un-pressurized state.

Function of Cylinder Seals

Wiper Seal	To prevent external contaminants from entering the system
Rod Seal	Installed and fixed on the cylinder bore to seal against the rod.
Piston Seal	Installed and fixed on the piston groove to seal against the cylinder
Wear Ring	To prevent metal-to-metal contact



INTRODUCTION

Sealing elements are vital in the hydraulic and pneumatic equipment. Their use is critical and they provide a way for the fluid power to be changed into linear motion. However, seal failure can be detrimental to the equipment, and it is important to have the right seals installed. Selection of proper hydraulic and pneumatic seals is depends on the following factors:

SIZE

- Shaft outer diameter (seal inner diameter);
bore diameter (seal outer diameter);
axial cross section and radial cross section.

TEMPERATURE

- Fluid temperature and operating temperature.
Both average and peak temperatures should be taken into consideration.

APPLICATION

- Hydraulic or pneumatic; piston or rod;
surface conditions of shaft and housing bore;
for use under dry running, lubrication or without lubrication.

MEDIA

- Type of media, its volume, and etc.

PRESSURE

- Static as well as dynamic pressure should be noticed.
Maximum operating pressure is also an important parameter.

SPEED

- Number of cylinder strokes, maximum speed, and etc.

MATERIAL

Item Material	Temp. Range (°C)		Property
	Low Temp.	High Temp.	
NBR	-40	100	Good resistance to alcohol, amines, petroleum oils, and gasoline over a wide range of temperatures. Also good resistance to caustic salts. Poor in strong oxidizing agents, chlorinated hydrocarbons, ketones, and esters.
FKM	-25	250	Excellent chemical resistance but ester and ketone. The cost is high. Good resistance to most oil and solvents
PU	-40	100	Extremely good tensile strength, abrasive resistance, tear strength, and extrusion strength but poor resistant to polar solvents, chlorinated hydrocarbons, aromatics, brake fluids, acids, and alkalis.

NBR

NBR is a polymer of butadiene and acrylonitrile. It is the most generally used material in all areas of seal technology due to its good physical and mechanical characteristics. The acrylonitrile (ACN) content affects physical property of NBR. When it contains high ACN, the hardness, tensile strength, oil resistance, heat resistance and etc. will increase. On the contrary, lower ACN contains will increase the elasticity and low temperature property. It is most frequently used in hydraulic and pneumatic system due to it is highly resistant to mineral oils, vegetable oils, many acids and improving the ozone resistance by recipe design.

FKM

FKM has various composition and fluorine contents from 66 to 70% is good characteristic. It has high resistance to temperature, weathering, ozone, and chemicals. It also has good tensile strength, resilience, low compression set. Moreover, it is highly resistant to almost all mineral oil based and synthetic-based hydraulic fluids. In contrast, it is not resistant to polar solvents, ketones, and brake fluid.

PU

Polyurethane is one of the groups of elastic thermoplastic materials. PU has been used in seal technology for many years because of their good physical characteristics. It is an organic material of high molecular weight whose chemical composition is characterized by a large number of urethane groups. In addition, it is characterized by extremely good mechanical properties such as high tensile strength, abrasive resistance, tear strength, and extrusion strength. However, it is not resistant to polar solvents, chlorinated hydrocarbons, aromatics, brake fluids, acids, and alkalis.

Material	Code	Hardness (Shore A)	Temperature range (°C)	Characteristic
NBR	NK761	71	-40 ~ 100	For resistant oils and friction
	NKC61	75	-40 ~ 100	For resistant oils and low temperature
	NK961	87	-35 ~ 100	For high hardness and high pressure
FKM	VN761	71	-25 ~ 200	For high temperature condition
	VNC06	75	-25 ~ 200	For reciprocating action and high temperature
	VN901	88	-20 ~ 200	For high hardness and high temperature
PU	UB903	94	-40 ~ 100	For high pressure and friction
	UB904	94	-40 ~ 100	For high pressure and friction
	UK908	95	-40 ~ 100	For high pressure and friction

Item Material	Oil Resistance	Base Resistance	Acid Resistance	Water Resistance	Weather Resistance	Abrasion Resistance
NBR	◎	○	○	○	△	◎
FKM	◎	△	○	○	◎	◎
PU	◎	△	△	△	◎	◎

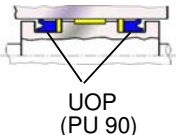
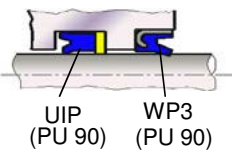
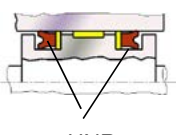
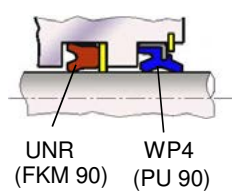
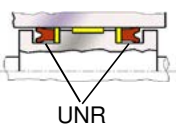
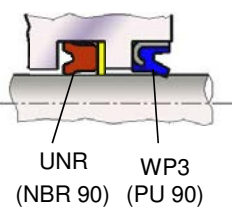
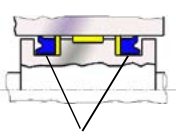
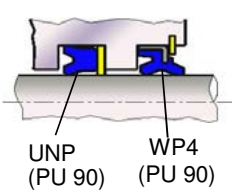
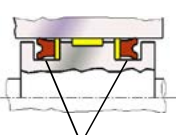
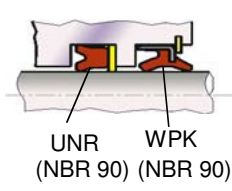
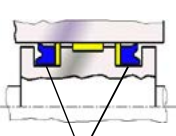
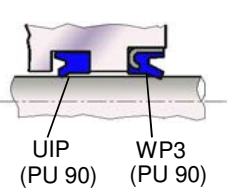
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○ : Fair

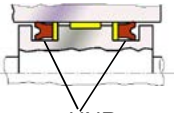
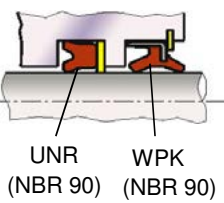
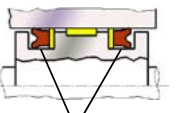
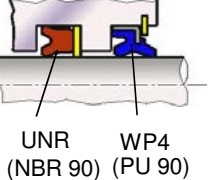
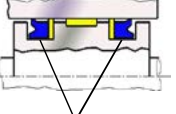
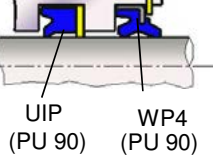
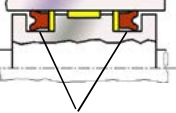
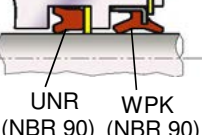
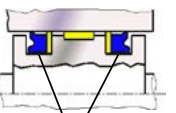
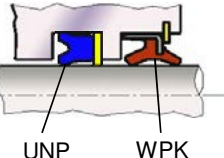
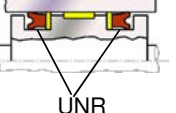
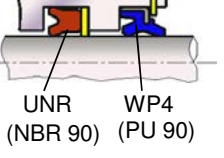
△ : Poor

✕ : Very Poor

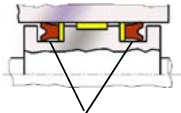
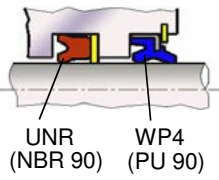
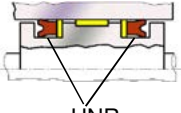
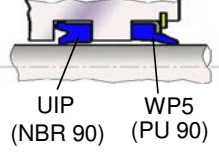
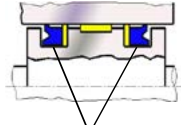
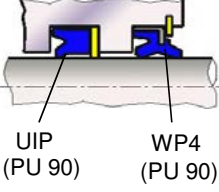
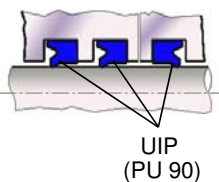
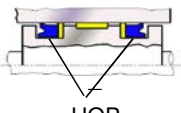
CLASSIFICATION OF ROD SEALS AND PISTON SEALS

Type	Application conditions	Piston seal	Characteristic	Rod seal	Characteristic
	0 ~ 20 Mpa -30 ~ 100°C	 UOP (PU 90)	UOP has excellent wear resistance and sealing ability.	 UIP (PU 90) WP3 (PU 90)	UIP has an asymmetrical profile of the sealing lips. WP3 protects cylinders against ingress of dirt and maintains sealing performance of packings.
	Heat-resistant 0 ~ 14 Mpa -25 ~ 200°C	 UNR (FKM 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (FKM 90) WP4 (PU 90)	UNR has a symmetrical profile of the sealing lip, and the height of both sealing lip is equal. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WP4 can prevent oil from scraping off.
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WP3 (PU 90)	UNR has a symmetrical profile of the sealing lip, and the height of both sealing lip is equal. WP3 protects cylinders against ingress of dirt and maintains sealing performance of packings.
	0 ~ 14 Mpa -30 ~ 100°C	 UO (PU 90)	UO has excellent wear resistance and sealing ability.	 UNP (PU 90) WP4 (PU 90)	UNP has a symmetrical profile of the sealing lip, excellent wear resistance, and sealing ability. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WP4 can prevent oil from scraping off.
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WPK (NBR 90)	UNR has a symmetrical profile of the sealing lip, and the height of both sealing lip is equal. WPK protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WPK can prevent oil from scraping off.
	0 ~ 20 Mpa -30 ~ 100°C	 UOP (PU 90)	UOP has excellent wear resistance and sealing ability.	 UIP (PU 90) WP3 (PU 90)	UIP has an asymmetrical profile of the sealing lips. WP3 protects cylinders against ingress of dirt and maintains sealing performance of packings.

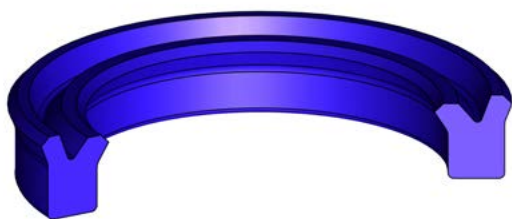
CLASSIFICATION OF ROD SEALS AND PISTON SEALS

Type	Application conditions	Piston seal	Characteristic	Rod seal	Characteristic
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WPK (NBR 90)	UNR has a symmetrical profile of the sealing lip, and the height of both sealing lip is equal. WPK protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WPK can prevent oil from scraping off.
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WP4 (PU 90)	UNR has a symmetrical profile of the sealing lip, and the height of both sealing lip is equal. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WP4 can prevent oil from scraping off.
	0 ~ 20 Mpa -30 ~ 100°C	 UOP (PU 90)	UOP has excellent wear resistance and sealing ability.	 UIP (PU 90) WP4 (PU 90)	UIP has an asymmetrical profile of the sealing lips. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, it can prevent oil from scraping off.
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WPK (NBR 90)	UNR has a symmetrical profile of the sealing lip, and the height of both sealing lip is equal. WPK protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WPK can prevent oil from scraping off.
	0 ~ 20 Mpa -30 ~ 100°C	 UOP (PU 90)	UOP has excellent wear resistance and sealing ability.	 UNP (PU 90) WPK (NBR 90)	UNP has a symmetrical profile of the sealing lip, excellent wear resistance, and sealing ability. WPK protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WPK can prevent oil from scraping off.
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WP4 (PU 90)	UNR has a symmetrical profile of the sealing lip, the height of both sealing lip is equal. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, it can prevent oil from scraping off.

CLASSIFICATION OF ROD SEALS AND PISTON SEALS

Type	Application conditions	Piston seal	Characteristic	Rod seal	Characteristic
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UNR (NBR 90) WP4 (PU 90)	UNR has a symmetrical profile of the sealing lip, the height of both sealing lip is equal. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, it can prevent oil from scraping off.
	0 ~ 14 Mpa -30 ~ 100°C	 UNR (NBR 90)	UNR has a symmetrical profile of the sealing lips, and the height of both sealing lip is equal.	 UIP (NBR 90) WP5 (PU 90)	UIP has an asymmetrical profile of the sealing lips. WP5 protects cylinders against ingress of dirt and maintains sealing performance of packings. WP5 has high dustproof performance.
	0 ~ 20 Mpa -30 ~ 100°C	 UOP (PU 90)	UOP has excellent wear resistance and sealing ability.	 UIP (PU 90) WP4 (PU 90)	UIP has an asymmetrical profile of the sealing lips. WP4 protects cylinders against ingress of dirt and maintains sealing performance of packings. Having a double lip, WP4 can prevent oil from scraping off.
	0 ~ 16 Mpa -30 ~ 100°C	—	—	 UIP (PU 90)	UIP has an asymmetrical profile of the sealing lips.
	0 ~ 16 Mpa -30 ~ 100°C	 UOP (PU 90)	UOP has excellent wear resistance and sealing ability.	—	—

NAK CODE	DRAWING	MATERIAL	PRESSURE (max)MPa	TEMPERATURE (max)°C	SLIDING SPEED(max) m/s	SIZE (mm)	PAGE
UIP		PU	40	-40 / +100	0.5	4-230	1.1 P14~P25 (METRIC)
UIP2		PU	40	-40 / +100	0.5	10-225	1.2 P26~P43 (METRIC)
UIP8		PU	40	-40 / +100	0.5	79-180	1.3 P44~P45 (METRIC)
UNP2		PU	40	-40 / +100	0.5	8-210	1.4 P46~P49 (METRIC)
CIP		PU	30	-40 / +100	0.5	38	1.5 P50~P51 (METRIC)
UIB 3		PU PA	50	-40 / +100	1	40-120	1.6 P52~P53 (METRIC)
HRO		NBR PTFE	35	-35 / +100	1.5	12-300	1.7 P54~P55 (METRIC)
UIR		NBR HNBR FKM	16	-35 / +100 -40 / +150 -25 / +250	0.5	6-90 24-32 9.5-16	1.8 P56~P59 (METRIC)
CIR		NBR FKM	10	-35 / +100 -25 / +250	0.5	7.9-33.5 20-34	1.9 P60~P61 (METRIC)



UIP

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. Usually used for rod hydraulic application.

TECHNICAL DATA

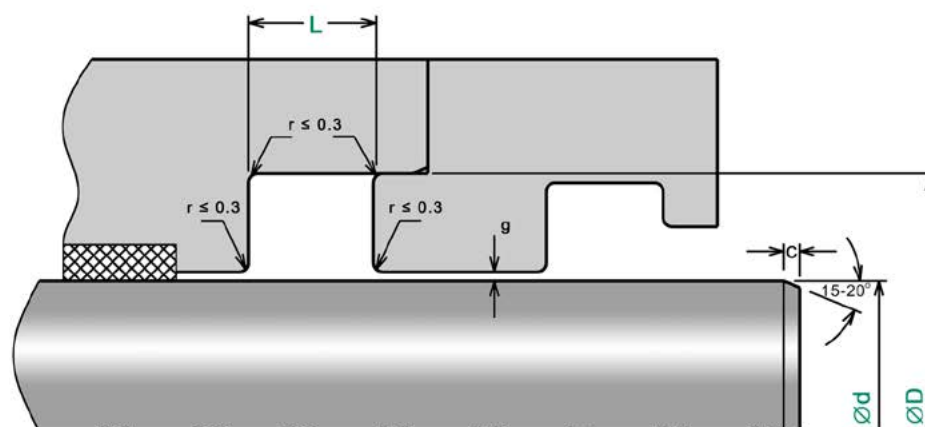
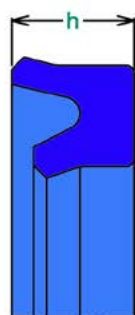
MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 40 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.27 mm	0.16 mm	0.06 mm

UIP Metric Size



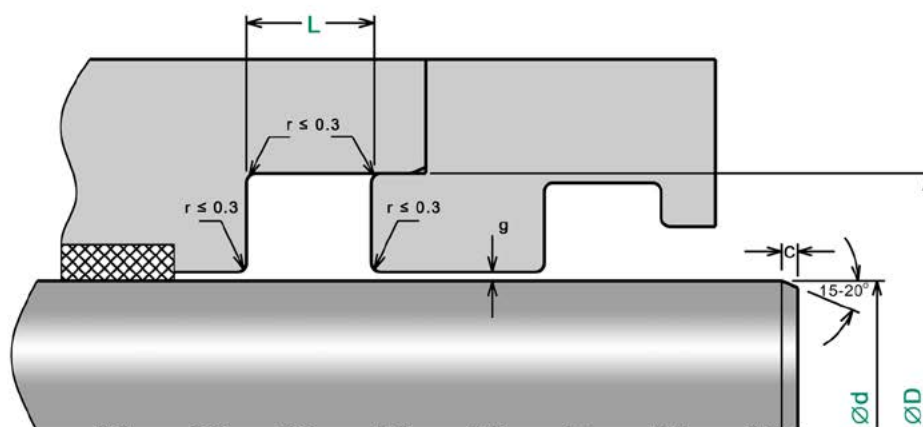
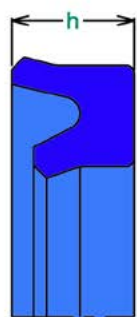
Ordering dimensions in green

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U2096	6	13	5.7	5	2.0
U4460	6	14	6.4	5.7	2.0
U2597	8	14	4.7	4	1.5
U3540	8	14	5.7	5	1.5
U0122	8	16	6.4	5.7	2.0
U4545	8	16	9	8	2.0
U6880	8	18	9	8	2.5
U7006	9	14.5	6.7	5.7	1.5
U2007	9.5	14	3.2	2.5	1.5
U7643	9.9	20	5.2	4.5	3.0
U3440	10	16	5.2	4.5	1.5
U2592	10	18	6.4	5.7	2.0
U7715	10	20	7	6	2.5
U1132	10	20	8.2	7.2	2.5
U6825	10	20	9	8	2.5
U9119	10	20	11	10	2.5
U8226	11.5	16	4.3	3.6	1.5
U7395	12	17	4.2	3.5	1.5
U9120	12	18	9	8	1.5
U7200	12	19	6.2	5.5	2.0
U3471	12	20	6.4	5.7	2.0
U4553	12	22	8.2	7.2	2.5
U2327	12	22	9	8	2.5
U1735	12.5	21	5.3	4.6	2.5
U4062	13	19	4.7	4	1.5
U7393	14	18.3	3.7	3	1.5
U4618	14	21	5.7	5	2.0
U3096	14	22	6.4	5.7	2.0
U9078	14	22	7	6	2.0
U2594	14	24	8.2	7.2	2.5
U4554	14	24	9	8	2.5
U4555	15	25	9	8	2.5
U6889	16	20.6	4	3.3	1.5
U2595	16	22	5.7	5	1.5
U8227	16	23	5.2	4.5	2.0
U6826	16	24	7.2	6.2	2.0
U1439	16	26	8.2	7.2	2.5

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UIP

Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U3280	16	26	9	8	2.5
U6834	18	24	5.4	4.7	1.5
U3447	18	24	5.7	5	1.5
U2119	18	26	6.4	5.7	2.0
U3063	18	26	9	8	2.0
U4208	18	28	8.2	7.2	2.5
U5871	19	27	7	6	2.0
U1643	20	26	5.7	5	1.5
U2054	20	27	7	6	2.0
U2319	20	28	6.4	5.7	2.0
U4558	20	28	7.2	6.2	2.0
U8441	20	28	7.6	6.6	2.0
U6833	20	28	8.2	7.2	2.0
U9085	20	30	5.7	5	2.5
U7550	20	30	6.4	5.7	2.5
U8553	20	30	8	7	2.5
U2055	20	30	8.2	7.2	2.5
U2426	20	30	9	8	2.5
U6914	20	32	8.5	7.5	3.0
U9046	22	28	4.7	4	1.5
U2160	22	30	6.4	5.7	2.0
U3083	22	30	8	7	2.0
U3064	22	30	9	8	2.0
U9045	22	32	5.7	5	2.5
U3388	22	32	8.2	7.2	2.5
U3573	22	32	9	8	2.5
U6912	22	32	10	9	2.5
U8378	22.22	28.57	5.46	4.76	2.0
U8248	23	31	5.7	5	2.0
U9146	24	32	5.7	5	2.0
U9113	24	32	7	6	2.0
U6831	24	34	8.2	7.2	2.5
U6830	24	34	9.5	8.5	2.5
U5567	25	32	7	6	2.0
U2081	25	33	6.4	5.7	2.0
U2520	25	33	7.2	6.2	2.0
U00335	25	33	7.5	6.5	2.0
U2659	25	33	8.2	7.2	2.0

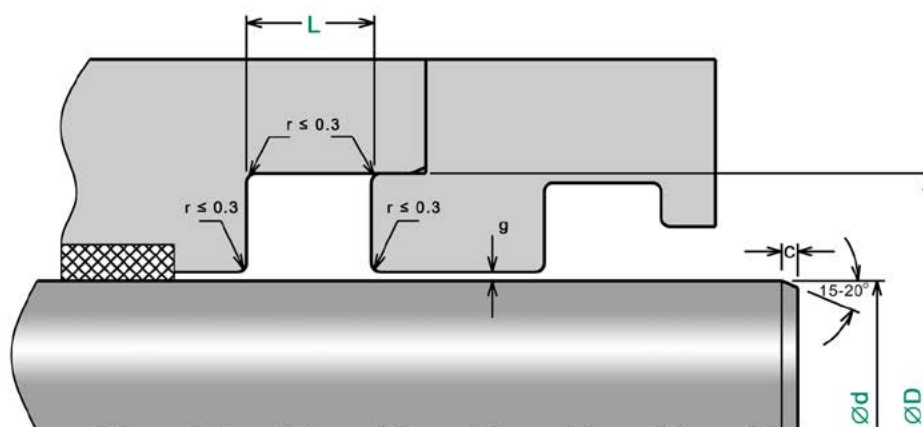
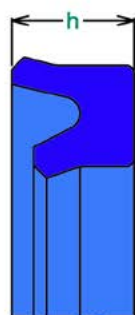
UIP Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U1835	25	33	9	8	2.0
U1193	25	33	11	10	2.0
U9048	25	35	5.7	5	2.5
U2258	25	35	8.2	7.2	2.5
U3818	25	35	9	8	2.5
U4559	25	35	11	10	2.5
U6854	25	38	10	9	3.5
U4560	25	40	11	10	4.0
U2211	28	36	6.4	5.7	2.0
U3013	28	38	8.2	7.2	2.5
U3339	28	38	9	8	2.5
U6853	28	38	11	10	2.5
U00236	28	39	9	8	3.0
U6852	28	40	9.5	8.5	3.0
U1836	28	40	11	10	3.0
U4561	28	43	12.5	11.5	4.0
U6851	29	35	5.5	4.8	1.5
U9047	30	37	5.7	5	2.0
U9077	30	37	7	6	2.0
U2085	30	38	6.4	5.7	2.0
U2122	30	38	9	8	2.0
U5719	30	40	6.4	5.7	2.5
U9076	30	40	7	6	2.5
U2661	30	40	8	7	2.5
U2077	30	40	8.2	7.2	2.5
U9084	30	40	9	8	2.5
U5720	30	40	10.5	9.5	2.5
U2328	30	40	11	10	2.5
U8077	30	42	9.5	8.5	3.0
U6926	30	43	10	9	3.3
U00060	30	43	11	10	3.3
U6570	30	45	9	8	3.8
U4562	30	45	11	10	3.8
U2521	32	40	6.4	5.7	2.0
U7379	32	40	7	6	2.0
U2848	32	40	9	8	2.0
U1837	32	42	8	7	2.5
U2640	32	42	8.2	7.2	2.5
U2542	32	42	9	8	2.5
U3789	32	42	11	10	2.5
U6192	32	42.53	9.5	8.5	3.0
U6199	32	45	10.5	9.5	3.5
U2017	32	45	11	10	3.5
U3799	32	47	11	10	4.0
U7375	32	48	11	10	4.0
U6222	33	43	11	10	2.5
U9097	35	42	7	6	2.0
U9114	35	42	8	7	2.0
U0013	35	43	6.4	5.7	2.0
U2042	35	43	9	8	2.0
U6221	35	45	7	6	2.5
U3424	35	45	8	7	2.5
U3196	35	45	8.2	7.2	2.5
U9157	35	45	9	8	2.5

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UIP

Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U3442	35	45	11	10	2.5
U6193	35	46	9	8	3.0
U6194	35	46	10	9	3.0
U6225	35	47	9.5	8.5	3.0
U2459	35	50	11	10	4.0
U9144	35	54	9	8	5.0
U5991	35	55	13	12	5.0
U2515	36	44	6.4	5.7	2.0
U6224	36	44	9	8	2.0
U2609	36	46	8.2	7.2	2.5
U3571	36	46	11	10	2.5
U6223	36	48	8.5	7.5	3.0
U2967	36	48	9	8	3.0
U6227	36	51	11	10	4.0
U4563	36	51	12.5	11.5	4.0
U6201	37	45	6.4	5.7	2.0
U1778	37	47	6.4	5.7	2.5
U3426	37	47	11	10	2.5
U6200	38	45	5.7	5	2.0
U00062	38	45	7	6	2.0
U6207	38	45	7.2	6.2	2.0
U9156	38	45	9	8	2.0
U9122	38	51	11	10	3.5
U00670	38	58	11	10	5.0
U2044	40	48	6.4	5.7	2.0
U1846	40	48	9	8	2.0
U6582	40	50	7	6	2.5
U2949	40	50	8.2	7.2	2.5
U1847	40	50	9	8	2.5
U6206	40	50	10	9	2.5
U3515	40	50	11	10	2.5
U6214	40	50	14.5	13.5	2.5
U2698	40	52	9	8	3.0
U2707	40	55	8.2	7.2	4.0
U6315	40	55	10	9	4.0
U4207	40	55	11	10	4.0
U3819	40	55	12.5	11.5	4.0
U9118	40	60	11	10	5.0

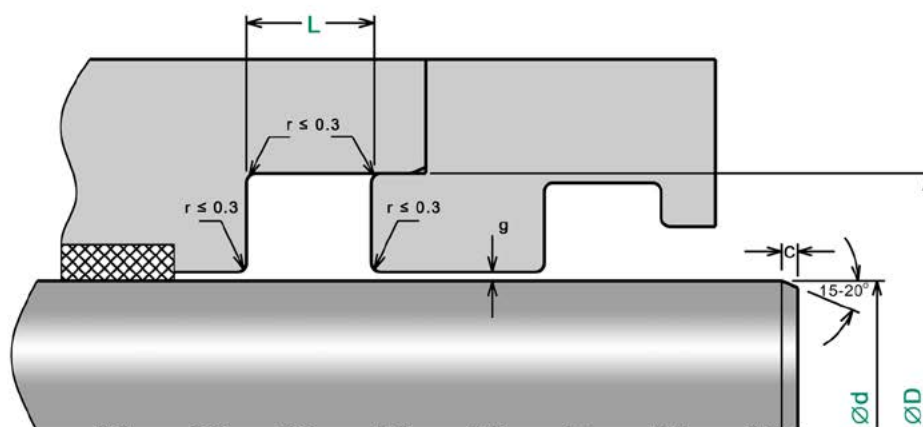
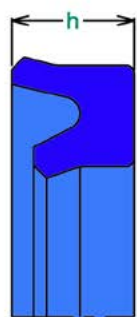
UIP Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U6213	40	60	13	12	5.0
U7174	42	50	6.7	5.7	2.0
U9145	42	50	7	6	2.0
U7245	42	52	8	7	2.5
U6215	42	62	11.5	10.5	5.0
U4841	42	62	13	12	5.0
U7751	43	52.5	8.6	7.6	2.5
U7090	44.4	54.03	7.53	6.53	2.5
U2102	45	53	6.4	5.7	2.0
U3443	45	53	9	8	2.0
U2212	45	53	11	10	2.0
U4564	45	53	13	12	2.0
U1848	45	55	6.4	5.7	2.5
U1956	45	55	6.5	5.8	2.5
U7094	45	55	7	6	2.5
U5731	45	55	7.2	6.2	2.5
U4565	45	55	8.2	7.2	2.5
U3201	45	55	8.3	7.3	2.5
U2193	45	55	11	10	2.5
U9092	45	56	8	7	3.0
U3032	45	57.7	10.5	9.5	3.5
U6129	45	58	10	9	3.5
U2867	45	60	11	10	4.0
U4566	45	60	12.5	11.5	4.0
U6211	45	65	11	10	5.0
U3522	45	65	13	12	5.0
U6216	46	56	8.5	7.5	2.5
U9067	48	60	9	8	3.0
U7368	50	60	7	6	2.5
U5732	50	60	7.4	6.4	2.5
U2681	50	60	8.2	7.2	2.5
U6730	50	60	9	8	2.5
U2194	50	60	11	10	2.5
U6217	50	60	13	12	2.5
U5990	50	62	9	8	3.0
U6220	50	62.7	11	10	3.5
U9143	50	65	10	9	4.0
U2460	50	65	11	10	4.0
U3195	50	65	12.4	11.4	4.0
U2593	50	65	12.5	11.5	4.0
U9061	50	65	13	12	4.0
U6203	50	68	10	9	4.5
U9099	50	68	11	10	4.5
U6219	50	70	11	10	5.0
U2644	50	70	13	12	5.0
U4507	50.8	63.5	10.52	9.52	3.5
U6935	51	61.4	7.4	6.4	2.6
U3285	52	62	11	10	2.5
U8086	53.97	66.67	10.52	9.52	3.5
U2805	55	63	9	8	2.0
U2298	55	63	13	12	2.0
U9112	55	65	5.7	5	2.5
U7361	55	65	7	6	2.5
U3281	55	65	8.2	7.2	2.5

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UIP

Metric Size



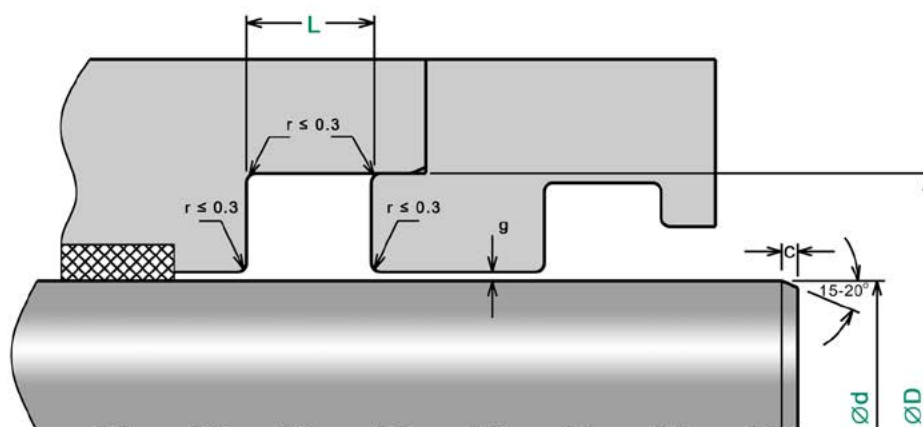
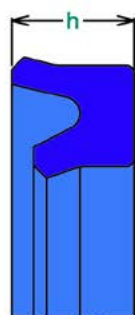
Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U3204	55	65	11	10	2.5
U2682	55	65	13	12	2.5
U6218	55	65	14.5	13.5	2.5
U2165	55	67	11	10	3.0
U6130	55	68	11.5	10.5	3.5
U3574	55	70	11	10	4.0
U2646	55	70	13	12	4.0
U6249	55	75	13	12	5.0
U6284	55	75	14.5	13.5	5.0
U9169	56	66	9	8	2.5
U3065	56	66	11	10	2.5
U3066	56	71	11	10	4.0
U2978	56	71	12.5	11.5	4.0
U6286	56	76	14.5	13.5	5.0
U6285	56	76	16	15	5.0
U8934	58	70.5	11	10	3.5
U9062	58	75	11	10	4.5
U9121	60	70	7	6	2.5
U2635	60	70	8.2	7.2	2.5
U4567	60	70	11	10	2.5
U2621	60	70	13	12	2.5
U0786	60	70	14	13	2.5
U6304	60	72	9	8	3.0
U6303	60	72	10	9	3.0
U9063	60	72	11	10	3.0
U7367	60	73	11	10	3.5
U4528	60	75	11	10	4.0
U3553	60	75	12.5	11.5	4.0
U2132	60	75	13	12	4.0
U6204	60	77	12	11	4.5
U2849	60	80	13	12	5.0
U6302	60	80	14.5	13.5	5.0
U6888	62	74	14	13	3.0
U6301	63	73	11	10	2.5
U2683	63	73	13	12	2.5
U2153	63	75	9.7	8.7	3.0
U3189	63	78	11	10	4.0
U2199	63	78	12.5	11.5	4.0

UIP Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U6247	63	78	13	12	4.0
U2616	63	83	13	12	5.0
U6246	63	83	14.5	13.5	5.0
U4546	63	83	16	15	5.0
U2026	65	75	11	10	2.5
U2113	65	75	13	12	2.5
U6245	65	75	14.5	13.5	2.5
U6208	65	77	10	9	3.0
U3272	65	78	11	10	3.5
U6242	65	80	11	10	4.0
U3067	65	80	13	12	4.0
U3832	65	85	13	12	5.0
U6241	65	85	14.5	13.5	5.0
U0824	69.85	82.55	10.52	9.52	3.5
U7036	70	80	7	6	2.5
U6619	70	80	8	7	2.5
U2898	70	80	8.2	7.2	2.5
U3302	70	80	11	10	2.5
U2286	70	80	13	12	2.5
U2259	70	82	10	9	3.0
U6185	70	83	11	10	3.5
U7025	70	85	10	9	4.0
U2965	70	85	11	10	4.0
U3274	70	85	12.5	11.5	4.0
U3206	70	85	13	12	4.0
U1849	70	90	13	12	5.0
U3550	70	90	16	15	5.0
U00377	72	80	13	12	2.0
U7366	75	85	7	6	2.5
U2878	75	85	8.2	7.2	2.5
U9091	75	85	11	10	2.5
U3157	75	85	13	12	2.5
U1952	75	87	9.6	8.6	3.0
U6835	75	90	10	9	4.0
U2280	75	90	11	10	4.0
U3055	75	90	13	12	4.0
U6270	75	95	13	12	5.0
U6269	75	95	14.5	13.5	5.0
U6887	76	84	8.5	7.5	2.0
U6624	76.2	92.07	10.52	9.52	4.0
U6622	76.2	95.25	10.52	9.52	5.0
U9155	80	90	7	6	2.5
U4508	80	90	8.2	7.2	2.5
U7360	80	90	9	8	2.5
U2070	80	90	13	12	2.5
U6288	80	90	15	14	2.5
U2128	80	92	9.6	8.6	3.0
U9692	80	92	11	10	3.0
U3027	80	93	11	10	3.5
U6552	80	95	10	9	4.0
U2699	80	95	11	10	4.0
U2733	80	95	12.5	11.5	4.0
U2135	80	95	13	12	4.0
U2398	80	100	13	12	5.0

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UIP Metric Size

Ordering dimensions in green

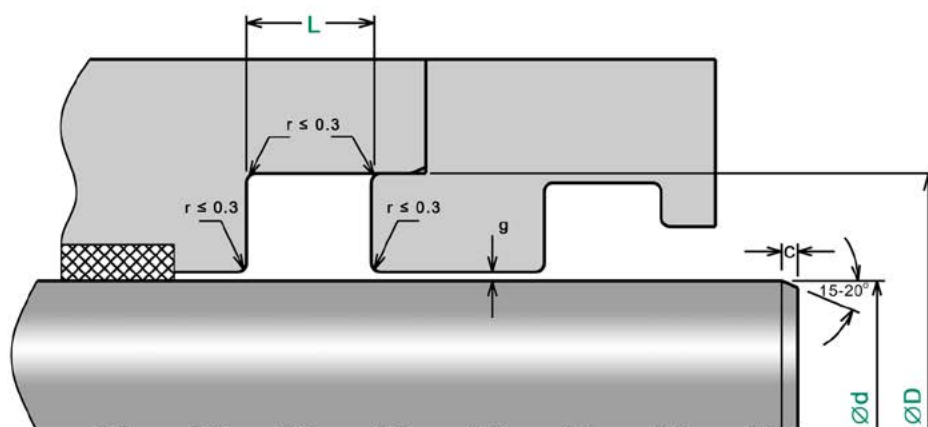
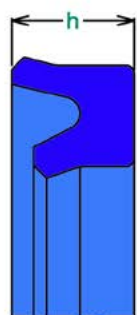
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U0897	80	100	14.5	13.5	5.0
U6287	80	100	16	15	5.0
U6635	82.55	98.42	8.93	7.93	4.0
U9224	85	94	15	14	2.5
U6925	85	95	8.2	7.2	2.5
U7031	85	100	10	9	4.0
U7359	85	100	11	10	4.0
U6924	85	100	12	11	4.0
U1851	85	100	12.5	11.5	4.0
U2710	85	100	13	12	4.0
U6898	85	105	13	12	5.0
U6886	85	105	14.5	13.5	5.0
U1839	90	100	11	10	2.5
U6923	90	100	12.5	11.5	2.5
U9123	90	100	13	12	2.5
U1957	90	102	10	9	3.0
U7854	90	102	15	14	3.0
U1852	90	105	9.5	8.5	4.0
U7358	90	105	10	9	4.0
U2648	90	105	11	10	4.0
U3014	90	105	12.5	11.5	4.0
U2149	90	105	13	12	4.0
U2450	90	110	13	12	5.0
U9781	90	110	14	13	5.0
U3422	90	110	16	15	5.0
U6921	93	104	11.5	10.5	3.0
U6906	95	105	11	10	2.5
U6930	95	105	13	12	2.5
U3092	95	110	10	9	4.0
U5721	95	110	10.5	9.5	4.0
U2700	95	110	13	12	4.0
U7262	98	112	9.5	8.5	3.5
U9098	100	110	5.7	5	2.5
U9066	100	110	13	12	2.5
U6929	100	110	15	14	2.5
U6928	100	113	14	13	3.5
U6398	100	115	10	9	4.0
U6821	100	115	10	9	4.0

UIP Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U7023	100	115	10	9	4.0
U6936	100	115	11	10	4.0
U0866	100	115	11.5	10.5	4.0
U2743	100	115	12	11	4.0
U2649	100	115	13	12	4.0
U2850	100	120	13	12	5.0
U6937	100	120	14.5	13.5	5.0
U1840	100	120	16	15	5.0
U6733	100	125	20	19	6.5
U6569	105	115	11	10	2.5
U6932	105	115	12.5	11.5	2.5
U6931	105	115	14.5	13.5	2.5
U6938	105	120	13	12	4.0
U6248	105	125	13	12	5.0
U1958	110	125	10.6	9.6	4.0
U9090	110	125	11	10	4.0
U2637	110	125	12	11	4.0
U6849	110	125	16	15	4.0
U3158	110	130	13	12	5.0
U6885	110	130	15.6	14.6	5.0
U2959	110	130	16	15	5.0
U6939	110	135	20	19	6.5
U7334	115	130	10	9	4.0
U3556	115	130	12	11	4.0
U6940	115	135	13	12	5.0
U6918	120	130	8.2	7.2	2.5
U1842	120	130	12.5	11.5	2.5
U6919	120	132	11.5	10.5	3.0
U6902	120	135	12.5	11.5	4.0
U5724	120	135	16	15	4.0
U2525	120	140	13	12	5.0
U2346	120	140	16	15	5.0
U6920	125	135	14.5	13.5	2.5
U4082	125	145	13	12	5.0
U2979	125	145	16	15	5.0
U5980	125	150	15	14	6.5
U6883	125	150	20	19	6.5
U6872	130	140	8.2	7.2	2.5
U4525	130	140	14.5	13.5	2.5
U3406	130	145	13	12	4.0
U6873	130	145	15	14	4.0
U6281	130	150	13	12	5.0
U1199	130	150	16	15	5.0
U3920	130	150	16	15	5.0
U9089	135	150	11	10	4.0
U7780	138	155	16	15	4.5
U00656	140	150	8	7	2.5
U5906	140	150	12.5	11.5	2.5
U2990	140	155	13	12	4.0
U4149	140	160	13	12	5.0
U2131	140	160	16	15	5.0
U6876	140	165	20	19	6.5
U9855	141	151	15	14	2.5
U6882	145	157.7	10	9	3.5

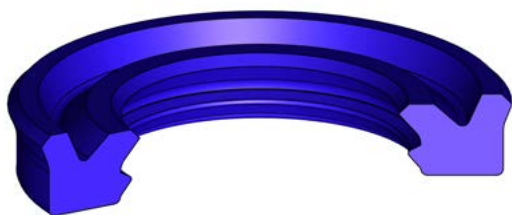
1.1

UIP Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U3301	150	170	13	12	5.0
U2966	150	170	16	15	5.0
U6875	160	170	13	12	2.5
U6863	160	180	13	12	5.0
U2141	160	180	16	15	5.0
U1843	160	185	16	15	6.5
U8258	160	185	20	19	6.5
U6862	160	190	19	18	7.5
U6865	170	190	13	12	5.0
U2142	170	190	16	15	5.0
U6861	180	200	13	12	5.0
U2140	180	200	16	15	5.0
U6857	180	200	20	19	5.0
U4547	180	205	16	15	6.5
U7855	180	205	19.2	18.2	6.5
U8703	185	195	14.5	13.5	2.5
U6860	190	210	13.5	12.5	5.0
U2734	190	210	16	15	5.0
U00613	195	215	13	12	5.0
U6859	200	220	13	12	5.0
U6848	200	220	16	15	5.0
U6847	200	225	20	19	6.5
U8272	200	230	19	18	7.5
U00630	204	224	13	12	5.0
U6856	210	230	16	15	5.0
U7816	220	240	13	12	5.0
U4895	220	240	16	15	5.0
U7817	230	250	13	12	5.0



UIP2

PRODUCT DESCRIPTION

Seal designed with symmetric lips. The secondary edge can reduce leakage, friction, and dirt ingress. Usually used for rod hydraulic application.

TECHNICAL DATA

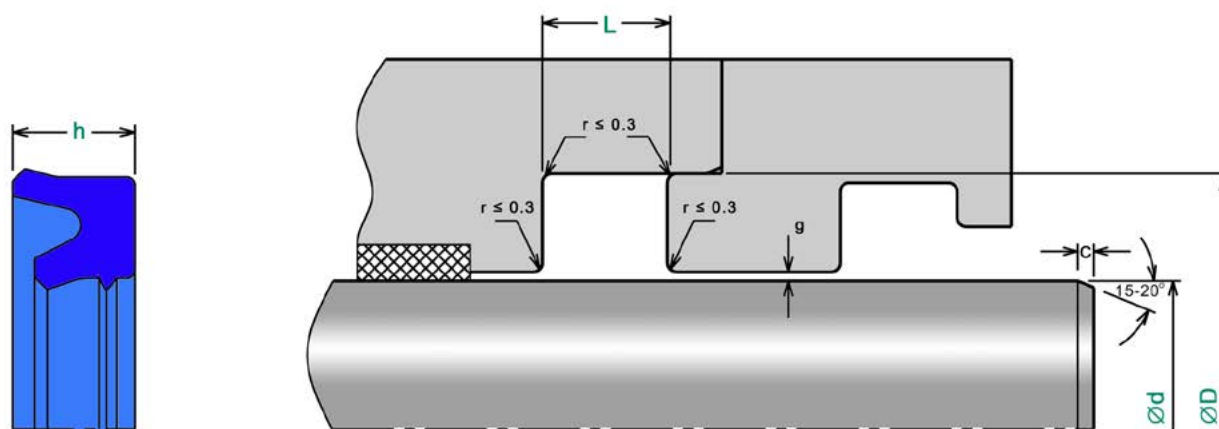
MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 40 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

UIP2 Metric Size

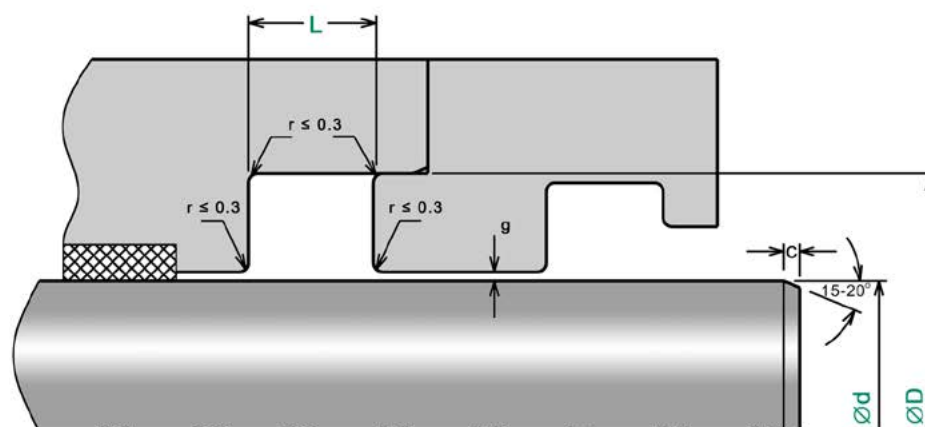
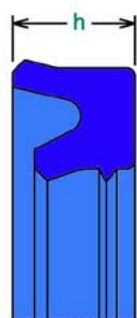


Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U9714	10	16	4.7	4	1.5
U1744	10	16	5.2	4.5	1.5
U9718	10	16	7	6	1.5
U0951	10	18	6.2	5.5	2.0
U1598	10	18	6.4	5.7	2.0
U2939	10	20	5.2	4.5	2.5
U1444	10	20	8	7	2.5
U1445	10	20	9	8	2.5
U9711	11	18	5.7	5	2.0
U9697	12	18	8	7	1.5
U9708	12	19	7	6	2.0
U1597	12	20	6.2	5.5	2.0
U3531	12	20	6.4	5.7	2.0
U1446	12	20	9	8	2.0
U7648	12	22	6.2	5.5	2.5
U9704	12	22	7	6	2.5
U3833	12	22	9	8	2.5
U4112	12.7	19.05	5.46	4.76	2.0
U8931	14	19	4.7	4	1.5
U9717	14	20	7	6	1.5
U1071	14	22	6.2	5.5	2.0
U2974	14	22	6.4	5.7	2.0
U1593	14	24	8.3	7.3	2.5
U2940	14	24	8.3	7.3	2.5
U00381	14	24	9	8	2.5
U1449	15	25	9	8	2.5
U4113	15.87	22.22	5.46	4.76	2.0
U4416	16	22	5.7	5	1.5
U7584	16	24	5.7	5	2.0
U1072	16	24	6.2	5.5	2.0
U1072	16	24	6.2	5.5	2.0
U1592	16	24	6.5	5.8	2.0
U1590	16	26	8.3	7.3	2.5
U2980	16	26	9	8	2.5
U9713	16	29	5.7	5	3.5
U5199	17	23	4.7	4	1.5
U8942	17	27	5.7	5	2.5
U4050	18	25	5.7	5	2.0

1.2

UIP2 Metric Size



Ordering dimensions in green

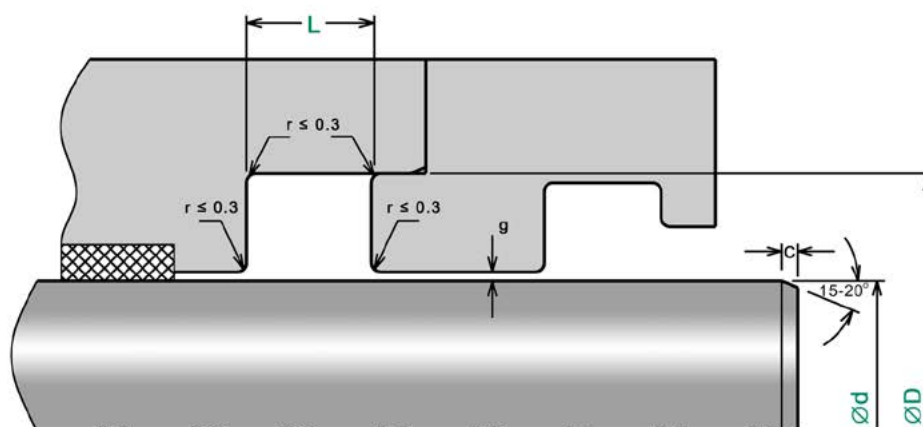
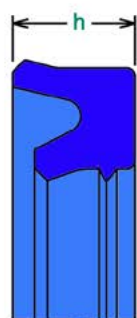
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U1588	18	26	6.2	5.5	2.0
U4821	18	26	6.2	5.5	2.0
U4051	18	26	7.3	6.3	2.0
U1589	18	28	8.3	7.3	2.5
U1450	18	28	9	8	2.5
U8939	19	27	5.7	5	2.0
U4115	19.05	25.4	5.46	4.76	2.0
U4103	19.05	25.4	7.35	6.35	2.0
U8930	20	25	4.7	4	1.5
U4052	20	26	5.5	4.8	1.5
U2291	20	28	6.2	5.5	2.0
U6959	20	28	6.4	5.7	2.0
U7641	20	30	7	6	2.5
U1622	20	30	8	7	2.5
U5872	20	30	8.3	7.3	2.5
U3228	20	30	9	8	2.5
U5278	20	30	11	10	2.5
U1073	20	35	11	10	4.0
U9709	20	38	8	7	4.5
U8929	22	28	4.7	4	1.5
U8940	22	30	5.7	5	2.0
U2975	22	30	6.4	5.7	2.0
U3917	22	30	7	6	2.0
U4154	22	32	8.3	7.3	2.5
U3810	22	32	9	8	2.5
U9486	22	32	9	8	2.5
U9696	22	34	7	6	3.0
U3232	22	35	11	10	3.5
U9720	22	38	7	6	4.0
U9721	22	40	9	8	4.5
U4106	22.22	28.57	5.46	4.76	2.0
U4107	22.22	28.57	7.35	6.35	2.0
U9695	25	32	5.7	5	2.0
U1605	25	32	11	10	2.0
U6125	25	33	5.5	4.8	2.0
U2429	25	33	6.2	5.5	2.0
U4382	25	33	6.4	5.7	2.0
U2539	25	33	7.5	6.5	2.0

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U3233	25	33	8	7	2.0
U6186	25	33	8.3	7.3	2.0
U4054	25	33	11	10	2.0
U4628	25	34	7.5	6.5	2.5
U3466	25	35	8	7	2.5
U3016	25	35	8.3	7.3	2.5
U2994	25	35	9	8	2.5
U2660	25	35	11	10	2.5
U3834	25	40	11	10	4.0
U3659	25	43	7.5	6.5	4.5
U0904	25	45	13	12	5.0
U4108	25.4	31.75	5.46	4.76	2.0
U4109	25.4	31.75	7.35	6.35	2.0
U4349	25.4	34.92	7.35	6.35	2.5
U4350	25.4	34.92	8.93	7.93	2.5
U4199	25.4	38.1	10.52	9.52	3.5
U7130	26	36	11	10	2.5
U9715	26	38	9	8	3.0
U9707	27	38	8	7	3.0
U4820	28	35.5	5.7	5	2.0
U00069	28	36	6.4	5.7	2.0
U2278	28	36	8	7	2.0
U3572	28	36	8.3	7.3	2.0
U00129	28	36	9	8	2.0
U9703	28	37	7	6	2.5
U3159	28	38	8.3	7.3	2.5
U3786	28	38	9	8	2.5
U4150	28	38	11	10	2.5
U4111	28.57	34.92	5.46	4.76	2.0
U4116	28.57	34.92	7.35	6.35	2.0
U4351	28.57	38.1	8.93	7.93	2.5
U4675	28.57	41.27	7.35	6.35	3.5
U6127	30	38	5.5	4.8	2.0
U5239	30	38	6.7	5.7	2.0
U5279	30	38	7	6	2.0
U4384	30	38	7.3	6.3	2.0
U9706	30	38	11	10	2.0
U8476	30	40	7	6	2.5
U7048	30	40	7.5	6.5	2.5
U4385	30	40	8	7	2.5
U6128	30	40	8	7	2.5
U3264	30	40	8.3	7.3	2.5
U3236	30	40	9	8	2.5
U2215	30	40	11	10	2.5
U3237	30	45	11	10	4.0
U4097	31.75	38.1	7.35	6.35	2.0
U4680	31.75	41.27	5.46	4.76	2.5
U4353	31.75	41.27	7.35	6.35	2.5
U4354	31.75	41.27	8.93	7.93	2.5
U7078	31.75	41.27	8.93	7.93	2.5
U4674	31.75	44.45	7.35	6.35	2.5
U4274	31.75	44.45	10.52	9.52	2.5
U9702	32	40	6.2	5.5	2.0
U9301	32	40	6.5	5.8	2.0

1.2

UIP2 Metric Size



Ordering dimensions in green

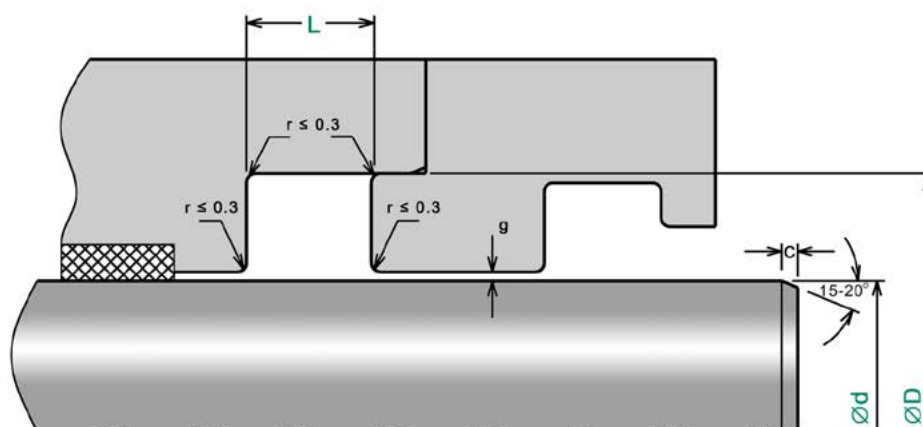
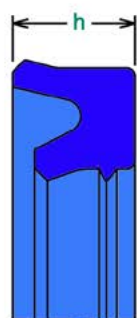
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U9716	32	40	7	6	2.0
U2184	32	40	7.7	6.7	2.0
U6018	32	40	8.2	7.2	2.0
U2662	32	40	9	8	2.0
U3238	32	40	11	10	2.0
U3283	32	41.5	8.9	7.9	2.5
U2874	32	42	8.3	7.3	2.5
U3508	32	42	11	10	2.5
U2163	32	45	11	10	3.5
U4078	32	47	11	10	4.0
U4098	34.92	41.27	7.35	6.35	2.0
U4355	34.92	44.45	8.93	7.93	2.5
U4275	34.92	47.62	10.52	9.52	3.5
U9701	35	42	7	6	2.0
U4429	35	43	6.4	5.7	2.0
U4429	35	43	6.4	5.7	2.0
U5994	35	43	8.2	7.2	2.0
U8475	35	45	7	6	2.5
U3472	35	45	8	7	2.5
U5560	35	45	8.3	7.3	2.5
U4055	35	45	9	8	2.5
U2924	35	45	11	10	2.5
U3217	35	50	11	10	4.0
U5559	36	44	6.4	5.7	2.0
U4681	36	44	7.3	6.3	2.0
U00148	36	44	8.6	7.6	2.0
U3084	36	46	8	7	2.5
U3017	36	46	8.3	7.3	2.5
U3420	36	46	8.5	7.5	2.5
U2641	36	46	11	10	2.5
U2879	36	51	11	10	4.0
U4356	36.51	46.04	8.93	7.93	2.5
U2111	37	47	9	8	2.5
U5502	37	47	11	10	2.5
U6125	25	33	5.5	4.8	2.0
U2429	25	33	6.2	5.5	2.0
U4382	25	33	6.4	5.7	2.0
U2539	25	33	7.5	6.5	2.0

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U5502	37	47	11	10	2.5
U6609	38	48	6.5	5.8	2.5
U5143	38	48	9	8	2.5
U1083	38	55	11	10	4.5
U4099	38.1	44.45	7.35	6.35	2.0
U4678	38.1	47.62	5.46	4.76	2.5
U4357	38.1	47.62	7.35	6.35	2.5
U4358	38.1	47.62	8.93	7.93	2.5
U4359	38.1	47.62	10.52	9.52	2.5
U7085	38.1	47.62	10.52	9.52	2.5
U4673	38.1	50.8	7.35	6.35	3.5
U4276	38.1	50.8	10.52	9.52	3.5
U4335	38.1	53.97	10.52	9.52	4.0
U4057	40	48	6.4	5.7	2.0
U3564	40	48	9	8	2.0
U9554	40	48	14.5	13.5	2.0
U4056	40	49.5	10.5	9.5	2.5
U8599	40	50	7	6	2.5
U9095	40	50	7	6	2.5
U9725	40	50	8	7	2.5
U5531	40	50	8.3	7.3	2.5
U2596	40	50	8.3	7.3	2.5
U6610	40	50	9	8	2.5
U2101	40	50	11	10	2.5
U7104	40	50	11	10	2.5
U6611	40	55	9	8	4.0
U3219	40	55	11	10	4.0
U5532	40	55	12.4	11.4	4.0
U1621	40	55	12.5	11.5	4.0
U5280	40	60	12.5	11.5	5.0
U3240	40	60	13	12	5.0
U4100	41.27	47.62	7.35	6.35	2.0
U4421	41.27	50.8	8.14	7.14	2.5
U4422	41.27	50.8	8.93	7.93	2.5
U4423	41.27	50.8	10.52	9.52	2.5
U6101	42	50	6.4	5.7	2.0
U6001	42	50	12.5	11.5	2.0
U3869	42	52	11	10	2.5
U8606	43	53	8.3	7.3	2.5
U4424	44.45	53.97	7.35	6.35	2.5
U4432	44.45	53.97	8.93	7.93	2.5
U4433	44.45	53.97	10.52	9.52	2.5
U4285	44.45	57.15	10.52	9.52	3.5
U4286	44.45	57.15	13.7	12.7	3.5
U3561	45	53	6.3	5.6	2.0
U8269	45	53	7.3	6.3	2.0
U00659	45	53	9.2	8.2	2.0
U4039	45	53	12.5	11.5	2.0
U9724	45	55	7	6	2.5
U1466	45	55	8	7	2.5
U3197	45	55	8.3	7.3	2.5
U5533	45	55	8.3	7.3	2.5
U1950	45	55	11	10	2.5
U00169	45	55	11	10	2.5

1.2

UIP2 Metric Size



Ordering dimensions in green

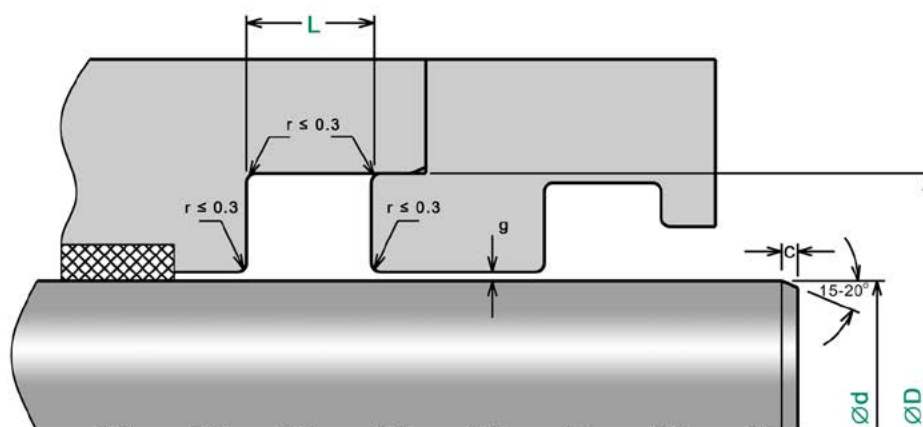
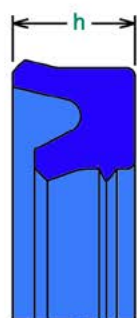
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U6002	45	55	12.5	11.5	2.5
U1610	45	57.7	10.5	9.5	3.5
U3559	45	60	11	10	4.0
U3241	45	60	11	10	4.0
U3407	45	60	12.5	11.5	4.0
U4058	45	65	11	10	5.0
U7250	47	56	10	9	2.5
U4434	47.62	57.15	10.52	9.52	2.5
U6003	48	56	12.5	11.5	2.0
U9699	48	60	9	8	3.0
U3276	48	63	11	10	4.0
U1468	48	68	13	12	5.0
U1468	48	68	13	12	5.0
U6021	50	57	11	10	2.0
U7587	50	58	7	6	2.0
U8155	50	58	11	10	2.0
U6006	50	58	12.5	11.5	2.0
U7059	50	58	14	13	2.0
U8490	50	60	7	6	2.5
U2381	50	60	8.3	7.3	2.5
U6884	50	60	8.3	7.3	2.5
U9712	50	60	9	8	2.5
U2256	50	60	11	10	2.5
U1208	50	62	11	10	3.0
U4041	50	62.7	10.5	9.5	3.5
U3243	50	63	11	10	3.5
U9722	50	65	7	6	4.0
U4042	50	65	10.5	9.5	4.0
U2820	50	65	11	10	4.0
U4040	50	65	12	11	4.0
U3327	50	65	12.5	11.5	4.0
U3539	50	65	16.5	15.5	4.0
U3631	50	70	13	12	5.0
U6007	50	70	15	14	5.0
U6125	25	33	5.5	4.8	2.0
U2429	25	33	6.2	5.5	2.0
U4382	25	33	6.4	5.7	2.0
U2539	25	33	7.5	6.5	2.0

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U6002	45	55	12.5	11.5	2.5
U1610	45	57.7	10.5	9.5	3.5
U3559	45	60	11	10	4.0
U3241	45	60	11	10	4.0
U3407	45	60	12.5	11.5	4.0
U4058	45	65	11	10	5.0
U7250	47	56	10	9	2.5
U4434	47.62	57.15	10.52	9.52	2.5
U6003	48	56	12.5	11.5	2.0
U9699	48	60	9	8	3.0
U3276	48	63	11	10	4.0
U1468	48	68	13	12	5.0
U1468	48	68	13	12	5.0
U6021	50	57	11	10	2.0
U7587	50	58	7	6	2.0
U8155	50	58	11	10	2.0
U6006	50	58	12.5	11.5	2.0
U7059	50	58	14	13	2.0
U8490	50	60	7	6	2.5
U2381	50	60	8.3	7.3	2.5
U6884	50	60	8.3	7.3	2.5
U9712	50	60	9	8	2.5
U2256	50	60	11	10	2.5
U1208	50	62	11	10	3.0
U4041	50	62.7	10.5	9.5	3.5
U3243	50	63	11	10	3.5
U9722	50	65	7	6	4.0
U4042	50	65	10.5	9.5	4.0
U2820	50	65	11	10	4.0
U4040	50	65	12	11	4.0
U3327	50	65	12.5	11.5	4.0
U3539	50	65	16.5	15.5	4.0
U3631	50	70	13	12	5.0
U6007	50	70	15	14	5.0
U4435	50.8	60.32	7.35	6.35	2.5
U4436	50.8	60.32	8.93	7.93	2.5
U4360	50.8	60.32	10.52	9.52	2.5
U4676	50.8	63.5	7.35	6.35	3.5
U4289	50.8	63.5	10.52	9.52	3.5
U4670	50.8	66.67	8.93	7.93	4.0
U4336	50.8	66.67	13.7	12.7	4.0
U1087	52	62	11	10	2.5
U7282	52	64	11	10	3.0
U4437	53.97	63.5	10.52	9.52	2.5
U6020	55	62.5	10	9	2.0
U6008	55	63	12.5	11.5	2.0
U0858	55	64	16	15	2.5
U7179	55	65	7.7	6.7	2.5
U6960	55	65	8.3	7.3	2.5
U9475	55	65	9	8	2.5
U6009	55	65	10	9	2.5
U3575	55	65	11	10	2.5
U2025	55	65	13	12	2.5
U3926	55	68	11	10	3.5

1.2

UIP2 Metric Size



Ordering dimensions in green

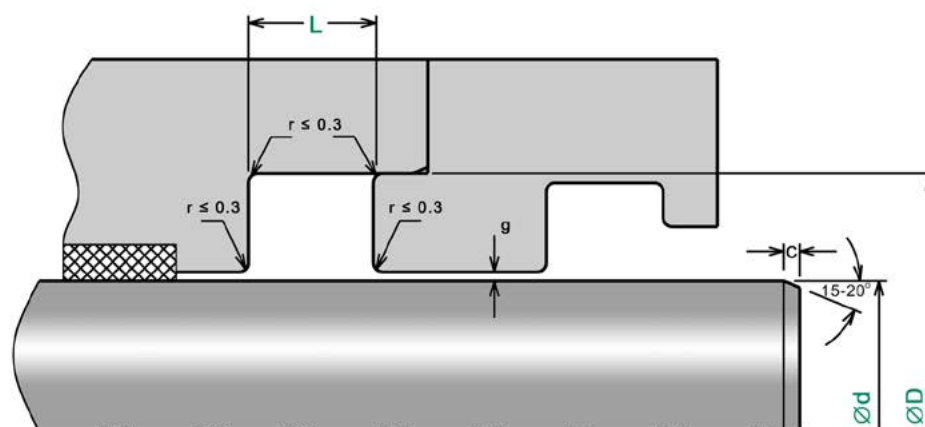
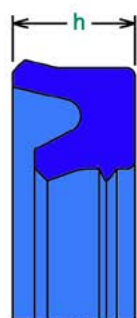
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U5227	55	70	10	9	4.0
U3427	55	70	11	10	4.0
U2806	55	70	13	12	4.0
U3246	55	75	13	12	5.0
U6961	56	66	8.3	7.3	2.5
U3836	56	66	11	10	2.5
U3837	56	71	11	10	4.0
U6957	56	71	12.4	11.4	4.0
U2811	56	71	12.5	11.5	4.0
U0979	56	76	13	12	5.0
U7681	57	66	14	13	2.5
U6079	57.1	66.7	11	10	2.5
U4446	57.15	66.67	10.52	9.52	2.5
U4277	57.15	69.85	10.52	9.52	3.5
U6019	58	68	12.5	11.5	2.5
U8268	60	68	7.3	6.3	2.0
U4053	60	68	12.5	11.5	2.0
U4156	60	68	14.5	13.5	2.0
U00070	60	70	8.3	7.3	2.5
U4669	60	70	8.5	7.5	2.5
U2321	60	70	11	10	2.5
U4174	60	70	12.5	11.5	2.5
U3085	60	70	13	12	2.5
U2393	60	72	11	10	3.0
U5228	60	75	10	9	4.0
U2708	60	75	11	10	4.0
U5354	60	75	12.4	11.4	4.0
U6024	60	75	12.5	11.5	4.0
U2995	60	75	13	12	4.0
U3808	60	75	16.5	15.5	4.0
U4024	60	80	11	10	5.0
U1659	60	80	13	12	5.0
U3203	60	80	13	12	5.0
U4438	60.32	69.85	10.52	9.52	2.5
U4290	60.32	73.02	10.52	9.52	3.5
U5229	63	73	7	6	2.5
U2647	63	73	13	12	2.5
U7017	63	75	11	10	3.0

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U9723	63	76	11	10	3.5
U3791	63	78	11	10	4.0
U6956	63	78	12.4	11.4	4.0
U1613	63	78	12.5	11.5	4.0
U9772	63	78	13	12	4.0
U2891	63	83	13	12	5.0
U4401	63.5	73.02	8.93	7.93	2.5
U4402	63.5	73.02	10.52	9.52	2.5
U4677	63.5	76.2	7.35	6.35	3.5
U3921	63.5	76.2	10.52	9.52	3.5
U6088	63.5	76.2	11	10	3.5
U4337	63.5	79.37	13.7	12.7	4.0
U4301	63.5	82.55	16.87	15.87	5.0
U8157	64	72	11	10	2.0
U6023	65	73	12.5	11.5	2.0
U3657	65	73	14	13	2.0
U6051	65	75	11	10	2.5
U6045	65	75	12.5	11.5	2.5
U2059	65	75	13	12	2.5
U4043	65	77.7	10.5	9.5	3.5
U5230	65	80	10	9	4.0
U3247	65	80	11	10	4.0
U2942	65	80	13	12	4.0
U3249	65	85	13	12	5.0
U4403	66.67	76.2	10.52	9.52	2.5
U4291	66.67	79.37	13.7	12.7	3.5
U6046	67	75	12.5	11.5	2.0
U7753	67	75	14	13	2.0
U7352	68	78	13	12	2.5
U6620	68	80	11	10	3.0
U7546	69.85	82.55	10.52	9.52	3.5
U4339	69.85	85.72	10.52	9.52	4.0
U6059	70	77.5	10	9	2.0
U6047	70	78	12.5	11.5	2.0
U1825	70	78	13	12	2.0
U6060	70	79	14	13	2.5
U0701	70	79	16	15	2.5
U5669	70	80	9	8	2.5
U6048	70	80	12.5	11.5	2.5
U2866	70	80	13	12	2.5
U5037	70	83	11	10	3.5
U3444	70	85	11	10	4.0
U6955	70	85	12.4	11.4	4.0
U2412	70	85	13	12	4.0
U3251	70	90	13	12	5.0
U4173	70	90	16	15	5.0
U4404	73.02	82.55	10.52	9.52	2.5
U8267	75	83	7.3	6.3	2.0
U3933	75	83	12.5	11.5	2.0
U00237	75	85	9.7	8.7	2.5
U6049	75	85	11.5	10.5	2.5
U1599	75	85	12.5	11.5	2.5
U3839	75	85	13	12	2.5
U8170	75	87	13.5	12.5	3.0

1.2

UIP2 Metric Size



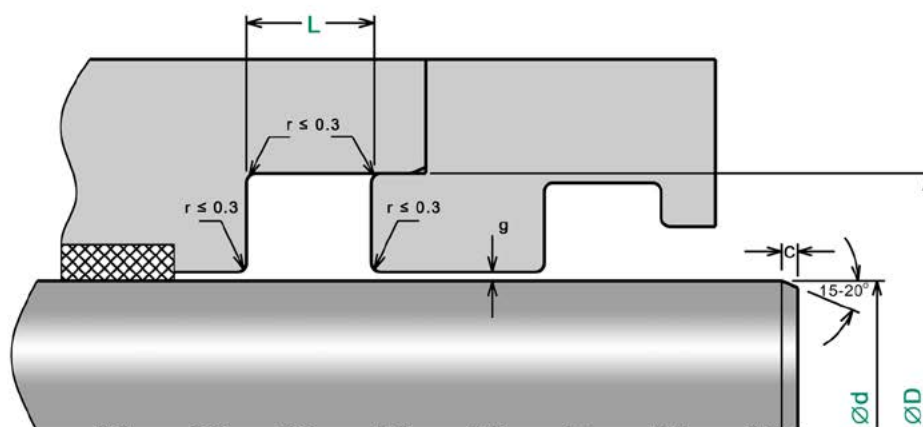
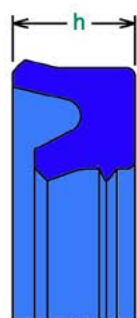
Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U5554	75	90	10	9	4.0
U3254	75	90	11	10	4.0
U2943	75	90	13	12	4.0
U5038	75	95	13	12	5.0
U8357	76	84	8.5	7.5	2.0
U4405	76.2	85.72	10.52	9.52	2.5
U4292	76.2	88.9	10.52	9.52	3.5
U4329	76.2	92.07	13.7	12.7	4.0
U4303	76.2	95.25	16.87	15.87	5.0
U8158	78	86	11	10	2.0
U1482	78	86	12.5	11.5	2.0
U6081	78	86	14.5	13.5	2.0
U4157	78	86	14.5	13.5	2.0
U6053	78	88	13	12	2.5
U7483	78	88	15	14	2.5
U8358	78	88	16	15	2.5
U9597	80	88	9	8	2.0
U6052	80	88	12.5	11.5	2.0
U5238	80	90	7	6	2.5
U8061	80	90	12.5	11.5	2.5
U2138	80	90	13	12	2.5
U9553	80	91	14.5	13.5	3.0
U5236	80	95	10	9	4.0
U2715	80	95	11	10	4.0
U6954	80	95	12.4	11.4	4.0
U2892	80	95	12.5	11.5	4.0
U2716	80	95	13	12	4.0
U3255	80	100	11	10	5.0
U2617	80	100	13	12	5.0
U3042	80	100	16	15	5.0
U6152	82.5	97.5	13.5	12.5	4.0
U3840	82.55	95.3	10.52	9.52	3.5
U4330	82.55	98.42	13.7	12.7	4.0
U6173	85	92.5	10	9	2.0
U4044	85	93	12.5	11.5	2.0
U7173	85	94	16	15	2.5
U9598	85	95	11	10	2.5
U9739	85	95	13	12	2.5

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U5488	85	100	10	9	4.0
U5240	85	100	11	10	4.0
U5536	85	100	12.4	11.4	4.0
U6153	85	100	12.5	11.5	4.0
U2147	85	100	13	12	4.0
U5552	85	105	13	12	5.0
U6625	86	98	11	10	3.0
U3656	87	95	14	13	2.0
U7674	87	96	14	13	2.5
U4409	88.9	98.42	10.52	9.52	2.5
U4294	88.9	101.6	8.93	7.93	3.5
U4193	88.9	101.6	10.52	9.52	3.5
U4304	88.9	107.95	16.87	15.87	5.0
U6154	90	98	12.5	11.5	2.0
U1826	90	98	13	12	2.0
U7586	90	100	7	6	2.5
U6155	90	100	8.5	7.5	2.5
U6156	90	100	10	9	2.5
U4096	90	100	11	10	2.5
U3103	90	100	12.5	11.5	2.5
U7062	90	102	13	12	3.0
U5039	90	105	11	10	4.0
U6941	90	105	12.5	11.5	4.0
U2737	90	105	12.5	11.5	4.0
U2322	90	105	13	12	4.0
U2738	90	110	13	12	5.0
U8359	91	99	8.5	7.5	2.0
U4194	92.07	104.77	10.52	9.52	3.5
U8424	93	105	11	10	3.0
U6121	95	103	12.5	11.5	2.0
U6136	95	105	10	9	2.5
U8060	95	105	12.5	11.5	2.5
U3510	95	105	13	12	2.5
U8171	95	107	13.5	12.5	3.0
U5241	95	110	10	9	4.0
U5242	95	110	11	10	4.0
U2739	95	115	13	12	5.0
U4195	95.25	107.95	10.52	9.52	3.5
U4196	95.25	107.95	13.7	12.7	3.5
U4197	95.25	107.95	15.28	14.28	3.5
U7520	96	104	13	12	2.0
U6150	97	105	12.5	11.5	2.0
U6151	97	105	14.5	13.5	2.0
U6175	97	106	15	14	2.5
U7482	97	107	16	15	2.5
U6990	99	109	15	14	2.5
U8353	99	109	16	15	2.5
U6142	100	108	12.5	11.5	2.0
U3072	100	108	13	12	2.0
U5524	100	109	16	15	2.5
U6177	100	110	11.5	10.5	2.5
U6143	100	110	12.5	11.5	2.5
U4091	100	110	15	14	2.5
U7233	100	112	15	14	3.0

1.2

UIP2 Metric Size

Ordering dimensions in green

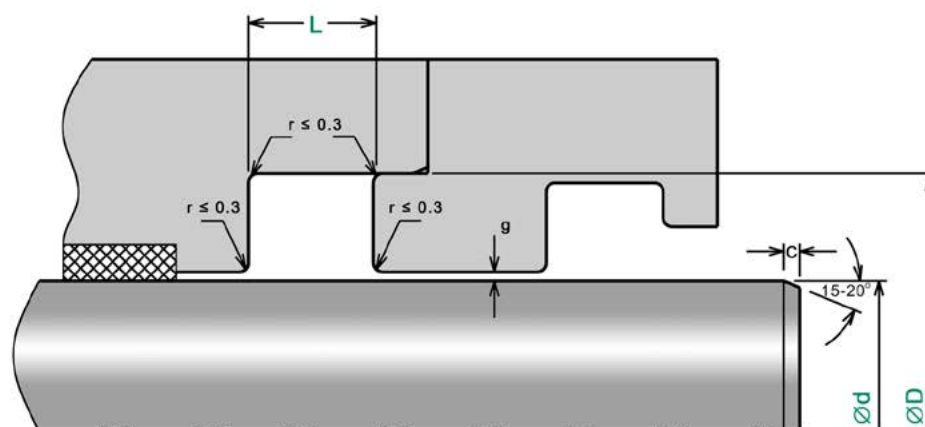
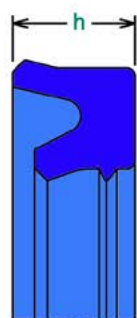
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U5252	100	115	10	9	4.0
U5255	100	115	11	10	4.0
U5457	100	115	11	10	4.0
U9478	100	115	12	11	4.0
U5515	100	115	12.4	11.4	4.0
U2257	100	115	13	12	4.0
U2336	100	120	13	12	5.0
U3936	100	120	15.5	14.5	5.0
U6958	100	120	15.6	14.6	5.0
U3636	100	120	16	15	5.0
U3214	100	125	16	15	6.5
U4198	101.6	111.12	10.52	9.52	2.5
U4279	101.6	114.3	10.52	9.52	3.5
U4331	101.6	117.47	10.52	9.52	4.0
U4295	101.6	120.65	13.7	12.7	5.0
U4296	101.6	120.65	16.87	15.87	5.0
U6144	105	113	12.5	11.5	2.0
U4341	105	115	12.5	11.5	2.5
U2740	105	120	10.9	9.9	4.0
U3215	105	125	11	10	5.0
U3216	105	125	16	15	5.0
U6626	106	118	11	10	3.0
U8352	107	115	8.5	7.5	2.0
U8160	110	118	11	10	2.0
U6146	110	118	12.5	11.5	2.0
U00468	110	120	11	10	2.5
U8164	110	120	12.5	11.5	2.5
U2893	110	120	13	12	2.5
U5253	110	125	10	9	4.0
U5281	110	125	13	12	4.0
U3256	110	125	16	15	4.0
U5042	110	130	11	10	5.0
U2825	110	130	13	12	5.0
U5254	110	130	14	13	5.0
U2575	110	130	15.5	14.5	5.0
U3043	110	130	16	15	5.0
U4045	110	135	16	15	6.5
U7668	112	121	14	13	2.5

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U00271	112	122	11	10	2.5
U6158	112	122	11.5	10.5	2.5
U7063	112	125	16	15	3.5
U6159	113	123	10	9	2.5
U8426	113	125	11	10	3.0
U4281	114.3	127	10.52	9.52	3.5
U4333	114.3	130.17	13.7	12.7	4.0
U4298	114.3	133.35	16.87	15.87	5.0
U6182	115	123	12.5	11.5	2.0
U5529	115	124	16	15	2.5
U1067	115	125	13	12	2.5
U6160	115	125	15	14	2.5
U6161	115	130	11.5	10.5	4.0
U3257	115	130	16	15	4.0
U1068	115	140	16	15	6.5
U8063	116	126	12.5	11.5	2.5
U7481	116	126	16	15	2.5
U6178	117	126	15	14	2.5
U9930	117	130	13	12	2.5
U8172	117	131	13.5	12.5	3.5
U2116	118	126	14.5	13.5	2.0
U6866	120	128	12.5	11.5	2.0
U4526	120	130	12.5	11.5	2.5
U6162	120	130	13	12	2.5
U3654	120	130	15	14	2.5
U8360	120	130	16	15	2.5
U5454	120	135	10	9	4.0
U9474	120	135	11	10	4.0
U4142	120	135	16	15	4.0
U3188	120	140	13	12	5.0
U5514	120	140	15.6	14.6	5.0
U2735	120	140	16	15	5.0
U4411	120.65	130.17	10.52	9.52	2.5
U6187	121	135	10	9	3.5
U6157	125	133	12.5	11.5	2.0
U6169	125	135	11	10	2.5
U4092	125	135	14.5	13.5	2.5
U7232	125	137	15	14	3.0
U3841	125	145	13	12	5.0
U2200	125	145	16	15	5.0
U3258	125	150	16	15	6.5
U2201	125	155	19	18	7.5
U8361	126	134	8.5	7.5	2.0
U4282	127	139.7	10.52	9.52	3.5
U4299	127	146.05	16.87	15.87	5.0
U6170	128	136	12.5	11.5	2.0
U6627	128	140	11	10	3.0
U8161	129	137	11	10	2.0
U00470	129	139	11	10	2.5
U8165	129	139	12.5	11.5	2.5
U6171	130	138	12.5	11.5	2.0
U7345	130	140	12.5	11.5	2.5
U9533	130	140	13	12	2.5
U7499	130	145	13	12	4.0

1.2

UIP2 Metric Size



Ordering dimensions in green

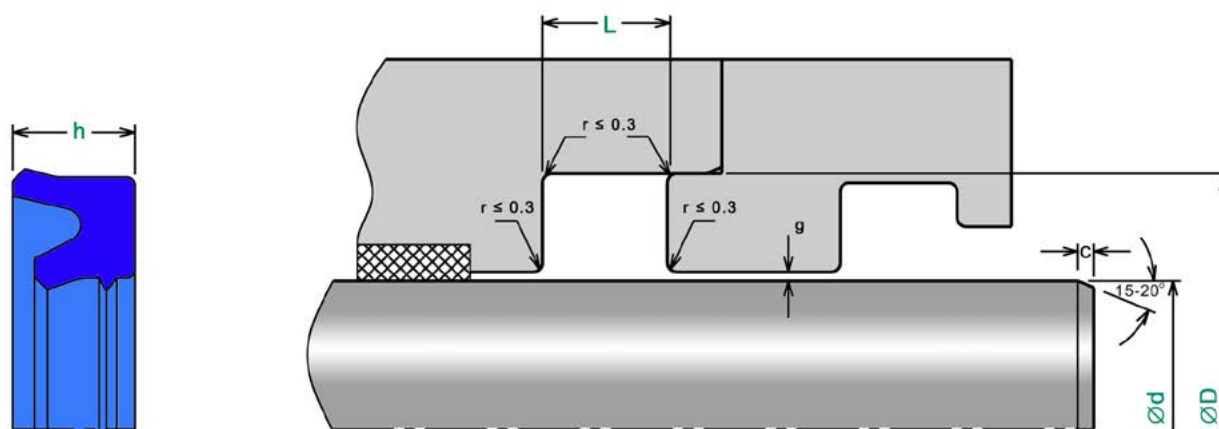
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U3919	130	150	13	12	5.0
U3259	130	150	16	15	5.0
U6188	131	150	14	13	5.0
U0859	132	142	16	15	2.5
U7512	133.35	146.05	16.7	15.7	3.5
U8037	135	145	12.5	11.5	2.5
U7504	135	145	13	12	2.5
U7480	135	145	16	15	2.5
U8425	135	147	11	10	3.0
U7064	135	148	16	15	3.5
U4047	135	150	12.5	11.5	4.0
U6189	137	146	15	14	2.5
U4305	139.7	158.75	16.87	15.87	5.0
U4527	140	150	12.5	11.5	2.5
U3655	140	150	13	12	2.5
U3842	140	160	13	12	5.0
U2204	140	160	16	15	5.0
U2205	140	165	16	15	6.5
U1125	140	170	19	18	7.5
U8362	141	151	5.7	5	2.5
U6989	141	151	15	14	2.5
U8398	141	151	16	15	2.5
U8173	142	156	13.5	12.5	3.5
U6881	143	151	12.5	11.5	2.0
U4159	143	151	14.5	13.5	2.0
U8102	145	153	12.5	11.5	2.0
U9636	145	155	13	12	2.5
U6237	145	155	13.5	12.5	2.5
U6168	145	160	13	12	4.0
U00469	149	159	11	10	2.5
U8166	149	159	12.5	11.5	2.5
U4094	150	160	14.5	13.5	2.5
U5530	150	160	16	15	2.5
U7231	150	162	15	14	3.0
U2894	150	170	16	15	5.0
U1502	150	180	19	18	7.5
U8162	151	159	11	10	2.0
U00683	152	160	12.5	11.5	2.0

UIP2 Metric Size

NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U9535	152	162	13	12	2.5
U6628	152	167	13	12	4.0
U8015	154	164	12.5	11.5	2.5
U7347	155	165	12.5	11.5	2.5
U7484	155	165	16	15	2.5
U0471	160	168	12.5	11.5	2.0
U5012	160	170	12.5	11.5	2.5
U6719	160	170	13	12	2.5
U7065	160	175	19	18	4.0
U3261	160	180	16	15	5.0
U1116	160	185	16	15	6.5
U1503	160	185	19.2	18.2	6.5
U6985	162	172	15	14	2.5
U8363	162	172	16	15	2.5
U8427	165	177	11	10	3.0
U00471	169	179	12	11	2.5
U8167	169	179	12.5	11.5	2.5
U7012	170	180	16	15	2.5
U2072	170	190	16	15	5.0
U00615	171	179	14	13	2.0
U4169	171	179	14.5	13.5	2.0
U7449	171.45	184.15	16.7	15.7	3.5
U7977	175	185	12.5	11.5	2.5
U7498	175	185	13	12	2.5
U7470	175	185	16	15	2.5
U7234	175	187	16	15	3.0
U8163	176	186	11	10	2.5
U4170	180	190	11	10	2.5
U7346	180	190	12.5	11.5	2.5
U6736	180	192	15	14	3.0
U7469	180	195	13.5	12.5	4.0
U1505	180	200	16	15	5.0
U2915	180	205	16	15	6.5
U2767	180	205	20	19	6.5
U6988	183	193	15	14	2.5
U8364	183	193	16	15	2.5
U7964	185	195	13	12	2.5
U7669	185	195	14.5	13.5	2.5
U7091	185	197	16	15	3.0
U8428	188	200	11	10	3.0
U8168	191	201	12.5	11.5	2.5
U00472	191	202	12	11	3.0
U9635	195	205	13	12	2.5
U7475	196	206	16	15	2.5
U7967	197	207	12.5	11.5	2.5
U9534	200	210	13	12	2.5
U7500	200	212	13	12	3.0
U8644	200	220	11.5	10.5	5.0
U2987	200	220	16	15	5.0
U5036	200	225	17	16	6.5
U1119	200	225	20	19	6.5
U1508	200	230	19	18	7.5
U6987	207	217	15	14	2.5
U8365	207	217	16	15	2.5

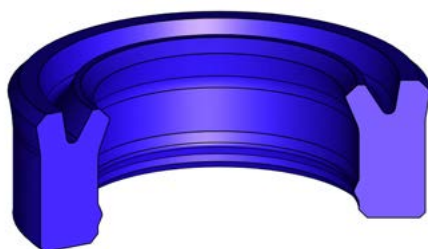
1.2

UIP2 Metric Size



Ordering dimensions in green

[illegible]



UIP8

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. The secondary edge can reduce leakage, friction, and dirt ingress. Usually used for rod hydraulic application.

TECHNICAL DATA

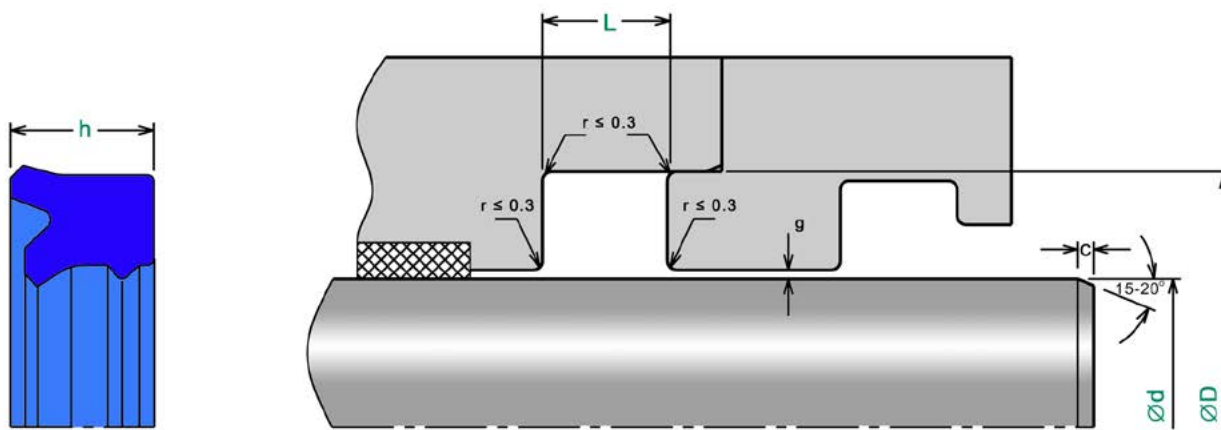
MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 40 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

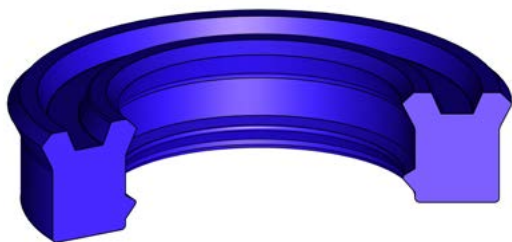
Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

UIP8 Metric Size



Ordering dimensions in green

[illegible]



UNP2

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. The secondary edge can reduce leakage, friction, and dirt ingress. Usually used for rod hydraulic application.

TECHNICAL DATA

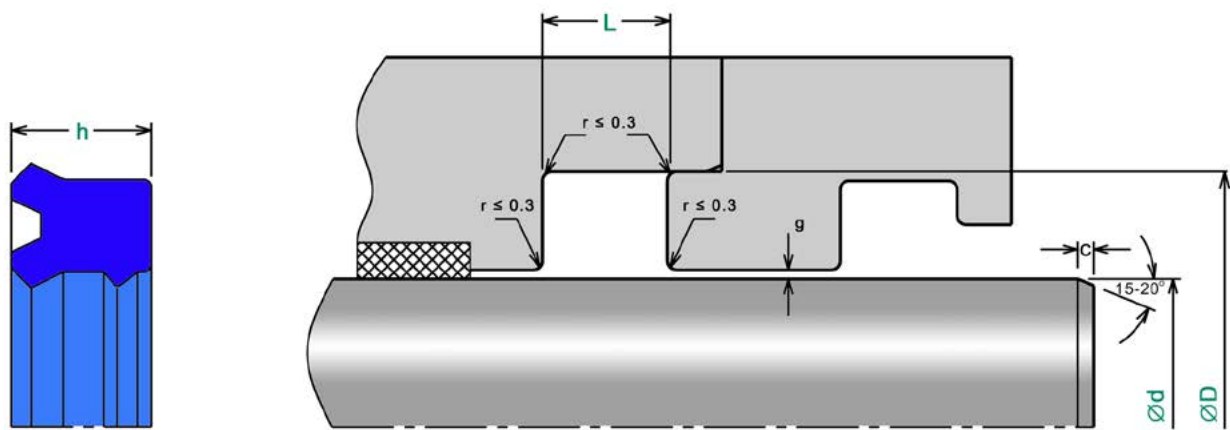
MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 40 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

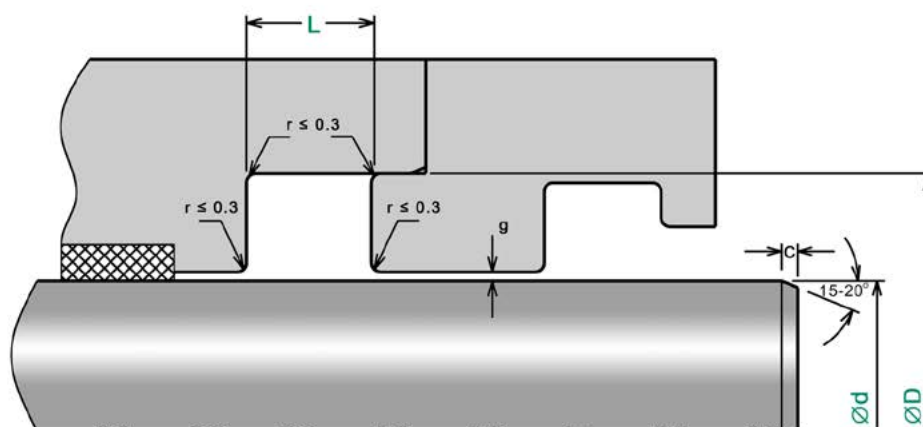
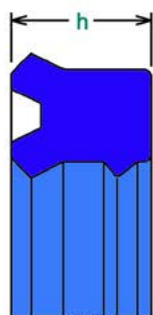
UNP2 Metric Size



Ordering dimensions in green

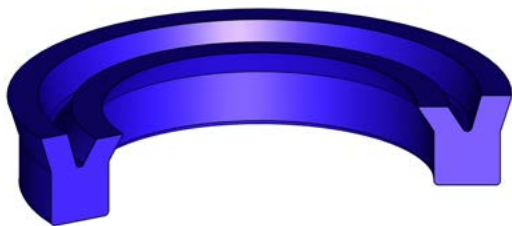
NAK NO.	Ød f8	ØD H10	L +/- 0.2	h	c
U7675	8	16	6.4	5.7	2
U8600	14	22	6.5	5.8	2
U6096	20	28	6.5	5.8	2
U7122	22	30	6.5	5.8	2
U6099	22	30	8	7	2
U2238	25	32	7	6	2
U2970	25	33	6.5	5.8	2
U6945	25	33	7.5	6.5	2
U00246	25	35	6.5	5.8	2.5
U00299	25	40	11	10	4
U3552	28	38	8	7	2.5
U5512	30	38	8	7	2
U2185	30	38	9	8	2
U2652	32	42	8	7	2.5
U00653	35	43	7	6	2
U00282	35	45	11	10	2.5
U2992	40	48	6.1	5.4	2
U00476	40	48	6.5	5.8	2
U2960	40	48	7	6	2
U2642	40	48	7.3	6.3	2
U2653	40	50	8	7	2.5
U8310	42	52	9	8	2.5
U2651	45	53	9	8	2
U3835	45	55	6.5	5.8	2.5
U4583	45	55	8	7	2.5
U3351	50	58	9	8	2
U3147	50	60	8	7	2.5
U7800	50	60	8.3	7.3	2.5
U6986	55	63	9	8	2
U7377	60	68	7.3	6.3	2
U7740	62	67.5	5.7	5	1.5
U3533	65	73	9	8	2
U00154	66.67	76.2	10.5	9.5	2.5
U7380	75	83	7.3	6.3	2
U00145	75	83	9	8	2
U7464	78	90	13	12	3
U7463	82	94	9	8	3
U7371	85	93	7.3	6.3	2

UNP2 Metric Size



Ordering dimensions in green

[illegible]



CIP

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. The secondary edge can reduce leakage, friction, and dirt ingress. Usually used for rod hydraulic application.

TECHNICAL DATA

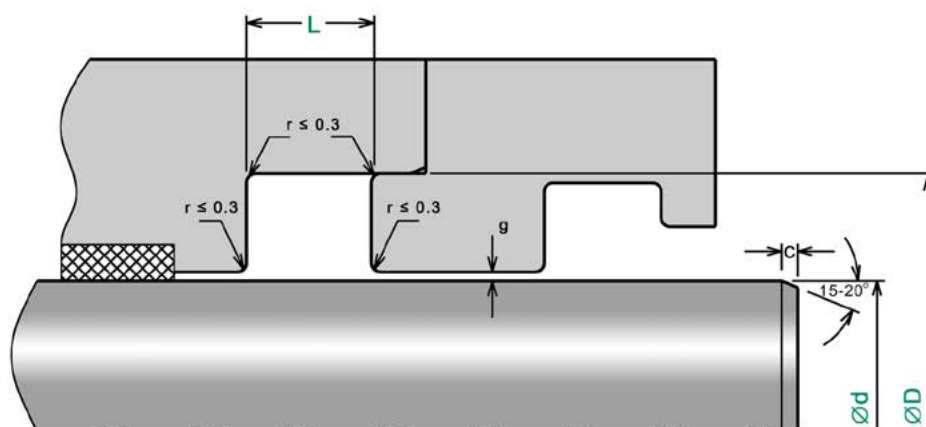
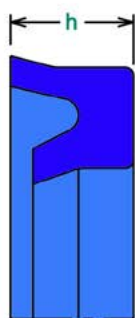
MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 30 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

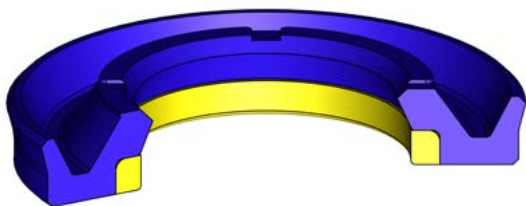
Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

Please contact our customer service for the dimensions out of our stock list.
service@mail.nak.com.tw



Ordering dimensions in green

[illegible]



UIB3

PRODUCT DESCRIPTION

Seal designed with asymmetric lip with back-up ring.
Usually used for rod higher pressure hydraulic application.

TECHNICAL DATA

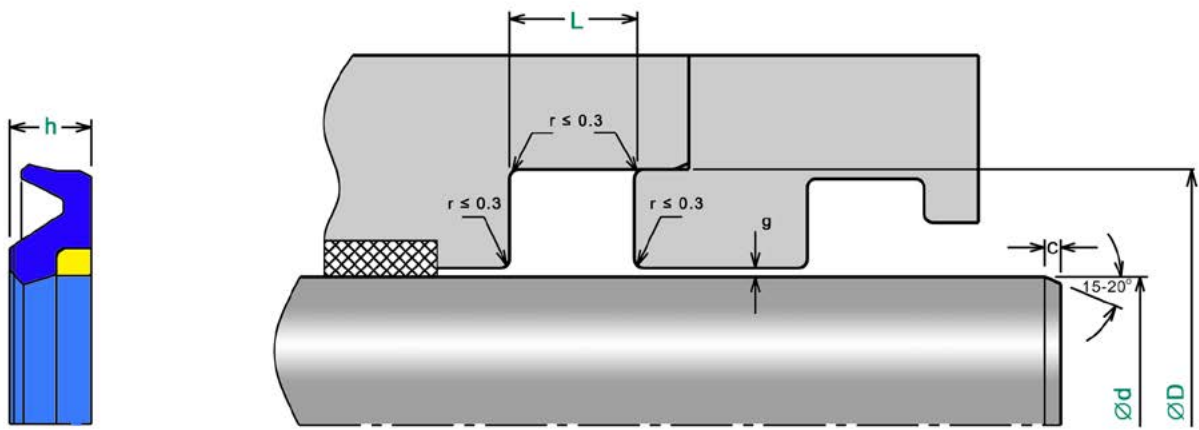
MATERIAL	PU + PA
MEDIA	Mineral oils
Pressure p	≤ 50 MPa
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

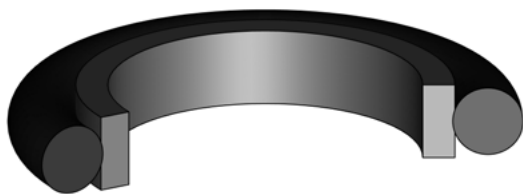
Admissible gap dimension			
	350 Bar	420 Bar	500 Bar
Profile dimension	0.4 mm	0.2 mm	0.125 mm

UIB3 Metric Size



Ordering dimensions in green

[illegible]



HRO

PRODUCT DESCRIPTION

Seal designed combine PTFE ring and O-ring.
Usually used for rod hydraulic applicaiton.

TECHNICAL DATA

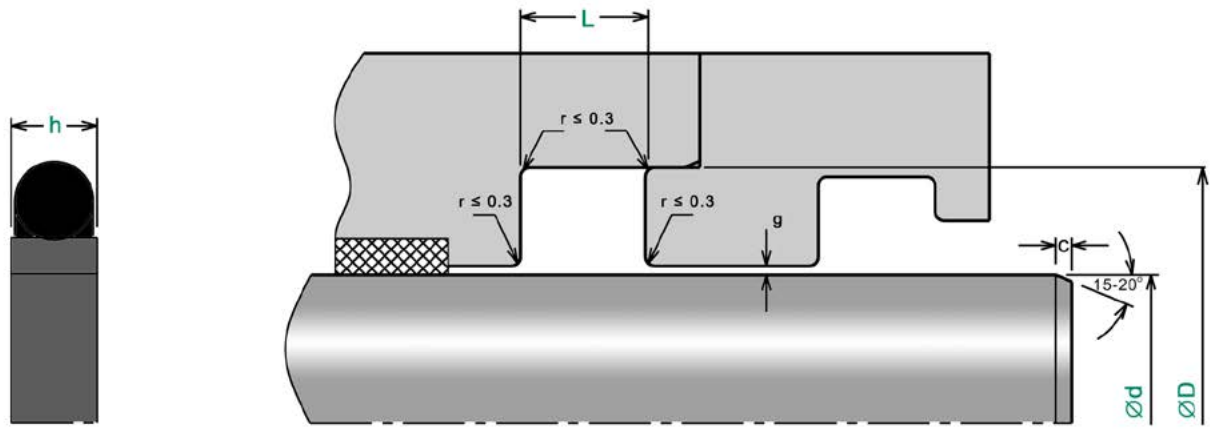
MATERIAL	NBR + PTFE
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 1.5 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	≤ 1.6 μm	≤ 6.3 μm
Groove flanks	L	≤ 3.2 μm	≤ 12.5 μm

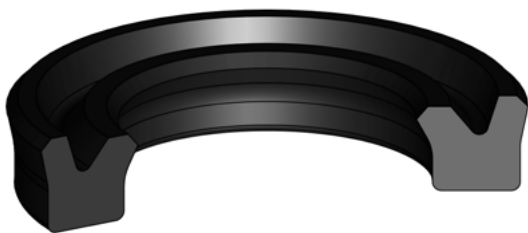
Admissible gap dimension			
	200 Bar	250 Bar	350 Bar
Profile dimension	0.6 mm	0.42 mm	0.25 mm

HRO Metric Size



Ordering dimensions in green

[illegible]



UIR

PRODUCT DESCRIPTION

Seal designed with asymmetric lips.
Usually used for rod hydraulic application.

TECHNICAL DATA

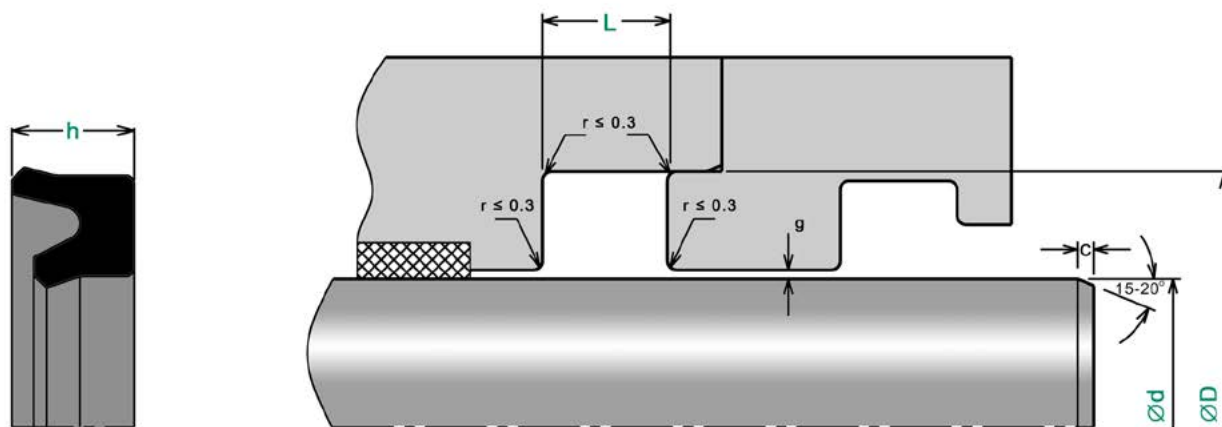
MATERIAL	NBR	HNBR	FKM
MEDIA	Mineral oils	Mineral oils	Mineral oils
Pressure p	≤ 16 MPa	≤ 16 MPa	≤ 16 MPa
Running speed v	≤ 0.5 m/s	≤ 0.5 m/s	≤ 0.5 m/s
Temperature	-35°C , +100°C	-40°C , +150°C	-25°C , +250°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension

	50 Bar	100 Bar	140 Bar
Profile dimension	0.3 mm	0.22 mm	0.15 mm

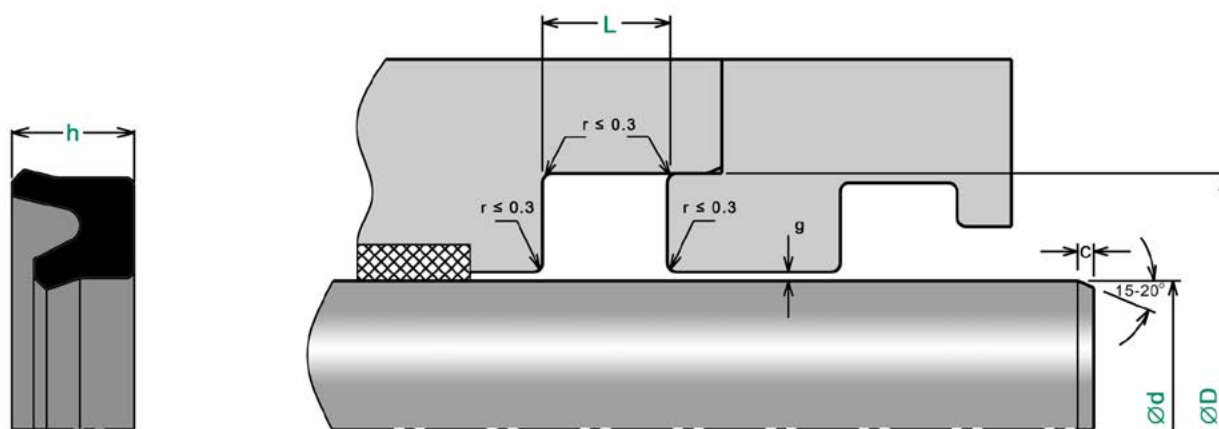
UIR Metric Size

Ordering dimensions in green

NAK NO.	NBR	HNBR	FKM	Ød f8	ØD 0/-0.2	L +/- 0.2	h	c
U7454	•			6	12	4.2	3.5	1.5
U7467	•			6.35	12.7	4.66	3.96	2.0
U8429	•			8	16		5.5/6	2.0
U8598	•			9.5	16.5	4.7	4	2.0
U8279	•			10	15	4.2	3.5	1.5
U0640	•			10	16	4.7	4	1.5
U8147	•			10	18	5.7	5	2.0
U00435	•			12.5	22	5.7	5	2.5
U00401	•			12.5	27	5.45	4.75	4.0
U8280	•			12.75	19.2	4.5	3.8	2.0
U8176	•			14	18.3	3.7	3	1.5
U00428	•			14	21.9	5.4	4.7	2.0
U00425	•			14	23.4	5.7	5	2.5
U00426	•			14	24.4	7	6	3.0
U00399	•			14	30	5.45	4.75	4.0
U8422	•			15	22	5	4.3	2.0
U7472	•			15.87	22.22	5.46	4.76	2.0
U8401	•			16	22	5.7	5	1.5
U7299	•			16	22.4	3.9	3.2	2.0
U00408	•			16	36	5.7	5	5.0
U7580	•			17	28	7.2	6.2	3.0
U7473	•			17.46	23.8	5.2	4.5	2.0
U7474	•			18	24	5.3	4.6	1.5
U5129	•			18	25	7	6	2.0
U9055	•			22	30	6.2	5.5	2.0
U8594	•			24	32		5.5/6	2.0
U7116	•			25	34	5.9	5.2	2.5
U8088	•			25	35	8	7	2.5
U7615	•			25.4	38.1	7.35	6.35	3.5
U7281	•			27	35	5.9	5.2	2.0
U0711	•			28	36	4.7	4	2.0
U7616	•			28	38	8	7	2.5
U1109	•			30	36	5.7	5	1.5
U00073	•			30.16	38.1	5.7	5	2.0
U7046	•			32	40	5.2	4.5	2.0
U9730	•			32	40	6.2	5.5	2.0
U7720	•			32.7	40.5	5.7	5	2.0
U7617	•			33.34	40.63	5.3	4.6	2.0

1.8

UIR Metric Size



Ordering dimensions in green

[illegible]



CIR

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. The lip seal may easily be snapped into the grooves. The seal should be taken carefully and not be pulled over sharp edges.

TECHNICAL DATA

MATERIAL	NBR	FKM
MEDIA	Mineral oils	Mineral oils
Pressure p	≤ 10 MPa	≤ 10 MPa
Running speed v	≤ 0.5 m/s	≤ 0.5 m/s
Temperature	-35°C , +100°C	-25°C , +250°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	50 Bar	100 Bar	140 Bar
Profile dimension	0.3 mm	0.22 mm	0.15 mm

1.9

NAK CODE	DRAWING	MATERIAL	PRESSURE (max)MPa	TEMPERATURE (max)°C	SLIDING SPEED(max) m/s	SIZE (mm)	PAGE
UOP		PU	40	-40 / +100	0.5	8-270 1.125-7.500	2.1 P64~P69 (METRIC) P70~P71 (INCH)
COP		PU	30	-40 / +100	0.5	49.29-108.36	2.2 P72~P73 (METRIC)
HOD		NBR PTFE PA	50	-35 / +100	1.5	55-207	2.3 P74~P75 (METRIC)
HPO		NBR PTFE	35	-35 / +100	1.5	51.5-300	2.4 P76~P77 (METRIC)
HPS		NBR PTFE	35	-35 / +100	1.5	40.5-85	2.5 P78~P79 (METRIC)
UOR		NBR FKM	16	-35 / +100 -25 / +250	0.5	1.8-165 42-90	2.6 P80~P83 (METRIC)
COR		NBR	10	-35 / +100	0.5	8-140	2.7 P84~P85 (METRIC)
PDV		NBR FKM	10	-35 / +100 -25 / +250	0.5	4-20 8-10	2.8 P86~P87 (METRIC)
PDC		NBR	10	-35 / +100	0.5	8-30	2.9 P88~P89 (METRIC)
PSV		NBR	10	-35 / +100	0.5		2.10 P90
PSV2		NBR	10	-35 / +100	0.5		2.11 P91



UOP

PRODUCT DESCRIPTION

Seal designed with asymmetric lips.
Usually used for piston hydraulic application.

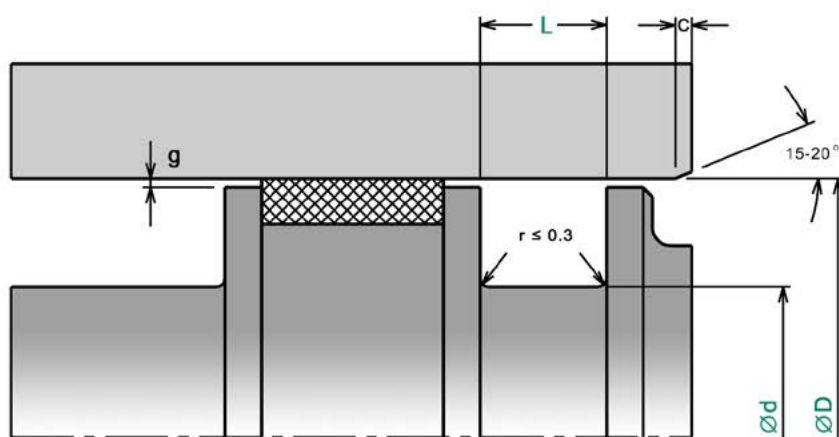
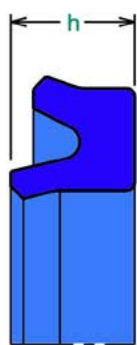
TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 40 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

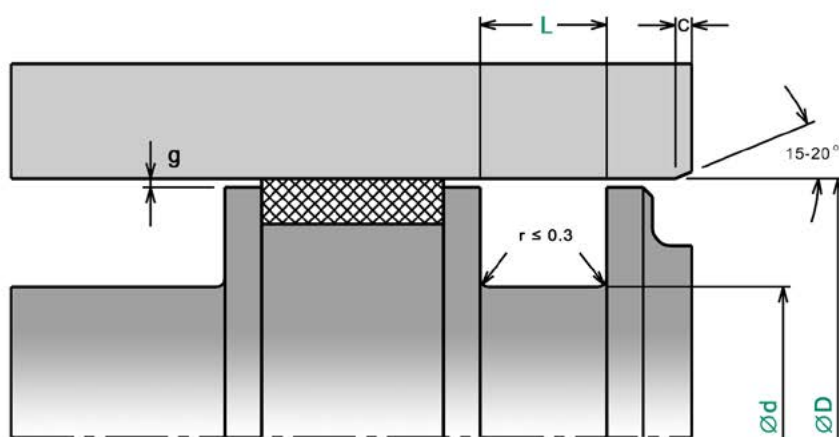
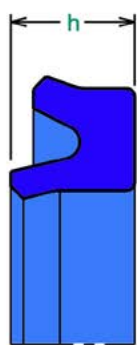
UOP Metric Size

Ordering dimensions in green

NAK NO.	Ød h10	ØD H9	L +/- 0.2	h	c
U3509	8	14	6.4	5.7	1.5
U7699	8	16	6.5	5.8	2.0
U4823	10	15.5	4.8	4.1	1.5
U4575	10	16	5.2	4.5	1.5
U7197	10	16	9	8	1.5
U2399	10	20	9	8	2.5
U4822	12	18	5.8	5.1	1.5
U2423	15	25	8.3	7.3	2.5
U2189	15	25	9	8	2.5
U9632	17	25	6.2	5.5	2.0
U2802	20	30	9	8	2.5
U7849	20.63	30.16	7.35	6.35	2.5
U2167	21	37	13	12	4.0
U2445	22	30	7.5	6.5	2.0
U8149	22	30	7.65	6.65	2.0
U2427	22	32	8.3	7.3	2.5
U6239	22	32	9	8	2.5
U5159	22	32	11	10	2.5
U6243	22	34	9.7	8.7	3.0
U9443	22	36	10.5	9.5	3.5
U4709	24	32	6.2	5.5	2.0
U2428	24	32	6.4	5.7	2.0
U2790	24	40	10	9	4.0
U2430	25	35	8.3	7.3	2.5
U2431	25	40	10	9	4.0
U2191	25	40	11	10	4.0
U6322	25	43	10	9	4.5
U6238	26	32	5.7	5	1.5
U9444	26.5	42	10.5	9.5	4.0
U7835	27	35	5.7	5	2.0
U2434	28	40	11	10	3.0
U6320	28.5	45	13	12	4.5
U4804	30	38.1	7	6	2.5
U7381	30	40	7	6	2.5
U6300	30	40	7.5	6.5	2.5
U2268	30	40	8	7	2.5
U00681	30	40	8.3	7.3	2.5
U7374	30	40	9	8	2.5

2.1

UOP Metric Size



Ordering dimensions in green

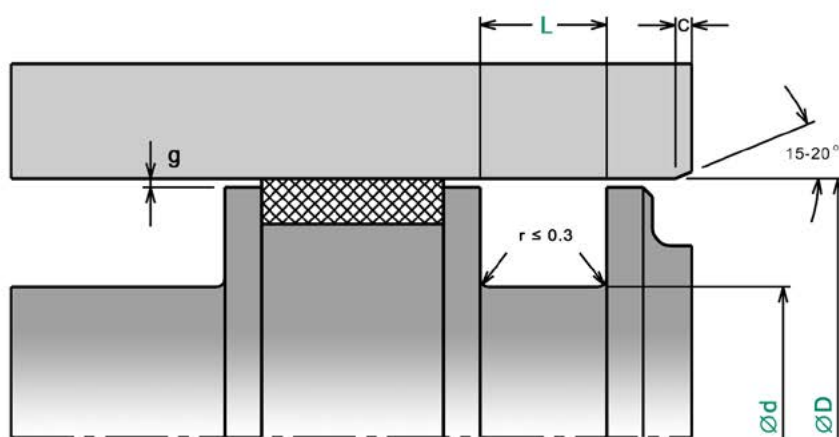
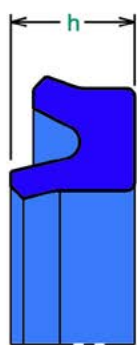
NAK NO.	Ød h10	ØD H9	L +/- 0.2	h	c
U2269	30	40	11	10	2.5
U6310	30	45	11	10	4.0
U2435	30	50	13	12	5.0
U6305	31	38	5.4	4.7	2.0
U3012	32	40	6.4	5.7	2.0
U6299	32	40	9	8	2.0
U2079	33	40	9	8	2.0
U2270	33	43	8.3	7.3	2.5
U6297	34.5	42	4.4	3.7	2.0
U5134	35	45	7	6	2.5
U3876	35	50	11	10	4.0
U8288	37	45	10	9	2.0
U8070	39.5	55	11	10	4.0
U8376	40	50	7	6	2.5
U2271	40	50	8	7	2.5
U2100	40	50	11	10	2.5
U6746	40	55	11	10	4.0
U0660	40	62	16.2	15.2	5.5
U6735	40.8	50.8	8.5	7.5	2.5
U6454	42	50	6.2	5.5	2.0
U00614	42	50	6.4	5.7	2.0
U2614	42	50	9	8	2.0
U6453	42	50	11	10	2.0
U7248	43	63	13	12	5.0
U2272	44	54	8.3	7.3	2.5
U7923	45	55		6	2.5
U6745	45	55	11	10	2.5
U2821	45	56	8	7	3.0
U2663	45	60	11	10	4.0
U6738	45	63	11	10	4.5
U3002	45	63	13	12	4.5
U6972	46	56	7	6	2.5
U6879	47	55	5.5	4.8	2.0
U4723	47.62	57.15	10.52	9.52	2.5
U2279	48	63	10	9	4.0
U9860	48	63	11	10	4.0
U2276	48	63	12.4	11.4	4.0
U2038	48	63	13	12	4.0

UOP Metric Size

NAK NO.	Ød h10	ØD H9	L +/- 0.2	h	c
U6747	48	68	11	10	5.0
U7369	50	60	7	6	2.5
U2255	50	60	8	7	2.5
U5160	50	60	9	8	2.5
U2323	50	60	11	10	2.5
U2277	50	65	11	10	4.0
U6030	50	70	14.5	13.5	5.0
U6796	50	75	15	14	6.5
U6741	52	60	9	8	2.0
U5774	53	63	4.95	4.25	2.5
U00061	53	63	7	6	2.5
U2105	53	63	8	7	2.5
U00651	53	63	9	8	2.5
U2395	53	63	13	12	2.5
U6969	53.98	69.85	8.93	7.93	4.0
U2282	55	63	6.4	5.7	2.0
U2283	55	65	8.3	7.3	2.5
U7378	55	70	10	9	4.0
U2803	55	70	13	12	4.0
U2281	57	67	8.3	7.3	2.5
U6795	58	72	13	12	3.5
U4874	60	70	7	6	2.5
U2804	60	70	13	12	2.5
U6793	60	70	14.5	13.5	2.5
U2401	60	75	13	12	4.0
U3156	60	80	13	12	5.0
U2402	60	80	15.5	14.5	5.0
U4012	61	71	7	6	2.5
U6794	62	70	8.5	7.5	2.0
U7331	63	73	7	6	2.5
U4722	63.5	76.2	7.35	6.35	3.5
U9890	65	75	7	6	2.5
U6996	65	75	8.2	7.2	2.5
U8629	65	75	9	8	2.5
U6414	65	75	14.5	13.5	2.5
U7743	65	80	9.6	8.6	4.0
U8325	65	80	11	10	4.0
U2732	65	80	12.4	11.4	4.0
U3872	65	80	13	12	4.0
U3674	65	85	11	10	5.0
U6802	65	85	14.5	13.5	5.0
U3004	65	90	13	12	6.5
U4978	70	80	7	6	2.5
U2403	70	80	7.7	6.7	2.5
U3547	70	80	8	7	2.5
U4620	70	80	9	8	2.5
U2636	70	80	13	12	2.5
U3003	70	85	13	12	4.0
U3284	70	90	13	12	5.0
U6808	70	90	14.5	13.5	5.0
U7370	71	80	7	6	2.5
U2067	72	80	13	12	2.0
U4979	75	85	7	6	2.5
U6807	75	85	11	10	2.5

2.1

UOP Metric Size



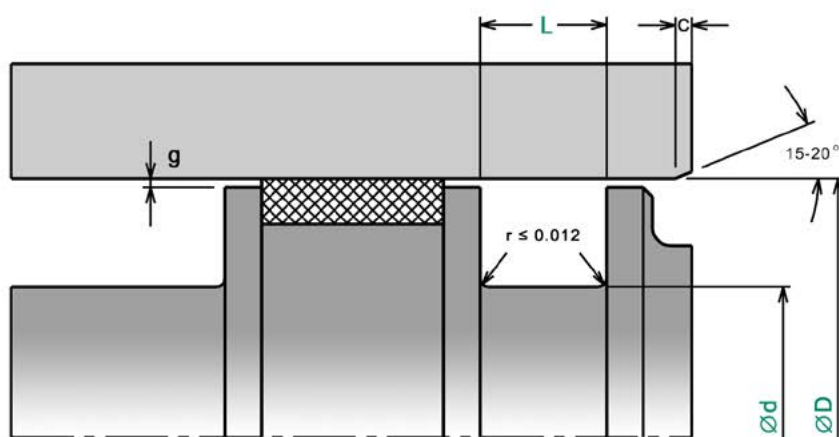
Ordering dimensions in green

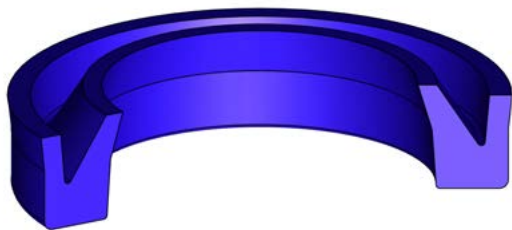
NAK NO.	Ød h10	ØD H9	L +/- 0.2	h	c
U7363	75	90	10	9	4.0
U3807	75	90	12.4	11.4	4.0
U2133	75	90	13	12	4.0
U6934	75	100	19	18	6.5
U3806	79.5	89	8	7	2.5
U7357	80	90	7	6	2.5
U6809	80	90	11	10	2.5
U6810	80	90	14	13	2.5
U7330	80	95	10	9	4.0
U2414	80	100	11	10	5.0
U2388	80	100	13	12	5.0
U2360	80	100	14.5	13.5	5.0
U2404	80	100	16	15	5.0
U6816	80	101.6	13	12	5.5
U9935	83	94	7	6	3.0
U7271	85	95	8	7	2.5
U7118	85	95	8.5	7.5	2.5
U7356	85	100	10	9	4.0
U4009	85	100	12.4	11.4	4.0
U2028	85	100	13	12	4.0
U5873	85	110	19	18	6.5
U6817	86	100	13	12	3.5
U6820	90	100	8	7	2.5
U6836	90	100	11.5	10.5	2.5
U7885	90	100	15	14	2.5
U1948	90	105	13	12	4.0
U2137	90	110	13	12	5.0
U2154	90	110	16	15	5.0
U2155	95	110	11	10	4.0
U4997	95	110	13	12	4.0
U6651	95	115	15.5	14.5	5.0
U3006	95	115	16	15	5.0
U6963	95	120	16	15	6.5
U2335	95	125	19	18	7.5
U3873	98	115	15.5	14.5	4.5
U6799	100	110	8	7	2.5
U6800	100	110	14.5	13.5	2.5
U6815	100	115	11.5	10.5	4.0

UOP Metric Size

NAK NO.	Ød h10	ØD H9	L +/- 0.2	h	c
U2031	100	120	13	12	5.0
U2853	100	125	16	15	6.5
U1630	105	120	13	12	4.0
U4007	105	125	13	12	5.0
U2152	105	125	16	15	5.0
U7355	106	120	9.5	8.5	3.5
U2405	110	125	10.6	9.6	4.0
U6823	110	125	11	10	4.0
U2406	110	130	13	12	5.0
U3652	110	130	16	15	5.0
U2897	115	125	16	15	2.5
U7354	115	130	10	9	4.0
U3801	115	145	19	18	7.5
U6538	120	130	14.5	13.5	2.5
U3425	120	140	13	12	5.0
U5740	120	140	16	15	5.0
U2989	120	150	19	18	7.5
U6309	125	140	10	9	4.0
U2872	125	140	16	15	4.0
U6308	125	150	14.5	13.5	6.5
U2176	130	150	16	15	5.0
U00478	130	150	17	16	5.0
U3870	130	160	19	18	7.5
U7351	135	150	10	9	4.0
U2203	135	160	19.5	18.5	6.5
U4712	139.7	152.4	15.28	14.28	3.5
U7802	140	155	10	9	4.0
U6643	140	160	14.5	13.5	5.0
U2828	140	160	16	15	5.0
U7638	140	165	20	19	6.5
U2832	150	170	16	15	5.0
U2827	150	180	19	18	7.5
U6325	160	180	14.5	13.5	5.0
U1879	160	180	16	15	5.0
U7165	165	180	10	9	4.0
U4717	165.1	177.8	7.05	6.35	3.5
U6326	166	180	16.5	15.5	3.5
U6327	170	190	16	15	5.0
U6366	170	200	19	18	7.5
U3643	175	200	19.2	18.2	6.5
U5687	175	200	20	19	6.5
U2361	180	200	16	15	5.0
U7667	180	200	19	18	5.0
U7096	190	210	13	12	5.0
U1988	190	220	19	18	7.5
U6558	195	220	17	16	6.5
U1756	200	220	16	15	5.0
U7815	220	240	17	16	5.0
U3863	220	250	19	18	7.5
U4642	225	250	19.2	18.2	6.5
U7320	230	250	16	15	5.0
U00109	270	300	19	18	7.5

2.1

[illegible]



COP

PRODUCT DESCRIPTION

Seal designed with asymmetric lips.
Usually used for piston hydraulic application.

TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 30 MPa
Running speed v	≤ 0.5 m/s
Temperature	-40°C , +100°C

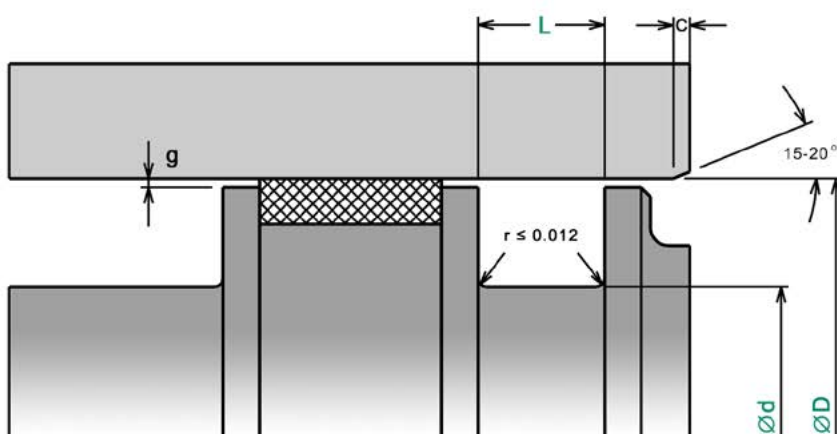
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

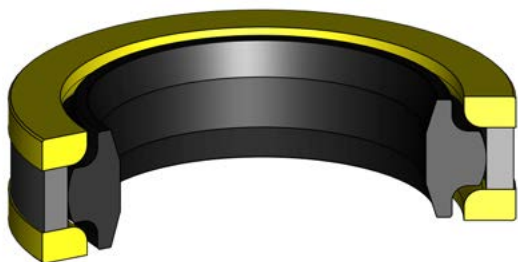
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COP



Ordering dimensions in green

[illegible]



HOD

PRODUCT DESCRIPTION

Seal designed combine with PTFE ring, rubber ring and both side back-up rings. Usually used for higher pressure piston hydraulic application.

TECHNICAL DATA

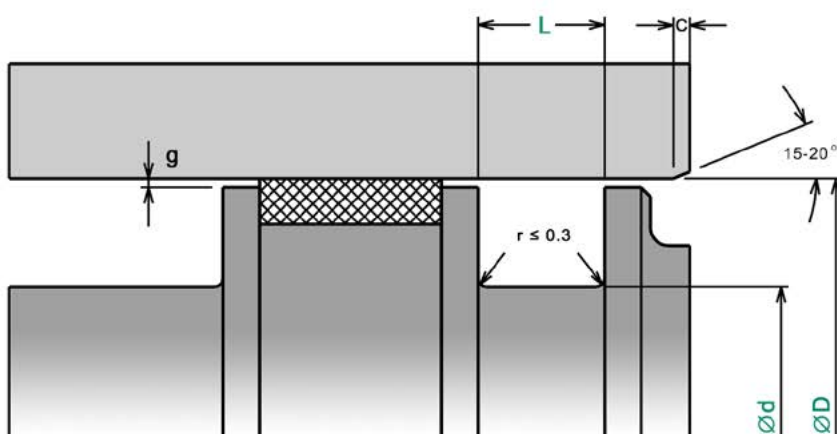
MATERIAL	NBR + PTFE + PA
MEDIA	Mineral oils
Pressure p	≤ 50 MPa
Running speed v	≤ 1.5 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	510 Bar	650 Bar	800 Bar
Profile dimension	1 mm	0.45 mm	0.19 mm

HOD



Ordering dimensions in green

[illegible]



HPO

PRODUCT DESCRIPTION

Seal designed combine with PTFE ring and O-ring.
Usually used for piston hydraulic application.

TECHNICAL DATA

MATERIAL	NBR + PTFE
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 1.5 m/s
Temperature	-35°C , +100°C

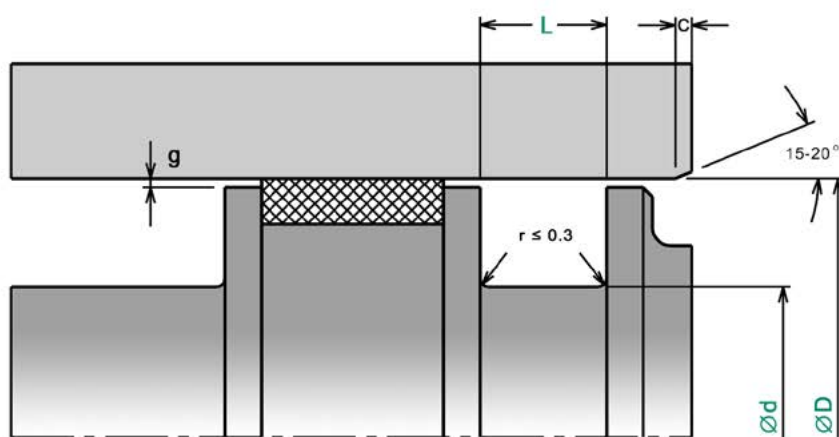
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	200 Bar	250 Bar	350 Bar
Profile dimension	0.6 mm	0.42 mm	0.25 mm

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HPO



Ordering dimensions in green

[illegible]



HPS

PRODUCT DESCRIPTION

Seal designed combine with PTFE ring and square ring. Usually used for piston hydraulic application.

TECHNICAL DATA

MATERIAL	NBR + PTFE
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 1.5 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

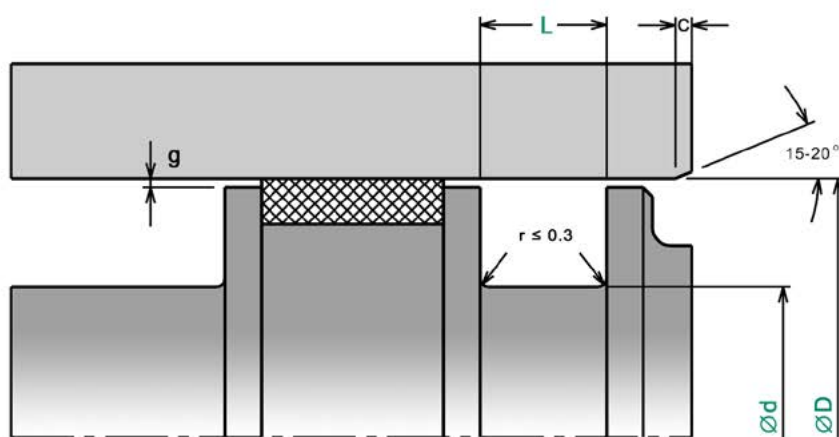
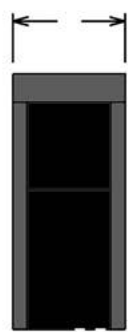
Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	300 Bar
Profile dimension	0.6 mm	0.6 mm	0.23 mm

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HPS

Metric Size



Ordering dimensions in green

[illegible]



UOR

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. Usually used for piston hydraulic and penumatic application.

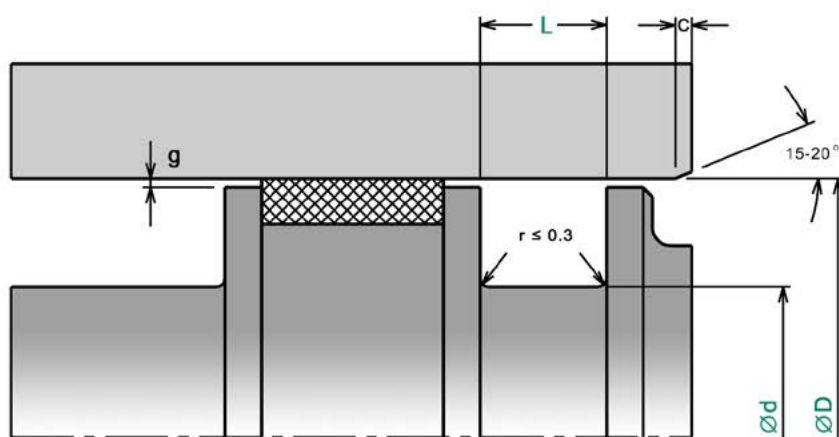
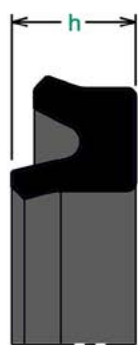
TECHNICAL DATA

	MATERIAL	
	NBR	
	FKM	
	MEDIA	
	Mineral oils	
Pressure p	≤ 16 MPa	≤ 16 MPa
Running speed v	≤ 0.5 m/s	≤ 0.5 m/s
Temperature	-35°C ,+100°C	-25°C ,+250°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

Admissible gap dimension			
	50 Bar	100 Bar	140 Bar
Profile dimension	0.3 mm	0.22 mm	0.15 mm

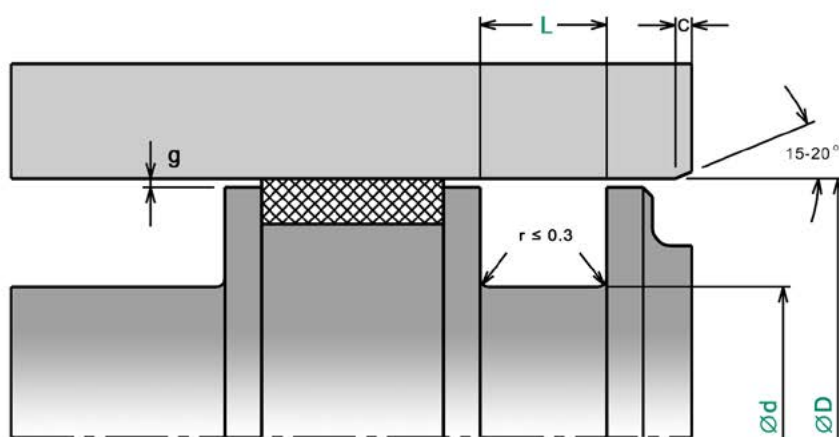
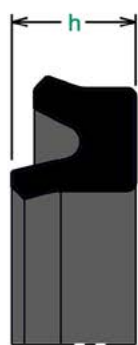
UOR Metric Size

Ordering dimensions in green

NAK NO.	NBR	FKM	Ød + 0.2/0	ØD H9	L +/- 0.2	h	c
U8450	•		1.8	4	2.8	2.1	1.0
U7435	•		4.76	11.11	4.66	3.96	2.0
U8451	•		5	8	2.9	2.2	1.0
U7947	•		5	9.5	4.7	4	1.5
U7949	•		5	10	4.7	4	1.5
U7948	•		5	10.5	4.7	4	1.5
U8343	•		6	9	2.8	2.1	1.0
U00598	•		6	10	3.6	2.9	1.0
U00597	•		8	12	3.2	2.5	1.0
U5622	•		10	14	3.6	2.9	1.0
U00596	•		10	15	4.4	3.7	1.5
U8665	•		11	15	3.5	2.8	1.0
U8420	•		11	22	8	7	3.0
U4921	•		12	16	3.9	3.2	1.0
U8419	•		12.5	25	7.6	6.6	3.5
U7660	•		12.7	19	3.87	3.17	2.0
U8561	•		14	20	3.25	2.55	1.5
U7731	•		14	23.5	4.7	4	2.5
U8666	•		18	22	3.5	2.8	1.0
U1137	•		18	27.4	5.9	5.2	2.5
U7436	•		19.05	29	7.35	6.35	2.5
U8120	•		20.5	30	7.35	6.35	2.5
U4922	•		21	25	3.9	3.2	1.0
U8232	•		22	35	7.35	6.35	3.5
U7437	•		23.8	36.5	8.92	7.92	3.5
U7619	•		27.3	40	7.35	6.35	3.5
U8276	•		28	34.3	5.5	4.8	2.0
U7607	•		28.57	44.45	10.52	9.52	4.0
U7439	•		30	38	7.35	6.35	2.0
U8993	•		30.16	38.1	5.7	5	2.0
U7043	•		31.5	38.5	6.2	5.5	2.0
U9173	•		32.5	40.5	5.9	5.2	2.0
U8690	•		33	42.5	8.2	7.2	2.5
U8324	•		35.2	45	9.5	7	2.5
U8286	•		37	44	6.2	5.5	2.0
U7440	•		40	50	8	7	2.5
U7438	•		41.27	57.15	7.35	6.35	4.0
U7441	•		41.3	54	10.52	9.52	3.5

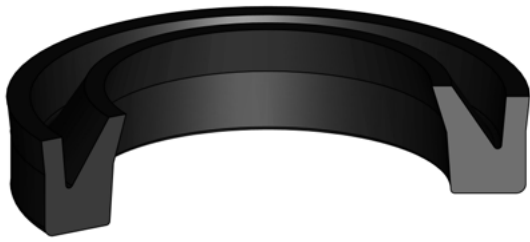
2.6

UOR Metric Size



Ordering dimensions in green

NAK NO.	NBR	FKM	Ød + 0.2/0	ØD H9	L +/- 0.2	h	c
U7442	•		44.3	57	7.35	6.35	3.5
U7608	•		45	55	8	7	2.5
U8458	•		47.62	63.5	10.5	9.5	4.0
U5032	•		50	60	4.95	4.25	2.5
U8247	•		53	63	8	7	2.5
U7662	•		54.1	70	8.93	7.93	4.0
U7429	•		55	65	8	7	2.5
U7661	•		58	70	9.5	8.5	3.0
U8373	•		63	75	9.5	8.5	3.0
U6916	•		65	75	8.5	7.5	2.5
U7022	•		65	75	8.5	7.5	2.5
U7430	•		67.3	80	7.35	6.35	3.5
U7431	•		68	80	9.5	8.5	3.0
U7609	•		73	85	9.5	8.5	3.0
U8146	•		73	86	10.5	9.5	3.5
U7428	•		76	95	8.4	7.4	5.0
U7611	•		76.1	92	10.5	9.5	4.0
U7610	•		78	90	9.5	8.5	3.0
U0253	•		83	95	6	5.3	3.0
U7009	•		86.3	99	10.52	9.52	3.5
U4970	•		87	99	9.5	8.5	3.0
U7010	•		88	100	9.5	8.5	3.0
U7612	•		89	102	13.5	12.5	3.5
U8083	•		98	110	9.5	8.5	3.0
U7613	•		110	125	11	10	4.0
U7629	•		120.6	146	16.9	15.9	6.5
U8291	•		124	137	10.5	9.5	3.5
U7422	•		127.6	153	16.9	15.9	6.5
U7423	•		133.6	159	16.9	15.9	6.5
U7614	•		159.3	191	20	19	8.0
U8290	•		160	177	12	11	4.5
U7421	•		165	184	12.11	11.11	5.0
U5299		•	42	50	3.95	3.25	2.0
U8300		•	45	55	8	5.5	2.5
U5302		•	80	90	4.95	4.25	2.5
U5821		•	82	100	10	9	4.5
U5303		•	90	100	4.95	4.25	2.5



COR

PRODUCT DESCRIPTION

Seal designed with asymmetric lips. Usually used for piston hydraulic and penumatic application.

TECHNICAL DATA

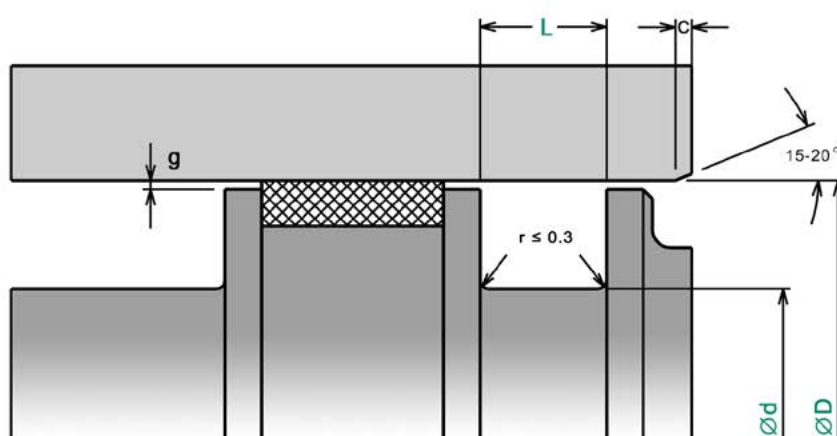
MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 10 MPa
Running speed v	≤ 0.5 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base	Ød	≤ 1.6 µm	≤ 6.3 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

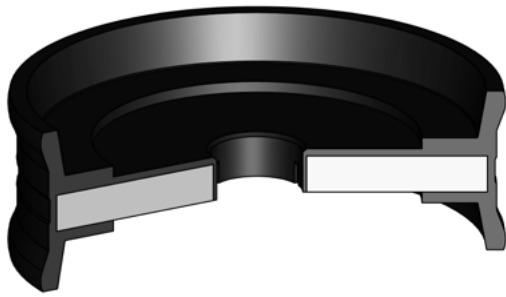
Admissible gap dimension			
	50 Bar	100 Bar	140 Bar
Profile dimension	0.3 mm	0.22 mm	0.15 mm

COR



Ordering dimensions in green

[illegible]



PDV

PRODUCT DESCRIPTION

Seal designed with a case body and rubber lips. Usually used for larger pressure condition than standard seal.

TECHNICAL DATA

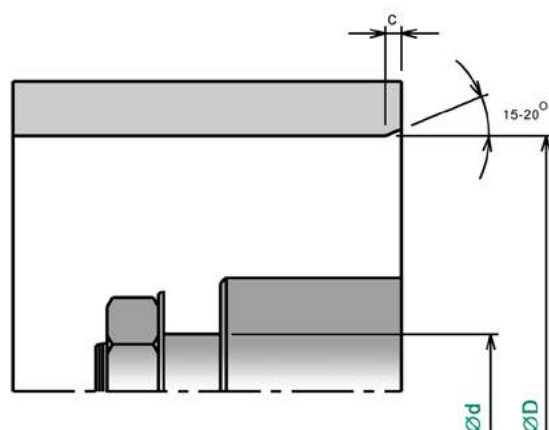
MATERIAL	NBR	FKM
MEDIA	Mineral oils	Mineral oils
Pressure p	≤ 10 MPa	≤ 10 MPa
Running speed v	≤ 0.5 m/s	≤ 0.5 m/s
Temperature	-35°C ,+100°C	-25°C ,+250°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm

PDV

Metric Size



2.8



PDC

PRODUCT DESCRIPTION

Seal designed with a case body, spring and rubber lip. Seals feature is stable for high pressure and slower speed.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 10 MPa
Running speed v	≤ 0.5 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

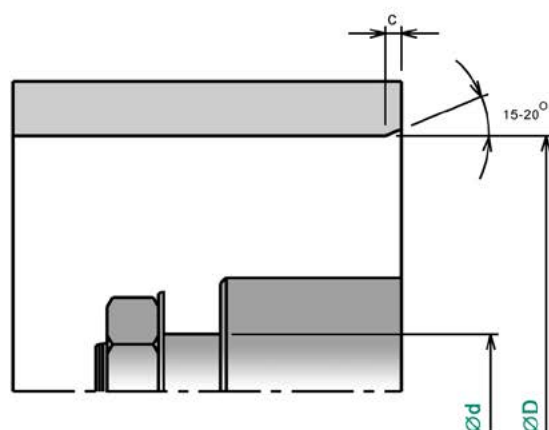
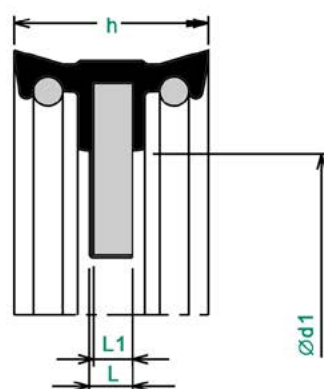
Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm

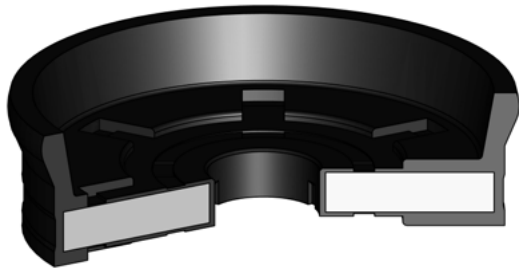
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PDC

Metric Size

[illegible]



PSV

PRODUCT DESCRIPTION

Seal designed with a case body and rubber lips. Usually used for larger pressure condition than standard seal.

TECHNICAL DATA

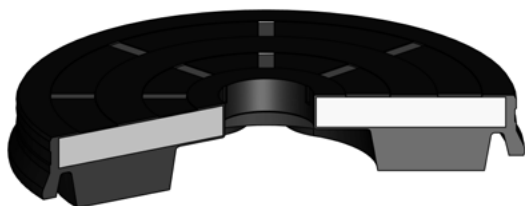
MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 10 MPa
Running speed v	≤ 0.5 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm

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PSV2

PRODUCT DESCRIPTION

Seal designed with a case body and rubber lips. Usually used for larger pressure condition than standard seal.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 10 MPa
Running speed v	≤ 0.5 m/s
Temperature	-35°C , +100°C

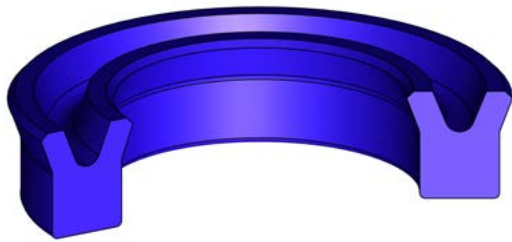
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	ØD	0.1~0.8 µm	0.4~3.2 µm

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SYMMETRICAL SEALS FOR ROD OR PISTON SEAL

NAK CODE	DRAWING	MATERIAL	PRESSURE (max)MPa	TEMPERATURE (max)°C	SLIDING SPEED(max) m/s	SIZE (mm)	PAGE
UNP		PU	30	-40 / +100	0.5	3-280 1.125-7.500	3.1 P94~P119 (METRIC) P120~P121 (INCH)
UNP1		PU	40	-40 / +100	0.5	6-180	3.2 P122~P127 (METRIC)
CNP		PU	20	-40 / +100	0.5	7-53	3.3 P128~P129 (METRIC)
UNR		NBR	16	-35 / +100	0.5	4-300	3.4 P130~P139 (METRIC)
CNR		NBR	10	-35 / +100	0.5	4-67 0.678-7.500	3.5 P140~P141 (METRIC) P142~P143 (INCH)
HB		PU NBR	16	-35 / +100 -25 / +250	0.5	1.8-165 42-90	3.6 P144~P145 (METRIC) P146~P153 (INCH)
HD		PU NBR	35	-35 / +100	0.5	0.250-9.250	3.7 P154~P157 (INCH)
HS		PU NBR	35	-35 / +100	0.5	0.187-10.000	3.8 P158~P163 (INCH)
HBX		PU NBR	35	-35 / +100	0.5	18-45 0.625-3.500	3.9 P164~P165 (METRIC) P166~P167 (INCH)
HSX		PU NBR	35	-35 / +100	0.5	2.000-3.375	3.10 P168-169 (METRIC)



UNP

PRODUCT DESCRIPTION

Seal designed with symmetric lips can be used for piston and rod application.

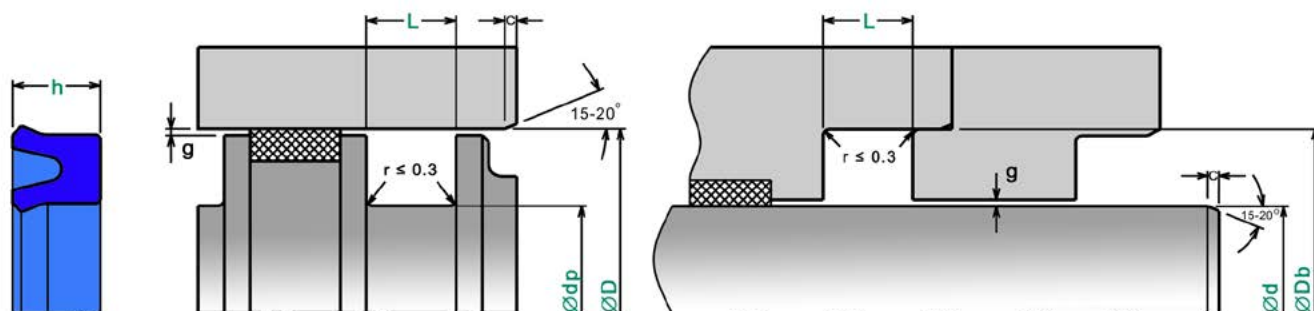
TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 30 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-40°C, +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

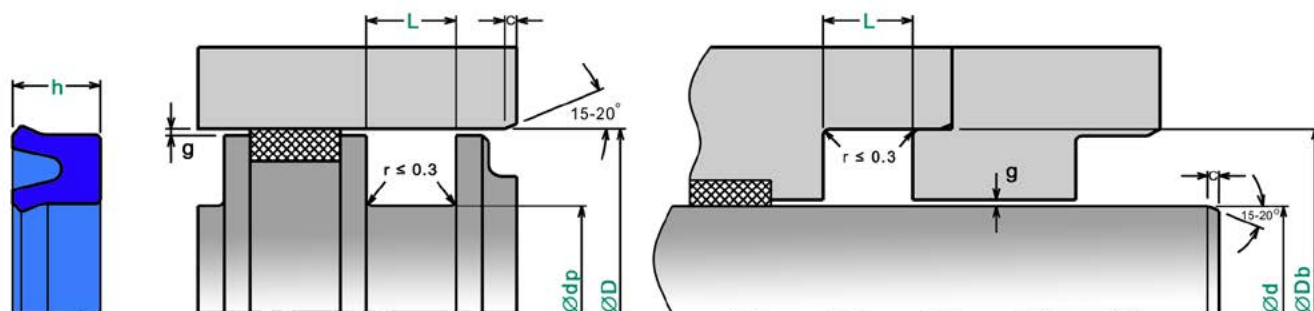
Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

UNP Metric Size


Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3788	3	9	5.2	4.5	3.0
U9788	4	9	4.7	4	2.5
U2711	4	10	4.7	4	3.0
U2444	4	10	5.2	4.5	3.0
U9674	4	10	5.7	5	3.0
U9795	4	12	4.7	4	4.0
U8308	4	12	6.2	5.5	4.0
U3029	4.5	11	5.7	5	3.5
U0994	4.5	12.5	5.2	4.5	4.0
U00618	5	10	5	4.3	2.5
U3829	5	10	5.7	5	2.5
U8994	5	10	7	6	2.5
U6276	5	11	5.7	5	3.0
U6275	5	11	8	7	3.0
U9802	5	11	9	8	3.0
U0458	5	12	5.7	5	3.5
U2948	5	12	7	6	3.5
U9808	5	12	9	8	3.5
U00620	5	13	7	6	4.0
U9810	5	15	9	8	5.0
U6011	5	17	10	9	6.0
U1927	6	10	4.9	4.2	2.0
U7413	6	11	8	7	2.5
U2882	6	12	4.7	4	3.0
U0925	6	12	5.2	4.5	3.0
U4398	6	12	5.7	5	3.0
U2394	6	12	6.2	5.5	3.0
U2159	6	12	7	6	3.0
U6312	6	12	8	7	3.0
U0273	6	12	9	8	3.0
U3995	6	13	9	8	3.5
U8417	6	14	6.4	5.7	4.0
U2650	6	14	7	6	4.0
U3477	6	15	4.7	4	4.5
U1779	6	15	7	6	4.5
U2419	6	15	9	8	4.5
U5553	6	16	7	6	5.0
U6255	6	16	7	6	5.0

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UNP Metric Size

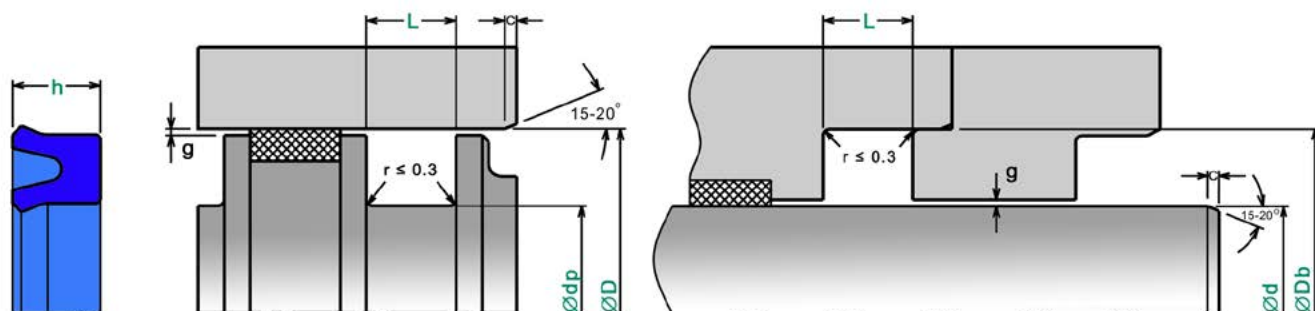
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U9892	6	17	7	6	5.5
U6562	6.30	12.80	3.9	3.20	3.5
U2301	6.3	16.3	9	8	5.0
U6253	7	14	3.9	3.2	3.5
U9177	7	14	4.2	3.5	3.5
U6261	7	15	5.7	5	4.0
U2206	7	15	8	7	4.0
U2207	8	14	4.7	4	3.0
U9349	8	14	5.7	5	3.0
U2613	8	14	7	6	3.0
U9854	8	15	4.7	4	3.5
U2420	8	15	6.4	5.7	3.5
U4582	8	15	7	6	3.5
U1775	8	15	7	6	3.5
U2241	8	15	9	8	3.5
U2097	8	16	6.4	5.7	4.0
U2976	8	16	7	6	4.0
U3271	8	16	9	8	4.0
U4840	8	18	7	6	5.0
U6260	8	18	9	8	5.0
U4143	8	18	9	8	5.0
U2557	8	18	10	9	5.0
U0892	8	22	9	8	7.0
U9865	9	15	5.7	5	3.0
U2249	9	15	9	8	3.0
U7812	9	16	5.7	5	3.5
U9908	9	16	7	6	3.5
U3038	9	19	7	6	5.0
U4784	9.95	22.22	7.85	6.85	6.5
U00340	10	14	3.7	3	2.0
U9891	10	14	4.7	4	2.0
U2098	10	16	4.7	4	3.0
U9402	10	16	5.2	4.5	3.0
U2922	10	16	5.5	4.8	3.0
U6258	10	16	5.7	5	3.0
U2210	10	16	7	6	3.0
U4084	10	18	4.7	4	4.0
U2237	10	18	5.7	5	4.0

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3554	10	18	6.2	5.5	4.0
U0249	10	18	7	6	4.0
U2816	10	18	9	8	4.0
U3421	10	20	7	6	5.0
U2703	10	20	9	8	5.0
U3673	10	20	11	10	5.0
U3672	10	22	5.7	5	6.0
U4271	10	22	7	6	6.0
U2051	10	22	9	8	6.0
U6257	10	25	11	10	7.5
U8870	10	32	11	10	11.0
U6256	11	17	4.7	4	3.0
U6264	11	20.5	7.5	6.5	5.0
U00105	11	26	9	8	7.5
U2914	11.2	21.5	9	8	5.5
U3828	12	18	4.7	4	3.0
U3343	12	18	5.7	5	3.0
U6319	12	18	5.7	5	3.0
U2242	12	18	7	6	3.0
U7663	12	18	9	8	3.0
U9829	12	20	4.7	4	4.0
U3865	12	20	5.7	5	4.0
U5270	12	20	6.2	5.5	4.0
U8465	12	20	7	6	4.0
U3337	12	20	8	7	4.0
U3056	12	20	9	8	4.0
U2632	12	22	5.7	5	5.0
U3785	12	22	8	7	5.0
U8151	12	22	8.5	7.5	5.0
U2117	12	22	9	8	5.0
U2186	12	24	7	6	6.0
U2846	12	24	9	8	6.0
U2951	12	24	10	9	6.0
U2289	12	25	9	8	6.5
U2633	12	25	11	10	6.5
U5525	12	28	7	6	8.0
U8397	12	28	9	8	8.0
U00678	12.5	18	4.2	3.5	3.0
U6561	12.7	19	3.9	3.2	3.5
U5093	12.7	19.05	4.66	3.96	3.5
U4783	12.7	22.22	5.95	5.25	5.0
U4902	12.70	25.40	7.35	6.35	6.5
U8685	13	20.2	9	8	4.0
U9799	13	25	8	7	6.0
U9801	13	25	11	10	6.0
U00512	13	26	6.2	5.5	6.5
U3347	14	20	4.7	4	3.0
U2052	14	20	5.5	4.8	3.0
U0279	14	20	5.7	5	3.0
U6266	14	20	7	6	3.0
U4127	14	20	9	8	3.0
U6615	14	22	4.7	4	4.0
U3990	14	22	5.2	4.5	4.0
U4020	14	22	5.7	5	4.0

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UNP Metric Size

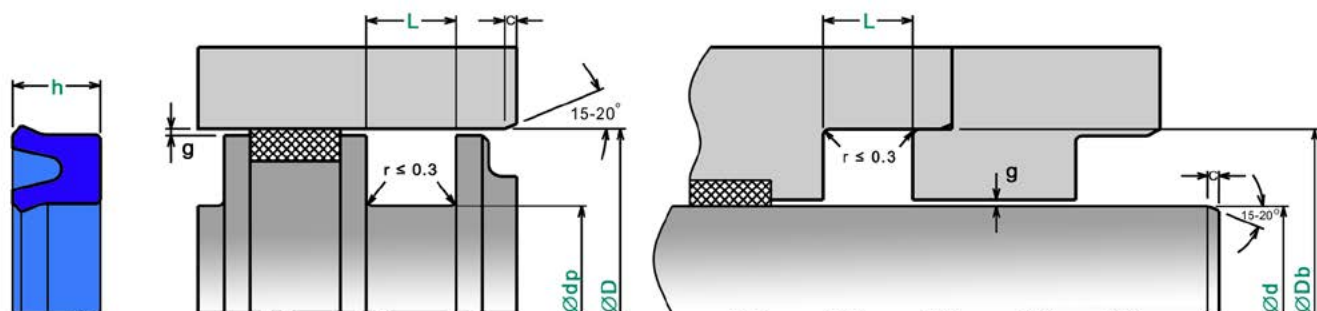
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3641	14	22	7	6	4.0
U3155	14	22	9	8	4.0
U2908	14	22	12	11	4.0
U9955	14	24	5.7	5	5.0
U3354	14	24	9	8	5.0
U2198	14	26	9	8	6.0
U6346	14	28	11	10	7.0
U3671	14	30	5.7	5	8.0
U4640	15	22	4.7	4	3.5
U7813	15	22	5.7	5	3.5
U2171	15	22	8	7	3.5
U2172	15	25	5.7	5	5.0
U8542	15	25	7	6	5.0
U2166	15	25	9	8	5.0
U3922	15	25	11	10	5.0
U3670	15	27	7	6	6.0
U9839	15	27	8	7	6.0
U6265	15	28	9	8	6.5
U2984	15	28	11	10	6.5
U3058	15	30	9	8	7.5
U7631	15	35	11	10	10.0
U4894	15.87	25.4	8.14	7.14	5.0
U4901	15.87	28.57	7.35	6.35	6.5
U4085	16	20	5.2	4.5	2.0
U2118	16	22	4.7	4	3.0
U9196	16	22	5.2	4.5	3.0
U2809	16	22	5.7	5	3.0
U8912	16	22	6.2	5.5	3.0
U9909	16	22	7	6	3.0
U3669	16	22	8	7	3.0
U9672	16	24	4.7	4	4.0
U2248	16	24	5.7	5	4.0
U4262	16	24	6.4	5.7	4.0
U7621	16	24	7	6	4.0
U9861	16	24	8	7	4.0
U3205	16	24	10	9	4.0
U7639	16	25	5.7	5	4.5
U9442	16	25	9	8	4.5

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3410	16	26	5.7	5	5.0
U6340	16	26	7	6	5.0
U1780	16	26	8	7	5.0
U2372	16	26	9	8	5.0
U1147	16	26	11	10	5.0
U2773	16	28	7	6	6.0
U9679	16	28	8	7	6.0
U00117	16	28	11	10	6.0
U3296	16	32	9	8	8.0
U9673	16.5	24	11	10	4.0
U9783	17	25	7	6	4.0
U2956	17	25	11	10	4.0
U7325	17	27	9	8	5.0
U00347	17	35	11	10	9.0
U9807	18	24	8	7	3.0
U9917	18	25	4.7	4	3.5
U2881	18	25	5.7	5	3.5
U2519	18	25	7	6	3.5
U9916	18	25	8	7	3.5
U3996	18	26	5.7	5	4.0
U7448	18	26	7	6	4.0
U2906	18	26	7.5	6.5	4.0
U2712	18	26	9	8	4.0
U4550	18	26	9.5	8.5	4.0
U9798	18	28	5.7	5	5.0
U3039	18	28	9	8	5.0
U2556	18	30	9	8	6.0
U6341	18	32	8	7	7.0
U2626	18	34	9	8	8.0
U3624	18	35	9	8	8.5
U2053	19	25	7	6	3.0
U8469	19	27	5.7	5	4.0
U8467	19	28	8	7	4.5
U8349	20	26	4	3.3	3.0
U2358	20	26	4.7	4	3.0
U5526	20	27	7	6	3.5
U9818	20	27	10	9	3.5
U2223	20	28	4.7	4	4.0
U3154	20	28	5.7	5	4.0
U6812	20	28	7	6	4.0
U1698	20	28	8	7	4.0
U2516	20	28	9	8	4.0
U3549	20	30	7	6	5.0
U2224	20	30	9	8	5.0
U3288	20	30	11	10	5.0
U6664	20	32	7	6	6.0
U4413	20	32	8.5	7.5	6.0
U2382	20	32	9	8	6.0
U9797	20	33	5.7	5	6.5
U5876	20	33	11	10	6.5
U9796	20	34	7	6	7.0
U9806	20	35	8	7	7.5
U0247	20	35	9	8	7.5
U8487	20	35	10	9	7.5

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UNP Metric Size

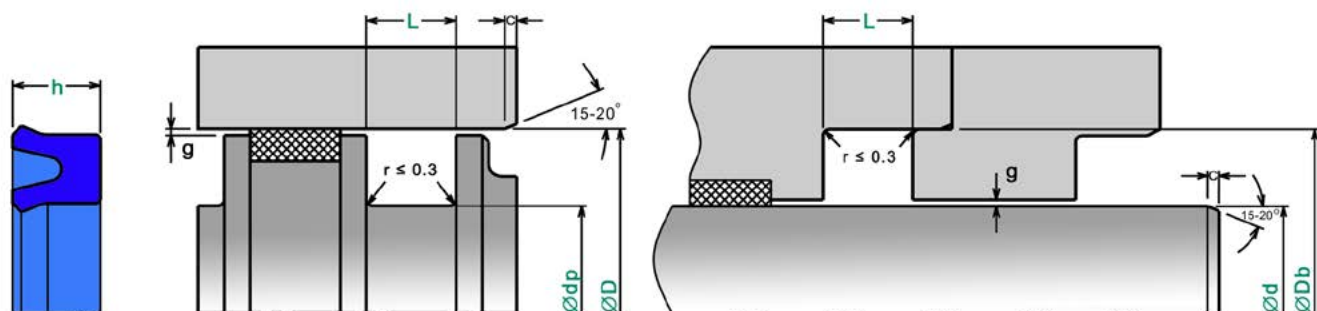
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3877	20	35	11	10	7.5
U3292	20	35	13	12	7.5
U6367	20	36	9	8	8.0
U5635	20	36	13	12	8.0
U9832	20	38	7	6	9.0
U6378	20	39	11.5	10.5	9.5
U9800	20	40	9	8	10.0
U2555	20	40	11	10	10.0
U3476	20	40	13	12	10.0
U9680	20	43	13	12	11.5
U6803	21	29.5	5.7	5	4.5
U2789	21	37	13	12	8.0
U2485	22	28	4.7	4	3.0
U6541	22	28	7	6	3.0
U2347	22	28	9	8	3.0
U2225	22	30	4.7	4	4.0
U4147	22	30	5.7	5	4.0
U6324	22	30	5.7	5	4.0
U2168	22	30	7	6	4.0
U2873	22	30	8	7	4.0
U6616	22	30	9	8	4.0
U1908	22	30	11	10	4.0
U2300	22	32	5.7	5	5.0
U5269	22	32	7	6	5.0
U2120	22	32	9	8	5.0
U2798	22	32	11	10	5.0
U7074	22	34	7	6	6.0
U4621	22	34	8	7	6.0
U00063	22	34	11	10	6.0
U00348	22	35	5.7	5	6.5
U9929	22	35	8	7	6.5
U9681	22	35	9	8	6.5
U2925	22	35	11	10	6.5
U9813	22	38	11	10	8.0
U2971	22	40	11	10	9.0
U3344	22	40	13	12	9.0
U9682	22	42	11	10	10.0
U2774	22.4	30	5.7	5	3.8

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U4768	22.4	32.4	9	8	5.0
U4908	23	30	7.5	6.5	3.5
U9820	23	33	8	7	5.0
U9925	23	35	7	6	6.0
U4263	23.5	31.5	5.7	5	4.0
U6345	24	32	4.7	4	4.0
U3660	24	32	5.7	5	4.0
U2868	24	32	8	7	4.0
U6344	24	32	8.5	7.5	4.0
U4842	24	32	9	8	4.0
U9828	24	34	7	6	5.0
U7664	24	34	8	7	5.0
U7148	24	36	8	7	6.0
U3831	24	40	9	8	8.0
U1522	24	50	13	12	13.0
U9895	25	30	7	6	2.5
U9849	25	31	7	6	3.0
U2464	25	32	4.7	4	3.5
U2464	25	32	4.7	4	3.5
U00337	25	32	5.7	5	3.5
U8295	25	32	7	6	3.5
U9838	25	32	8	7	3.5
U9225	25	32	11	10	3.5
U2080	25	33	4.7	4	4.0
U3000	25	33	5.7	5	4.0
U5792	25	33	7	6	4.0
U3816	25	33	8	7	4.0
U3475	25	33	9	8	4.0
U9835	25	33	11	10	4.0
U2679	25	35	5.7	5	5.0
U3599	25	35	7	6	5.0
U2810	25	35	8	7	5.0
U2161	25	35	9	8	5.0
U8263	25	35	9.6	8.6	5.0
U2082	25	35	11	10	5.0
U9827	25	36	9	8	5.5
U7635	25	37	7	6	6.0
U6372	25	37	8	7	6.0
U4960	25	37	10	9	6.0
U9833	25	38	7	6	6.5
U7172	25	38	8	7	6.5
U1781	25	38	9	8	6.5
U2457	25	38	10.5	9.5	6.5
U2162	25	38	11	10	6.5
U9805	25	40	7	6	7.5
U9804	25	40	7.5	6.5	7.5
U4843	25	40	9	8	7.5
U2304	25	40	11	10	7.5
U2432	25	40	11	10	7.5
U9831	25	42	9	8	8.5
U2487	25	45	11	10	10.0
U6371	25	45	11	10	10.0
U7394	25	50	13	12	12.5
U9894	25.4	31.75	4.66	3.96	3.5

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UNP Metric Size

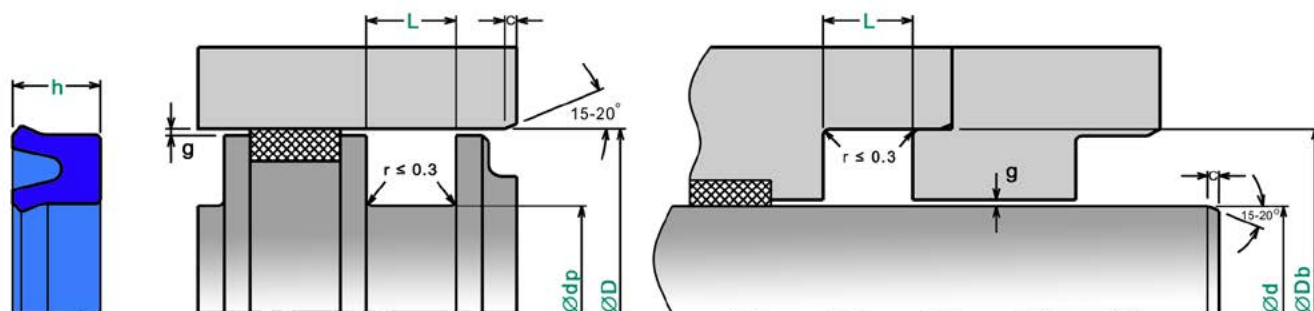
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U4994	25.4	31.75	5.46	4.76	3.5
U4900	25.4	38.1	7.35	6.35	6.5
U00208	25.4	41.27	12.11	11.11	8.0
U00509	26	34	9	8	4.0
U9819	26	36	7	6	5.0
U6680	26	38	7	6	6.0
U2083	26	38	9	8	6.0
U9691	26	42	9	8	8.0
U9919	26	43	8	7	8.5
U7703	26	47	13	12	10.5
U5727	27	35	4.7	4	4.0
U2488	27	35	5.7	5	4.0
U3040	27	36	7	6	4.5
U6370	27	38	7	6	5.5
U1204	27	43	11	10	8.0
U2299	28	35	5.7	5	3.5
U3988	28	35.5	5.7	5	4.0
U2354	28	36	4.7	4	4.0
U3338	28	36	5.7	5	4.0
U6369	28	36	7	6	4.0
U2390	28	36	7.5	6.5	4.0
U9787	28	36	8	7	4.0
U8573	28	36	9	8	4.0
U9822	28	38	5.7	5	5.0
U6687	28	38	7	6	5.0
U9842	28	38	8	7	5.0
U2290	28	38	9	8	5.0
U2887	28	38	11	10	5.0
U2173	28	40	9	8	6.0
U0220	28	40	11	10	6.0
U6377	28	43	11	10	7.5
U3506	28	44	9	8	8.0
U00349	28	46	11	10	9.0
U8488	28	50	11	10	11.0
U7646	28.57	38.1	8.93	7.93	5.0
U4899	28.57	41.27	10.52	9.52	6.5
U8282	30	36	4.7	4	3.0
U9826	30	36	7	6	3.0

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U2704	30	37	6.5	5.8	3.5
U2465	30	37	7	6	3.5
U00507	30	37	7	6	3.5
U6376	30	38	4.7	4	4.0
U2458	30	38	5.7	5	4.0
U2247	30	38	7	6	4.0
U2817	30	38	8	7	4.0
U5523	30	38	9	8	4.0
U2123	30	40	5.7	5	5.0
U3987	30	40	6.2	5.5	5.0
U2174	30	40	7	6	5.0
U2075	30	40	7.5	6.5	5.0
U2697	30	40	8	7	5.0
U4016	30	40	9	8	5.0
U2124	30	40	11	10	5.0
U8264	30	40	11.2	10.2	5.0
U9413	30	40	12	11	5.0
U5550	30	42	7	6	6.0
U9823	30	42	9	8	6.0
U2818	30	42	10	9	6.0
U2329	30	42	11	10	6.0
U4839	30	43	7.5	6.5	6.5
U5548	30	43	9	8	6.5
U5547	30	43	11	10	6.5
U9899	30	44	8.5	7.5	7.0
U9898	30	44	9	8	7.0
U2467	30	45	9	8	7.5
U2342	30	45	10	9	7.5
U2078	30	45	11	10	7.5
U9920	30	47	7	6	8.5
U9812	30	47	9	8	8.5
U9918	30	47	11	10	8.5
U2058	30	50	11	10	10.0
U2295	30	50	13	12	10.0
U8074	30	55	11	10	12.5
U4897	30.16	42.86	8.93	7.93	6.5
U3313	31.5	41.5	7	6	5.0
U4893	31.75	41.27	8.93	7.93	5.0
U4898	31.75	44.45	7.35	6.35	6.5
U4687	31.75	44.45	10.52	9.52	6.5
U8572	32	38	7	6	3.0
U2216	32	40	4.7	4	4.0
U2926	32	40	5.7	5	4.0
U2977	32	40	6.2	5.5	4.0
U2680	32	40	7	6	4.0
U2870	32	40	8	7	4.0
U2705	32	40	9	8	4.0
U4576	32	40	11	10	4.0
U3548	32	42	7	6	5.0
U8130	32	42	8	7	5.0
U5549	32	42	9	8	5.0
U2217	32	42	11	10	5.0
U0686	32	44	4.7	4	6.0
U9791	32	44	7	6	6.0

3.1

UNP Metric Size

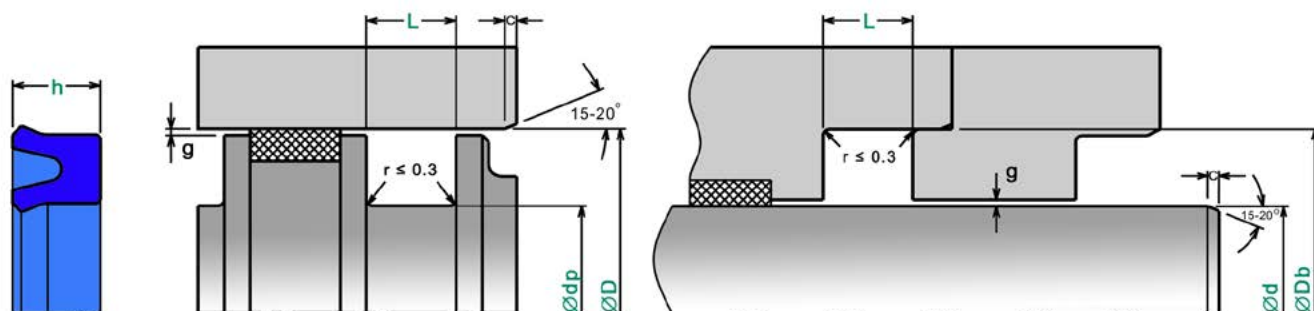
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U5874	32	44	9	8	6.0
U9927	32	45	5.7	5	6.5
U9840	32	45	7	6	6.5
U8540	32	45	9	8	6.5
U2292	32	45	11	10	6.5
U2706	32	47	11	10	7.5
U7261	32	48	11	10	8.0
U2522	32	50	11	10	9.0
U2819	32	50	13	12	9.0
U9809	32	52	11	10	10.0
U4004	33	43	11	10	5.0
U9792	34	44	8.5	7.5	5.0
U6374	34	45	8	7	5.5
U2957	34	45	10	9	5.5
U7487	34	46	9	8	6.0
U5909	34	50	15	14	8.0
U4777	34.92	44.45	10.52	9.52	5.0
U7110	34.92	50.8	10.52	9.52	8.0
U4887	34.92	50.80	13.7	12.70	8.0
U8174	35	42	9	8	3.5
U9830	35	42	11	10	3.5
U3153	35	43	7	6	4.0
U3057	35	43	8	7	4.0
U9874	35	43	9	8	4.0
U9944	35	45	5.7	5	5.0
U2169	35	45	7	6	5.0
U8281	35	45	8	7	5.0
U00029	35	45	8.5	7.5	5.0
U2375	35	45	9	8	5.0
U2109	35	45	11	10	5.0
U6725	35	47	7.2	6.2	6.0
U9786	35	47	10	9	6.0
U3938	35	47	11	10	6.0
U9931	35	48	9	8	6.5
U2927	35	48	11	10	6.5
U2446	35	50	9	8	7.5
U2047	35	50	11	10	7.5
U9793	35	50	13	12	7.5

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U9803	35	52	9	8	8.5
U9821	35	52	11	10	8.5
U2330	35	55	11	10	10.0
U2824	35	55	13	12	10.0
U2713	35.5	43.5	9	8	4.0
U3997	35.5	45	7	6	5.0
U2048	36	44	4.7	4	4.0
U2110	36	44	6.4	5.7	4.0
U6086	36	44	7	6	4.0
U5775	36	45	11	10	4.5
U2608	36	46	8	7	5.0
U2359	36	46	11	10	5.0
U00644	36	48	9	8	6.0
U5932	36	50	9	8	7.0
U9789	36	51	11	10	7.5
U00477	36	52	11	10	8.0
U5283	36	70	11	10	17.0
U2871	38	45	7	6	3.5
U2963	38	45	9	8	3.5
U9868	38	46	7	6	4.0
U4086	38	46	7.5	6.5	4.0
U3315	38	48	7	6	5.0
U6717	38	48	9	8	5.0
U2491	38	48	11	10	5.0
U4584	38	50	7	6	6.0
U6661	38	50	9	8	6.0
U2043	38	50	10	9	6.0
U2610	38	50	11	10	6.0
U2218	38	55	11	10	8.5
U3041	38	58	11	10	10.0
U9785	38	60	11	10	11.0
U9896	38	62	13	12	12.0
U00511	38.1	47.62	7.35	6.35	5.0
U4684	38.10	47.62	8.93	7.93	5.0
U7447	38.1	50.8	10.52	9.52	6.5
U4891	38.10	57.15	12.91	11.91	10.0
U00622	40	48	5.7	5	4.0
U7590	40	48	7	6	4.0
U1786	40	48	9	8	4.0
U2112	40	48	9	8	4.0
U2192	40	48	12	11	4.0
U9816	40	49	11	10	4.5
U2219	40	50	5.7	5	5.0
U2954	40	50	7	6	5.0
U2045	40	50	7.5	6.5	5.0
U2799	40	50	8	7	5.0
U9817	40	50	8.5	7.5	5.0
U2046	40	50	9	8	5.0
U00338	40	50	10	9	5.0
U2099	40	50	11	10	5.0
U7100	40	50	11.2	10.2	5.0
U8871	40	50	13	12	5.0
U7260	40	52	9	8	6.0
U9811	40	52	10	9	6.0

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UNP Metric Size

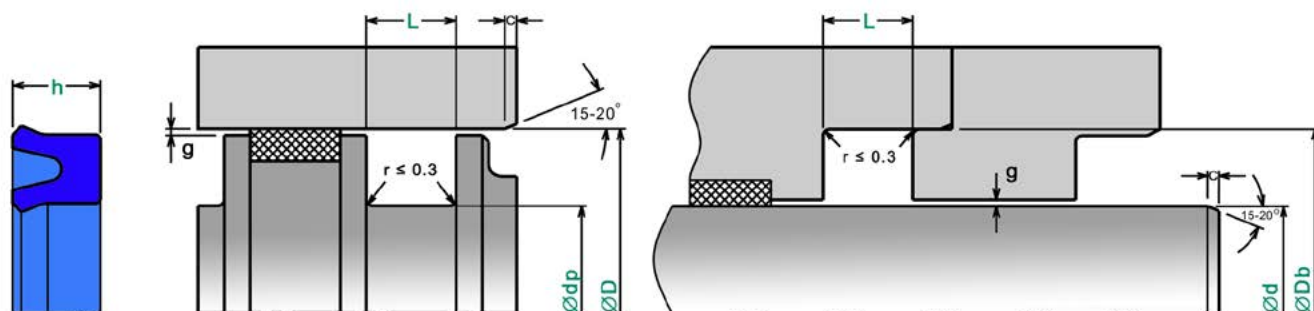
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U9824	40	52	12	11	6.0
U6612	40	55	9	8	7.5
U2035	40	55	11	10	7.5
U5933	40	56	9	8	8.0
U3098	40	56	11	10	8.0
U7060	40	56	13	12	8.0
U8570	40	58	11	10	9.0
U5546	40	60	9	8	10.0
U2800	40	60	11	10	10.0
U00350	40	60	12	11	10.0
U3860	40	60	13	12	10.0
U2643	40	60	14	13	10.0
U3026	40	60	19	18	10.0
U4087	40	62	16.2	15.2	11.0
U5934	40	65	11	10	12.5
U3059	40	65	13	12	12.5
U5935	40	70	16	15	15.0
U7485	41	51	8	7	5.0
U4896	41.27	57.15	12.11	11.11	8.0
U4892	41.27	60.32	10.52	9.52	10.0
U2380	42	50	9	8	4.0
U8543	42	50	11	10	4.0
U6373	42	52	7	6	5.0
U5937	42	52	10	9	5.0
U2002	42	52	11	10	5.0
U8080	42	55	11	10	6.5
U7671	42	62	11	10	10.0
U7001	42	62	13	12	10.0
U3661	44	50	9.5	8.5	3.0
U00341	44	54	7	6	5.0
U4844	44.45	53.97	10.52	9.52	5.0
U9633	44.45	57.15	8.93	7.93	6.5
U4688	44.45	57.15	10.52	9.52	6.5
U4885	44.45	60.32	12.11	11.11	8.0
U2847	45	53	6.4	5.7	4.0
U7175	45	53	7	6	4.0
U3570	45	53	9	8	4.0
U3316	45	53	11	10	4.0

UNP Metric Size

Ø	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U0566	45	53	13	12	4.0
U2036	45	55	5.7	5	5.0
U2862	45	55	7	6	5.0
U2037	45	55	7.5	6.5	5.0
U5305	45	55	8	7	5.0
U5521	45	55	9	8	5.0
U2331	45	55	11	10	5.0
U3998	45	56	8	7	5.5
U6689	45	57	7	6	6.0
U2447	45	57.7	10.5	9.5	6.5
U6540	45	58	11	10	6.5
U9952	45	60	8.5	7.5	7.5
U9961	45	60	10	9	7.5
U2220	45	60	11	10	7.5
U8541	45	60	13	12	7.5
U7656	45	62	11	10	8.5
U9825	45	63	9	8	9.0
U3045	45	63	11	10	9.0
U3663	45	63	13	12	9.0
U1644	45	65	7.5	6.5	10.0
U2438	45	65	11	10	10.0
U3100	45	65	13	12	10.0
U4585	45	70	13	12	12.5
U6082	45	70	13	12	12.5
U7592	45	70	16.5	15.5	12.5
U5002	45	75	16	15	15.0
U9794	46	56	9	8	5.0
U7632	47	55	7	6	4.0
U7683	47	57	10	9	5.0
U5728	47	65	13	12	9.0
U4886	47.62	63.50	8.93	7.93	8.0
U4779	47.62	73.02	17.37	16.37	13.0
U3046	48	58	11	10	5.0
U5938	48	60	10	9	6.0
U2103	48	63	11	10	7.5
U7737	48	65	11	10	8.5
U1710	48	68	11	10	10.0
U00343	48	70	11	10	11.0
U4586	50	57	11	10	3.5
U2213	50	58	9	8	4.0
U2164	50	60	5.7	5	5.0
U3999	50	60	7	6	5.0
U8129	50	60	8	7	5.0
U2371	50	60	8.2	7.2	5.0
U4155	50	60	9	8	5.0
U2865	50	60	11	10	5.0
U2888	50	60	12	11	5.0
U2187	50	60	13	12	5.0
U2188	50	60	14	13	5.0
U9922	50	62	7	6	6.0
U9923	50	62	7.5	6.5	6.0
U9924	50	62	8	7	6.0
U2104	50	62	10	9	6.0
U9928	50	62	11	10	6.0

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UNP Metric Size

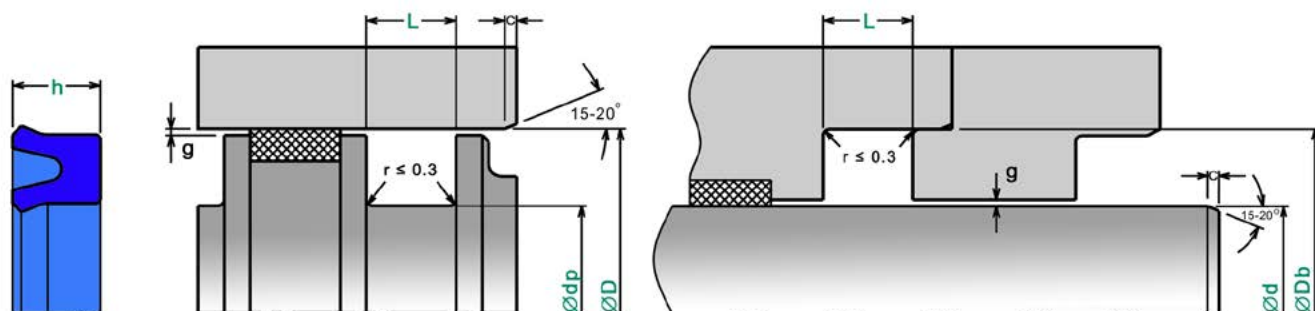
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U1885	50	62	13	12	6.0
U2869	50	63	7	6	6.5
U3070	50	63	7.5	6.5	6.5
U3473	50	63	9	8	6.5
U5907	50	63	10	9	6.5
U2332	50	63	11	10	6.5
U7382	50	63	13	12	6.5
U9897	50	64	10.5	9.5	7.0
U9960	50	65	8	7	7.5
U9959	50	65	8.5	7.5	7.5
U9576	50	65	9	8	7.5
U2023	50	65	11	10	7.5
U5908	50	65	13	12	7.5
U2473	50	68	11	10	9.0
U2087	50	70	11	10	10.0
U2022	50	70	13	12	10.0
U9974	50	70	14	13	10.0
U2293	50	70	19	18	10.0
U3864	50	75	13	12	12.5
U00344	50	75	16	15	12.5
U00032	50	80	11	10	15.0
U00272	50.6	66.67	8.93	7.93	8.5
U5729	50.8	60.32	5.46	4.76	5.0
U4685	50.8	60.32	7.35	6.35	5.0
U4682	50.8	60.32	8.93	7.93	5.0
U4686	50.8	60.32	10.52	9.52	5.0
U5008	50.8	63.5	8.93	7.93	6.5
U4689	50.8	63.5	10.52	9.52	6.5
U6634	50.8	66.67	10.52	9.52	8.0
U4907	50.8	66.67	15.28	14.28	8.0
U6633	50.8	69.85	10.52	9.52	10.0
U8645	50.8	69.85	13.7	12.7	10.0
U8468	51	66	13	12	7.5
U3862	51	66	15	14	7.5
U3648	51	76	15	14	12.5
U3507	52	62	13	12	5.0
U9905	52	70	11	10	9.0
U8079	52	72	11	10	10.0

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U2221	53	63	5.7	5	5.0
U3664	53	63	7	6	5.0
U2351	53	63	7.5	6.5	5.0
U00664	53	63	8.5	7.5	5.0
U9473	53	63	11	10	5.0
U9940	53	63	13	12	5.0
U8256	53	65	7	6	6.0
U8249	53	65	10	9	6.0
U9900	53.97	63.5	7.35	6.35	5.0
U00601	53.97	66.67	10.52	9.52	6.5
U9939	54	64	7	6	5.0
U2474	54	70	13	12	8.0
U7259	54	72	11	10	9.0
U3662	55	60	11	10	2.5
U8292	55	63	9	8	4.0
U2024	55	65	5.7	5	5.0
U4000	55	65	7	6	5.0
U6716	55	65	9	8	5.0
U2645	55	65	11	10	5.0
U5804	55	65	12	11	5.0
U2991	55	65	13	12	5.0
U7756	55	70	8.5	7.5	7.5
U8943	55	70	10	9	7.5
U2801	55	70	11	10	7.5
U2260	55	70	13	12	7.5
U2554	55	75	11	10	10.0
U2523	55	75	13	12	10.0
U2634	55	80	13	12	12.5
U1649	55	80	19	18	12.5
U00034	55	85	13	12	15.0
U3318	56	66	7	6	5.0
U2475	56	66	7.5	6.5	5.0
U2214	56	66	7.7	6.7	5.0
U5886	56	66	9	8	5.0
U7492	56	66	11	10	5.0
U5636	56	70	13	12	7.0
U4117	56	71	11	10	7.5
U3319	56	76	13	12	10.0
U9912	57	63	9	8	3.0
U9911	57	63	11	10	3.0
U8294	57	73	9	8	8.0
U9904	57	73	12	11	8.0
U4690	57.15	69.85	10.52	9.52	6.5
U6637	57.15	73.02	10.52	9.52	8.0
U4890	57.15	76.20	10.52	9.52	10.0
U4778	57.16	66.75	6.25	5.25	5.0
U9947	58	68	7	6	5.0
U9977	58	78	13	12	10.0
U2349	60	68	9	8	4.0
U9914	60	68	11	10	4.0
U5730	60	68	13	12	4.0
U2222	60	70	5.7	5	5.0
U2310	60	70	7	6	5.0
U7412	60	70	8	7	5.0

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UNP Metric Size

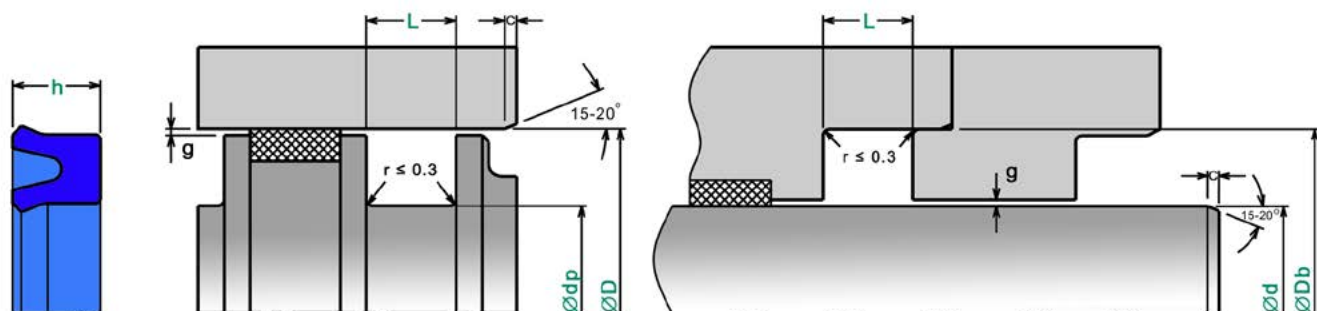
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3101	60	70	9	8	5.0
U2294	60	70	11	10	5.0
U2021	60	70	13	12	5.0
U4264	60	71	8	7	5.5
U8481	60	72	10	9	6.0
U7249	60	72	12	11	6.0
U1105	60	72	14	13	6.0
U8075	60	75	9	8	7.5
U00339	60	75	10	9	7.5
U2288	60	75	11	10	7.5
U2148	60	75	13	12	7.5
U9980	60	80	7	6	10.0
U9881	60	80	9	8	10.0
U2088	60	80	11	10	10.0
U2139	60	80	13	12	10.0
U2261	60	80	19	18	10.0
U2896	60	85	11	10	12.5
U3817	60	85	13	12	12.5
U5915	60	85	13.5	12.5	12.5
U00035	60	90	11	10	15.0
U3047	60	90	16	15	15.0
U4998	60.32	73.02	10.52	9.52	6.5
U4961	62	72	13.5	12.5	5.0
U9926	62	75	11	10	6.5
U5716	62	78	13	12	8.0
U2178	63	73	5.7	5	5.0
U3320	63	73	7	6	5.0
U9946	63	73	8	7	5.0
U9945	63	73	9.5	8.5	5.0
U9943	63	73	11	10	5.0
U1056	63	75	10	9	6.0
U2744	63	75	11	10	6.0
U2151	63	75	13	12	6.0
U2666	63	78	11	10	7.5
U6399	63	80	8	7	8.5
U2275	63	80	10	9	8.5
U3300	63	80	11	10	8.5
U2476	63	80	13	12	8.5

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U6901	63	80	17	16	8.5
U5004	63	83	13	12	10.0
U2972	63	83	16	15	10.0
U4884	63.50	76.20	10.52	9.52	6.5
U6638	63.5	79.37	8.93	7.93	8.0
U9642	63.5	82.55	10.52	9.52	10.0
U4877	63.50	88.90	20.05	19.05	13.0
U9933	64	77	13	12	6.5
U3666	64	80	9	8	8.0
U9903	64	80	13	12	8.0
U3991	65	75	7	6	5.0
U5519	65	75	9	8	5.0
U2130	65	75	11	10	5.0
U2320	65	75	13	12	5.0
U6617	65	77	10	9	6.0
U9932	65	78	11	10	6.5
U9958	65	80	8.5	7.5	7.5
U8944	65	80	10	9	7.5
U2928	65	80	11	10	7.5
U2411	65	80	12	11	7.5
U2768	65	80	13	12	7.5
U2353	65	85	11	10	10.0
U2684	65	85	13	12	10.0
U8065	65	85	16	15	10.0
U2714	65	90	13	12	12.5
U6723	65	90	16	15	12.5
U2823	65	90	19	18	12.5
U5914	66	76	9	8	5.0
U6748	66	77	11	10	5.5
U6749	66	77	13	12	5.5
U4906	66.67	82.55	8.93	7.93	8.0
U9910	67	74	11	10	3.5
U9913	67	75	11	10	4.0
U3322	67	77	7	6	5.0
U3665	67	77	11	10	5.0
U2182	67	77	13	12	5.0
U4883	69.85	82.55	10.52	9.52	6.5
U4905	69.85	85.72	10.52	9.52	8.0
U00188	69.85	88.9	10.52	9.52	10.0
U4889	69.85	88.9	13.7	12.7	10.0
U2844	70	80	5.7	5	5.0
U2527	70	80	7	6	5.0
U2999	70	80	8	7	5.0
U3102	70	80	9	8	5.0
U2262	70	80	11	10	5.0
U2183	70	80	13	12	5.0
U6614	70	82	10	9	6.0
U9954	70	85	8	7	7.5
U8257	70	85	9	8	7.5
U8562	70	85	10	9	7.5
U5913	70	85	11	10	7.5
U2180	70	85	13	12	7.5
U2066	70	90	11	10	10.0
U2263	70	90	13	12	10.0

3.1

UNP Metric Size

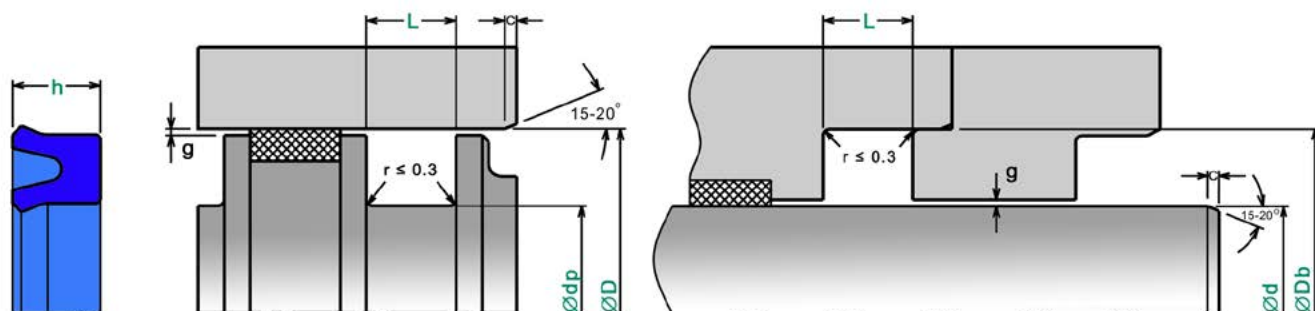
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U6542	70	90	16	15	10.0
U2027	70	90	19	18	10.0
U00021	70	95	11	10	12.5
U2929	70	95	13	12	12.5
U4962	70	100	16	15	15.0
U3323	71	80	7	6	4.5
U4882	73.02	85.72	10.52	9.52	6.5
U6798	75	85	7	6	5.0
U6715	75	85	9	8	5.0
U2243	75	85	11	10	5.0
U2553	75	85	13	12	5.0
U2068	75	90	8.5	7.5	7.5
U2831	75	90	10	9	7.5
U2709	75	90	11	10	7.5
U2069	75	90	13	12	7.5
U2795	75	95	11	10	10.0
U1776	75	95	11	10	10.0
U2745	75	95	13	12	10.0
U7186	75	95	16	15	10.0
U00015	75	100	10	9	12.5
U6721	75	100	13	12	12.5
U2413	75	100	16	15	12.5
U3649	76	91	15	14	7.5
U4881	76.2	88.9	10.52	9.52	6.5
U4904	76.20	92.07	15.28	14.28	8.0
U00665	76.2	95.25	13.7	12.7	10.0
U5518	76.2	95.25	15.2	14.2	10.0
U00014	76.2	101.6	13.7	12.7	13.0
U9836	76.2	101.6	18	17	13.0
U5912	78	93	11.5	10.5	7.5
U9969	78	98	11	10	10.0
U9901	78	100	10	9	11.0
U9906	78	100	13	12	11.0
U2449	78	100	16	15	11.0
U9915	80	88	9	8	4.0
U2177	80	90	5.7	5	5.0
U3994	80	90	7	6	5.0
U3993	80	90	9	8	5.0

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U2264	80	90	11	10	5.0
U2889	80	90	13	12	5.0
U4267	80	95	10	9	7.5
U5911	80	95	10	9	7.5
U3667	80	95	11	10	7.5
U2334	80	95	13	12	7.5
U2089	80	100	11	10	10.0
U2287	80	100	13	12	10.0
U9975	80	100	14	13	10.0
U4026	80	100	16	15	10.0
U3346	80	105	13	12	12.5
U1035	82	92	11	10	5.0
U4880	82.55	95.25	10.52	9.52	6.5
U6974	82.55	101.6	10.52	9.52	10.0
U9921	83	100	11	10	8.5
U7306	85	95	7	6	5.0
U7591	85	95	9	8	5.0
U3062	85	95	9.5	8.5	5.0
U6722	85	95	11	10	5.0
U3093	85	95	13	12	5.0
U00067	85	100	8.5	7.5	7.5
U2146	85	100	9	8	7.5
U6867	85	100	9	8	7.5
U2145	85	100	10	9	7.5
U6618	85	100	11	10	7.5
U1024	85	100	12	11	7.5
U2415	85	100	13	12	7.5
U7666	85	105	11	10	10.0
U2930	85	105	13	12	10.0
U1782	85	105	16	15	10.0
U2134	85	105	19	18	10.0
U2344	85	110	13	12	12.5
U6665	85	110	16	15	12.5
U6097	85	115	16	15	15.0
U4775	85.72	104.77	15.78	14.78	10.0
U4963	88	96	9	8	4.0
U4879	88.90	101.60	10.52	9.52	6.5
U4903	88.90	104.77	10.52	9.52	8.0
U6629	88.9	107.95	13.7	12.7	10.0
U4865	88.90	114.30	10.52	9.52	13.0
U2030	90	100	5.7	5	5.0
U1977	90	100	8	7	5.0
U2746	90	100	9	8	5.0
U8470	90	100	10	9	5.0
U3584	90	100	11	10	5.0
U2461	90	100	13	12	5.0
U8354	90	102	10	9	6.0
U7650	90	102	11	10	6.0
U2861	90	105	10	9	7.5
U2136	90	105	13	12	7.5
U2618	90	110	11	10	10.0
U2029	90	110	13	12	10.0
U9976	90	110	14	13	10.0
U5003	90	110	16	15	10.0

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UNP Metric Size

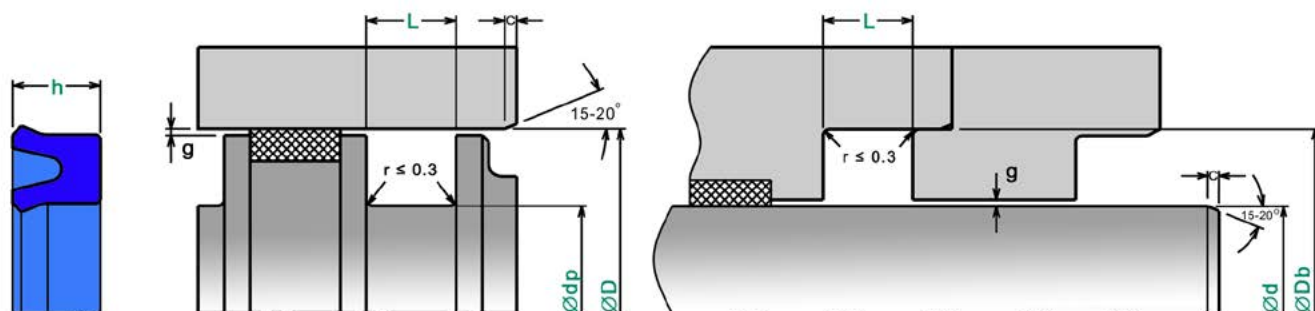
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3514	90	110	19	18	10.0
U00027	90	115	10	9	12.5
U7647	90	115	13	12	12.5
U2741	90	115	16	15	12.5
U5970	90	120	16	15	15.0
U4999	92.07	117.47	13.7	12.7	13.0
U1777	95	105	11	10	5.0
U5633	95	105	12	11	5.0
U2416	95	105	13	12	5.0
U4265	95	110	10	9	7.5
U2769	95	110	13	12	7.5
U3208	95	110	16	15	7.5
U5928	95	112	12	11	8.5
U0692	95	112	13	12	8.5
U9979	95	115	10	9	10.0
U2717	95	115	13	12	10.0
U9981	95	115	14	13	10.0
U2736	95	115	19	18	10.0
U00013	95	118	13	12	11.5
U7568	96	105	12	11	4.5
U5517	100	108	9	8	4.0
U2020	100	110	5.7	5	5.0
U5929	100	110	9	8	5.0
U0691	100	110	12	11	5.0
U3939	100	115	10	9	7.5
U2129	100	115	13	12	7.5
U9987	100	120	10	9	10.0
U2179	100	120	11	10	10.0
U2090	100	120	13	12	10.0
U9986	100	120	14	13	10.0
U2742	100	120	16	15	10.0
U6868	100	120	19	18	10.0
U6667	100	125	11	10	12.5
U2796	100	125	13	12	12.5
U2418	100	125	16	15	12.5
U00036	100	130	10	9	15.0
U3048	100	130	13	12	15.0
U3011	100	130	16	15	15.0

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U7076	101.6	114.3	10.52	9.52	6.5
U00224	104.77	130.17	17.27	16.27	13.0
U3001	105	120	9	8	7.5
U9964	105	120	10	9	7.5
U6666	105	120	11	10	7.5
U0405	105	120	13	12	7.5
U2181	105	120	16	15	7.5
U2114	105	125	11	10	10.0
U2032	105	125	13	12	10.0
U5282	105	125	15	14	10.0
U2462	105	125	16	15	10.0
U4598	106	120	9.5	8.5	7.0
U5007	107.95	127	13.7	12.7	10.0
U5001	107.95	133.35	13.7	12.7	13.0
U4995	108.36	123.82	15.28	14.28	8.0
U9951	110	120	10	9	5.0
U7652	110	122	10	9	6.0
U5579	110	125	10	9	7.5
U6229	110	125	11	10	7.5
U2931	110	125	13	12	7.5
U2552	110	125	16	15	7.5
U6668	110	130	11	10	10.0
U2091	110	130	13	12	10.0
U2144	110	130	16	15	10.0
U0970	110	130	19	18	10.0
U00022	110	135	10	9	12.5
U00026	110	135	14	13	12.5
U00037	110	140	10	9	15.0
U6673	110	140	13	12	15.0
U6839	110	140	16	15	15.0
U2317	112	125	10	9	6.5
U2932	113	125	10.5	9.5	6.0
U5005	114.3	133.35	10.52	9.52	10.0
U5926	115	125	16	15	5.0
U5925	115	126	16	15	5.5
U7658	115	127	10	9	6.0
U7752	115	130	10	9	7.5
U5285	115	130	11	10	7.5
U6669	115	130	13	12	7.5
U2127	115	130	16	15	7.5
U9985	115	135	10	9	10.0
U3474	115	135	13	12	10.0
U2619	115	135	16	15	10.0
U00024	115	140	10	9	12.5
U6671	115	140	13	12	12.5
U3049	115	140	16	15	12.5
U7344	115	140	19	18	12.5
U6672	115	145	13	12	15.0
U6679	115	145	16	15	15.0
U3822	115	145	19	18	15.0
U9950	117	127	9	8	5.0
U6660	118	133	10	9	7.5
U2175	120	130	5.7	5	5.0
U9949	120	130	10	9	5.0

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UNP Metric Size

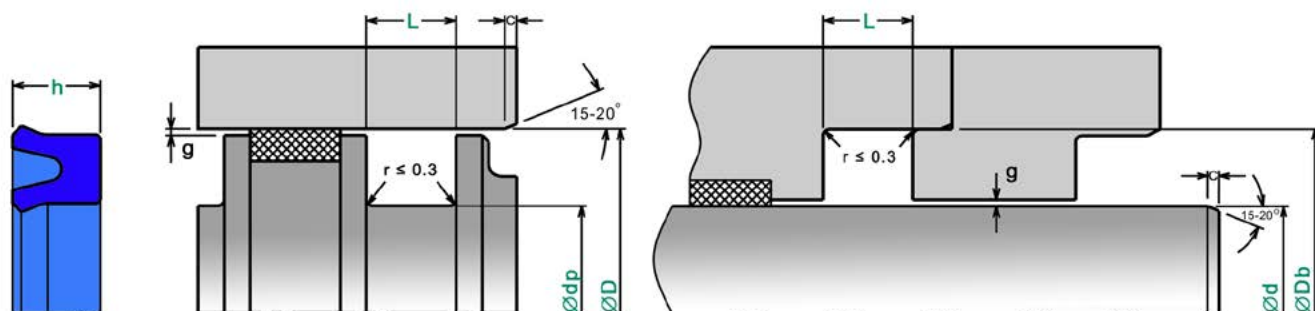
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U5923	120	130	15	14	5.0
U7649	120	132	10	9	6.0
U9963	120	135	10	9	7.5
U6670	120	135	13	12	7.5
U2265	120	135	16	15	7.5
U2383	120	140	11	10	10.0
U2092	120	140	13	12	10.0
U2451	120	140	16	15	10.0
U3909	120	140	17	16	10.0
U7383	120	145	16	15	12.5
U2274	120	150	16	15	15.0
U7657	123	135	10	9	6.0
U5632	125	135	11	10	5.0
U7653	125	137	10	9	6.0
U2318	125	140	10	9	7.5
U6933	125	140	11	10	7.5
U2125	125	140	12	11	7.5
U3114	125	140	13	12	7.5
U2143	125	140	16	15	7.5
U8064	125	141	19	18	8.0
U9983	125	145	10	9	10.0
U5658	125	145	13	12	10.0
U2337	125	145	16	15	10.0
U00025	125	150	10	9	12.5
U3265	125	150	13	12	12.5
U6677	125	150	16	15	12.5
U00038	125	155	13	12	15.0
U1790	125	155	16	15	15.0
U9948	130	140	10	9	5.0
U3325	130	145	10	9	7.5
U6678	130	145	13	12	7.5
U9962	130	145	14	13	7.5
U4964	130	145	16	15	7.5
U9984	130	150	10	9	10.0
U00608	130	150	11	10	10.0
U6100	130	150	13	12	10.0
U2126	130	150	16	15	10.0
U1279	130	150	17	16	10.0

UNP Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U00023	130	155	10	9	12.5
U4990	130	160	16	15	15.0
U6375	130	160	19	18	15.0
U9956	135	145	9	8	5.0
U9790	135	145	11	10	5.0
U5791	135	150	10	9	7.5
U6676	135	150	13	12	7.5
U3502	135	150	16	15	7.5
U9994	135	155	9	8	10.0
U9993	135	155	13	12	10.0
U2202	135	155	16	15	10.0
U00039	135	160	10	9	12.5
U00040	135	160	13	12	12.5
U3050	135	160	16	15	12.5
U3326	136	150	10	9	7.0
U4773	139.7	155.57	11.49	10.49	8.0
U2170	140	150	5.7	5	5.0
U9938	140	150	10	9	5.0
U0693	140	150	16	15	5.0
U8355	140	152	10	9	6.0
U4266	140	155	10	9	7.5
U2958	140	155	16	15	7.5
U5634	140	160	7	6	10.0
U7107	140	160	10	9	10.0
U6656	140	160	11	10	10.0
U2340	140	160	13	12	10.0
U2826	140	160	16	15	10.0
U1281	140	160	17	16	10.0
U00041	140	165	10	9	12.5
U6675	140	165	13	12	12.5
U0245	140	165	16	15	12.5
U00042	140	170	10	9	15.0
U00044	140	170	13	12	15.0
U0578	140	170	16	15	15.0
U00431	140	170	18	17	15.0
U6685	140	170	19	18	15.0
U2863	145	160	10	9	7.5
U9953	145	160	16	15	7.5
U9991	145	165	10	9	10.0
U6686	145	165	13	12	10.0
U2370	145	165	16	15	10.0
U00045	145	170	10	9	12.5
U00046	145	170	13	12	12.5
U3583	146	160	10	9	7.0
U9990	150	170	10	9	10.0
U2339	150	170	16	15	10.0
U1283	150	170	17	16	10.0
U4529	150	170	19	18	10.0
U00033	150	180	10	9	15.0
U00047	150	180	13	12	15.0
U0579	150	180	16	15	15.0
U6909	150	180	19	18	15.0
U8066	155	170	10	9	7.5
U2829	155	175	16	15	10.0

3.1

UNP Metric Size

Ordering dimensions in green

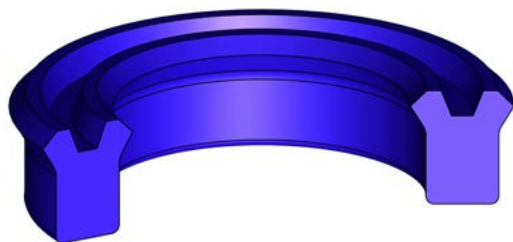
NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U00048	155	180	13	12	12.5
U1785	155	180	16	15	12.5
U4001	155	180	16	15	12.5
U3651	160	175	8.5	7.5	7.5
U6196	160	175	9	8	7.5
U1284	160	175	10.5	9.5	7.5
U8273	160	180	10	9	10.0
U2341	160	180	16	15	10.0
U00049	160	185	13	12	12.5
U4002	160	185	16	15	12.5
U1285	160	185	17	16	12.5
U2854	160	190	16	15	15.0
U3543	160	190	20	19	15.0
U8081	160	200	19	18	20.0
U6980	165	180	10	9	7.5
U9998	165	185	13	12	10.0
U9997	165	185	16	15	10.0
U4003	165	190	16	15	12.5
U4861	165.10	190.50	15.28	14.28	13.0
U8356	168	180	10	9	6.0
U9393	170	185	10	9	7.5
U9988	170	190	10	9	10.0
U2611	170	190	16	15	10.0
U00043	170	195	13	12	12.5
U4988	170	195	16	15	12.5
U6654	170	200	16	15	15.0
U2830	170	200	19	18	15.0
U5751	171.45	184.15	10.13	9.13	6.5
U9784	175	195	13	12	10.0
U00011	175	195	16	15	10.0
U2955	175	200	16	15	12.5
U4860	177.80	203.20	13.7	12.70	13.0
U00010	180	200	10	9	10.0
U7642	180	200	13	12	10.0
U2338	180	200	16	15	10.0
U7850	180	200	19	18	10.0
U00051	180	205	13	12	12.5
U1286	180	205	17	16	12.5

UNP

Metric Size

3.1

[illegible]



UNP1

PRODUCT DESCRIPTION

Seal designed with symmetric lips can be used for piston and rod application.

TECHNICAL DATA

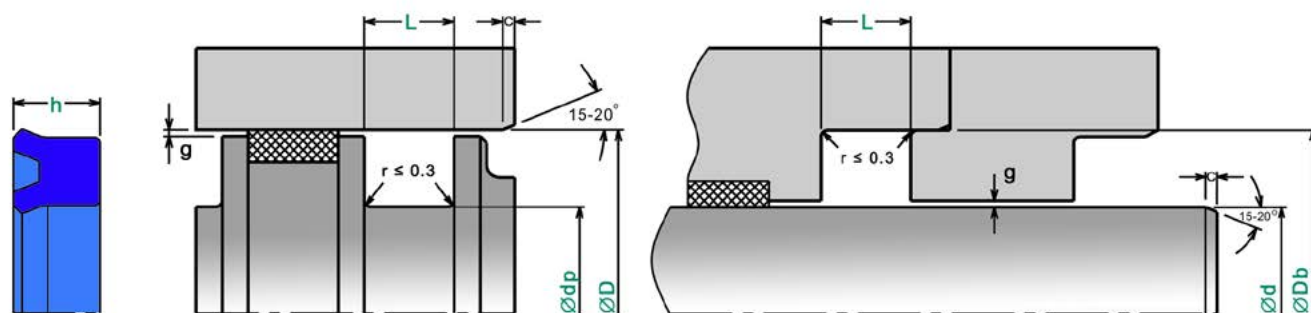
MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 40 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-40°C, +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

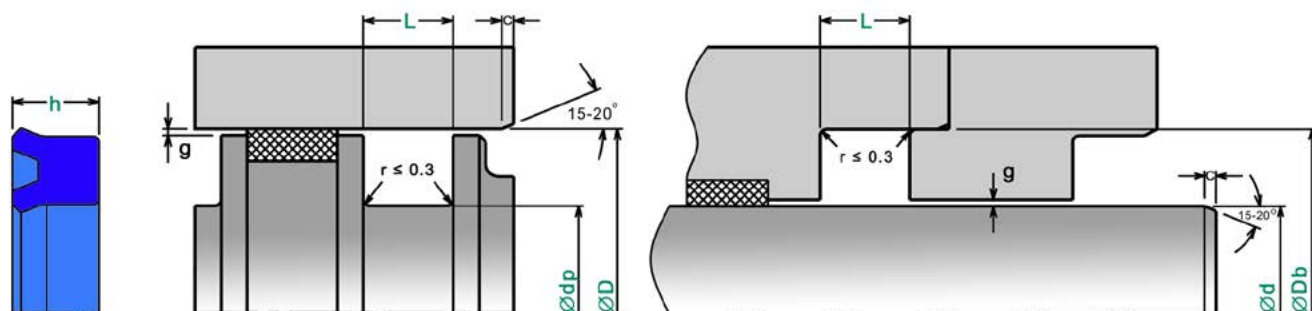
UNP1 Metric Size



Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U2909	6	14	6.4	5.7	4.0
U2910	8	16	6.4	5.7	4.0
U2912	10	18	6.4	5.7	4.0
U3437	12	18	5.2	4.5	3.0
U2913	12	20	6.2	5.5	4.0
U6108	14	22	6.2	5.5	4.0
U6122	16	22	4.7	4	3.0
U2993	16	24	7	6	4.0
U6012	18	25	5.8	5.1	3.5
U4203	18	26	7	6	4.0
U6379	18	28	6.5	5.8	5.0
U6137	18	28	8	7	5.0
U3010	20	26	5.7	5	3.0
U6138	20	26	6.2	5.5	3.0
U6139	20	27	6.4	5.7	3.5
U00430	20	28	6.4	5.7	4.0
U2778	20	28	6.5	5.8	4.0
U3562	20	28	7	6	4.0
U7095	20	28	8	7	4.0
U6005	20	30	8.2	7.2	5.0
U6140	20	30	9	8	5.0
U6109	20	30	11	10	5.0
U6107	21	27	4.7	4	3.0
U00351	22	28	5.2	4.5	3.0
U6105	22	28	5.7	5	3.0
U6104	22	29	5.7	5	3.5
U1218	22	30	7	6	4.0
U4445	22	30	8	7	4.0
U4080	22	32	8	7	5.0
U6103	22	32	9	8	5.0
U1149	24	30	9	8	3.0
U4023	24	32	7	6	4.0
U5989	24	34	7	6	5.0
U6102	25	32	4.7	4	3.5
U4991	25	33	6.5	5.8	4.0
U3289	25	33	7	6	4.0
U2121	25	33	9	8	4.0
U6110	25	35	5.7	5	5.0

3.2

UNP1 Metric Size

Ordering dimensions in green

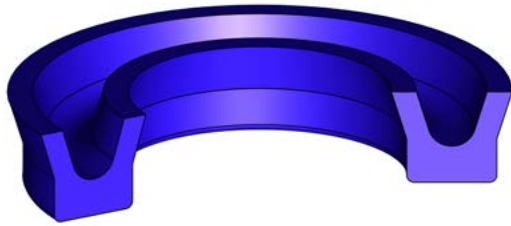
NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U6111	25	35	8	7	5.0
U6112	25	35	9	8	5.0
U6113	25	35	10	9	5.0
U6114	25	35	11	10	5.0
U6115	25	36	5.7	5	5.5
U6116	25	38	11	10	6.5
U3305	28	36	6.5	5.8	4.0
U2084	28	36	9	8	4.0
U6117	28	38	8	7	5.0
U6124	30	38	6.5	5.8	4.0
U2606	30	38	9	8	4.0
U2076	30	40	8	7	5.0
U5962	30	40	11	10	5.0
U5969	32	40	6.5	5.8	4.0
U2541	32	40	7	6	4.0
U2230	32	40	9	8	4.0
U5967	32	42	8	7	5.0
U2701	32	42	9	8	5.0
U5966	32	42	11	10	5.0
U2108	35	43	7	6	4.0
U2607	35	45	8	7	5.0
U5968	35	45	11	10	5.0
U00551	36	44	6.4	5.7	4.0
U2049	36	44	7	6	4.0
U00684	36	44	8	7	4.0
U2208	36	44	9	8	4.0
U5961	36	46	8	7	5.0
U5954	36	46	11	10	5.0
U2543	38	45	7	6	3.5
U3282	40	48	6.5	5.8	4.0
U1655	40	48	7	6	4.0
U2933	40	48	9	8	4.0
U5987	40	50	7	6	5.0
U5953	40	50	8	7	5.0
U3551	40	50	11	10	5.0
U5947	42	50	7	6	4.0
U8012	42	52	9.5	8.5	5.0
U5946	42	53	10	9	5.5

UNP1 Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U5945	45	53	7	6	4.0
U2615	45	53	9	8	4.0
U5944	45	53	11	10	4.0
U2702	45	55	8	7	5.0
U5739	45	55	11	10	5.0
U5943	46	54	9	8	4.0
U2324	50	58	9	8	4.0
U2039	50	60	8	7	5.0
U3985	50	60	11	10	5.0
U5942	50	62	9	8	6.0
U5951	50	62	11	10	6.0
U5952	50	65	11	10	7.5
U2664	55	63	9	8	4.0
U5940	55	65	8.2	7.2	5.0
U3417	55	65	8.5	7.5	5.0
U1219	55	65	11	10	5.0
U3935	56	66	7.5	6.5	5.0
U5955	56	66	11	10	5.0
U2544	56	71	9.5	8.5	7.5
U00365	59.5	67.5	9	8	4.0
U3478	60	68	7	6	4.0
U4013	60	68	9	8	4.0
U2545	60	68	14	13	4.0
U3015	60	70	8	7	5.0
U3207	60	70	8.2	7.2	5.0
U2665	60	70	8.5	7.5	5.0
U5986	60	70	11	10	5.0
U5985	60	70	13	12	5.0
U0423	60	72	10	9	6.0
U5995	61	69	9	8	4.0
U1691	63	71	9	8	4.0
U5924	63	75	11	10	6.0
U00504	64	72	9	8	4.0
U6026	65	73	9	8	4.0
U6025	65	75	13	12	5.0
U1553	65	77	10	9	6.0
U00436	66	80	11	10	7.0
U6028	68	76	9	8	4.0
U6029	70	78	9	8	4.0
U6035	70	80	8	7	5.0
U6036	70	80	11	10	5.0
U6037	70	80	13	12	5.0
U3558	70	82	10	9	6.0
U3112	71	79	9	8	4.0
U00437	72	78	7	6	3.0
U6010	72	78	7.5	6.5	3.0
U6038	75	85	8.2	7.2	5.0
U6054	75	85	13	12	5.0
U6061	76	84	9	8	4.0
U6062	78	86	14	13	4.0
U6055	78	93	11.5	10.5	7.5
U2918	79.75	85.99	7.5	6.5	3.5
U9677	80	88	9	8	4.0
U5939	80	90	8.2	7.2	5.0

3.2

126 • NAK SEALING SOLUTIONS



CNP

PRODUCT DESCRIPTION

Seal designed with symmetric lips can be used for piston and rod application.

TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Pressure p	≤ 20 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-40°C, +100°C

DESIGN NOTES

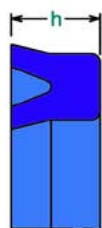
Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	140 Bar	210 Bar	350 Bar
Profile dimension	0.25 mm	0.15 mm	0.08 mm

Please contact our customer service for the dimensions out of our stock list.
service@mail.nak.com.tw

CNP

Metric Size



Ordering dimensions in green

3.3



UNR

PRODUCT DESCRIPTION

Seal designed with symmetric lips can be used for piston and rod application.

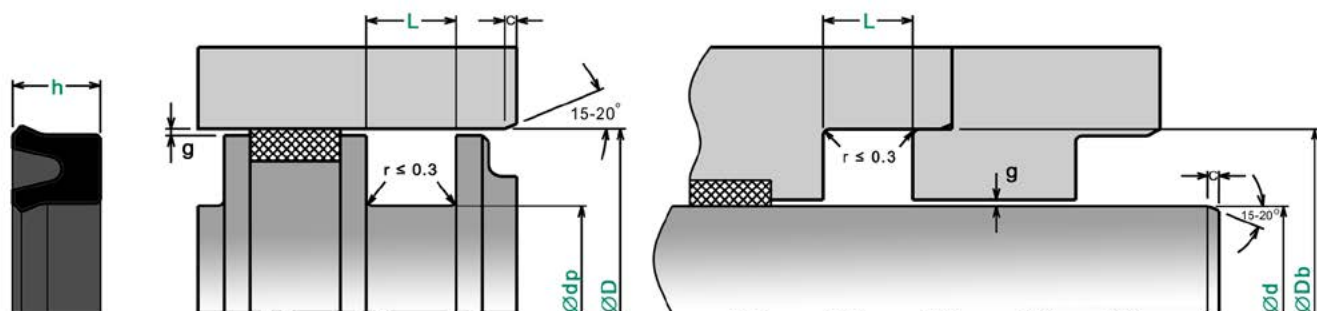
TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 16 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-35°C, +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	1.6 µm	6.3 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	1.6 µm	6.3 µm
Groove flanks	L	3.2 µm	12.5 µm

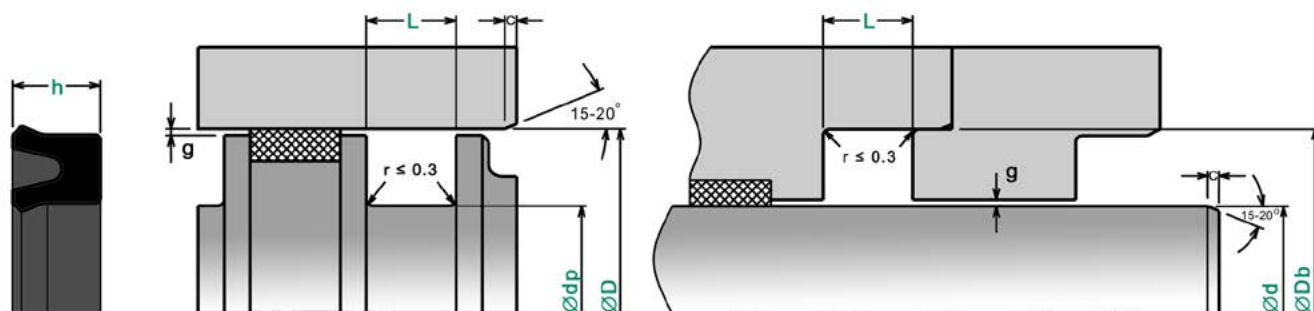
Admissible gap dimension			
	50 Bar	100 Bar	140 Bar
Profile dimension	0.3 mm	0.26 mm	0.15 mm

UNR Metric Size


Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U00273	4	8	3.7	3	2.0
U8117	4	12	4.7	4	4.0
U8118	4.76	9.53	3.08	2.38	2.5
U1797	5	12	5.7	5	3.5
U00321	6	10	4.7	4	2.0
U1520	6	12	4.7	4	3.0
U8113	6	16	5.7	5	5.0
U9777	6	16	9	8	5.0
U8436	6.1	10.8	3.16	2.46	2.5
U7443	6.35	11.11	3.87	3.17	2.5
U7444	7.93	12.7	5.46	4.76	2.5
U00316	8	14	3.7	3	3.0
U8379	8	14	4.7	4	3.0
U5175	8	14	5.2	4.5	3.0
U00407	8	15	5.2	4.5	3.5
U00310	8	16	5.2	4.5	4.0
U8114	8	16	7	6	4.0
U8115	8	18	5.7	5	5.0
U8328	8	18	8.5	7.5	5.0
U9039	8	18	9	8	5.0
U5068	9	17	4.7	4	4.0
U00314	10	14	3.7	3	2.0
U5115	10	15	3.7	3	2.5
U00313	10	16	4.2	3.5	3.0
U0386	10	18	5.7	5	4.0
U5684	10	18	7	6	4.0
U5684	10	18	7	6	4.0
U00654	10	19	5.7	5	4.5
U8590	10	20	5.7	5	5.0
U9847	10	20	9	8	5.0
U00315	10	20	9	8	5.0
U00317	10	22	7	6	6.0
U9418	10	25	9	8	7.5
U5114	10.1	14.9	3.7	3	2.5
U7670	11	17.46	5.46	4.76	3.5
U00119	12	18	4.7	4	3.0
U9408	12	18	5.7	5	3.0
U00120	12	18	7	6	3.0

3.4

UNR Metric Size

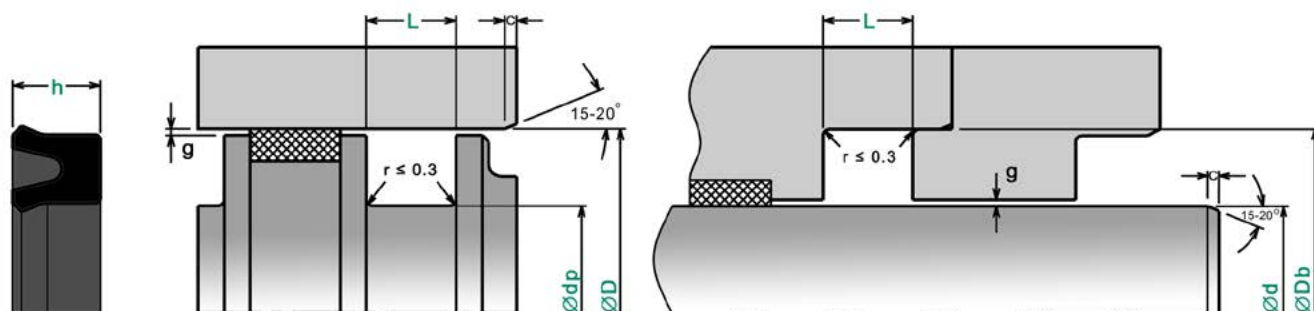
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U00562	12	18	9	8	3.0
U1138	12	19	4.7	4	3.5
U9726	12	19	5.7	5	3.5
U8116	12	20	4.7	4	4.0
U5916	12	20	5.7	5	4.0
U00318	12	20	7	6	4.0
U9038	12	20	8	7	4.0
U6040	12	22	6	5	5.0
U9040	12	22	8	7	5.0
U8658	12	22	11	10	5.0
U7191	12	24	7	6	6.0
U00329	12	25	8	7	6.5
U00522	12	26	5.7	5	7.0
U9017	12.7	19	3.9	3.2	3.5
U00536	13	20	5.7	5	3.5
U5069	13	23	5.7	5	5.0
U5071	13	25	7	6	6.0
U00523	14	20	4.7	4	3.0
U00531	14	20	5.7	5	3.0
U00533	14	20	7	6	3.0
U1802	14	22	5.7	5	4.0
U00319	14	22	5.7	5	4.0
U00320	14	22	7	6	4.0
22886	14	22.6	5.7	5	4.5
U8235	14	24	5.7	5	5.0
U00534	14	25	7	6	5.5
U00445	14	25.5	5.9	5.2	6.0
U00132	14	26	5.7	5	6.0
U5072	14	26	7	6	6.0
U00488	15	20	3.7	3	2.5
U00532	15	20	5.7	5	2.5
U00524	15	22	5.7	5	3.5
U7446	15	25	5.7	5	5.0
U00525	15	25	7	6	5.0
U6566	15	25	9	8	5.0
U9417	15	30	9	8	7.5
U7456	15	35	11	10	10.0
U9643	16	22	4.7	4	3.0

UNR Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U8231	16	24	4.7	4	4.0
U1803	16	24	5.7	5	4.0
U00543	16	24	5.7	5	4.0
U1803	16	24	5.7	5	4.0
U1803	16	24	5.7	5	4.0
U7445	16	24	6.2	5.5	4.0
U7445	16	24	6.2	5.5	4.0
U7622	16	26	5.7	5	5.0
U9398	16	26	9	8	5.0
U00131	16	28	5.7	5	6.0
U8589	16	28	7	6	6.0
U7461	17	25	4.7	4	4.0
U9406	17	27	5.7	5	5.0
U00535	18	24	5.7	5	3.0
U5747	18	24	7.2	6.2	3.0
U1937	18	25	5.7	5	3.5
U1937	18	25	5.7	5	3.5
U0745	18	26	5.7	5	4.0
U00450	18	26	5.7	5	4.0
U6962	18	26	6.2	5.5	4.0
U00491	18	27	5.7	5	4.5
U5748	18	28	6	5.3	5.0
03855	18	28	7	6	5.0
03855	18	28	7	6	5.0
U1990	18	28	9	8	5.0
U7628	18	40	11	10	11.0
U7300	19	25	5.7	5	3.0
U8313	19	25.5	6	5.3	3.5
U1936	19	27	5.7	5	4.0
U7455	19.05	31.76	7.35	6.35	6.5
U00526	20	25	5.7	5	2.5
U00121	20	26	5.7	5	3.0
U8377	20	26	7.5	6.5	3.0
U9404	20	27	5.7	5	3.5
U7507	20	28	4.7	4	4.0
U0716	20	28	5.7	5	4.0
U00449	20	28	5.7	5	4.0
U00537	20	28	8	7	4.0
U8055	20	30	5.7	5	5.0
U00680	20	30	7	6	5.0
U9753	20	30	8	7	5.0
U1991	20	30	9	8	5.0
U7508	20	32	5.7	5	6.0
U7514	20	32	7	6	6.0
U9037	20	32	8	7	6.0
U00530	20	32	9	8	6.0
U7192	20	40	11	10	10.0
U1993	20	40	13	12	10.0
U1568	22	30	5.7	5	4.0
U6520	22	30	7.5	6.5	4.0
U00538	22	30	9	8	4.0
U3827	22	31	5.7	5	4.5
U6039	22	32	5.7	5	5.0
U1995	22	32	7	6	5.0

3.4

UNR Metric Size

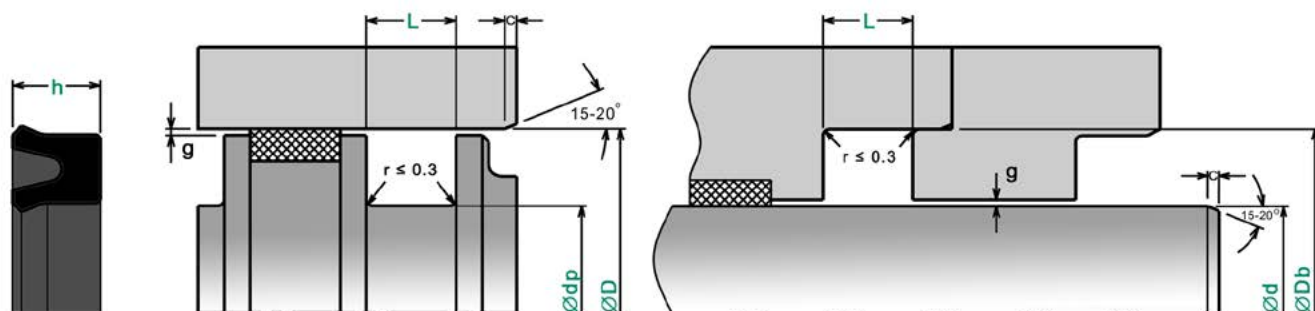
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U9427	22	32	8	7	5.0
U3009	22	32	9	8	5.0
U1996	22	40	13	12	9.0
U4603	22.4	30	5.7	5	4.0
U4603	22.4	30	5.7	5	4.0
U1938	22.4	32.4	9	8	5.0
U9982	22.7	30.5	5.7	5	4.0
U9965	23	30.5	5.7	5	4.0
U1701	23.5	31.5	5.7	5	4.0
U7462	24	32	4.7	4	4.0
U00527	24	32	5.7	5	4.0
U8368	24	34	5.7	5	5.0
U00489	25	31	5.2	4.5	3.0
U00529	25	32	4.7	4	3.5
U00487	25	32	5.7	5	3.5
U8236	25	33	4.7	4	4.0
U0715	25	33	5.7	5	4.0
U9399	25	33	6.2	5.5	4.0
U8961	25	35	5.46	4.76	5.0
U5811	25	35	5.7	5	5.0
U00322	25	35	7	6	5.0
U9401	25	35	8	7	5.0
U1997	25	35	9	8	5.0
U7457	25	37	7	6	6.0
U9751	25	38	8	7	6.5
U1998	25	38	11	10	6.5
U2385	25	40	11	10	7.5
U00323	25	45	11	10	10.0
U7458	26	38	7	6	6.0
U00490	27	35	5.7	5	4.0
U8056	27	40	9	8	6.5
U9752	28	34	5.7	5	3.0
U9750	28	34	7	6	3.0
U9773	28	34	8	7	3.0
U1934	28	35.5	5.7	5	4.0
U00122	28	36	5.7	5	4.0
U00128	28	36	6.2	5.5	4.0
U3811	28	38	5.7	5	5.0

UNR Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U9428	28	38	8	7	5.0
U8237	28	40	7	6	6.0
U7278	28	40	11	10	6.0
U7530	28	48	11	10	10.0
U9403	30	37	8	7	3.5
U9729	30	38	6.1	5.4	4.0
U00123	30	38	8	7	4.0
U8432	30	39	6.2	5.5	4.5
U7459	30	40	5.7	5	5.0
U0713	30	40	7	6	5.0
U0713	30	40	7	6	5.0
U0713	30	40	7	6	5.0
U8519	30	40	8	7	5.0
U8520	30	40	9	8	5.0
U9429	30	42	9	8	6.0
U8423	30	43	10	9	6.5
U4038	30	43	11	10	6.5
U00324	30	45	8	7	7.5
U1933	30	45	11	10	7.5
U8253	30	50	11	10	10.0
U9485	31.5	41.5	7	6	5.0
U8246	32	40	4.7	4	4.0
U7810	32	40	5.7	5	4.0
U9409	32	40	9	8	4.0
U00679	32	42	7	6	5.0
U7391	32	42	8	7	5.0
U6731	32	48	11	10	8.0
U7451	32	52	11	10	10.0
U7460	35	45	5.7	5	5.0
U2106	35	45	7	6	5.0
U7246	35	45	8	7	5.0
U8521	35	45	9	8	5.0
U1867	35	45	11	10	5.0
U8234	35	47	7	6	6.0
U1910	35	50	11	10	7.5
U8125	35.5	45	7	6	5.0
U8960	37.79	49.47	7.98	6.98	6.0
U00325	38	48	9	8	5.0
U7471	38	50	7	6	6.0
U0714	40	50	7	6	5.0
U7450	40	50	8	7	5.0
U8504	40	50	9	8	5.0
U8503	40	50	11	10	5.0
U8073	40	52	7	6	6.0
U5627	40	52.5	9.3	8.3	6.5
U1819	40	52.75	9.3	8.3	6.5
U1911	40	55	11	10	7.5
U7453	40	60	11	10	10.0
U2001	40	60	13	12	10.0
U1939	42	55	9	8	6.5
U7624	44	56	7	6	6.0
U7452	45	55	5.7	5	5.0
U1804	45	55	7	6	5.0
U8505	45	55	8	7	5.0

3.4

UNR Metric Size

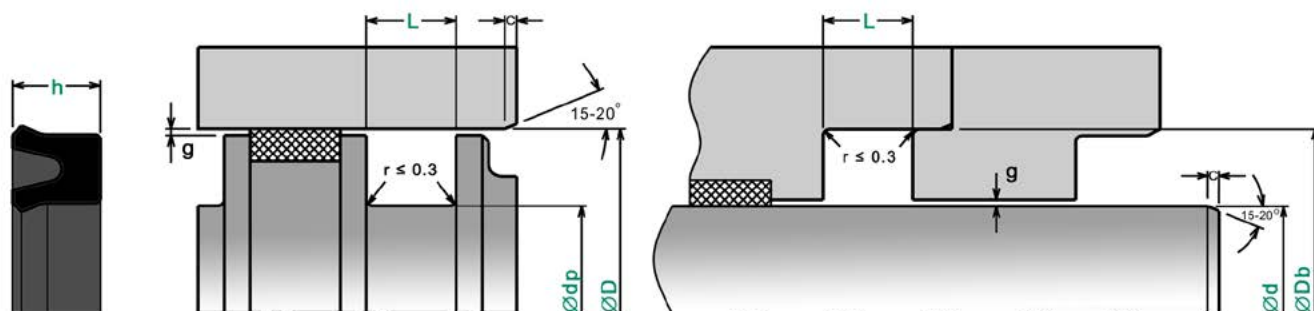
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U8508	45	55	9	8	5.0
U1796	45	55	11	10	5.0
U00071	45	56	8	7	5.5
U7618	45	57	7	6	6.0
U9410	45	57	9	8	6.0
U9411	45	57	11	10	6.0
U1912	45	60	11	10	7.5
U7531	46	56	5.7	5	5.0
U8252	46	58	7	6	6.0
U8244	47.63	63.50	8.43	7.43	8.0
U8245	48	60	7	6	6.0
U8240	49.21	69.85	10.52	9.52	10.5
U1389	50	60	7	6	5.0
U8507	50	60	8	7	5.0
U8506	50	60	11	10	5.0
U3866	50	63	9	8	6.5
U8515	50	63	11	10	6.5
U8369	50	65	8.5	7.5	7.5
U1913	50	65	11	10	7.5
U7620	52	62	8	7	5.0
U8250	52	72	11	10	10.0
U7432	53	63	5.7	5	5.0
U1390	53	63	7	6	5.0
U1390	53	63	7	6	5.0
U7433	53	65	7	6	6.0
U5073	54	70	13	12	8.0
U1818	55	65	7	6	5.0
U3121	55	70	10	9	7.5
U8517	55	70	11	10	7.5
U8513	55	70	13	12	7.5
U8329	56	66	7	6	5.0
U8283	57	68	7.5	6.5	5.5
U1391	60	70	7	6	5.0
U8512	60	70	8	7	5.0
U00655	60	70	8.2	7.2	5.0
U8374	60	75	8.5	7.5	7.5
U8577	60	75	11	10	7.5
U9416	60	80	9	8	10.0

UNR Metric Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U5081	60	80	11	10	10.0
U1808	60	80	13	12	10.0
U00124	63	73	7	6	5.0
U8372	65	75	5.7	5	5.0
U1805	65	75	7	6	5.0
U9414	65	78	11	10	6.5
U8241	65	80	8.5	7.5	7.5
U1935	65	80	11	10	7.5
U4909	65	85	11	10	10.0
U6952	70	80	5.7	5	5.0
U1392	70	80	7	6	5.0
U9779	70	80	9	8	5.0
U00326	70	80	11	10	5.0
U00125	70	83	10	9	6.5
U1813	70	85	10	9	7.5
U8576	70	85	11	10	7.5
U8578	70	85	13	12	7.5
U7497	70	90	11	10	10.0
U1809	70	90	13	12	10.0
U8514	70	95	10	9	12.5
U8579	70	95	11	10	12.5
U8580	70	95	13	12	12.5
U2376	71	80	7	6	4.5
U1806	75	85	7	6	5.0
U00327	75	90	10	9	7.5
U7426	75	95	11	10	10.0
U7415	75	100	16	15	12.5
U1807	80	90	7	6	5.0
U1814	80	95	10	9	7.5
U8586	80	95	11	10	7.5
U8585	80	95	13	12	7.5
U7427	80	100	11	10	10.0
U1810	80	100	13	12	10.0
U8483	82	102	11	10	10.0
U1393	85	95	7	6	5.0
U8511	85	95	8	7	5.0
U7434	85	100	8.5	7.5	7.5
U1895	85	100	10	9	7.5
U6567	85	105	13	12	10.0
U7284	88.7	101	7.5	6.5	6.5
U7623	90	100	7	6	5.0
U7763	90	100	8	7	5.0
U3122	90	105	10	9	7.5
U7496	90	110	11	10	10.0
U1811	90	110	13	12	10.0
U9689	90	110	16	15	10.0
U7606	90	120	16	15	15.0
U6640	95	110	10	9	7.5
U8523	95	115	13	12	10.0
U8593	100	115	10	9	7.5
U3123	100	115	10	9	7.5
U1812	100	120	13	12	10.0
U8532	100	120	16	15	10.0
U7417	100	125	13	12	12.5

3.4

UNR Metric Size


Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U9690	100	125	16	15	12.5
U7735	101.6	120.65	12.11	11.11	10.0
U3728	105	120	10	9	7.5
U7332	105	125	16	15	10.0
U2730	106	120	9.5	8.5	7.0
U2377	110	125	10	9	7.5
U8584	110	130	13	12	10.0
U1394	112	125	9.5	8.5	6.5
U4601	115	130	10	9	7.5
U8522	115	135	13	12	10.0
U8551	118	133	10	9	7.5
U8866	120	135	10	9	7.5
U1815	120	140	13	12	10.0
U1816	120	140	16	15	10.0
U7490	120	150	16	15	15.0
U3727	125	140	10	9	7.5
U3536	125	150	20	19	12.5
U8587	127	146.05	10.53	9.53	10.0
U3726	130	145	10	9	7.5
U9415	130	150	13	12	10.0
U8554	130	150	16	15	10.0
U7418	130	160	16	15	15.0
U7495	133.35	158.8	13.7	12.7	13.0
U00517	135	150	10	9	7.5
U8555	135	150	11	10	7.5
U9774	135	155	16	15	10.0
U1395	136	150	9.5	8.5	7.0
U4600	140	155	10	9	7.5
U00127	140	160	13	12	10.0
U8699	140	170	19	18	15.0
U3725	145	160	10	9	7.5
U8575	145	175	16	15	15.0
U7764	149	160	10	9	5.5
U8552	150	170	13	12	10.0
U7419	150	180	16	15	15.0
U8556	155	170	10	9	7.5
U8550	155	180	17	16	12.5
U00126	160	185	16	15	12.5

UNR

Metric Size

3.4



CNR

PRODUCT DESCRIPTION

Seal designed with symmetric lips can be used for piston and rod application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Pressure p	≤ 10 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-35°C, +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	1.6 µm	6.3 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	1.6 µm	6.3 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	50 Bar	100 Bar	140 Bar
Profile dimension	0.3 mm	0.26 mm	0.15 mm

Please contact our customer service for the dimensions out of our stock list.
service@mail.nak.com.tw

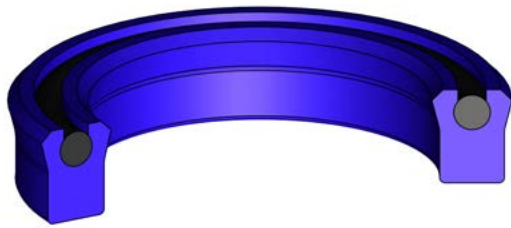
Metric Size



Ordering dimensions in green

3.5

[illegible]



HB

PRODUCT DESCRIPTION

A loaded U-packing with a fitted O-ring, and the O-ring provides extreme loads for effective sealing at low or zero pressure.

TECHNICAL DATA

MATERIAL	PU + NBR
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-30°C, +100°C

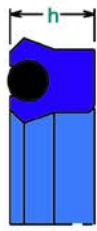
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	100 Bar	200 Bar	300 Bar
Profile dimension	0.36 mm	0.18 mm	0.09 mm

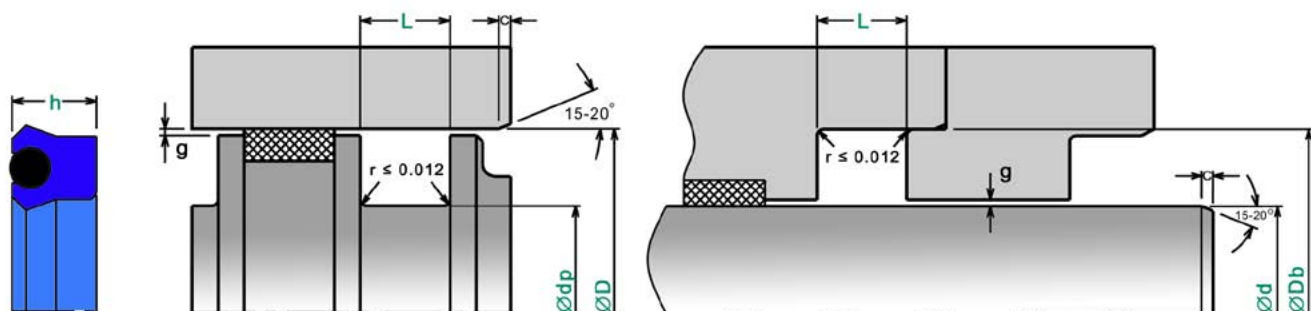
HB

Metric Size



Ordering dimensions in green

3.6

HB Inch Size


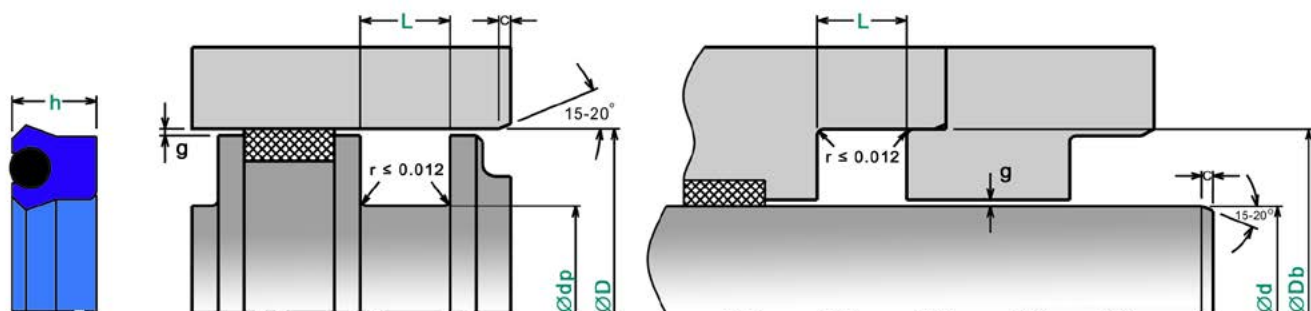
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U9440	0.187	0.437	0.289	0.250	0.138
U6506	0.250	0.500	0.289	0.250	0.138
U4757	0.250	0.624	0.215	0.187	0.197
U5058	0.312	0.562	0.289	0.250	0.138
U3499	0.375	0.625	0.153	0.125	0.138
U6966	0.375	0.625	0.289	0.250	0.138
U3647	0.375	0.749	0.215	0.187	0.197
U3931	0.375	0.749	0.289	0.250	0.197
U3868	0.437	0.687	0.289	0.250	0.138
U5031	0.437	0.811	0.215	0.187	0.197
U3396	0.500	0.750	0.153	0.125	0.138
U5217	0.500	0.750	0.289	0.250	0.138
U6485	0.500	0.875	0.351	0.312	0.197
U5655	0.625	0.875	0.153	0.125	0.138
U00291	0.625	0.875	0.215	0.187	0.138
U5608	0.625	0.875	0.289	0.250	0.138
U5288	0.625	1.000	0.351	0.312	0.197
U7605	0.625	1.125	0.289	0.250	0.256
U3941	0.687	0.937	0.289	0.250	0.138
U4770	0.750	1.000	0.153	0.125	0.138
U5152	0.750	1.000	0.215	0.187	0.138
U5654	0.750	1.000	0.289	0.250	0.138
U5461	0.750	1.125	0.351	0.312	0.197
U4512	0.750	1.500	0.414	0.375	0.394
U3527	0.812	1.186	0.351	0.312	0.197
U3634	0.875	1.125	0.153	0.125	0.138
U9864	0.875	1.125	0.215	0.187	0.138
U5785	0.875	1.125	0.289	0.250	0.138
U4606	0.875	1.249	0.215	0.187	0.197
U4192	0.875	1.375	0.414	0.375	0.256
U3374	0.937	1.437	0.289	0.250	0.256
U5030	1.000	1.250	0.153	0.125	0.138
U5988	1.000	1.250	0.184	0.156	0.138
U5023	1.000	1.250	0.215	0.187	0.138
U5204	1.000	1.250	0.289	0.250	0.138
U3497	1.000	1.312	0.184	0.156	0.157
U4873	1.000	1.312	0.258	0.230	0.157
U4911	1.000	1.374	0.215	0.187	0.197

HB Inch Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U4166	1.000	1.374	0.289	0.250	0.197
U7585	1.000	1.374	0.351	0.312	0.197
U9222	1.000	1.374	0.414	0.375	0.197
U6417	1.000	1.500	0.289	0.250	0.256
U4912	1.000	1.500	0.414	0.375	0.256
U3618	1.000	1.624	0.351	0.312	0.315
U3487	1.000	1.750	0.414	0.375	0.394
U3951	1.062	1.562	0.414	0.375	0.256
U9601	1.125	1.375	0.164	0.125	0.138
U8975	1.125	1.375	0.215	0.187	0.138
U5057	1.125	1.375	0.289	0.250	0.138
U9030	1.125	1.499	0.289	0.250	0.197
U9262	1.125	1.499	0.351	0.312	0.197
U4589	1.125	1.499	0.414	0.375	0.197
U5375	1.125	1.500	0.215	0.187	0.197
U3612	1.125	1.625	0.289	0.250	0.256
U3386	1.125	1.625	0.414	0.375	0.256
U9426	1.187	1.561	0.289	0.250	0.197
U8535	1.250	1.500	0.215	0.187	0.138
U9334	1.250	1.624	0.215	0.187	0.197
U3501	1.250	1.624	0.289	0.250	0.197
U7086	1.250	1.624	0.351	0.312	0.197
U5440	1.250	1.625	0.414	0.375	0.197
U5027	1.250	1.750	0.289	0.250	0.256
U5292	1.250	1.750	0.414	0.375	0.256
U4372	1.250	1.874	0.414	0.375	0.315
U3500	1.250	1.874	0.539	0.500	0.315
U5601	1.250	2.000	0.539	0.500	0.394
U3913	1.250	2.000	0.664	0.625	0.394
U6899	1.312	1.562	0.153	0.125	0.138
U3942	1.312	1.562	0.414	0.375	0.138
U9735	1.375	1.625	0.153	0.125	0.138
U9441	1.375	1.625	0.215	0.187	0.138
U3128	1.375	1.625	0.289	0.250	0.138
U4920	1.375	1.749	0.215	0.187	0.197
U4031	1.375	1.749	0.414	0.375	0.197
U5808	1.375	1.750	0.289	0.250	0.197
U4982	1.375	1.750	0.351	0.312	0.197
U3365	1.375	1.875	0.289	0.250	0.256
U5171	1.375	1.875	0.414	0.375	0.256
U3615	1.375	1.999	0.414	0.375	0.315
U6461	1.375	2.000	0.539	0.500	0.315
U5141	1.417	1.732	0.351	0.312	0.177
U3943	1.437	1.687	0.289	0.250	0.138
U5168	1.500	1.750	0.153	0.125	0.138
U3934	1.500	1.750	0.215	0.187	0.138
U3129	1.500	1.750	0.289	0.250	0.138
U7000	1.500	1.812	0.351	0.312	0.157
U5139	1.500	1.874	0.215	0.187	0.197
U5024	1.500	1.874	0.289	0.250	0.197
U3134	1.500	1.874	0.351	0.312	0.197
U4799	1.500	1.874	0.414	0.375	0.197
U5436	1.500	2.000	0.289	0.250	0.256
U5620	1.500	2.000	0.351	0.312	0.256

3.6

HB Inch Size


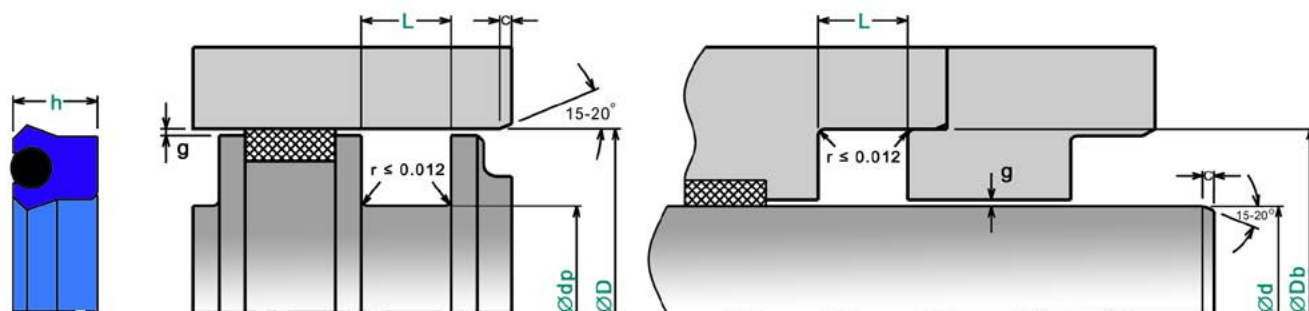
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U5409	1.500	2.000	0.414	0.375	0.256
U7911	1.500	2.000	0.539	0.500	0.256
U3800	1.500	2.124	0.351	0.312	0.315
U3616	1.500	2.124	0.414	0.375	0.315
U6388	1.500	2.125	0.539	0.500	0.315
U5795	1.500	2.500	0.789	0.750	0.512
U6065	1.625	1.875	0.289	0.250	0.138
U3404	1.625	1.999	0.289	0.250	0.197
U5656	1.625	2.000	0.215	0.187	0.197
U5496	1.625	2.000	0.351	0.312	0.197
U5605	1.625	2.000	0.414	0.375	0.197
U6513	1.625	2.125	0.414	0.375	0.256
U3594	1.625	2.249	0.539	0.500	0.315
U7177	1.625	2.250	0.414	0.375	0.315
U6508	1.750	2.000	0.289	0.250	0.138
U5059	1.750	2.124	0.215	0.187	0.197
U3524	1.750	2.124	0.289	0.250	0.197
U3368	1.750	2.124	0.351	0.312	0.197
U3586	1.750	2.124	0.414	0.375	0.197
U3613	1.750	2.250	0.289	0.250	0.256
U9220	1.750	2.250	0.351	0.312	0.256
U9369	1.750	2.250	0.414	0.375	0.256
U3387	1.750	2.374	0.414	0.375	0.315
U5445	1.750	2.375	0.539	0.500	0.315
U4075	1.750	2.500	0.414	0.375	0.394
U5463	1.750	2.500	0.664	0.625	0.394
U4755	1.875	2.125	0.153	0.125	0.138
U4167	1.875	2.249	0.215	0.187	0.197
U4593	1.875	2.249	0.351	0.312	0.197
U6395	1.875	2.250	0.414	0.375	0.197
U3598	1.875	2.375	0.414	0.375	0.256
U5432	1.875	2.500	0.414	0.375	0.315
U3969	1.937	2.311	0.320	0.281	0.197
U9024	2.000	2.374	0.289	0.250	0.197
U4798	2.000	2.374	0.351	0.312	0.197
U5082	2.000	2.374	0.414	0.375	0.197
U5169	2.000	2.375	0.215	0.187	0.197
U3382	2.000	2.500	0.289	0.250	0.256

HB Inch Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U5890	2.000	2.500	0.351	0.312	0.256
U4200	2.000	2.500	0.414	0.375	0.256
U3620	2.000	2.624	0.414	0.375	0.315
U3535	2.000	2.624	0.539	0.500	0.315
U3633	2.000	2.624	0.601	0.562	0.315
U5677	2.000	2.625	0.351	0.312	0.315
U6900	2.125	2.375	0.289	0.250	0.138
U9782	2.125	2.499	0.289	0.250	0.197
U9430	2.125	2.499	0.351	0.312	0.197
U5408	2.125	2.500	0.215	0.187	0.197
U5574	2.125	2.500	0.414	0.375	0.197
U3375	2.125	2.625	0.289	0.250	0.256
U6503	2.125	2.625	0.414	0.375	0.256
U3690	2.125	3.125	0.539	0.500	0.512
U3495	2.250	2.500	0.289	0.250	0.138
U3614	2.250	2.624	0.215	0.187	0.197
U3957	2.250	2.624	0.351	0.312	0.197
U5350	2.250	2.625	0.414	0.375	0.197
U5609	2.250	2.750	0.414	0.375	0.256
U00589	2.250	2.874	0.414	0.375	0.315
U6384	2.250	2.875	0.539	0.500	0.315
U3463	2.250	3.000	0.414	0.375	0.394
U6472	2.250	3.000	0.664	0.625	0.394
U3971	2.375	2.749	0.215	0.187	0.197
U3972	2.375	2.749	0.305	0.266	0.197
U5798	2.375	2.750	0.289	0.250	0.197
U5647	2.375	2.750	0.414	0.375	0.197
U3383	2.375	2.875	0.414	0.375	0.256
U3681	2.375	2.999	0.539	0.500	0.315
U5607	2.375	3.000	0.351	0.312	0.315
U4076	2.375	3.125	0.414	0.375	0.394
U3457	2.500	2.874	0.215	0.187	0.197
U4570	2.500	2.874	0.351	0.312	0.197
U5648	2.500	2.875	0.289	0.250	0.197
U5346	2.500	2.875	0.414	0.375	0.197
U5061	2.500	3.000	0.289	0.250	0.256
U3139	2.500	3.000	0.351	0.312	0.256
U5062	2.500	3.000	0.414	0.375	0.256
U4987	2.500	3.000	0.539	0.500	0.256
U7794	2.500	3.125	0.351	0.312	0.315
U7604	2.500	3.125	0.414	0.375	0.315
U6493	2.500	3.125	0.539	0.500	0.315
U3364	2.500	3.250	0.414	0.375	0.394
U5460	2.500	3.250	0.539	0.500	0.394
U3914	2.500	3.500	0.539	0.500	0.512
U3397	2.625	2.999	0.320	0.281	0.197
U5403	2.625	3.000	0.215	0.187	0.197
U6467	2.625	3.000	0.289	0.250	0.197
U6500	2.625	3.000	0.414	0.375	0.197
U6514	2.625	3.125	0.414	0.375	0.256
U5650	2.625	3.375	0.664	0.625	0.394
U9520	2.750	3.000	0.414	0.375	0.138
U3458	2.750	3.124	0.320	0.281	0.197
U6352	2.750	3.125	0.414	0.375	0.197

3.6

HB Inch Size


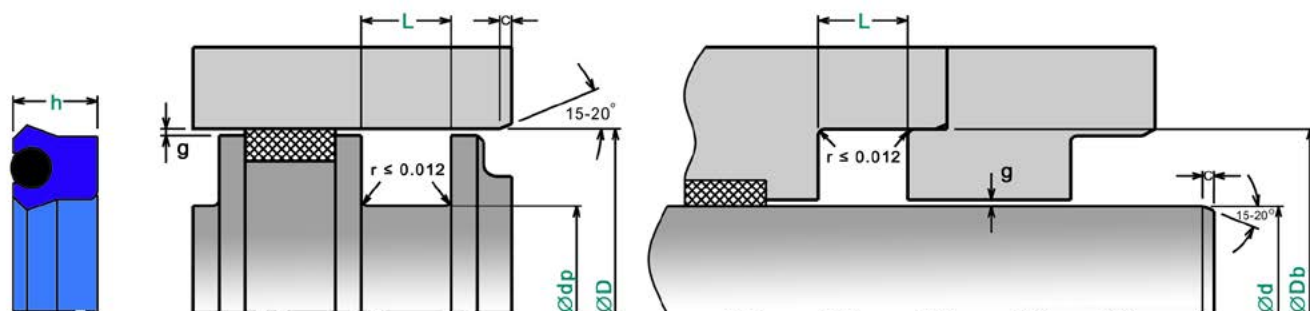
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U5657	2.750	3.250	0.289	0.250	0.256
U5334	2.750	3.250	0.414	0.375	0.256
U5120	2.750	3.250	0.539	0.500	0.256
U3956	2.750	3.374	0.351	0.312	0.315
U5596	2.750	3.375	0.414	0.375	0.315
U6494	2.750	3.375	0.539	0.500	0.315
U5573	2.750	3.500	0.539	0.500	0.394
U6497	2.750	3.500	0.664	0.625	0.394
U4588	2.875	3.249	0.289	0.250	0.197
U5336	2.875	3.250	0.414	0.375	0.197
U4590	2.875	3.375	0.414	0.375	0.256
U8348	2.875	3.500	0.351	0.312	0.315
U6495	2.875	3.500	0.539	0.500	0.315
U4645	3.000	3.250	0.289	0.250	0.138
U3389	3.000	3.374	0.215	0.187	0.197
U3135	3.000	3.374	0.351	0.312	0.197
U9744	3.000	3.375	0.289	0.250	0.197
U5455	3.000	3.375	0.414	0.375	0.197
U3693	3.000	3.500	0.289	0.250	0.256
U5476	3.000	3.500	0.414	0.375	0.256
U5442	3.000	3.500	0.539	0.500	0.256
U3385	3.000	3.624	0.539	0.500	0.315
U9219	3.000	3.750	0.539	0.500	0.394
U4471	3.000	3.750	0.601	0.562	0.394
U8538	3.000	3.750	0.664	0.625	0.394
U3983	3.000	4.000	0.539	0.500	0.512
U8977	3.125	3.499	0.215	0.187	0.197
U4984	3.125	3.499	0.414	0.375	0.197
U00200	3.125	3.500	0.289	0.250	0.197
U6064	3.125	3.500	0.351	0.312	0.197
U4032	3.125	3.625	0.414	0.375	0.256
U6066	3.125	3.750	0.539	0.500	0.315
U4072	3.250	3.500	0.167	0.139	0.138
U3136	3.250	3.624	0.414	0.375	0.197
U6492	3.250	3.750	0.414	0.375	0.256
U4808	3.250	3.750	0.539	0.500	0.256
U6337	3.250	3.875	0.539	0.500	0.315
U3141	3.250	4.000	0.539	0.500	0.394

HB Inch Size

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U4870	3.250	4.000	0.601	0.562	0.394
U6382	3.250	4.000	0.664	0.625	0.394
U4246	3.250	4.250	0.539	0.500	0.512
U3449	3.375	3.749	0.215	0.187	0.197
U3371	3.375	3.749	0.289	0.250	0.197
U4591	3.375	3.875	0.414	0.375	0.256
U3959	3.375	3.999	0.664	0.625	0.315
U5511	3.375	4.000	0.351	0.312	0.315
U6496	3.375	4.000	0.539	0.500	0.315
U9014	3.500	3.875	0.414	0.375	0.197
U5065	3.500	4.000	0.289	0.250	0.256
U3974	3.500	4.000	0.351	0.312	0.256
U3695	3.500	4.000	0.414	0.375	0.256
U9031	3.500	4.000	0.539	0.500	0.256
U5790	3.500	4.000	0.664	0.625	0.256
U3686	3.500	4.250	0.539	0.500	0.394
U3679	3.500	4.250	0.601	0.562	0.394
U6389	3.500	4.250	0.664	0.625	0.394
U9446	3.625	3.875	0.215	0.187	0.138
U3450	3.625	3.999	0.289	0.250	0.197
U3082	3.625	3.999	0.414	0.375	0.197
U9612	3.625	4.000	0.215	0.187	0.197
U9234	3.625	4.250	0.539	0.500	0.315
U3961	3.625	4.375	0.414	0.375	0.394
U5434	3.625	4.375	0.664	0.625	0.394
U3723	3.750	4.250	0.289	0.250	0.256
U5892	3.750	4.250	0.414	0.375	0.256
U4128	3.750	4.250	0.601	0.562	0.256
U3143	3.750	4.500	0.664	0.625	0.394
U3716	3.875	4.249	0.414	0.375	0.197
U8927	4.000	4.374	0.289	0.250	0.197
U4379	4.000	4.374	0.414	0.375	0.197
U9484	4.000	4.500	0.289	0.250	0.256
U5443	4.000	4.500	0.414	0.375	0.256
U00170	4.000	4.500	0.539	0.500	0.256
U6392	4.000	4.500	0.601	0.562	0.256
U9867	4.000	4.624	0.539	0.500	0.315
U6353	4.000	4.625	0.601	0.562	0.315
U9043	4.000	4.750	0.539	0.500	0.394
U3734	4.000	4.750	0.664	0.625	0.394
U3907	4.000	4.874	0.664	0.625	0.453
U4594	4.000	5.000	0.539	0.500	0.512
U3137	4.125	4.499	0.414	0.375	0.197
U3731	4.125	4.625	0.601	0.562	0.256
U00133	4.125	4.749	0.601	0.562	0.315
U3733	4.125	4.875	0.789	0.750	0.394
U4753	4.250	4.624	0.320	0.281	0.197
U6334	4.250	4.750	0.414	0.375	0.256
U9235	4.250	4.750	0.601	0.562	0.256
U3962	4.250	5.000	0.414	0.375	0.394
U9053	4.250	5.000	0.539	0.500	0.394
U4568	4.250	5.000	0.601	0.562	0.394
U6385	4.250	5.000	0.664	0.625	0.394
U3775	4.375	4.875	0.289	0.250	0.256

3.6

HB Inch Size

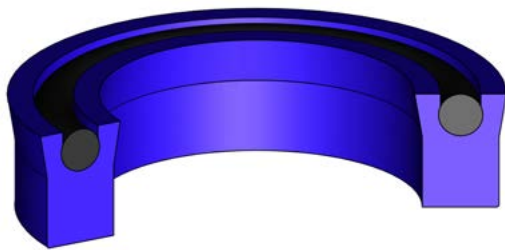
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U4021	4.375	4.875	0.414	0.375	0.256
U3952	4.375	4.875	0.601	0.562	0.256
U00134	4.375	4.999	0.351	0.312	0.315
U9054	4.375	4.999	0.601	0.562	0.315
U3948	4.500	4.874	0.414	0.375	0.197
U3776	4.500	5.000	0.289	0.250	0.256
U3732	4.500	5.000	0.414	0.375	0.256
U4129	4.500	5.000	0.539	0.500	0.256
U5338	4.500	5.000	0.601	0.562	0.256
U9013	4.500	5.000	0.664	0.625	0.256
U6383	4.500	5.250	0.664	0.625	0.394
U3777	4.625	5.125	0.289	0.250	0.256
U00424	4.625	5.249	0.664	0.625	0.315
U3144	4.625	5.375	0.664	0.625	0.394
U3795	4.750	5.250	0.601	0.562	0.256
U6381	4.750	5.500	0.664	0.625	0.394
U4244	4.875	5.625	0.664	0.625	0.394
U9198	5.000	5.374	0.414	0.375	0.197
U8979	5.000	5.500	0.414	0.375	0.256
U3741	5.000	5.500	0.601	0.562	0.256
U3912	5.000	5.624	0.664	0.625	0.315
U3451	5.000	5.750	0.539	0.500	0.394
U6481	5.000	5.750	0.664	0.625	0.394
U3454	5.000	6.000	0.539	0.500	0.512
U3124	5.000	6.000	0.789	0.750	0.512
U8976	5.250	5.750	0.414	0.375	0.256
U4071	5.250	5.874	0.664	0.625	0.315
U6482	5.250	6.000	0.539	0.500	0.394
U4472	5.250	6.000	0.601	0.562	0.394
U8889	5.250	6.000	0.664	0.625	0.394
U4037	5.375	5.999	0.351	0.312	0.315
U9025	5.500	5.874	0.414	0.375	0.197
U9065	5.500	6.000	0.289	0.250	0.256
U3910	5.500	6.000	0.414	0.375	0.256
U5602	5.500	6.000	0.539	0.500	0.256
U3763	5.500	6.000	0.601	0.562	0.256
U7267	5.500	6.125	0.664	0.625	0.315
U3764	5.500	6.250	0.664	0.625	0.394

HB

Inch Size

3.6



HD

PRODUCT DESCRIPTION

A loaded U-packing with a fitted O-ring, and the O-ring provides extreme loads for effective sealing at low or zero pressure.

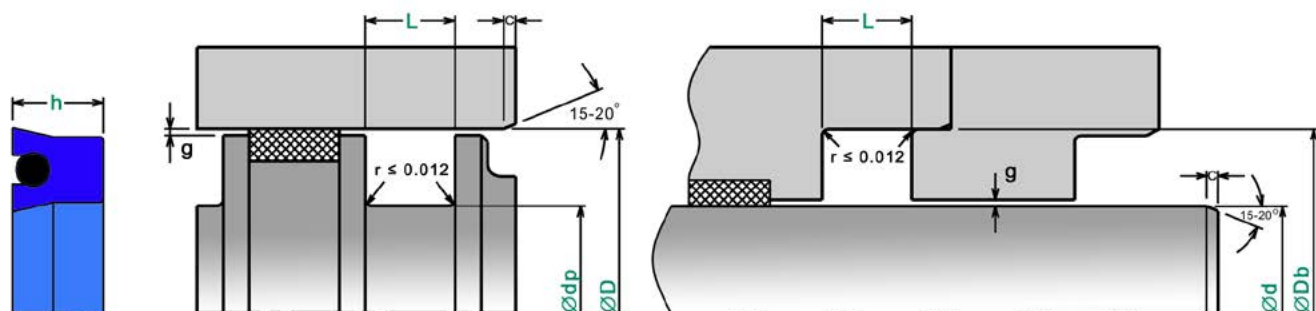
TECHNICAL DATA

MATERIAL	PU + NBR
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-35°C, +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

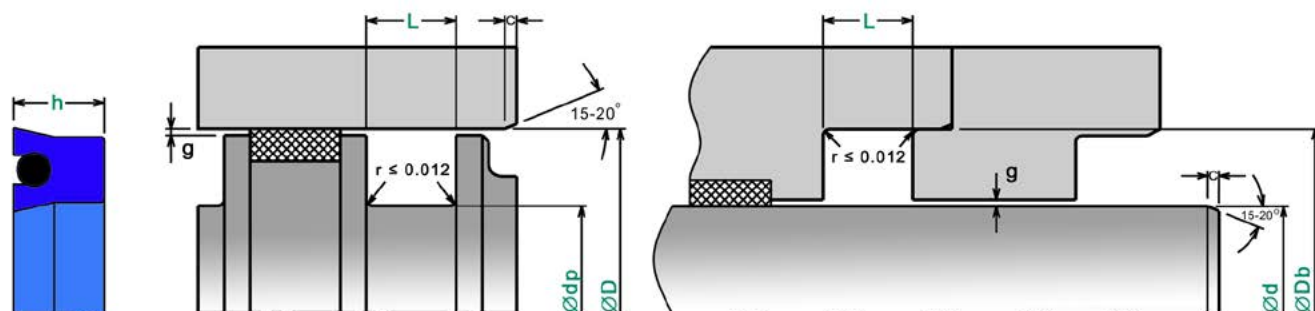
Admissible gap dimension			
	100 Bar	200 Bar	300 Bar
Profile dimension	0.36 mm	0.18 mm	0.09 mm

HD Inch Size


Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U6463	0.250	0.500	0.289	0.250	0.138
U6973	0.375	0.875	0.414	0.375	0.256
U5148	0.500	0.750	0.289	0.250	0.138
U4571	0.625	0.875	0.289	0.250	0.138
U6908	0.750	1.000	0.289	0.250	0.138
U5226	0.875	1.125	0.289	0.250	0.138
U3761	1.000	1.374	0.289	0.250	0.197
U8974	1.125	1.375	0.289	0.250	0.138
U5766	1.125	1.500	0.351	0.312	0.197
U5370	1.250	1.500	0.289	0.250	0.138
U3133	1.250	1.624	0.351	0.312	0.197
U3349	1.250	1.750	0.414	0.375	0.256
U9009	1.375	1.625	0.289	0.250	0.138
U3489	1.375	1.749	0.351	0.312	0.197
U7047	1.375	1.750	0.414	0.375	0.197
U3609	1.375	1.875	0.414	0.375	0.256
U3080	1.500	1.874	0.320	0.281	0.197
U6489	1.500	1.875	0.414	0.375	0.197
U3350	1.500	2.000	0.414	0.375	0.256
U7406	1.625	1.875	0.289	0.250	0.138
U3373	1.625	1.999	0.351	0.312	0.197
U9547	1.625	1.999	0.414	0.375	0.197
U5289	1.625	2.000	0.289	0.250	0.197
U00423	1.750	2.124	0.351	0.312	0.197
U9026	1.750	2.124	0.414	0.375	0.197
U00080	1.750	2.250	0.414	0.375	0.256
U6907	1.750	2.250	0.539	0.500	0.256
U4644	1.875	2.249	0.289	0.250	0.197
U4162	1.875	2.249	0.414	0.375	0.197
U9010	1.875	2.625	0.664	0.625	0.394
U5619	2.000	2.375	0.414	0.375	0.197
U3632	2.000	2.500	0.414	0.375	0.256
U9170	2.000	2.624	0.539	0.500	0.315
U5094	2.000	2.750	0.664	0.625	0.394
U3970	2.125	2.499	0.414	0.375	0.197
U3611	2.125	2.625	0.414	0.375	0.256
U6460	2.250	2.750	0.414	0.375	0.256
U5063	2.500	3.000	0.414	0.375	0.256

3.7

HD Inch Size

Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U00141	2.625	3.000	0.289	0.250	0.197
U6347	2.625	3.000	0.414	0.375	0.197
U3698	2.750	3.250	0.414	0.375	0.256
U6364	3.000	3.375	0.414	0.375	0.197
U5362	3.000	3.500	0.414	0.375	0.256
U6354	3.125	3.500	0.414	0.375	0.197
U3720	3.125	3.625	0.414	0.375	0.256
U3696	3.500	4.000	0.351	0.312	0.256
U5498	3.500	4.000	0.414	0.375	0.256
U3717	3.625	4.125	0.414	0.375	0.256
U3393	3.750	4.250	0.414	0.375	0.256
U5497	3.750	4.500	0.664	0.625	0.394
U3739	4.000	4.500	0.414	0.375	0.256
U3770	4.000	4.500	0.601	0.562	0.256
U3771	4.250	4.750	0.414	0.375	0.256
U3737	4.500	5.000	0.351	0.312	0.256
U6386	4.500	5.000	0.414	0.375	0.256
U3736	4.500	5.000	0.601	0.562	0.256
U3743	4.500	5.250	0.664	0.625	0.394
U3745	5.000	5.374	0.320	0.281	0.197
U5503	5.000	5.500	0.414	0.375	0.256
U3394	5.000	6.000	0.789	0.750	0.512
U9382	5.250	6.000	0.539	0.500	0.394
U5568	5.500	6.000	0.414	0.375	0.256
U3754	5.500	6.000	0.601	0.562	0.256
U4648	5.875	6.375	0.601	0.562	0.256
U3772	6.000	6.500	0.601	0.562	0.256
U5151	6.000	6.750	0.664	0.625	0.394
U3774	6.500	7.000	0.601	0.562	0.256
U00183	6.500	7.500	0.789	0.750	0.512
U6727	6.750	7.500	0.601	0.562	0.394
U3760	7.000	7.500	0.414	0.375	0.256
U5499	7.000	8.000	0.789	0.750	0.512
U9199	8.750	9.500	0.664	0.625	0.394
U9568	9.250	10.000	0.539	0.500	0.394



HS

PRODUCT DESCRIPTION

A loaded U-packing with a fitted O-ring, and the O-ring provides extreme loads for effective sealing at low or zero pressure.

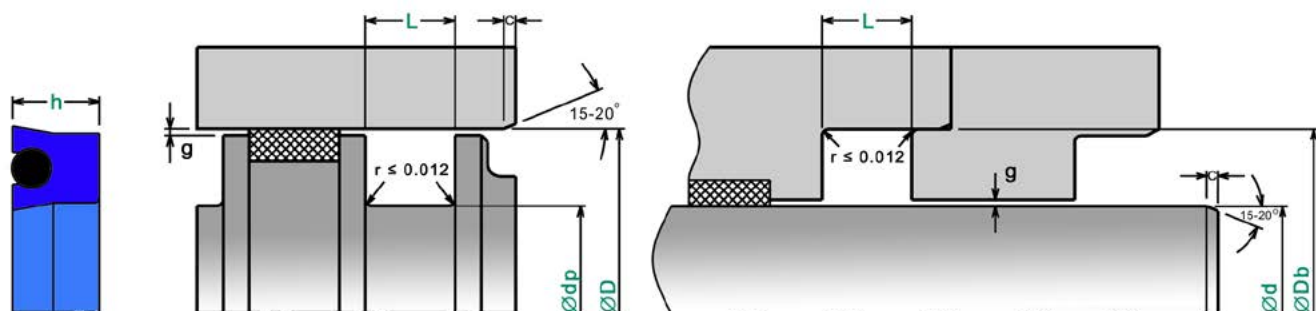
TECHNICAL DATA

MATERIAL	PU + NBR
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-35°C, +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	100 Bar	200 Bar	300 Bar
Profile dimension	0.36 mm	0.18 mm	0.09 mm

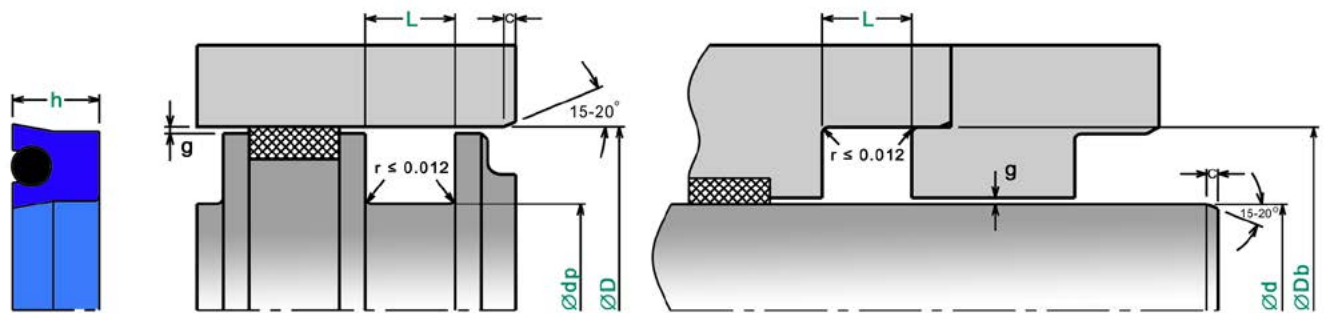
HS Inch Size


Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.2	h	c
U3490	0.187	0.437	0.153	0.125	0.138
U6819	0.189	0.433	0.162	0.134	0.138
U6499	0.250	0.500	0.153	0.125	0.138
U3131	0.250	0.624	0.215	0.187	0.197
U4510	0.250	0.750	0.289	0.250	0.256
U5431	0.312	0.562	0.153	0.125	0.138
U7089	0.312	0.748	0.246	0.218	0.236
U3126	0.375	0.625	0.153	0.125	0.138
U3491	0.375	0.749	0.215	0.187	0.197
U4190	0.375	0.875	0.289	0.250	0.256
U3395	0.437	0.687	0.153	0.125	0.138
U3132	0.437	0.811	0.215	0.187	0.197
U5649	0.500	0.750	0.153	0.125	0.138
U5165	0.500	0.874	0.215	0.187	0.197
U3646	0.500	1.000	0.289	0.250	0.256
U5095	0.562	0.812	0.153	0.125	0.138
U5060	0.625	0.875	0.153	0.125	0.138
U6486	0.625	1.000	0.215	0.187	0.197
U6518	0.687	1.062	0.215	0.187	0.197
U00220	0.687	1.187	0.289	0.250	0.256
U6507	0.750	1.000	0.153	0.125	0.138
U7405	0.750	1.062	0.184	0.156	0.157
U4977	0.750	1.124	0.215	0.187	0.197
U9315	0.750	1.250	0.289	0.250	0.256
U4609	0.812	1.062	0.153	0.125	0.138
U7634	0.812	1.562	0.414	0.375	0.394
U5439	0.875	1.125	0.153	0.125	0.138
U6487	0.875	1.250	0.215	0.187	0.197
U7037	0.875	1.500	0.351	0.312	0.315
U00235	0.937	1.249	0.184	0.156	0.157
U9027	0.937	1.437	0.289	0.250	0.256
U9941	1.000	1.250	0.153	0.125	0.138
U6965	1.000	1.312	0.184	0.156	0.157
U3528	1.000	1.374	0.215	0.187	0.197
U3602	1.000	1.500	0.289	0.250	0.256
U3127	1.125	1.375	0.153	0.125	0.138
U7268	1.125	1.437	0.184	0.156	0.157
U7049	1.125	1.500	0.215	0.187	0.197

3.8

HS Inch Size



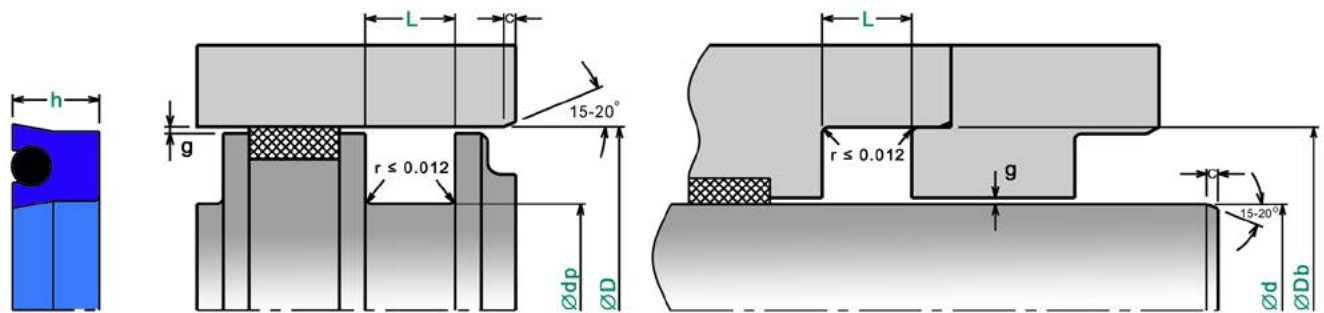
Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U6349	1.125	1.625	0.289	0.250	0.256
U4074	1.125	1.875	0.414	0.375	0.394
U4656	1.187	1.437	0.153	0.125	0.138
U5847	1.187	1.500	0.184	0.156	0.157
U9289	1.187	1.561	0.215	0.187	0.197
U5462	1.187	1.687	0.289	0.250	0.256
U5121	1.250	1.500	0.153	0.125	0.138
U6465	1.250	1.562	0.184	0.156	0.157
U3376	1.250	1.624	0.215	0.187	0.197
U5173	1.250	1.750	0.289	0.250	0.256
U5618	1.250	2.000	0.414	0.375	0.394
U9052	1.375	1.625	0.153	0.125	0.138
U6466	1.375	1.687	0.184	0.156	0.157
U6488	1.375	1.750	0.215	0.187	0.197
U9333	1.375	1.875	0.289	0.250	0.256
U3595	1.375	1.999	0.351	0.312	0.315
U6251	1.437	1.937	0.289	0.250	0.256
U9288	1.500	1.750	0.153	0.125	0.138
U5415	1.500	1.875	0.215	0.187	0.197
U5138	1.500	2.000	0.289	0.250	0.256
U6471	1.500	2.250	0.414	0.375	0.394
U3589	1.625	1.999	0.215	0.187	0.197
U3603	1.625	2.125	0.289	0.250	0.256
U3496	1.625	2.249	0.351	0.312	0.315
U6329	1.625	2.375	0.414	0.375	0.394
U6718	1.750	2.124	0.215	0.187	0.197
U3138	1.750	2.250	0.289	0.250	0.256
U6350	1.750	2.375	0.351	0.312	0.315
U4245	1.750	2.624	0.476	0.437	0.453
U5891	1.875	2.125	0.153	0.125	0.138
U3488	1.875	2.249	0.215	0.187	0.197
U3604	1.875	2.375	0.289	0.250	0.256
U3400	1.875	2.499	0.351	0.312	0.315
U6818	1.929	2.283	0.225	0.197	0.197
U00674	2.000	2.250	0.153	0.125	0.138
U6511	2.000	2.375	0.215	0.187	0.197
U4806	2.000	2.500	0.289	0.250	0.256
U5889	2.000	2.625	0.351	0.312	0.315

HS Inch Size

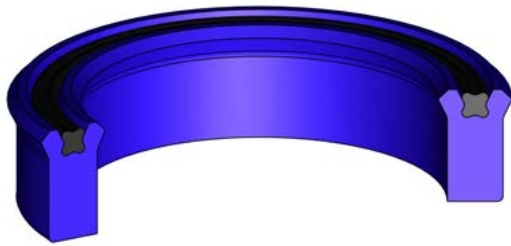
NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U6363	2.000	2.750	0.414	0.375	0.394
U6359	2.000	3.000	0.539	0.500	0.512
U8989	2.125	2.499	0.215	0.187	0.197
U3605	2.125	2.625	0.289	0.250	0.256
U4035	2.125	2.749	0.351	0.312	0.315
U3688	2.125	2.875	0.414	0.375	0.394
U6391	2.250	2.625	0.215	0.187	0.197
U6491	2.250	2.750	0.289	0.250	0.256
U6333	2.250	2.875	0.351	0.312	0.315
U3715	2.250	3.000	0.414	0.375	0.394
U9175	2.375	2.750	0.215	0.187	0.197
U3704	2.375	2.875	0.289	0.250	0.256
U3077	2.375	3.125	0.414	0.375	0.394
U5025	2.500	2.874	0.215	0.187	0.197
U9273	2.500	3.000	0.289	0.250	0.256
U5571	2.500	3.125	0.351	0.312	0.315
U5846	2.500	3.250	0.414	0.375	0.394
U9023	2.625	2.999	0.215	0.187	0.197
U3703	2.625	3.125	0.289	0.250	0.256
U3493	2.625	3.375	0.414	0.375	0.394
U9176	2.750	3.124	0.215	0.187	0.197
U5651	2.750	3.250	0.289	0.250	0.256
U4610	2.750	3.374	0.351	0.312	0.315
U4122	2.750	3.500	0.414	0.375	0.394
U5126	2.812	3.248	0.246	0.218	0.236
U9019	2.875	3.250	0.215	0.187	0.197
U5600	2.875	3.500	0.351	0.312	0.315
U4034	3.000	3.374	0.215	0.187	0.197
U3456	3.000	3.500	0.289	0.250	0.256
U5064	3.000	3.750	0.414	0.375	0.394
U6338	3.000	4.000	0.539	0.500	0.512
U7750	3.125	3.500	0.215	0.187	0.197
U5083	3.125	3.625	0.289	0.250	0.256
U4377	3.125	3.875	0.414	0.375	0.394
U3702	3.250	3.750	0.289	0.250	0.256
U6498	3.250	4.000	0.414	0.375	0.394
U9652	3.375	4.000	0.351	0.312	0.315
U6335	3.500	3.875	0.215	0.187	0.197
U3718	3.500	4.000	0.289	0.250	0.256
U7050	3.500	4.250	0.414	0.375	0.394
U9880	3.500	4.500	0.539	0.500	0.512
U5441	3.625	4.000	0.215	0.187	0.197
U4796	3.625	4.375	0.414	0.375	0.394
U4029	3.625	4.499	0.414	0.375	0.453
U3719	3.750	4.250	0.289	0.250	0.256
U9274	3.750	4.500	0.414	0.375	0.394
U4069	3.875	4.249	0.215	0.187	0.197
U3081	3.875	4.499	0.351	0.312	0.315
U4164	4.000	4.374	0.215	0.187	0.197
U6365	4.000	4.500	0.289	0.250	0.256
U6360	4.000	4.750	0.414	0.375	0.394
U6361	4.000	5.000	0.539	0.500	0.512
U3794	4.125	4.499	0.215	0.187	0.197
U3390	4.125	5.125	0.539	0.500	0.512

3.8

HS Inch Size

Ordering dimensions in green

NAK NO.	Ød/Ødp f8/h10	ØDb/ØD H10/H9	L +/- 0.008	h	c
U9221	4.250	4.624	0.215	0.187	0.197
U9740	4.250	4.750	0.289	0.250	0.256
U3363	4.250	5.000	0.414	0.375	0.394
U5569	4.375	4.875	0.289	0.250	0.256
U4059	4.437	4.999	0.320	0.281	0.295
U5572	4.500	4.875	0.215	0.187	0.197
U5404	4.500	5.000	0.289	0.250	0.256
U3744	4.500	5.250	0.414	0.375	0.394
U6330	4.500	5.500	0.539	0.500	0.512
U5965	4.500	5.750	0.664	0.625	0.630
U00238	4.750	5.124	0.215	0.187	0.197
U3992	4.750	5.500	0.414	0.375	0.394
U5492	5.000	5.500	0.289	0.250	0.256
U4036	5.000	5.750	0.414	0.375	0.394
U9051	5.000	6.000	0.539	0.500	0.512
U5444	5.250	5.750	0.289	0.250	0.256
U9050	5.250	6.000	0.414	0.375	0.394
U6459	5.375	5.750	0.215	0.187	0.197
U4514	5.375	5.999	0.351	0.312	0.315
U4488	5.437	5.999	0.320	0.281	0.295
U5447	5.500	6.000	0.289	0.250	0.256
U5435	5.500	6.250	0.414	0.375	0.394
U5410	5.500	6.500	0.539	0.500	0.512
U3767	5.750	6.250	0.289	0.250	0.256
U5494	5.875	6.375	0.289	0.250	0.256
U9016	6.000	6.500	0.289	0.250	0.256
U3768	6.250	6.750	0.289	0.250	0.256
U8966	6.250	7.000	0.414	0.375	0.394
U5448	6.500	7.000	0.289	0.250	0.256
U7014	7.125	7.625	0.289	0.250	0.256
U5407	7.250	8.000	0.414	0.375	0.394
U6355	7.375	8.000	0.351	0.312	0.315
U5405	7.500	8.000	0.289	0.250	0.256
U7013	8.000	8.500	0.289	0.250	0.256
U6726	8.250	8.750	0.289	0.250	0.256
U9608	8.250	9.000	0.414	0.375	0.394
U9015	9.250	10.000	0.414	0.375	0.394
U9659	9.750	10.500	0.414	0.375	0.394
U9290	10.000	11.000	0.539	0.500	0.512



HBX

PRODUCT DESCRIPTION

A loaded U-packing with a fitted O-ring, and the O-ring provides extreme loads for effective sealing at low or zero pressure.

TECHNICAL DATA

MATERIAL	PU + NBR
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-35°C, +100°C

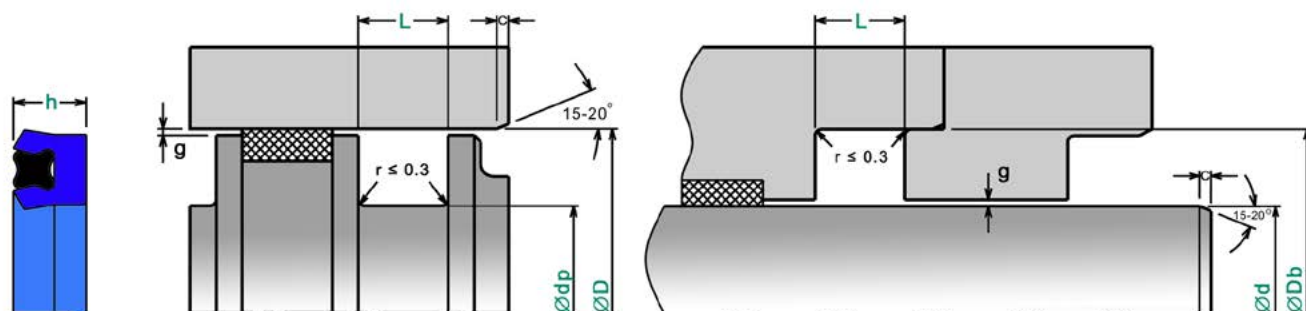
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	100 Bar	200 Bar	300 Bar
Profile dimension	0.36 mm	0.18 mm	0.09 mm

Please contact our customer service for the dimensions out of our stock list.
service@mail.nak.com.tw

Metric Size



Ordering dimensions in green

[illegible]

[illegible]



HSX

PRODUCT DESCRIPTION

A loaded U-packing with a fitted X-ring, and the X-Ring under the U-packing provides positive wiping for effective sealing at low or zero pressure.

TECHNICAL DATA

MATERIAL	PU + NBR
MEDIA	Mineral oils
Pressure p	≤ 35 MPa
Running speed v	≤ 0.5 m/s
Medium/ Temperature	-35°C, +100°C

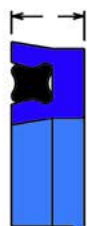
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface (Rod)	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base (Rod)	ØD	0.8 µm	3.2 µm
Sliding surface (Piston)	ØD	0.1~0.8 µm	0.4~3.2 µm
Groove base (Piston)	Ød	0.8 µm	3.2 µm
Groove flanks	L	3.2 µm	12.5 µm

Admissible gap dimension			
	100 Bar	200 Bar	300 Bar
Profile dimension	0.36 mm	0.18 mm	0.09 mm

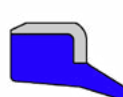
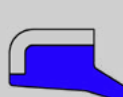
Please contact our customer service for the dimensions out of our stock list.
service@mail.nak.com.tw

Inch Size

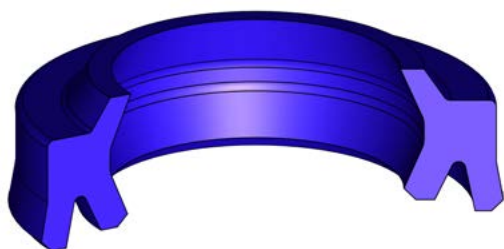


Ordering dimensions in green

3.10

NAK CODE	DRAWING	MATERIAL	TEMPERATURE (max)°C	SLIDING SPEED(max) m/s	SIZE (mm)	PAGE
WP1		PU	-40 / +100	1	10-200 0.375-7.000	4.1 P172~P175 (METRIC) P176~P177 (INCH)
WP2		PU	-40 / +100	1	12.5-160 1.000-6.969	4.2 P178~P179 (METRIC) P180~P181 (INCH)
WP3		PU	-40 / +100	1	16-230 0.625-8.000	4.3 P182~P188 (METRIC) P189~P191 (INCH)
WP4		PU	-40 / +100	1	18-230	4.4 P192~P195 (METRIC)
WP5		PU	-40 / +100	1	8-220	4.5 P196~201 (METRIC)
WP6		PU	-40 / +100	1	48-66	4.6 P202~P203 (METRIC)
WP7		PU	-40 / +100	1	12-113	4.7 P204~P205 (METRIC)
WP8		PU	-40 / +100	1	30-40	4.8 P206~P207 (METRIC)
WP10		PU	-40 / +100	1	35	4.9 P208~P209 (METRIC)
WP11		PU	-40 / +100	1	40-200 1.250-6.500	4.10 P210~P212 (METRIC) P213 (INCH)
WP12		NBR	-35 / +100	1	12-250 0.250-1.000	4.11 P214~P216 (METRIC) P217 (INCH)
WP13		NBR	-35 / +100	1	9.5-25	4.12 P218~P219 (METRIC)
WP14		NBR	-35 / +100	1	6-70	4.13 P220~P223 (METRIC)
WP15		NBR	-35 / +100	1	18-238	4.14 P224~P225 (METRIC)
WP16		PU	-40 / +100	1	25-100	4.15 P226~P227 (METRIC)

NAK CODE	DRAWING	MATERIAL	TEMPERATURE (max)°C	SLIDING SPEED(max) m/s	SIZE (mm)	PAGE
WP17		NBR	-35 / +100	1	12-200	4.16 P228~P229 (METRIC)
WP19		NBR	-35 / +100	1	12-50	4.17 P230~P231 (METRIC)
WP21		PU	-40 / +100	1	20-140	4.18 P232~P233 (METRIC)
WP22		PU	-40 / +100	1	20-36	4.19 P234~P235 (METRIC)
WP24		NBR	-35 / +100	1	25-35	4.20 P236~P237 (METRIC)
WP32		PU	-40 / +100	1	79-191	4.21 P238~P239 (METRIC)
WPB		NBR	-35 / +100	1	9-133 1.063-4.250	4.22 P240~P241 (METRIC) P242~P243 (INCH)
WPC		NBR	-35 / +100	1	17-30.16	4.23 P244~P245 (METRIC)
WPK		NBR	-35 / +100	1	10-180	4.24 P246~P247 (METRIC)
WPK1		NBR	-35 / +100	1	30-260	4.25 P248~P249 (METRIC)
WPK3		NBR	-35 / +100	1	35-155	4.26 P250~P251 (METRIC)
WPM		NBR FKM	-35 / +100 -25 / +250	1	6-160	4.27 P252~P259 (METRIC)
WPR		NBR	-35 / +100	1	6-180	4.28 P260~P263 (METRIC)
WPV		NBR	-35 / +100	1	16-150	4.29 P264~P265 (METRIC)



WP1

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

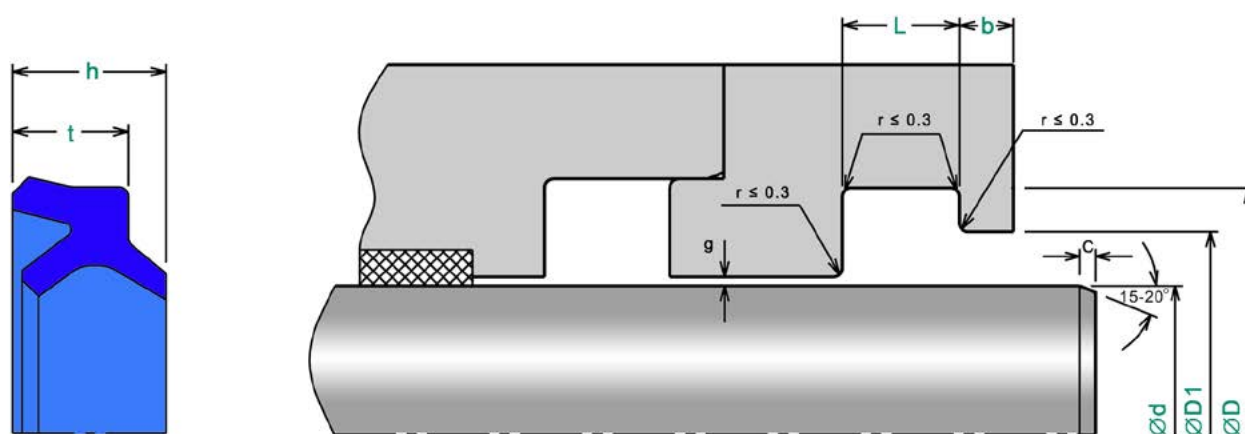
TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

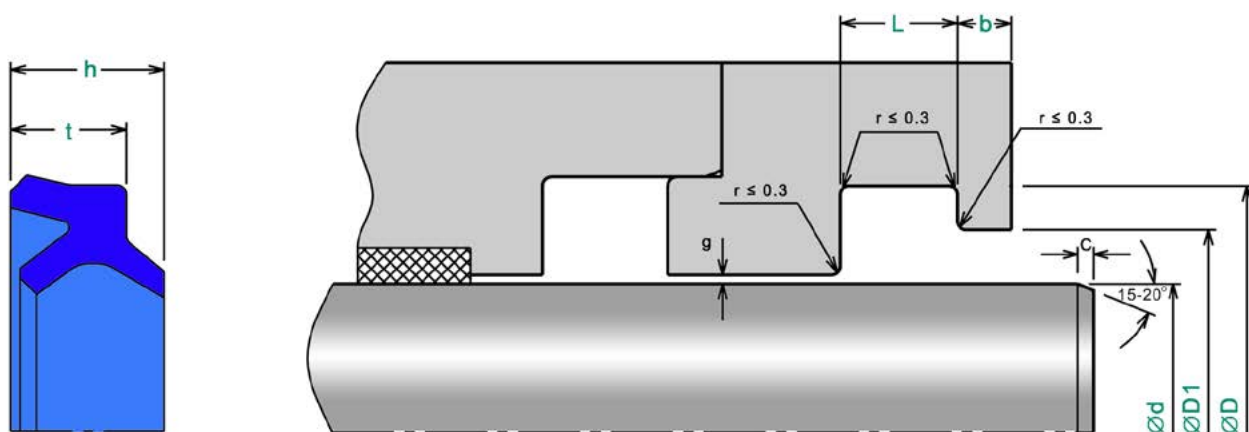
WP1 Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD 0/-0.008	ØD1 +/-0.004	L +/- 0.2	b	t	h	c
U6590	10	16	13	4.1	2	3.6	4.8	1.5
U2920	12	18	15	4.1	2	3.6	4.8	1.5
U7270	13	20	17	5.5	2	5	6.5	2.0
U3858	14	20	17	4.1	2	3.6	4.8	1.5
U6131	14	22	18	5	2	4.5	6	2.0
U1660	16	22	19	4.1	2	3.6	4.8	1.5
U6278	16	24	20	5	2	4.5	5.8	2.0
U4145	16	24	20	5	2	4.5	6	2.0
U2719	18	24	21	4.1	2	3.6	4.8	1.5
U5484	18	26	22	5	2	4.5	6	2.0
U6279	20	25	22.5	5	2	4.5	5.8	1.5
U1949	20	26	23	4.1	2	3.6	4.8	1.5
U3307	20	28	24	5	2	4.5	6	2.0
U1162	20	28	24	5.5	2	5	6.8	2.0
U2285	22	28	25	4.1	2	3.6	4.8	1.5
U5671	22	30	26	5	2	4.5	5.8	2.0
U5685	22	30	26	5	2	4.5	6	2.0
U6603	22	30	26	5.5	2	5	6.8	2.0
U7776	22	30.6	26.3	4.9	2	4.4	5.8	2.5
U1289	22.4	30.4	26.4	5	2	4.5	6	2.0
U6750	24	32	28	5.3	2	4.8	6	2.0
U3031	25	31	28	4.1	2	3.6	4.8	1.5
U2315	25	33	29	5	2	4.5	6	2.0
U6149	25	33	29	5.3	2	4.8	6	2.0
U6592	25	33	29	5.5	2	5	6.8	2.0
U2463	28	36	32	5	2	4.5	5.8	2.0
U3308	28	36	32	5	2	4.5	6	2.0
U6757	28	36	32	5.3	2	4.8	6	2.0
U3279	28	36	32	5.5	2	5	6.8	2.0
U3532	30	38	34	5	2	4.5	5.8	2.0
U5879	30	38	34	5.3	2	4.8	6	2.0
U0981	30	38	34	5.5	2	5	6.5	2.0
U6606	30	38	34	5.5	2	5	6.8	2.0
U8473	30	38	34	5.7	2	5.2	6.8	2.0
U6262	31.5	39.5	35.5	5.5	2	5	6.5	2.0
U2374	32	40	36	5	2	4.5	5.8	2.0
U6751	32	40	36	5.3	2	4.8	6	2.0
U4616	32	40	36	5.5	2	5	6.5	2.0

4.1

WP1 Metric Size

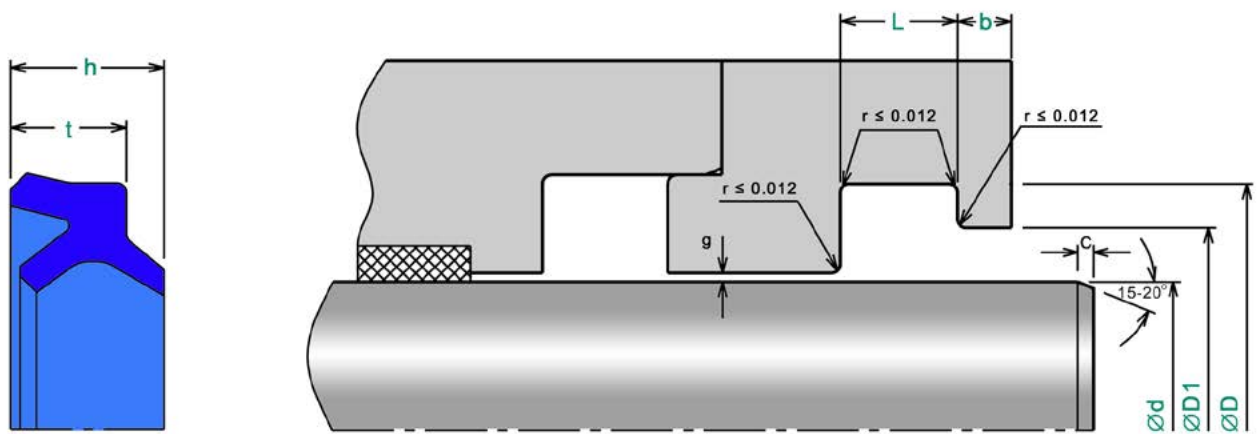
Ordering dimensions in green

NAK NO.	Ød f8	ØD 0/-0.008	ØD1 +/-0.004	L +/- 0.2	b	t	h	c
U2921	32	40	36	5.5	2	5	6.8	2.0
U7809	35	43	39.5	4	2	3.5	5	2.0
U2630	35	43	39	5.5	2	5	6.8	2.0
U3118	35	43	39	5.5	2	5	6.5	2.0
U8474	35	43	39	5.7	2	5.2	6.8	2.0
U5485	35.5	43.5	39.5	5.5	2	5	6.5	2.0
U2851	36	44	40	5	2	4.5	5.8	2.0
U00576	36	44	40	5	3	4.5	6.5	2.0
U3859	36	44	40	5.5	2	5	6.8	2.0
U7324	38	46	42	5.5	2	5	6.5	2.0
U2296	40	48	44	5	2	4.5	5.8	2.0
U7553	40	48	44	5.15	2	4.65	5.95	2.0
U2314	40	48	44	5.5	2	5	6.5	2.0
U4919	40	48	44	5.5	2	5	6.5	2.0
U2720	40	48	44	5.5	2	5	6.8	2.0
U8156	45	52.5	49.25	5.5	3	5	7.2	2.0
U3467	45	53	49	5	2	4.5	5.8	2.0
U8480	45	53	49	5	2	4.5	5.9	2.0
U3852	45	53	49	5.5	2	5	6.5	2.0
U2333	45	55	50	5.5	2	5	6.8	2.5
U6589	50	58	54	5	2	4.5	5.8	2.0
U2313	50	58	54	5.5	2	5	6.5	2.0
U6604	50	58	54	5.5	2	5	6.8	2.0
U1421	50	60	55	5.5	2	5	6.8	2.5
U6313	53	61	57	5.5	2	5	6.5	2.0
U6314	55	63	59	5.5	2	5	6.5	2.0
U6596	55	63	59	5.5	2	5	6.8	2.0
U3309	56	64	60	5.5	2	5	6.5	2.0
U6607	56	64	60	5.5	2	5	6.8	2.0
U3113	56	66	61	5.8	2	5.3	6.8	2.5
U4617	60	68	64	5.5	2	5	6.5	2.0
U6044	60	68	64	5.5	2	5	6.8	2.0
U3210	60	70	65	5.5	2	5	6.8	2.5
U8464	60	70	65	5.8	2	5.3	6.8	2.5
U1296	63	71	67	5.5	2	5	6.5	2.0
U6593	63	71	67	5.5	2	5	6.8	2.0
U6588	63	73	68	5.5	2	5	6.8	2.5
U6636	65	73	69	5.5	2	5	6.8	2.0

WP1 Metric Size

NAK NO.	Ød f8	ØD 0/-0.008	ØD1 +/-0.004	L +/- 0.2	b	t	h	c
U8144	65	73	69	5.5	2	5	6.45	2.0
U2721	65	75	70	5.5	2	5	6.8	2.5
U7353	68	78	73	5.5	2	5	6.8	2.5
U5777	70	78	74	5.5	2	5	6.8	2.0
U3008	70	80	75	5.5	2	5	6.8	2.5
U2312	70	80	75	6.5	3	6	8	2.5
U5793	71	81	76	6.5	3	6	8	2.5
U6644	75	83	79	5.5	2	5	6.8	2.0
U6282	75	85	80	5.5	3	5	7	2.5
U5623	75	85	80	6.5	3	6	8	2.5
U2384	75	87	81	7.5	3	7	9	3.0
U7466	78	88	83	7.5	3	7	9	2.5
U2722	80	90	85	5.8	2	5.3	6.8	2.5
U3311	80	90	85	6.5	3	6	8	2.5
U4152	80	92	86	7.5	3	7	9	3.0
U7465	82	94	88	7.5	3	7	9	3.0
U6752	85	95	90	6.5	3	6	8	2.5
U7727	85	97.2	91.1	7.6	3	7.1	10	3.5
U2668	90	100	95	5.8	2	5.3	6.8	2.5
U3310	90	100	95	6.5	3	6	8	2.5
U1158	90	102	96	7.5	3	7	9	3.0
U7922	95	105	100	6.5	3	6	8	2.5
U1156	95	107	101	7.3	3	6.8	9	3.0
U1514	100	110	105	6.5	3	6	8	2.5
U6605	100	112	106	7.5	3	7	9	3.0
U7317	104	116	110	7.5	3	7	9	3.0
U6599	110	122	116	7.5	4	7	10	3.0
U3087	110	125	117.5	8	3	7.5	9.5	4.0
U7061	112	124	119	7.5	3	7	9	3.0
U1423	115	127	121	7.5	3	7	9	3.0
U8443	120	132.2	126.1	5.8	2	5.3	6.8	3.5
U7905	120	135	127.5	8.8	3	8.3	10.5	4.0
U7238	125	137	131	7.5	3	7	9	3.0
U1303	125	138	131.5	7.5	3	7	9.5	3.5
U7314	129	141	135	7.5	3	7	9	3.0
U6557	130	143	136.5	7.5	3	7	9.5	3.5
U6571	135	147	141	7.5	3	7	9	3.0
U3200	140	155	147.5	8	3	7.5	9.5	4.0
U7252	150	162	156	7.5	3	7	9.5	3.0
U7315	154	166	160	7.5	3	7	9.5	3.0
U1426	155	167	161	7.5	3	7	9.5	3.0
U1427	160	172	166	7.5	3	7	9.5	3.0
U1516	160	173	166.5	7.5	3	7	9.5	3.5
U1517	170	183	176.5	7.5	3	7	9.5	3.5
U7251	175	187	181	7.5	3	7	9.5	3.0
U7316	180	192	186	7.5	3	7	9.5	3.0
U1428	185	197	191	7.5	3	7	9.5	3.0
U7965	200	220	210	13.5	3	13	15.5	5.0

4.1

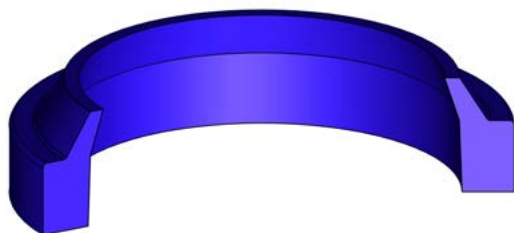
WP1 Inch Size

Ordering dimensions in green

NAK NO.	Ø d f8	Ø D 0/-0.008	Ø D1 +/-0.004	L +/- 0.008	b	t	h	c
U5329	0.375	0.687	0.570	0.214	0.079	0.194	0.245	0.079
U5331	0.500	0.812	0.695	0.214	0.079	0.194	0.245	0.079
U5333	0.625	0.937	0.820	0.214	0.079	0.194	0.245	0.079
U5335	0.750	1.062	0.926	0.214	0.079	0.194	0.245	0.079
U5337	0.875	1.250	1.082	0.225	0.079	0.205	0.265	0.098
U9617	0.937	1.312	1.125	0.225	0.079	0.205	0.265	0.098
U5339	1.000	1.375	1.188	0.225	0.079	0.205	0.265	0.098
U5341	1.125	1.500	1.313	0.225	0.079	0.205	0.265	0.098
U5343	1.250	1.625	1.438	0.244	0.079	0.205	0.265	0.098
U5345	1.375	1.750	1.563	0.244	0.079	0.205	0.265	0.098
U5347	1.500	1.875	1.688	0.244	0.079	0.205	0.265	0.098
U5349	1.625	2.000	1.813	0.244	0.079	0.205	0.265	0.098
U5351	1.750	2.125	1.938	0.244	0.079	0.205	0.265	0.098
U5353	1.875	2.250	2.063	0.244	0.079	0.205	0.265	0.098
U5355	2.000	2.375	2.188	0.244	0.079	0.205	0.265	0.098
U5359	2.250	2.750	2.500	0.307	0.079	0.268	0.325	0.138
U5361	2.375	2.875	2.625	0.307	0.079	0.268	0.325	0.138
U5363	2.500	3.000	2.750	0.307	0.079	0.268	0.325	0.138
U5367	2.750	3.250	3.000	0.307	0.079	0.268	0.325	0.138
U5369	2.875	3.375	3.125	0.307	0.079	0.268	0.325	0.138
U5371	3.000	3.500	3.250	0.307	0.079	0.268	0.325	0.138
U5372	3.125	3.625	3.375	0.307	0.079	0.268	0.325	0.138
U5373	3.250	3.750	3.500	0.307	0.079	0.268	0.325	0.138
U5374	3.375	3.875	3.625	0.307	0.079	0.268	0.325	0.138
U5376	3.500	4.000	3.750	0.307	0.079	0.268	0.325	0.138
U5377	3.625	4.125	3.875	0.307	0.079	0.268	0.325	0.138
U5379	3.750	4.250	4.000	0.307	0.079	0.268	0.325	0.138
U5380	3.875	4.375	4.125	0.307	0.079	0.268	0.325	0.138
U5381	4.000	4.500	4.250	0.307	0.079	0.268	0.325	0.138
U5382	4.125	4.625	4.375	0.307	0.079	0.268	0.325	0.138
U5383	4.250	4.750	4.500	0.307	0.079	0.268	0.325	0.138
U5384	4.375	4.875	4.625	0.307	0.079	0.268	0.325	0.138
U5386	4.500	5.000	4.750	0.307	0.079	0.268	0.325	0.138
U5387	4.625	5.125	4.875	0.307	0.079	0.268	0.325	0.138
U5388	4.750	5.250	5.000	0.307	0.079	0.268	0.325	0.138
U5389	4.875	5.375	5.125	0.307	0.079	0.268	0.325	0.138
U5390	5.000	5.500	5.250	0.307	0.079	0.268	0.325	0.138
U5391	5.250	5.750	5.500	0.307	0.079	0.268	0.325	0.138

WP1

4.1



WP2

PRODUCT DESCRIPTION

Popular dust seals.

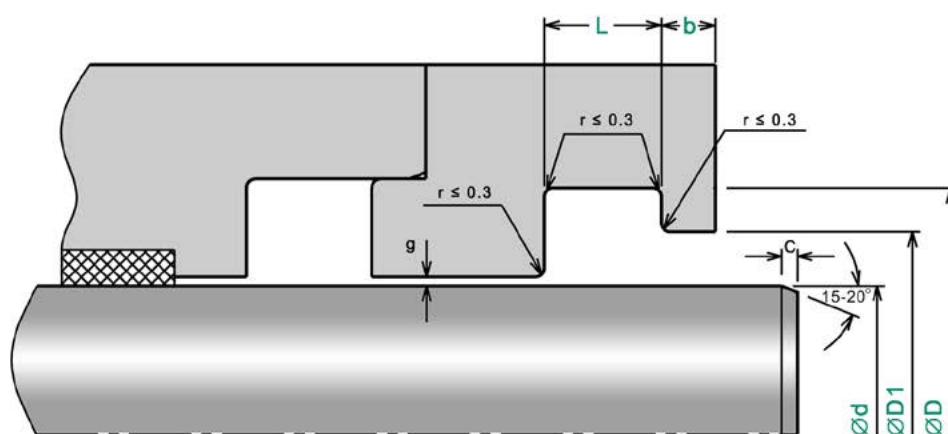
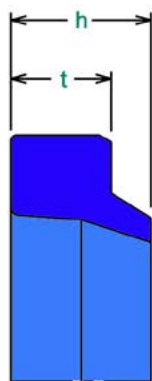
TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	$\leq 0.8 \text{ µm}$	$\leq 3.2 \text{ µm}$
Groove flanks	L	$\leq 3.2 \text{ µm}$	$\leq 12.5 \text{ µm}$

WP2 Metric Size

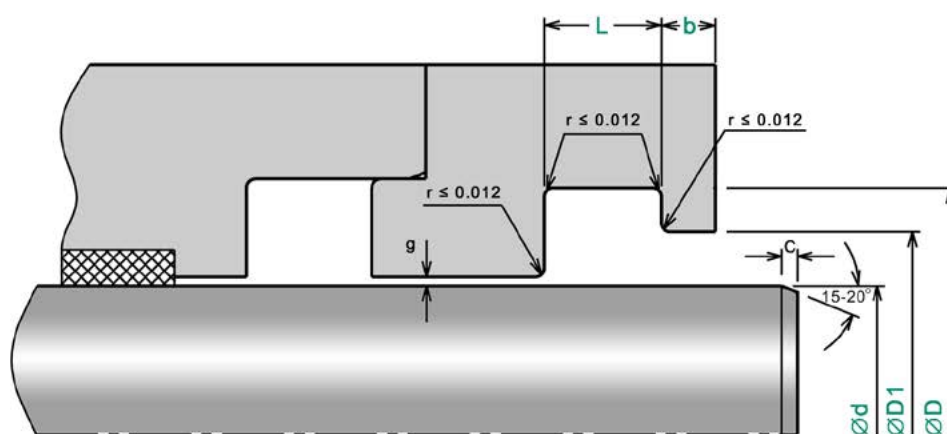
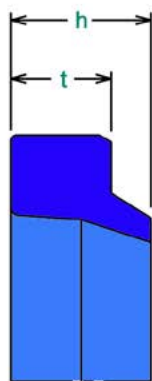


Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	ØD1 +/-0.1	L +/- 0.2	b	t	h	c
U7279	12.5	20.5	16.5	5	2	4.5	6	2.0
U00227	15	21	19	4	2	3.5	5	1.5
U5910	15	23	19	5	2	4.5	6	2.0
U4255	16	24	20.5	5.5	3	5	7	2.0
U00385	19	25.7	23.35	3	2	2.5	4.4	2.0
U4256	20	28	24.5	5.5	3	5	7	2.0
U7698	22	30	27	5.5	4	5	8.5	2.0
U8138	25	33	30.5	5.5	4	5	8	2.0
U4259	30	38	34.5	6	3	5	7	2.0
U8139	36	44	41	6	4	5	8	2.0
U9585	40	50	48	8	6	7	12	2.5
U9676	44.45	50.8	49.13	7.35	4	6.35	9.53	2.0
U5011	46	54.5	50.25	3.9	3	2.9	5	2.5
U7254	47	55.6	51.3	6.3	2	5.3	7	2.5
U00259	50	62	56	6	3	5	7	3.0
U7589	55	63	59	6	2	5	6.5	2.0
U4981	60	68	64	6	2	5	6.5	2.0
U5717	60	68.5	65.25	3.8	3	2.8	5	2.5
U7725	60	70	68	8	6	7	12	2.5
U1252B	63	73	68	7.1	3	6.1	8.3	2.5
U4809	69.5	78	75.75	3.8	3	2.8	5	2.5
U5718	80	89	85.5	4.8	3	3.8	6	2.5
U5084	90	98.5	96.25	4.8	3	3.8	6	2.5
U5581	90	100	96	7.8	3	6.8	10	2.5
U6904	100	109	105.5	4.8	3	3.8	6	2.5
U9086	100	110	105	7	3	6	8	2.5
U8516	100	112.2	107.6	8.2	5	7.2	12	3.5
U6905	110	119.5	115.75	4.8	3	3.8	6	2.5
U5207	117.5	127	124.25	4.8	3	3.8	6	2.5
U8262	140	148.6	145.8	7	4	6	9	2.5
U6903	140	149.5	145.75	4.8	3	3.8	6	2.5
U7706	160	172.2	167.1	8.1	4	7.1	11	3.5

4.2

WP2 Metric Size



Ordering dimensions in green

[illegible]



WP3

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

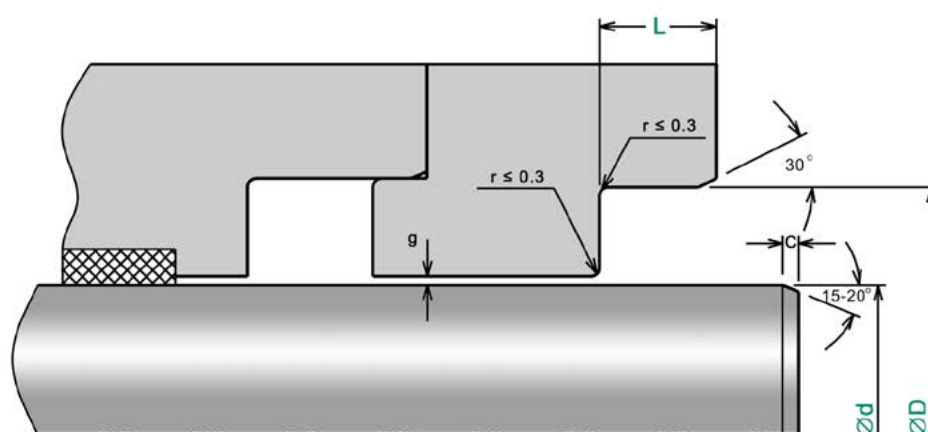
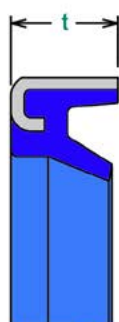
TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	≤ 3.2 μm	≤ 12.5 μm
Groove flanks	L	≤ 3.2 μm	≤ 12.5 μm

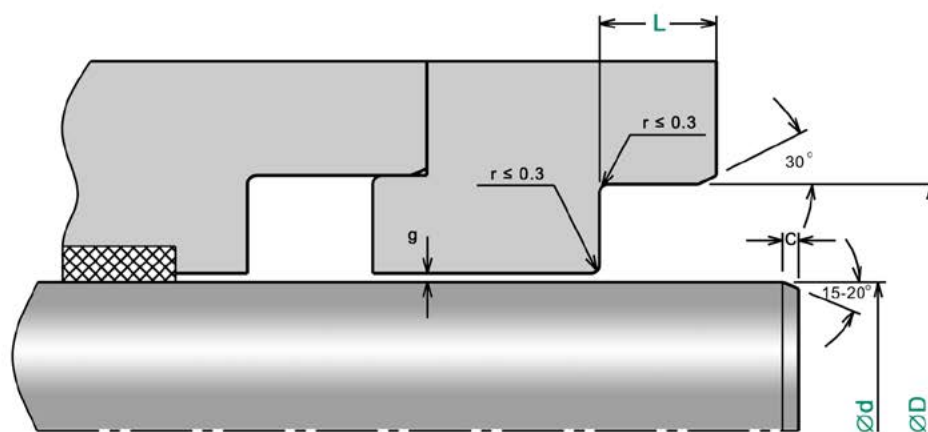
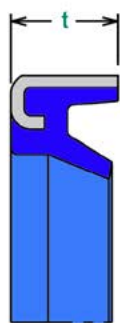
WP3 Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	c
U2727	16	23	5	5	2.0
U4596	16	26	7	7	2.5
U7011	20	28	5	5	2.0
U4597	20	30	7	7	2.5
U00189	25	32	5	5	2.0
U1246	25	32	7	7	2.0
U5673	25	35	4	4	2.5
U8602	25	37	6	6	3.0
U2947	25	38	7.5	7.5	3.5
U7532	30	38	5	5	2.0
U4211	30	40	4	4	2.5
U5101	30	40	7.0	7.0	2.5
U0883	30	43	7.5	7.5	3.5
U5897	30	45	4	4	4.0
U4454	35	45	4	4	2.5
U9737	35	45	7	7	2.5
U0882	35	50	7.5	7.5	4.0
U2724	36	46	7	7	2.5
U9623	36	48	6	6	3.0
U3861	38	50	7.5	7.5	3.0
U5902	40	50	3.2	3.2	2.5
U4455	40	50	4	4	2.5
U5614	40	50	5	5	2.5
U4462	40	50	7	7	2.5
U7410	40	50	8	8	2.5
U4474	40	50	9	9	2.5
U6559	40	52	6	6	3.0
U4394	40	52	7	7	3.0
U4463	40	55	9	9	4.0
U3821	44	61.9	10	10	4.5
U6878	45	55	3.18	3.2	2.5
U4522	45	55	3.2	3.2	2.5
U4456	45	55	4	4	2.5
U5662	45	55	4.5	4.5	2.5
U0217	45	55	5	5	2.5
U5756	45	55	6	6	2.5
U8611	45	55	7	7	2.5
U4518	45	57	7	7	3.0

4.3

WP3 Metric Size

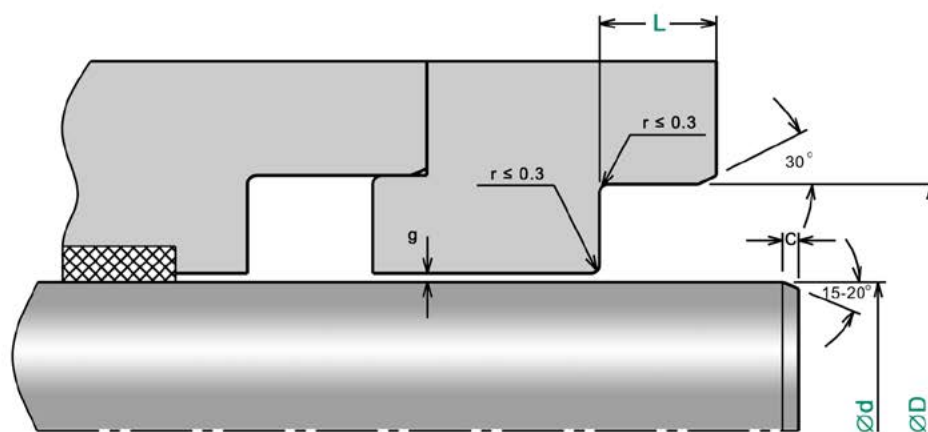
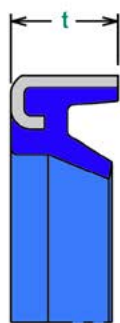
Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	c
U5757	45	60	5	5	4.0
U3517	45	60	7.5	7.5	4.0
U8700	47.62	60.32	6.4	6.4	3.5
U4548	50	60	4	4	2.5
U3521	50	60	4.2	4.2	2.5
U5882	50	60	4.5	4.5	2.5
U3105	50	60	5	5	2.5
U5900	50	60	6	6	2.5
U2725	50	60	7	7	2.5
U3853	50	62	7	7	3.0
U9637	50	63	4	4	3.5
U5610	50	63	5	5	3.5
U7511	50	63	7.5	7.5	3.5
U5247	50	64	8	8	3.5
U5675	50	65	5	5	4.0
U5901	50	65	6	6	4.0
U2997	50	65	7.5	7.5	4.0
U4475	50	65	9	9	4.0
U9244	50.8	59.06	4.75	4.8	2.5
U4769	55	65	3.2	3.2	2.5
U4461	55	65	5	5	2.5
U4492	55	65	7	7	2.5
U9856	55	68	4	4	3.5
U5629	55	68	6	6	3.5
U3854	55	69	8	8	3.5
U5895	55	70	5	5	4.0
U5674	55	70	6	6	4.0
U3419	55	70	7.5	7.5	4.0
U6477	56	68	7	7	3.0
U5898	56	70	7.5	7.5	3.5
U4483	56	71	9	9	4.0
U3266	60	70	5	5	2.5
U6581	60	70	7	7	2.5
U00396	60	72	8	8	3.0
U5245	60	73	6	6	3.5
U8978	60	73	7	7	3.5
U6515	60	73	8	8	3.5
U3044	60	74	8	8	3.5

WP3 Metric Size

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	c
U5653	60	75	4	4	4.0
U5422	60	75	5	5	4.0
U5642	60	75	6	6	4.0
U4536	60	75	7	7	4.0
U3106	60	75	7.5	7.5	4.0
U2833	60	75	8	8	4.0
U4466	63	75	7	7	3.0
U4312	63	78	7.5	7.5	4.0
U4484	63	78	9	9	4.0
U6516	63	83	10	10	5.0
U3855	65	75	7	7	2.5
U3856	65	79	8	8	3.5
U5644	65	80	5	5	4.0
U2250	65	80	7.5	7.5	4.0
U5881	65	80	8	8	4.0
U7847	70	80	5	5	2.5
U3857	70	80	7	7	2.5
U7839	70	80	8	8	2.5
U8607	70	84	4	4	3.5
U1728	70	84	7.5	7.5	3.5
U0936	70	84	8	8	3.5
U00199	70	85	4	4	4.0
U5643	70	85	5	5	4.0
U4543	70	85	7	7	4.0
U2251	70	85	7.5	7.5	4.0
U3639	70	85	8	8	4.0
U8106	70	85	8	8	4.0
U5464	70	85	10	10	4.0
U5628	71	86	5	5	4.0
U5880	71	86	6	6	4.0
U00514	74.5	90	7.5	7.5	4.0
U4201	75	89	8	8	3.5
U00492	75	90	5	5	4.0
U5963	75	90	6	6	4.0
U2441	75	90	7.5	7.5	4.0
U5631	75	90	8	8	4.0
U8655	75	95	10	10	5.0
U00434	76.5	96.5	10	10	5.0
U4523	80	90	3.18	3.2	2.5
U5741	80	90	4.5	4.5	2.5
U7273	80	90	6	6	2.5
U3445	80	90	7	7	2.5
U7838	80	90	8	8	2.5
U2391	80	94	8	8	3.5
U3303	80	95	5	5	4.0
U5742	80	95	6	6	4.0
U2397	80	95	7.5	7.5	4.0
U3642	80	95	8	8	4.0
U4046	80	100	10	10	5.0
U00515	84	100	4	4	4.0
U6742	85	95	5	5	2.5
U7742	85	95	7	7	2.5
U7837	85	95	8	8	2.5
U2852	85	99	8	8	3.5

4.3

WP3 Metric Size

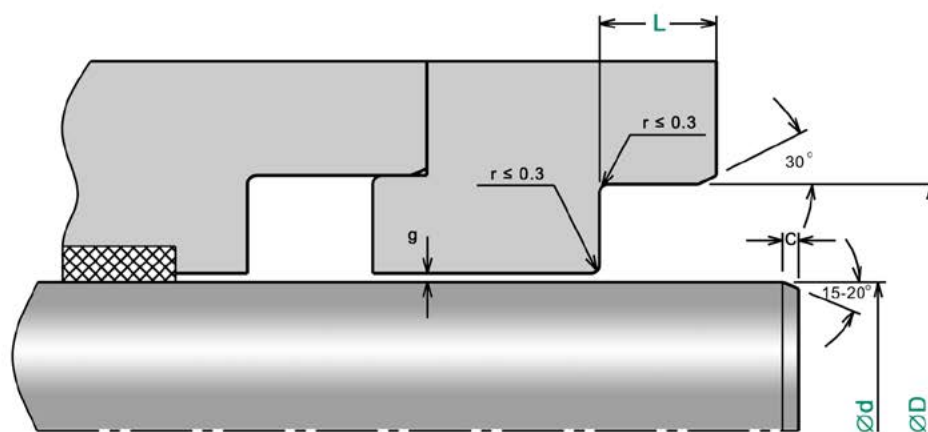
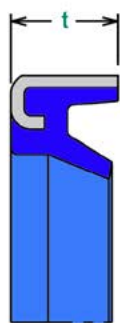
Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	c
U00652	85	99	12	12	3.5
U8229	85	100	4	4	4.0
U5097	85	100	6	6	4.0
U5419	85	100	8	8	4.0
U7603	85	100	10	10	4.0
U3518	85	105	10	10	5.0
U7208	88.9	101.6	9.5	9.5	3.5
U5641	88.9	107.95	7.9	7.9	5.0
U2787	90	100	7	7	2.5
U7836	90	100	8	8	2.5
U3088	90	104	8	8	3.5
U2981	90	105	6	6	4.0
U3418	90	105	8	8	4.0
U2784	90	106	8	8	4.0
U00599	90	108	8	8	4.5
U4506	90	110	7	7	5.0
U0734	90	110	8	8	5.0
U2442	90	110	10	10	5.0
U5423	95	109	8	8	3.5
U6176	95	110	6	6	4.0
U4549	95	110	8	8	4.0
U9424	95	115	10	10	5.0
U7005	95.25	107.95	9.52	9.5	3.5
U6533	100	110	7	7	2.5
U7832	100	110	8	8	2.5
U3328	100	114	8	8	3.5
U9671	100	115	4	4	4.0
U5984	100	115	6	6	4.0
U4532	100	115	6.5	6.5	4.0
U4496	100	115	7	7	4.0
U2982	100	115	7.5	7.5	4.0
U4095	100	115	8	8	4.0
U1987	100	115	9	9	4.0
U2785	100	116	8	8	4.0
U8686	100	118	8	8	4.5
U6000	100	120	4	4	5.0
U8107	100	120	6.5	6.5	5.0
U7749	100	120	7	7	5.0

WP3 Metric Size

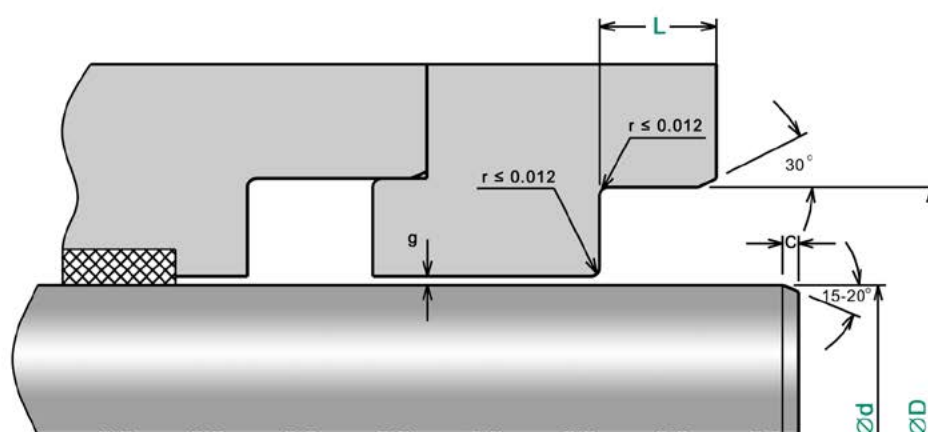
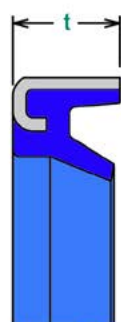
NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	c
U0736	100	120	8	8	5.0
U2325	100	120	10	10	5.0
U7019	101.6	114.3	9.5	9.5	3.5
U6608	105	120	7	7	4.0
U9857	105	120	7.5	7.5	4.0
U5420	105	120	8	8	4.0
U7335	110	120	7	7	2.5
U7833	110	120	8	8	2.5
U5904	110	125	4	4	4.0
U4500	110	125	8	8	4.0
U4019	110	125	9	9	4.0
U2783	110	126	8	8	4.0
U2694	110	126	9	9	4.0
U8612	110	128	8	8	4.5
U9217	110	130	7.5	7.5	5.0
U5096	110	130	8	8	5.0
U3523	110	130	9	9	5.0
U2983	110	130	10	10	5.0
U7505	110	140	8	8	7.5
U4314	115	130	8	8	4.0
U5850	115	135	8	8	5.0
U7114	120	130	7	7	2.5
U7834	120	130	8	8	2.5
U00257	120	135	4	4	4.0
U9611	120	135	6.5	6.5	4.0
U2365	120	135	8	8	4.0
U0939	120	136	9	9	4.0
U2655	120	140	8	8	5.0
U4519	120	140	10	10	5.0
U4395	120	150	12	12	7.5
U5421	125	140	8	8	4.0
U4535	125	140	9	9	4.0
U7008	125	145	8	8	5.0
U00258	125	145	10	10	5.0
U00186	130	140	7	7	2.5
U2678	130	145	7.5	7.5	4.0
U0937	130	146	9	9	4.0
U3803	130	150	8	8	5.0
U4425	130	150	10	10	5.0
U3268	130	154	8	8	6.0
U8108	130	155	8	8	6.5
U5849	130	155	10	10	6.5
U4396	130	160	12	12	7.5
U7848	140	155	8	8	4.0
U7829	140	160	8	8	5.0
U3468	140	160	10	10	5.0
U5122	140	162	7.5	7.5	5.5
U7789	140	164	8	8	6.0
U7280	140	165	8	8	6.5
U5266	140	170	8	8	7.5
U7781	140	170	12	12	7.5
U7830	145	160	10	10	4.0
U5276	150	165	8	8	4.0
U5132	150	170	10	10	5.0

4.3

WP3 Metric Size

Ordering dimensions in green

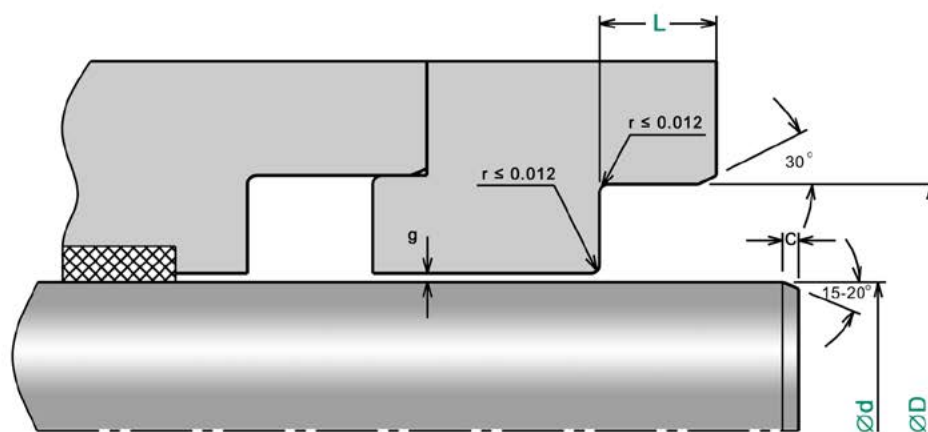
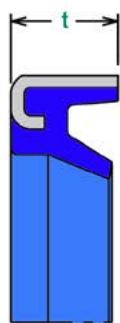
NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	c
U5267	150	175	8	8	6.5
U2366	150	175	9	9	6.5
U9646	150	180	8	8	7.5
U7215	150	180	10	10	7.5
U8696	160	175	8	8	4.0
U7007	160	175	10	10	4.0
U8071	160	180	5	5	5.0
U7831	160	180	8	8	5.0
U3267	160	180	10	10	5.0
U5277	160	184	8	8	6.0
U2367	160	190	9	9	7.5
U5268	160	190	10	10	7.5
U5630	170	190	8	8	5.0
U2368	170	190	9	9	5.0
U2369	170	200	10	10	7.5
U5667	175	195	10	10	5.0
U6280	180	195	9	9	4.0
U4397	180	200	10	10	5.0
U9872	180	210	15	15	7.5
U00397	190	210	10	10	5.0
U8289	200	220	10	10	5.0
U6729	200	225	12	12	6.5
U00195	205	225	10	10	5.0
U5676	215	240	10	10	6.5
U5209	220	240	16	16	5.0
U5317	230	250	12	12	5.0

WP3 Inch Size

Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.008	t	c
U5102	0.625	1.125	0.312	0.312	0.138
U5017	0.750	1.250	0.312	0.312	0.138
U4537	0.875	1.375	0.312	0.312	0.138
U5797	1.000	1.250	0.125	0.125	0.079
U5664	1.000	1.375	0.187	0.187	0.098
U4541	1.000	1.500	0.312	0.312	0.138
U4485	1.125	1.625	0.312	0.312	0.138
U4652	1.250	1.687	0.187	0.187	0.118
U8089	1.250	1.750	0.187	0.187	0.138
U5877	1.250	1.750	0.250	0.250	0.138
U4542	1.250	1.750	0.312	0.312	0.138
U5100	1.250	2.062	0.312	0.312	0.217
U5033	1.250	2.062	0.375	0.375	0.217
U5465	1.375	1.750	0.187	0.187	0.098
U4464	1.375	1.875	0.312	0.312	0.138
U5665	1.500	1.875	0.187	0.187	0.098
U5818	1.500	1.875	0.250	0.250	0.098
U2761	1.500	2.000	0.250	0.250	0.138
U3823	1.500	2.000	0.312	0.312	0.138
U4538	1.500	2.125	0.312	0.312	0.157
U5466	1.500	2.250	0.250	0.250	0.197
U4477	1.500	2.250	0.375	0.375	0.197
U5262	1.625	2.125	0.250	0.250	0.138
U5263	1.625	2.125	0.312	0.312	0.138
U5836	1.750	2.125	0.125	0.125	0.098
U4482	1.750	2.250	0.187	0.187	0.138
U3762	1.750	2.250	0.250	0.250	0.138
U5264	1.750	2.250	0.312	0.312	0.138
U4478	1.750	2.437	0.375	0.375	0.177
U4465	1.750	2.500	0.375	0.375	0.197
U5265	1.875	2.375	0.312	0.312	0.138
U5467	1.875	2.562	0.375	0.375	0.177
U5250	2.000	2.375	0.187	0.187	0.098
U5261	2.000	2.375	0.250	0.250	0.098
U4746	2.000	2.500	0.250	0.250	0.138
U3304	2.000	2.500	0.312	0.312	0.138
U5817	2.000	2.625	0.250	0.250	0.157
U4479	2.000	2.686	0.375	0.375	0.177

4.3

WP3 Inch Size

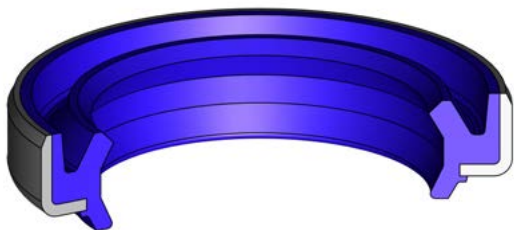
Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.008	t	c
U3007	2.000	2.688	0.313	0.313	0.177
U4480	2.000	2.750	0.375	0.375	0.197
U5816	2.125	2.500	0.125	0.125	0.098
U5837	2.125	2.625	0.250	0.250	0.138
U5838	2.125	2.625	0.313	0.313	0.138
U4615	2.125	2.875	0.375	0.375	0.197
U5815	2.250	2.750	0.187	0.187	0.138
U5260	2.250	2.750	0.250	0.250	0.138
U4539	2.250	2.750	0.312	0.312	0.138
U5246	2.250	3.000	0.250	0.250	0.197
U4643	2.250	3.000	0.375	0.375	0.197
U1374	2.345	2.727	0.210	0.210	0.098
U4481	2.375	2.875	0.312	0.312	0.138
U5814	2.375	3.125	0.375	0.375	0.177
U4654	2.500	3.000	0.187	0.187	0.138
U5360	2.500	3.000	0.250	0.250	0.138
U2348	2.500	3.000	0.312	0.312	0.138
U5813	2.500	3.250	0.250	0.250	0.197
U4505	2.500	3.250	0.375	0.375	0.197
U5803	2.500	3.500	0.375	0.375	0.256
U5106	2.625	3.125	0.250	0.250	0.138
U4067	2.734	3.250	0.313	0.313	0.138
U5827	2.750	3.250	0.187	0.187	0.138
U5258	2.750	3.250	0.312	0.312	0.138
U4467	2.750	3.500	0.375	0.375	0.197
U5820	2.750	3.750	0.500	0.500	0.256
U4516	3.000	3.500	0.312	0.312	0.138
U5822	3.000	3.750	0.250	0.250	0.197
U4493	3.000	3.750	0.375	0.375	0.197
U4501	3.000	4.000	0.500	0.500	0.256
U5865	3.125	3.750	0.375	0.375	0.157
U4502	3.250	3.750	0.312	0.312	0.138
U4668	3.250	3.875	0.312	0.312	0.157
U4540	3.250	4.250	0.500	0.500	0.256
U4745	3.375	4.000	0.312	0.312	0.157
U4494	3.500	4.000	0.312	0.312	0.138
U3333	3.500	4.125	0.312	0.312	0.157
U5034	3.500	4.500	0.500	0.500	0.256

WP3

Inch Size

[illegible]



WP4

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

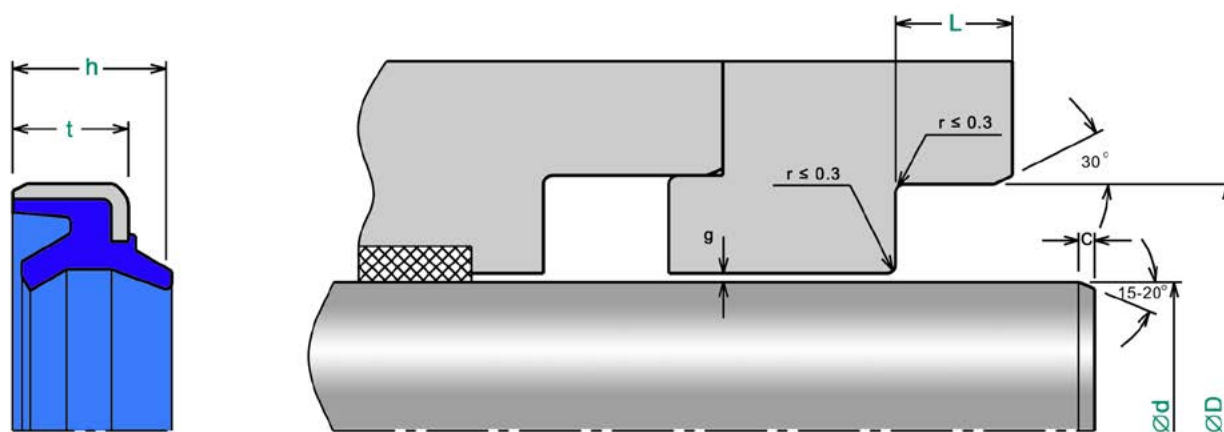
TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

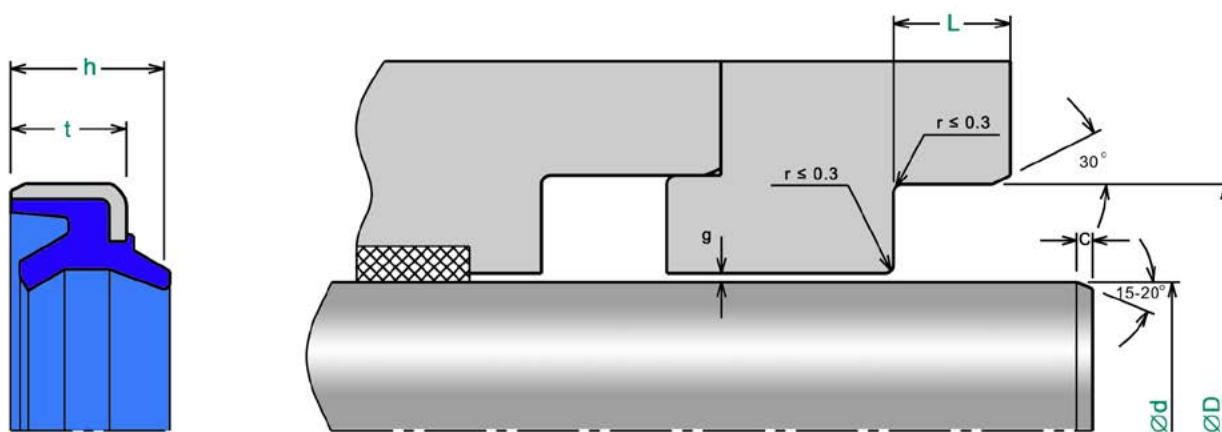
WP4 Metric Size



Ordering dimensions in green

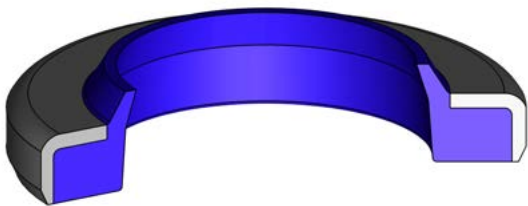
NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U5762	18	30	6.0	6	9	3.0
U1834	19	26	5.0	5	7	2.0
U4614	19	27	4.0	4	6	2.0
U3149	20	32	6.0	6	9	3.0
U00647	25	32	4.0	4	6	2.0
U2695	25	37	6.0	6	9	3.0
U4476	25.4	38	6.0	6	9	3.5
U7301	25.6	34.8	4.9	4.9	5.6	2.5
U2558	30	42	6.0	6	9	3.0
U2548	32	44	7.0	7	10	3.0
U0523	34	46	7.0	7	10	3.0
U2252	35	47	7.0	7	10	3.0
U5293	35	55	7.0	7	10	5.0
U6600	36	48	7.0	7	10	3.0
U4414	38	50	7.0	7	10	3.0
U9445	40	50	7.0	7	10	2.5
U2396	40	52	7.0	7	10	3.0
U8131	40	55	7.0	7	10	4.0
U5291	40	60	7.0	7	10	5.0
U2297	45	57	7.0	7	10	3.0
U5294	45	65	7.0	7	10	5.0
U3107	50	62	7.0	7	10	3.0
U7673	50	62	8.0	8	12	3.0
U8133	50	65	7.0	7	10	4.0
U6601	50	72	8.0	8	11	5.5
U8085	53.97	66.67	6.6	6.6	9.9	3.5
U3097	55	69	8.0	8	11	3.5
U7390	55	69	8.0	8	12	3.5
U5295	55	75	7.0	7	10	5.0
U00583	60	70	7.0	7	10	2.5
U00078	60	73	7.0	7	10	3.5
U3022	60	74	8.0	8	11	3.5
U7387	60	74	8.0	8	12	3.5
U8134	60	75	7.0	7	10	4.0
U5308	63	77	8.0	8	11	3.5
U2676	65	79	8.0	8	11	3.5
U7388	65	79	8.0	8	12	3.5
U6977	65	79	8.0	8	13	3.5

4.4

WP4 Metric Size

Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U5296	65	85	7.0	7	10	5.0
U3278	70	83	8.0	8	11	3.0
U2443	70	84	8.0	8	11	2.0
U7392	70	84	8.0	8	12	2.0
U6894	70	84	8.0	8	13	3.0
U8132	70	85	7.0	7	10	2.0
U2392	75	89	8.0	8	11	3.0
U7389	75	89	8.0	8	12	3.5
U8266	80	94	8.0	8	11	2.5
U2350	80	94	8.0	8	12	3.0
U6895	80	94	8.0	8	13	3.0
U8339	80	95	7.0	7	10	3.0
U2677	85	99	8.0	8	11	3.0
U7384	85	99	8.0	8	12	5.0
U6896	85	99	8.0	8	13	3.0
U3023	90	104	8.0	8	11	3.0
U7748	90	104	8.0	8	12	2.5
U6976	90	104	8.0	8	13	3.0
U00231	90	110	8.0	8	11	4.0
U0514	95	109	8.0	8	11	5.0
U7385	95	109	8.0	8	12	3.0
U6893	95	109	8.0	8	13	5.0
U0515	100	114	8.0	8	11	3.0
U7678	100	114	8.0	8	12	3.0
U6870	100	114	8.0	8	13	4.0
U7679	105	121	8.0	8	12	5.5
U2998	105	121	9.0	9	12	3.5
U2950	110	126	9.0	9	12	3.5
U7680	110	126	9.0	9	12.5	3.5
U3569	115	131	9.0	9	12	5.0
U7672	115	131	9.0	9	12.5	2.5
U3150	120	136	9.0	9	12	3.5
U3151	130	146	9.0	9	12	3.5
U2835	140	160	10.0	10	14	3.5
U8396	145	165	10.0	10	14	4.0
U00600	150	170	10.0	10	14	3.5
U00493	170	190	10.0	10	14	3.5
U5040	200	216	8.0	8	11	3.5
U5810	230	255	12.0	12	17	6.5



WP5

Wiper Seal 防塵密封件

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

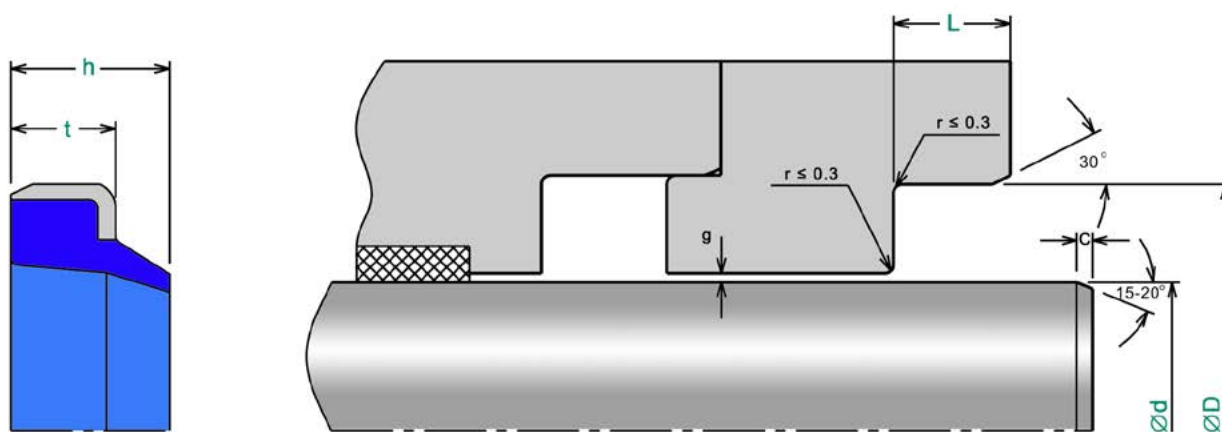
TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

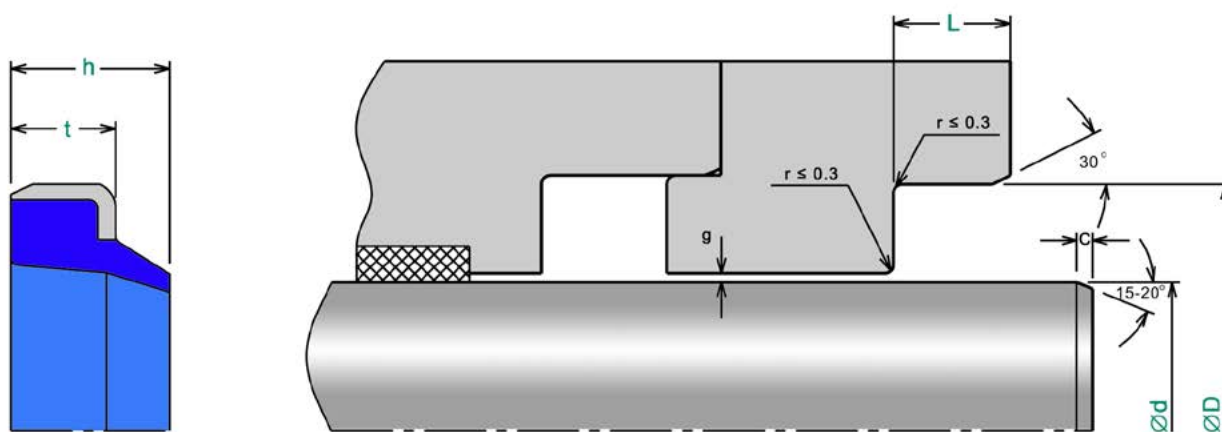
WP5 Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U6578	8	14	3	3	5	1.5
U3820	8	14	3.5	3.5	5	1.5
U00494	8	18	5	5	8	2.5
U8293	9	16	5	5	7	2.0
U5979	10	16	3.5	3.5	5	1.5
U2551	10	20	5	5	8	2.5
U6577	12	18	3	3	5	1.5
U7079	12	18	3.5	3.5	5	1.5
U00346	12	18	4	4	6	1.5
U2352	12	20	4	4	6	2.0
U3086	12	22	5	5	8	2.5
U2689	14	20	3.5	3.5	5	1.5
U7275	14	22	3	3	4	2.0
U4580	14	24	5	5	7	2.5
U7503	14	25	5	5	8	3.0
U4399	15	21	3.5	3.5	5	1.5
U0638	15	25	5	5	8	2.5
U9907	15	25	5	5	7	2.5
U2667	16	22	3	3	4	1.5
U1694	16	22	3.5	3.5	5	1.5
U2907	16	26	5	5	8	2.5
U2673	17	24	3.5	3.5	5	2.0
U1729	18	28	5	5	7	2.5
U3843	18	28	5	5	8	2.5
U3937	18	28	7	7	10	2.5
U2759	18	30	6	6	9	3.0
U3033	19	26	5	5	8	2.0
U5539	19	26	5	5	7	2.0
U00546	19	27	4	4	5.2	2.0
U2781	20	28	3.5	3.5	5	2.0
U2674	20	30	4	4	6	2.5
U2326	20	30	5	5	8	2.5
U2240	20	30	7	7	10	2.5
U2373	20	32	6	6	9	3.0
U3290	20	32	7	7	10	3.0
U5978	22	28	5	5	8	1.5
U4792	22	32	5	5	8	2.5
U3928	22	32	7	7	10	2.5

4.5

WP5 Metric Size

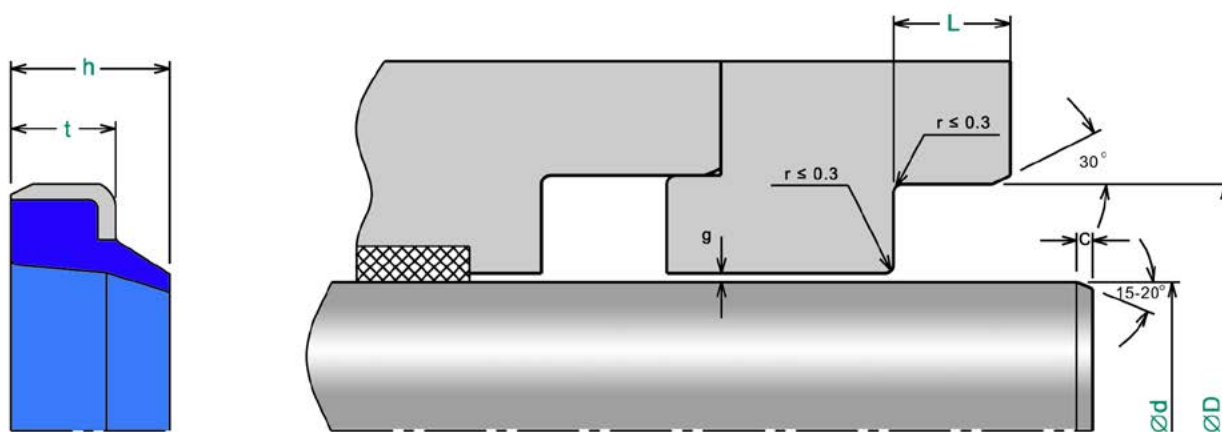
Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U2229	25	32	5	5	8	2.0
U00077	25	32	5	5	7	2.0
U4273	25	35	5	5	8	2.5
U7181	25	35	7	7	9	2.5
U2690	25	35	7	7	10	2.5
U4392	25	37	6	6	9	3.0
U5540	25.4	38	6	6	9	3.5
U3844	28	38	5	5	8	2.5
U3804	28	38	7	7	10	2.5
U1326	28	40	6	6	9	3.0
U5325	28	40	7	7	10	3.0
U5604	28	42	7	7	10	3.5
U9853	28.57	38.1	5.56	5.56	7.14	2.5
U3211	30	40	5	5	8	2.5
U4252	30	42	6	6	9	3.0
U0064	30	45	5	5	8	4.0
U0967	30	47	7	7	10	4.5
U5723	32	40	4	4	7	2.0
U3845	32	42	5	5	8	2.5
U2786	32	42	7	7	10	2.5
U3198	32	44	7	7	10	3.0
U2934	32	45	4	4	6	3.5
U00568	32	45	4	4	8	3.5
U2691	32	45	7	7	10	3.5
U5977	33	43	7	7	10	2.5
U00624	35	43	5	5	8	2.0
U3277	35	45	7	7	10	2.5
U3798	35	47	7	7	10	3.0
U2245	36	45	7	7	10	2.5
U5976	36	46	5	5	7	2.5
U2935	36	46	7	7	10	2.5
U3846	36	50	7	7	10	3.5
U4393	38	48	7	7	10	2.5
U00244	38	50	7	7	10	3.0
U00270	38.1	47.62	5.56	5.56	7.14	2.5
U3630	40	50	5	5	8	2.5
U2454	40	50	7	7	10	2.5
U3470	40	50	7	7	10	2.5

WP5 Metric Size

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U2229	25	32	5	5	8	2.0
U00077	25	32	5	5	7	2.0
U4273	25	35	5	5	8	2.5
U7181	25	35	7	7	9	2.5
U2690	25	35	7	7	10	2.5
U4392	25	37	6	6	9	3.0
U5540	25.4	38	6	6	9	3.5
U3844	28	38	5	5	8	2.5
U3804	28	38	7	7	10	2.5
U1326	28	40	6	6	9	3.0
U5325	28	40	7	7	10	3.0
U5604	28	42	7	7	10	3.5
U9853	28.57	38.1	5.56	5.56	7.14	2.5
U3211	30	40	5	5	8	2.5
U4252	30	42	6	6	9	3.0
U0064	30	45	5	5	8	4.0
U0967	30	47	7	7	10	4.5
U5723	32	40	4	4	7	2.0
U3845	32	42	5	5	8	2.5
U2786	32	42	7	7	10	2.5
U3198	32	44	7	7	10	3.0
U2934	32	45	4	4	6	3.5
U00568	32	45	4	4	8	3.5
U2691	32	45	7	7	10	3.5
U5977	33	43	7	7	10	2.5
U00624	35	43	5	5	8	2.0
U3277	35	45	7	7	10	2.5
U3798	35	47	7	7	10	3.0
U2245	36	45	7	7	10	2.5
U5976	36	46	5	5	7	2.5
U2935	36	46	7	7	10	2.5
U3846	36	50	7	7	10	3.5
U4393	38	48	7	7	10	2.5
U00244	38	50	7	7	10	3.0
U00270	38.1	47.62	5.56	5.56	7.14	2.5
U3630	40	50	5	5	8	2.5
U2454	40	50	7	7	10	2.5
U3470	40	50	7	7	10	2.5
U2517	40	52	7	7	10	3.0
U00079	40	57	7	7	10	4.5
U0202	40	60	7	7	10	5.0
U5975	42	52	5	5	8	2.5
U0843	42	52	7	7	10	2.5
U8482	44.45	53.97	5.56	5.56	7.14	2.5
U5794	45	55	5	5	8	2.5
U2755	45	55	7	7	10	2.5
U2232	45	57	7	7	10	3.0
U3104	45	60	7	7	10	4.0
U3790	48	60	7	7	10	3.0
U7077	48	62	7	7	10	3.5
U2996	50	56	5	5	8	1.5
U4992	50	60	5	5	8	2.5
U2228	50	60	7	7	10	2.5
U9599	50	62	5	5	8	3.0

4.5

WP5 Metric Size

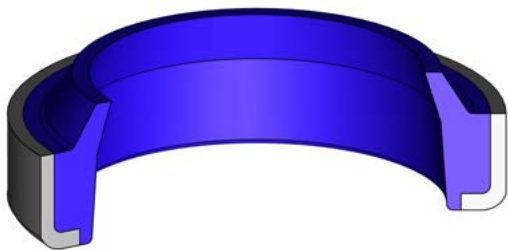
Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U3199	50	62	7	7	10	3.0
U0965	50	65	7	7	10	4.0
U0853	52	62	7	7	10	2.5
U1331	53	67	8	8	11	3.5
U8496	53.97	63.5	5.56	5.56	7.14	2.5
U3428	55	63	7	7	10	2.0
U2057	55	65	5	5	8	2.5
U2233	55	65	7	7	10	2.5
U2669	55	69	8	8	11	3.5
U5973	56	66	5	5	8	2.5
U2550	56	66	7	7	10	2.5
U1333	56	70	8	8	11	3.5
U9226	57.15	66.67	5.56	5.56	7.14	2.5
U5972	60	70	5	5	8	2.5
U2253	60	70	7	7	10	2.5
U2670	60	74	5	5	8	3.5
U2936	60	74	8	8	11	3.5
U0964	60	75	7	7	10	4.0
U8869	60	80	7	7	10	5.0
U3516	63	73	5	5	8	2.5
U1887	63	73	7	7	10	2.5
U2631	63	75	7	7	10	3.0
U2937	63	77	8	8	11	3.5
U5896	63	78	7	7	10	4.0
U1336	64	78	8	8	11	3.5
U5971	65	75	5	5	8	2.5
U2455	65	75	7	7	10	2.5
U3847	65	79	8	8	11	3.5
U9247	66	76	7	7	10	2.5
U1337	67	81	8	8	11	3.5
U5981	70	80	5	5	8	2.5
U2760	70	80	7	7	10	2.5
U2671	70	84	8	8	11	3.5
U9693	70	85	7	7	10	4.0
U9850	70	86	7	7	10	4.0
U5983	75	83	7	7	10	2.0
U2456	75	85	7	7	10	2.5
U5500	75	89	8	8	11	3.5

WP5 Metric Size

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U9733	77	87	7	7	10	2.5
U5982	80	88	7	7	10	2.0
U2254	80	90	7	7	10	2.5
U2672	80	94	8	8	11	3.5
U4253	82	92	7	7	10	2.5
U2675	85	95	7	7	10	2.5
U3187	85	99	8	8	11	3.5
U2094	90	100	7	7	10	2.5
U1341	90	104	8	8	11	3.5
U3148	90	104	8	8	11	3.5
U2757	90	110	10	10	14	5.0
U1342	92	106	8	8	11	3.5
U2095	95	105	7	7	10	2.5
U5430	95	109	8	8	11	3.5
U2834	100	110	7	7	10	2.5
U2518	100	114	8	8	11	3.5
U2693	100	115	8	8	11	4.0
U2410	105	115	7	7	10	2.5
U1344	105	121	9	9	12	4.0
U2880	110	120	7	7	10	2.5
U2244	110	126	9	9	12	4.0
U6583	110	130	7	7	10	5.0
U1347	112	128	9	9	12	4.0
U4254	115	125	7	7	10	2.5
U5045	115	131	9	9	12	4.0
U1348	118	134	9	9	12	4.0
U2756	120	130	7	7	10	2.5
U2822	120	136	9	9	12	4.0
U3295	125	140	9	9	12	4.0
U3560	125	141	9	9	12	4.0
U3848	130	145	9	9	12	4.0
U1351	130	146	9	9	12	4.0
U1352	132	148	9	9	12	4.0
U2345	135	150	9	9	12	4.0
U1353	135	155	10	10	14	5.0
U2961	140	155	9	9	12	4.0
U3849	140	160	10	10	16	5.0
U1355	145	165	10	10	14	5.0
U2246	150	165	9	9	12	4.0
U3850	150	170	10	10	14	5.0
U2227	160	175	9	9	12	4.0
U3851	160	180	10	10	14	5.0
U9742	165	185	10	10	14	5.0
U2780	170	190	10	10	14	5.0
U2549	180	195	10	10	14	4.0
U2895	190	215	12	12	17	6.5
U3629	200	220	12	12	16	5.0
U3411	200	225	12	12	17	6.5
U1364	210	235	12	12	17	6.5
U1366	220	245	12	12	17	6.5

4.5



WP6

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

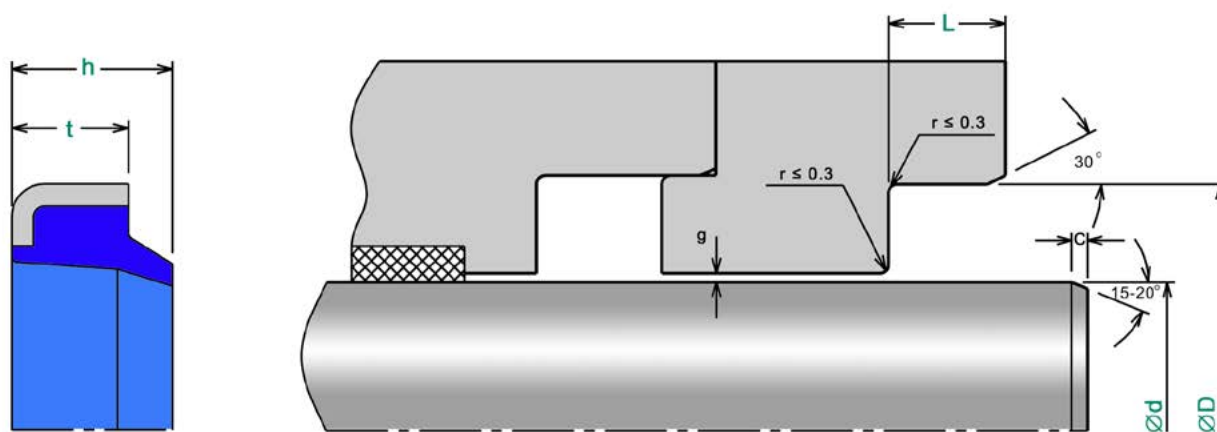
MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 3.2 µm	≤ 12.5 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

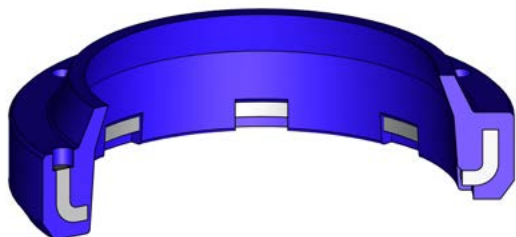
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WP6



Ordering dimensions in green

[illegible]



WP7

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

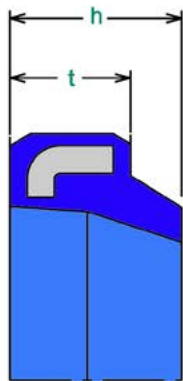
MATERIAL	PU
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	$\leq 0.8 \text{ µm}$	$\leq 3.2 \text{ µm}$
Groove flanks	L	$\leq 3.2 \text{ µm}$	$\leq 12.5 \text{ µm}$

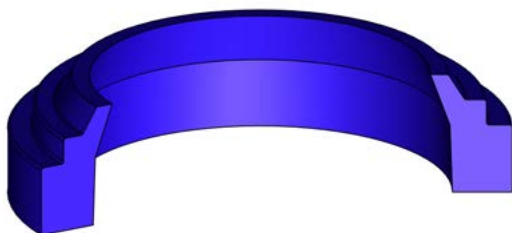
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WP7



Ordering dimensions in green

[illegible]



WP8

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

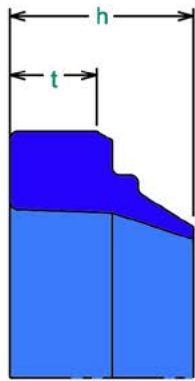
MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

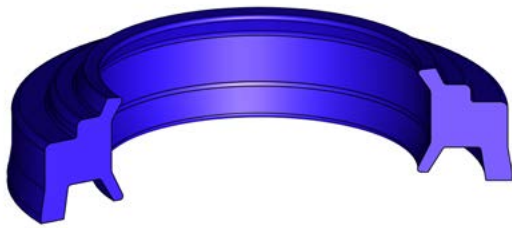
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WP8



Ordering dimensions in green

[illegible]



WP10

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

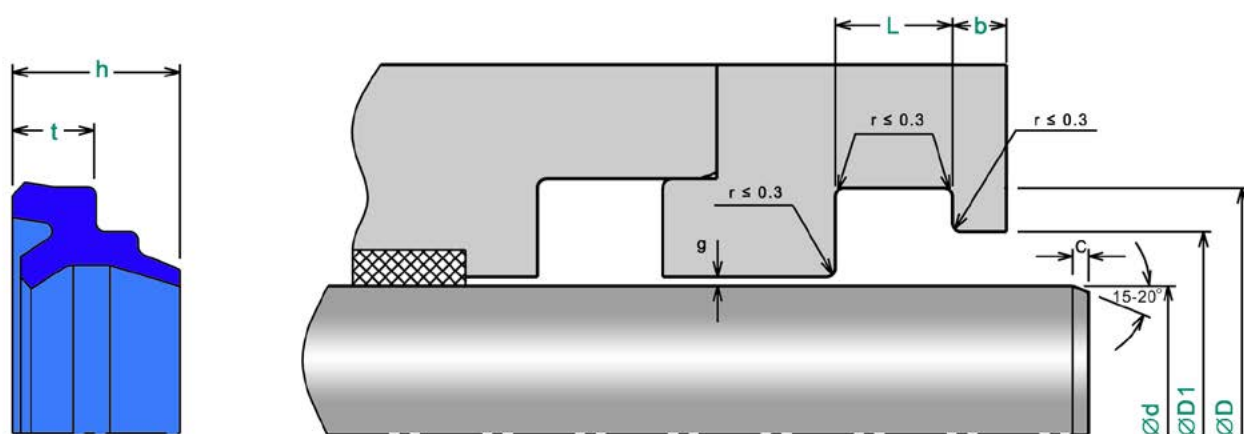
MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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WP10



Ordering dimensions in green

[illegible]



WP11

Wiper Seal 防塵密封件

PRODUCT DESCRIPTION

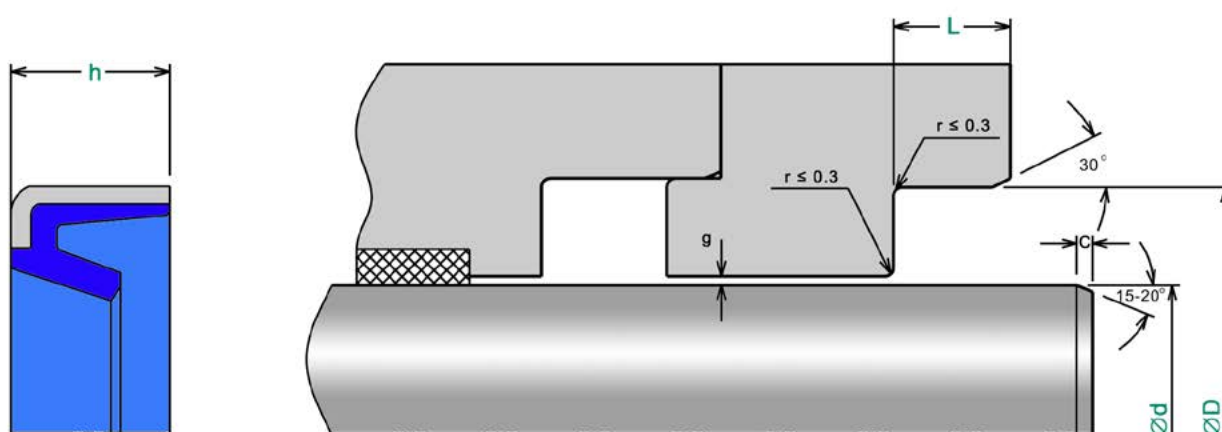
Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

WP11 Metric Size

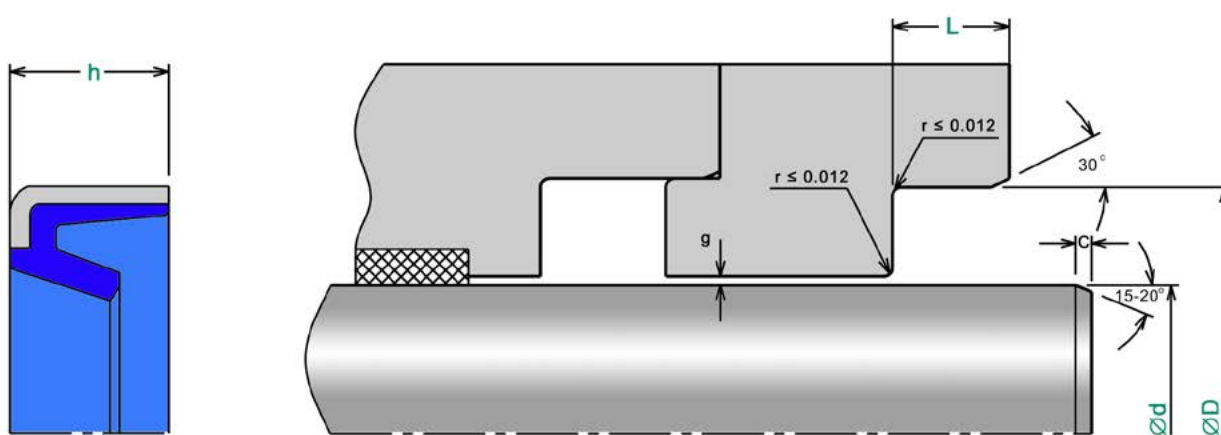
Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	h	c
U2953	40	50	4.2	4.2	2.5
U3897	40	50	7	7	2.5
U4459	40	55	9	9	4.0
U9628	45	55	7.5	7.5	2.5
U9626	45	60	7.5	7.5	4.0
U4213	50	60	7	7	2.5
U7747	50	65	7.5	7.5	4.0
U2453	52	68	6	6	4.0
U4221	56	71	9	9	4.0
U0371	60	74	8	8	3.5
U7808	60	75	7.5	7.5	4.0
U9627	63	75	8	8	3.0
U4942	70	84	8	8	3.5
U8592	80	93	8	8	3.5
U7807	80	95	6	6	4.0
U7109	80	95	8	8	4.0
U7707	85	99	8	8	3.5
U7786	85	100	8	8	4.0
U7723	90	105	8	8	4.0
U2841	90	106	8	8	4.0
U4229	90	110	7	7	5.0
U8539	90	110	10	10	5.0
U8528	90	115	6	6	6.5
U5714	100	114	8	8	3.5
U8530	100	115	6	6	4.0
U4218	100	115	7	7	4.0
U7112	100	120	8	8	5.0
U4241	110	125	8	8	4.0
U7806	110	130	10	10	5.0
U7688	115	130	7.5	7.5	4.0
U7687	115	135	8	8	5.0
U7087	120	150	12	12	7.5
U7686	125	140	8	8	4.0
U9658	130	140	7	7	2.5
U8489	130	145	7.5	7.5	4.0
U7082	130	154	8	8	6.0
U2156	130	154	8.5	8.5	6.0
U7083	130	160	12	12	7.5

4.10

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WP11 Inch Size



Ordering dimensions in green

[illegible]



WP12

PRODUCT DESCRIPTION

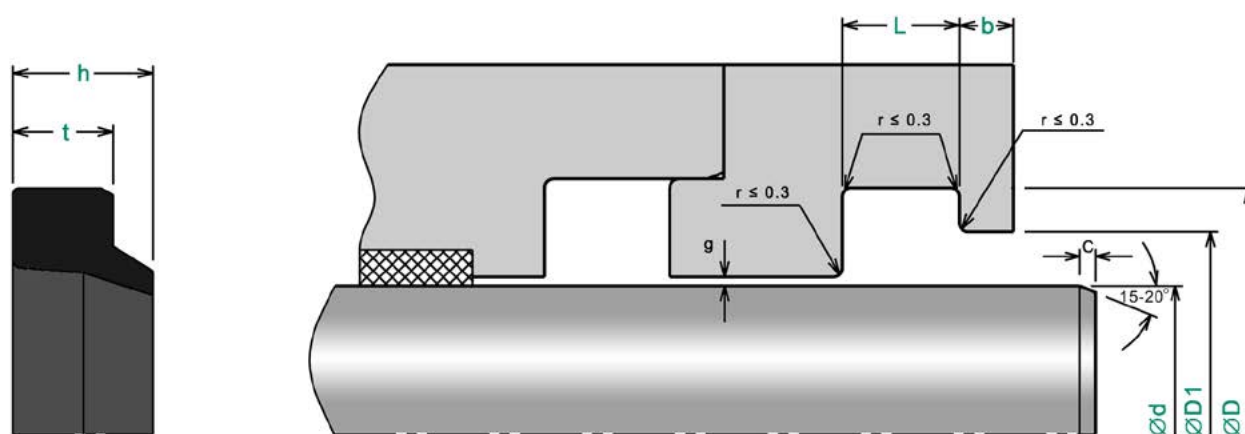
Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

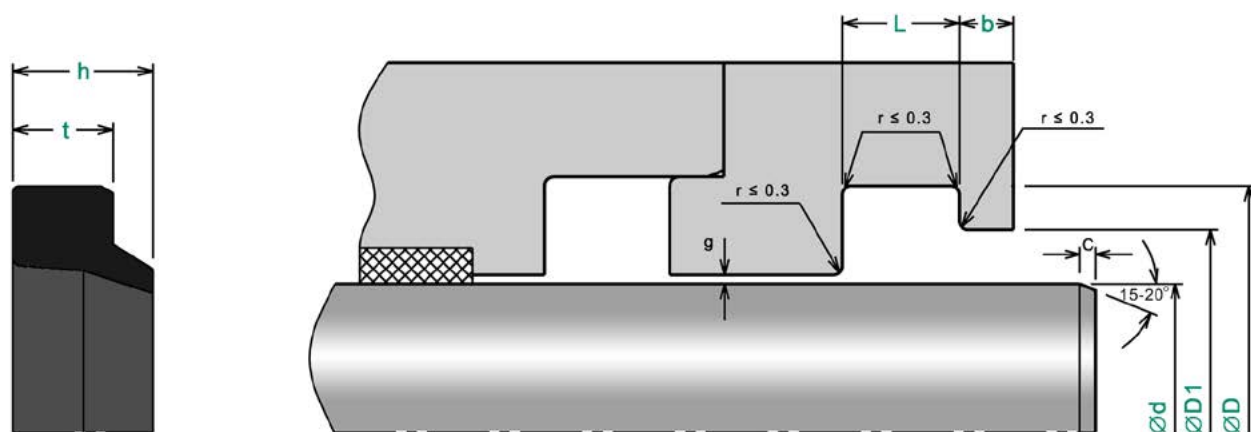
Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

WP12 Metric Size

Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	ØD1 +/- 0.1	L +/- 0.2	b	t	h	c
U7536	12	18	20.5	3.5	3	3	5.3	1.5
U5700	16	20	18	3.4	2	2.9	3.9	1.0
U7153	16	22	20.5	3.5	3	3	5.3	1.5
U5768	16	22	20	5.5	3	5	7	1.5
U7134	17	25	22.5	4.4	4	3.9	7	2.0
U5576	18	22	20.5	3.4	2	2.9	3.9	1.0
U7125	18	24	22.5	3.5	3	3	5.3	1.5
U8444	18	26	24	4.4	4	3.9	7	2.0
U4833	22	30	26	5.5	3	5	7	2.0
U8255	24	30	27	4.8	2	4.3	5.8	1.5
U7535	24	32	28	5.5	3	5	7	2.0
U00474	25	34	30	4.6	3	4.1	6.5	2.5
U0846	27.95	35	31.48	4	2	3.5	5.3	2.0
U4832	30	38	34.5	6	3	5	7	2.0
U7168	33	41	37	6	3	5	7	2.0
U7159	35	43	39	6	3	5	7	2.0
U9686	35.5	41.5	38.5	5.7	2	4.7	5.8	1.5
U5222	36	44	40.5	6	3	5	7	2.0
U8239	38	46	42	6	3	5	7	2.0
U5781	40	48	44	6	3	5	7	2.0
U4831	40	50	45	8	3	7	9	2.5
U7154	42	50	46	6	3	5	7	2.0
U7537	44	52	49.5	4.9	4	3.9	7	2.0
U7149	45	53	49	6	3	5	7	2.0
U7150	45	53.5	51.25	6	4	5	8.5	2.5
U5223	45	55	50	6	3	5	7	2.5
U5780	50	58	54	6	3	5	7	2.0
U7169	50	60	55	6	3	5	7	2.5
U7135	51	59	56.5	4.9	4	3.9	7	2.0
U7152	52	60	57.5	4.9	4	3.9	7	2.0
U7170	53	61	57	6	3	5	7	2.0
U7171	55	63	59	6	3	5	7	2.0
U7160	55	65	60	6	3	5	7	2.5
U5224	56	64	60.5	6	3	5	7	2.0
U5243	56	66	61	6	3	5	7	2.5
U4830	60	70	65	6	3	5	7	2.5
U5232	63	71	67.5	6	3	5	7	2.0
U7533	63	73	68	6	3	5	7	2.5

4.11

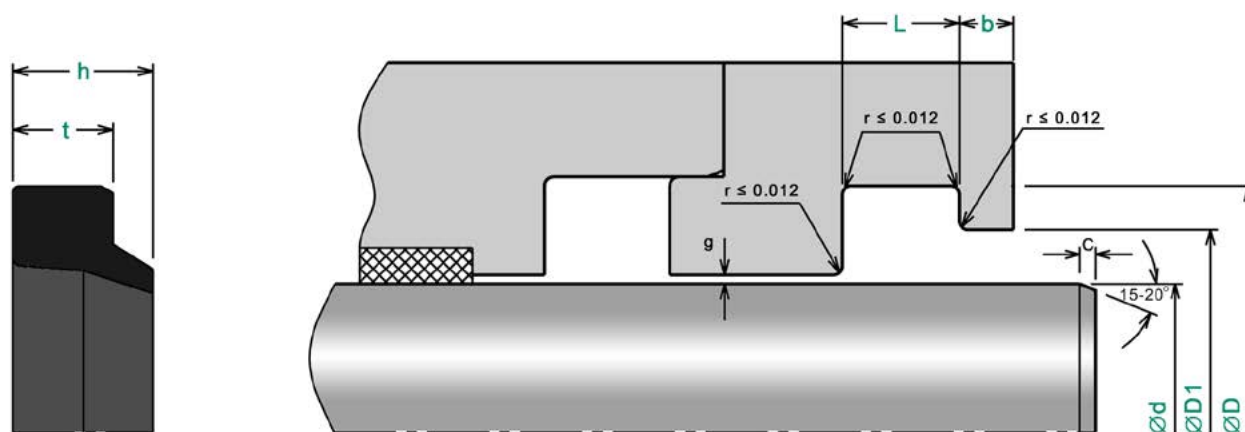
WP12 Metric Size

Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	ØD1 +/- 0.1	L +/- 0.2	b	t	h	c
U7161	65	73	69	6	3	5	7	2.0
U7162	70	78	74	6	3	5	7	2.0
U5233	70	80	75	6	3	5	7	2.5
U7166	70	82	77	8	5	7	11.5	3.0
U7136	73	81	77	6	3	5	7	2.0
U7528	75	83	79	6	3	5	7	2.0
U7163	75	87	83	8	6	7	12	3.0
U7164	80	88	84	6	3	5	7	2.0
U4827	85	97	93	8	6	7	12	3.0
U7155	90	102	98	8	6	7	12	3.0
U7156	93	101	98.5	4.9	4	3.9	7	2.0
U7529	95	103	100.5	4.9	4	3.9	7	2.0
U7151	95	107	104	8	6	7	12.5	3.0
U8230	100	108	105.5	5	4	4	7	2.0
U7167	100	112	108.5	8	6	7	12	3.0
U5235	110	122	117	8	6	7	12	3.0
U7157	115	127	123.5	8	6	7	12	3.0
U7158	125	137	134	8	6	7	12.5	3.0
U7138	125	140	134.5	11	7	10	16	4.0
U7146	130	142	139	8	6	7	12.5	3.0
U7147	135	149	143	8	6	7	12	3.5
U7131	140	152	149	8	6	7	12.5	3.0
U7526	150	162	159	8	6	7	12.5	3.0
U7132	160	175	170	11	7	10	16	4.0
U7139	178	198	188	6	4	5	8	5.0
U8111	185	197	194	8	6	7	12	3.0
U8112	212	224	221	8	6	7	12	3.0
U7527	240	255	251.5	7.35	7	6.35	13	4.0
U7593	250	265	261.5	7.35	7	6.35	13	4.0

WP12

Inch Size



Ordering dimensions in green



WP13

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

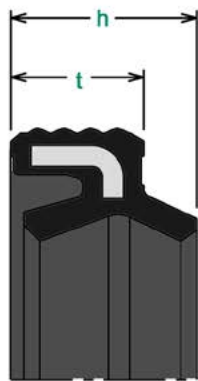
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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WP13

Metric Size



Ordering dimensions in green

[illegible]



WP14

PRODUCT DESCRIPTION

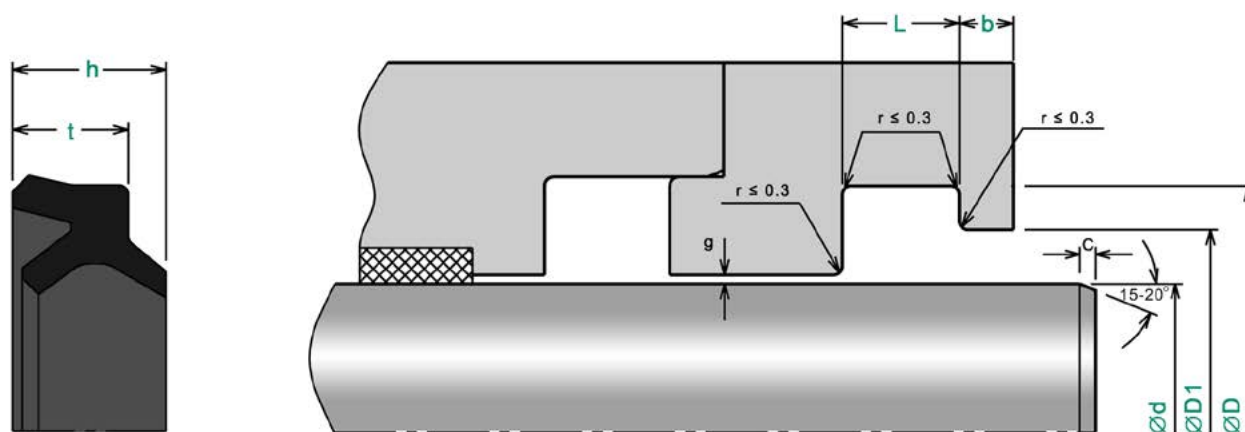
Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

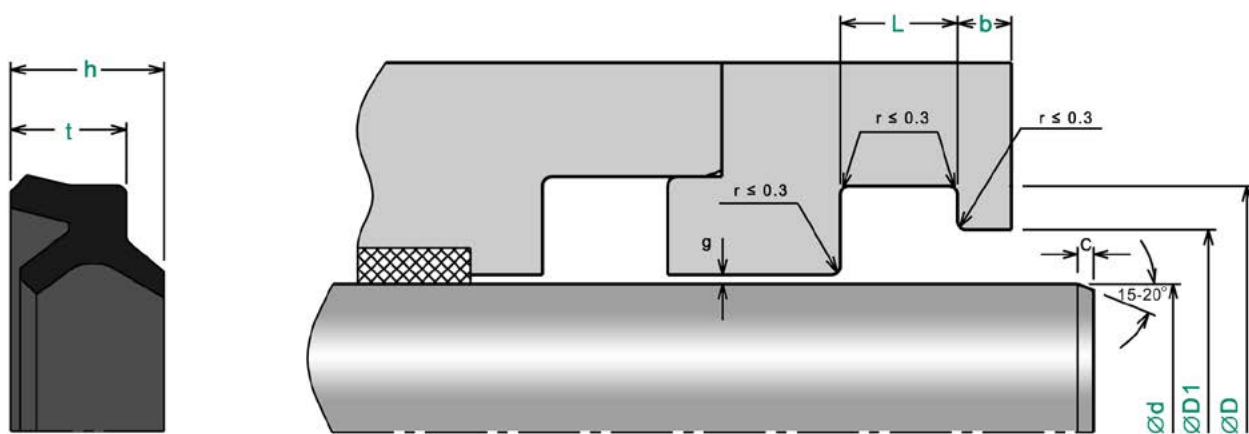
WP14 Metric Size

Ordering dimensions in green

NAK NO.	Ød f8	ØD 0/- 0.2	ØD1 +/- 0.1	L +/- 0.2	b	t	h	c
U5694	6	10	8	3.5	2	3	4.4	1.0
U6416	8	12	10	4	2	3.5	4.9	1.0
U6017	10	14	12	4	2	3.5	5	1.0
U8278	10	16	13	5.5	2	5	6.5	1.5
U2752	10	18	14	5.5	3	5	7	2.0
U5206	10.5	16	14.25	4.7	2	4.2	5.8	1.5
U7133	12	16	14	4	2	3.5	4.7	1.0
U5734	12	17	15.5	4.7	2	4.2	6	1.5
U3896	12	18	16	4.5	3	4	6	1.5
U7097	12.5	20	16.25	4.5	3	4	6	2.0
U7341	13	20	16.5	5.5	2	5	6.5	2.0
U5625	14	18	16	4	2	3.5	4.8	1.0
U3895	14	20	17.5	4.5	3	4	6	1.5
U5736	16	24	20	6	3	5.5	7.5	2.0
U3894	18	24	22	4.5	3	4	6	1.5
14833	18	26	22	5	2	4.5	6	2.0
14906	20	28	24	5	2	4.5	6	2.0
U5996	20	28	24	6	3	5.5	7.5	2.0
U6639	20	30	25	7	4	6.5	9.5	2.5
U3893	22	28	25	4.5	3	4	6	1.5
U5598	22	28	25	5.8	2	5.3	6.3	1.5
U5807	25	31	28	6.2	2	5.7	7	1.5
14907	25	33	29	5	2	4.5	6	2.0
U6236	25	33	29	6	3	5.5	7.5	2.0
U5599	25	35	30	5	3	4.5	6.5	2.5
14924	27.3	35	31.15	4	2	3.5	5	2.0
U3545	30	38	34	6	2	5	6.5	2.0
U5624	30	40	35	7.5	2	6.5	8.2	2.5
U7051	31	40	37.5	5.5	2	4.5	6.4	2.5
U9472	31	44.5	39.75	6	5	5	9	3.5
U4772	32	40	36	6	2	5	6.5	2.0
U6724	32	42	37	7.5	3	6.5	8.5	2.5
U7053	34	43	40.5	5.5	2	4.5	6.2	2.3
U7736	35	43	39	4.5	2	3.5	5	2.0
U1397	35	43	39	6	2	5	6.5	2.0
U6688	35	45	40	7.5	3	6.5	8.5	2.5
U8123	35.5	43.5	39.5	6	2	5	6.5	2.0
U7236	36.5	47	41.75	7.5	3	6.5	8.5	3.0

4.13

WP14 Metric Size



Ordering dimensions in green

[illegible]



WP15

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

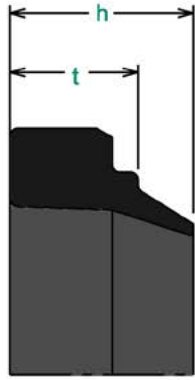
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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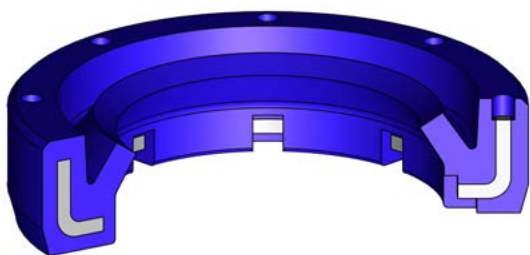
WP15

Metric Size



Ordering dimensions in green

[illegible]



WP16

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

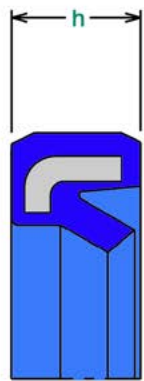
MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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WP16



Ordering dimensions in green

[illegible]



WP17

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

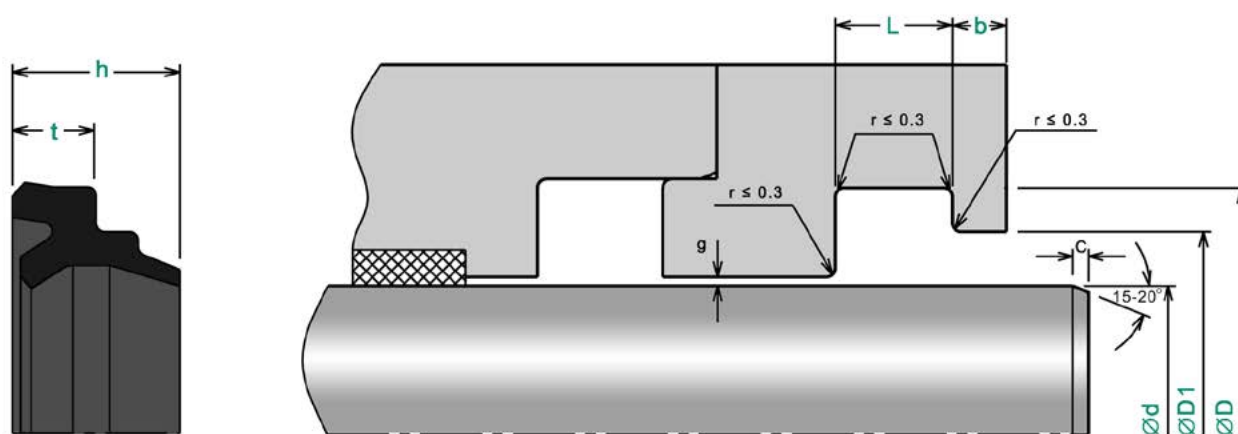
TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

WP17 Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD 0/- 0.2	ØD1 +/- 0.1	L +/- 0.2	b	t	h	c
U7326	12	20	18.2	4	5	4	8	2.0
U7327	14	22	20.2	4	5	4	8	2.0
U5179	16	24	22.2	4	5	4	8	2.0
U5505	18	26	24	4	5	4	8	2.0
U5180	20	28	26.2	4	5	4	8	2.0
U6572	22	30	28.3	4	5	4	8	2.0
U6519	25	33	30.8	4	5	4	8	2.0
U5181	28	36	34.3	4	5	4	8	2.0
U5182	30	38	36.3	4	5	4	8	2.0
U5178	32	40	38.3	4	5	4	8	2.0
U5177	35	43	41.4	4	5	4	8	2.0
U5191	36	44	42.4	4	5	4	8	2.0
U7685	38	46	44.4	4	5	4	8	2.0
U5192	40	48	46.4	4	5	4	8	2.0
U5193	45	53	51.4	4	5	4	8	2.0
U5183	50	58	56.6	4	5	4	8	2.0
U7654	50	58	56.4	6	7	6	12	2.0
U6951	55	63	61.6	4	5	4	8	2.0
U5184	56	64	62.6	4	5	4	8	2.0
U6573	60	68	66.6	4	5	4	8	2.0
U5185	63	71	69.6	4	5	4	8	2.0
U5186	65	73	72	4	5	4	8	2.0
U5195	70	78	76.6	4	5	4	8	2.0
U6521	75	83	81.6	4	5	4	8	2.0
U5557	80	88	86.6	3.8	5	3.8	8	2.0
U5194	90	98	96.6	4	5	4	8	2.0
U5196	100	108	106.6	4	5	4	8	2.0
U5558	110	122	120	5.4	6	5.4	11	3.0
U5543	120	132	129.8	5.4	6	5.4	11	3.0
U5544	125	137	134.8	5.4	6	5.4	11	3.0
U5545	140	152	149.8	5.4	6	5.4	11	3.0
U7321	160	172	169.8	5.5	6	5.5	11	3.0
U7322	180	192	189.8	5.5	6	5.5	11	3.0
U7323	200	212	210	5.5	6	5.5	11	3.0

4.16



WP19

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

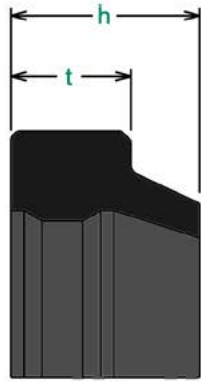
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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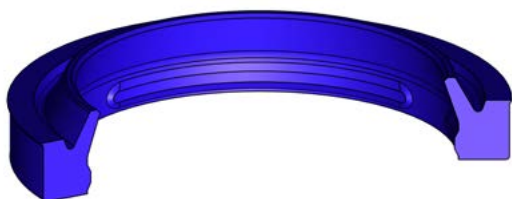
WP19

Metric Size



Ordering dimensions in green

[illegible]



WP21

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

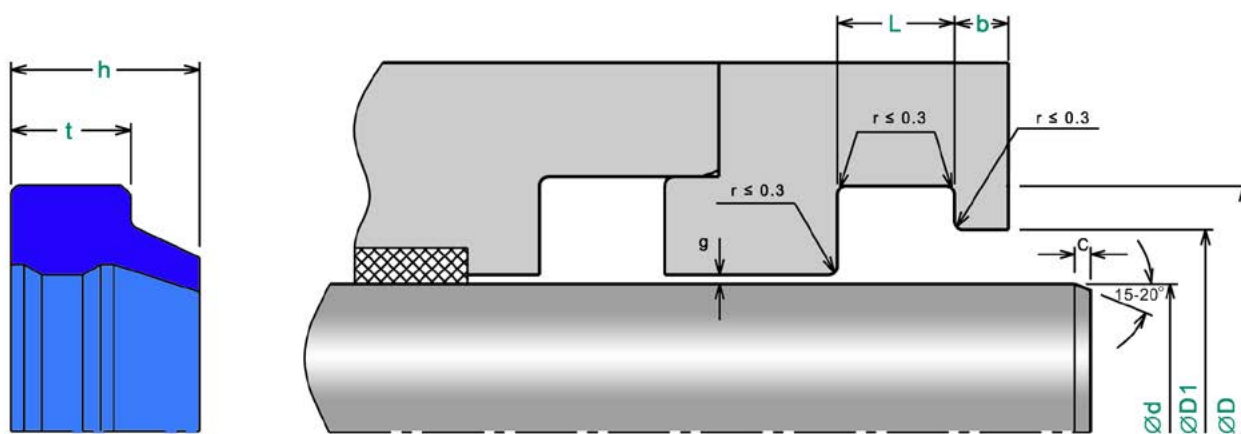
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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WP21

Metric Size



Ordering dimensions in green

[illegible]



WP22

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

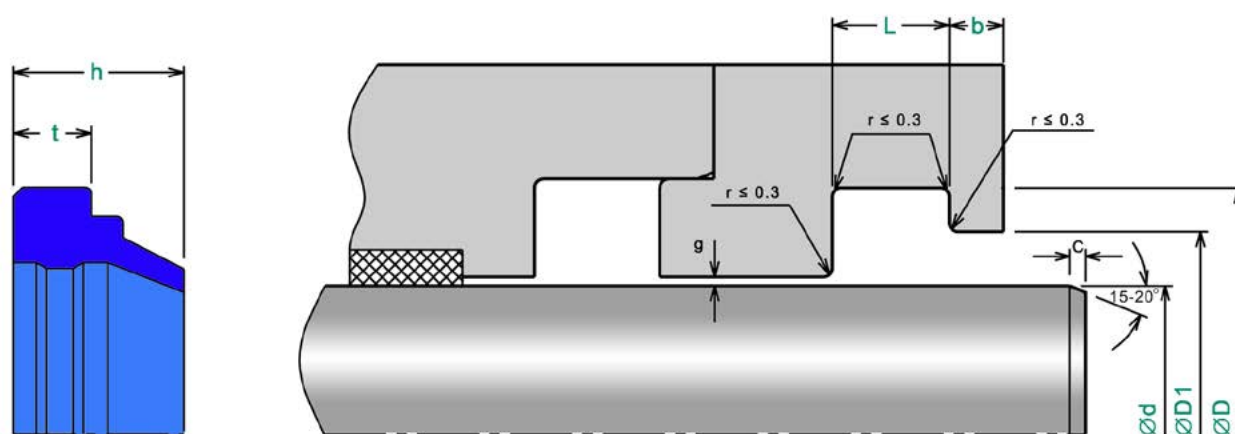
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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WP22

Metric Size



Ordering dimensions in green

[illegible]



WP24

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

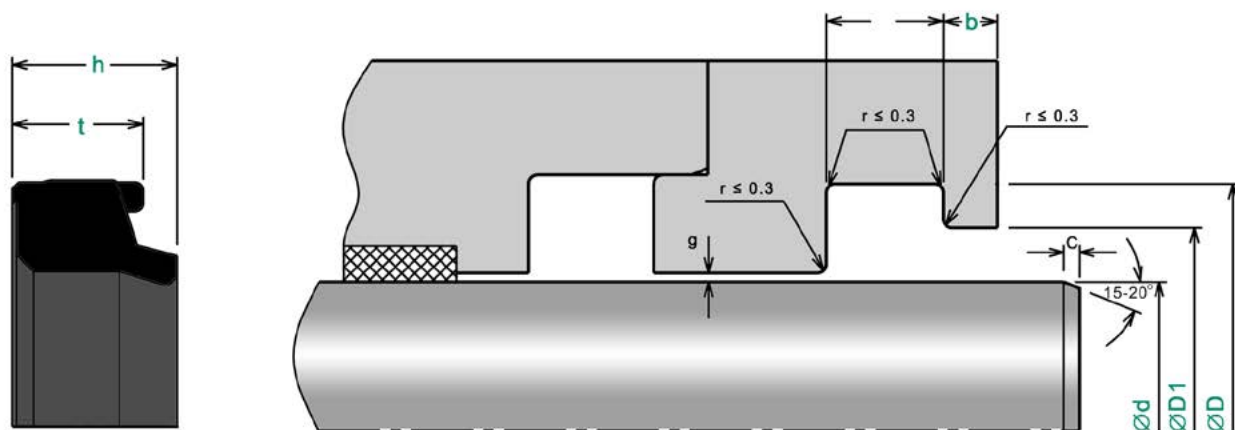
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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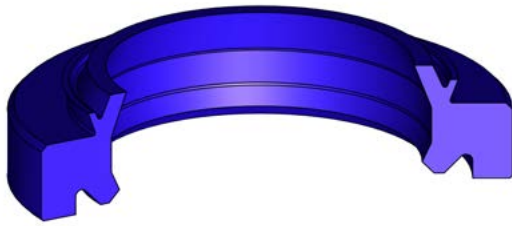
WP24

Metric Size



Ordering dimensions in green

[illegible]



WP32

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	PU
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-40°C , +100°C

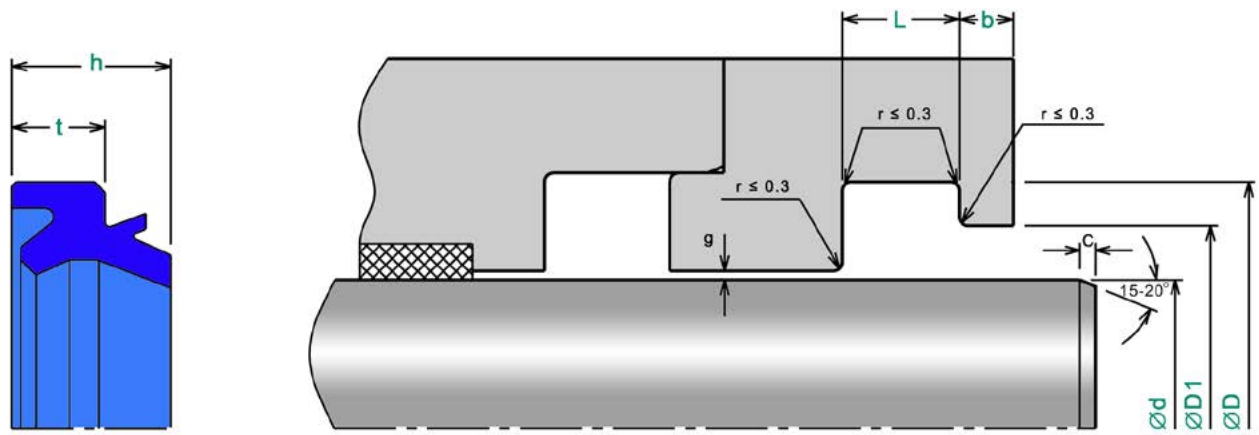
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

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WP32

Metric Size



Ordering dimensions in green

[illegible]



WPB

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

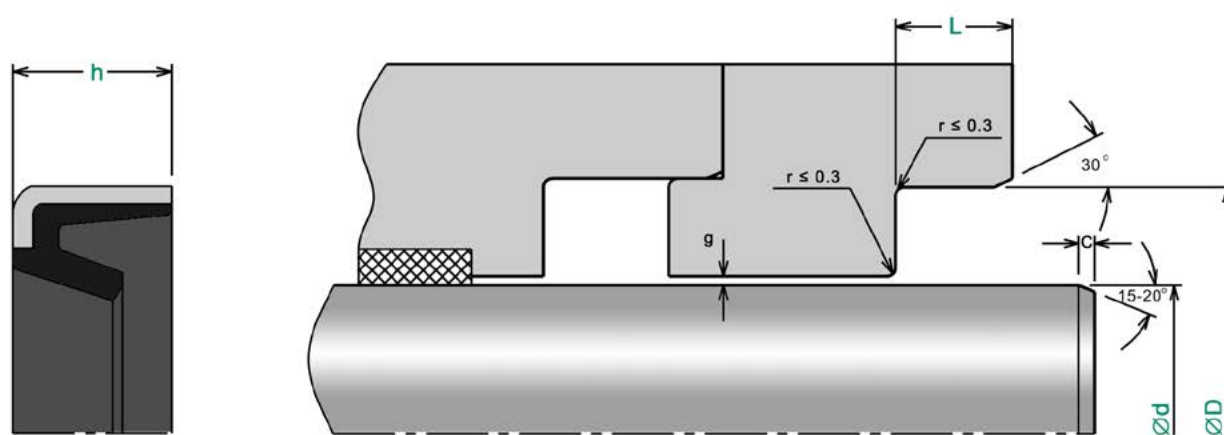
TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

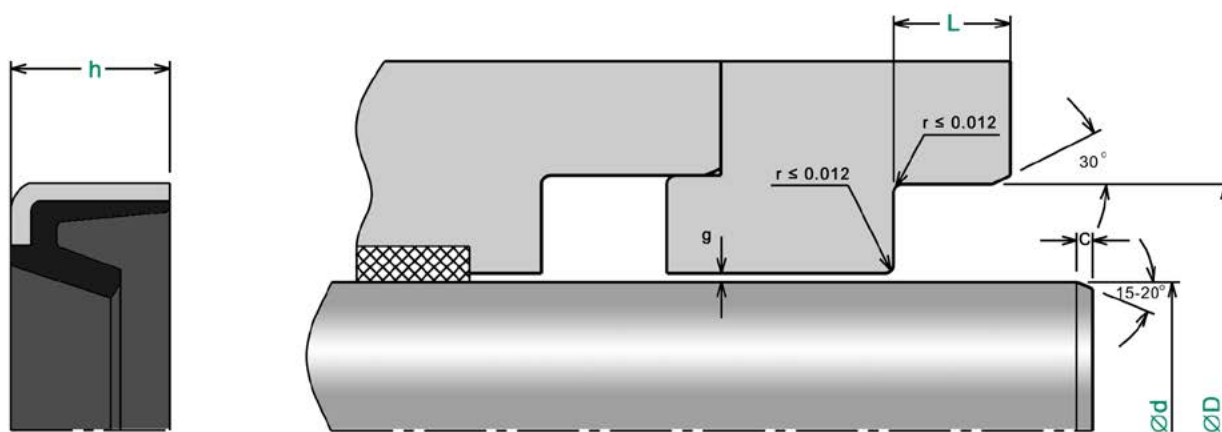
Please contact our customer service for the dimensions out of our stock list.
service@mail.nak.com.tw



Ordering dimensions in green

[illegible]

WPB Metric Size



Ordering dimensions in green

[illegible]



WPC

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

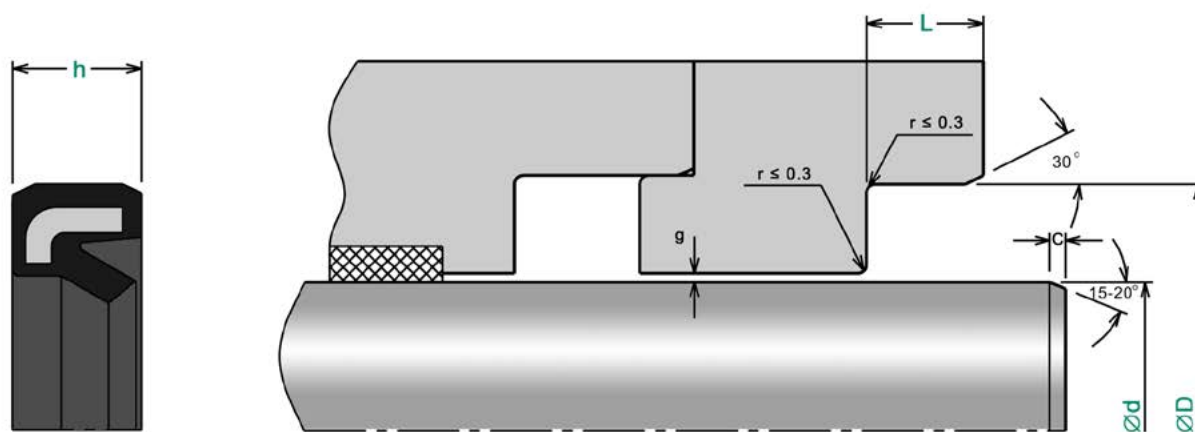
MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

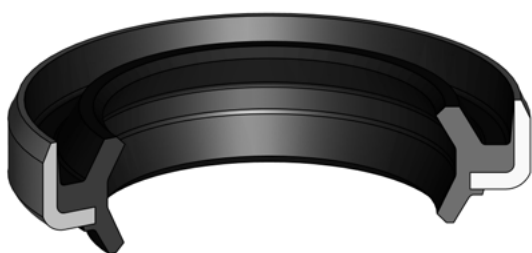
Please contact our customer service for the dimensions out of our stock list.
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WPC



Ordering dimensions in green

[illegible]



WPK

PRODUCT DESCRIPTION

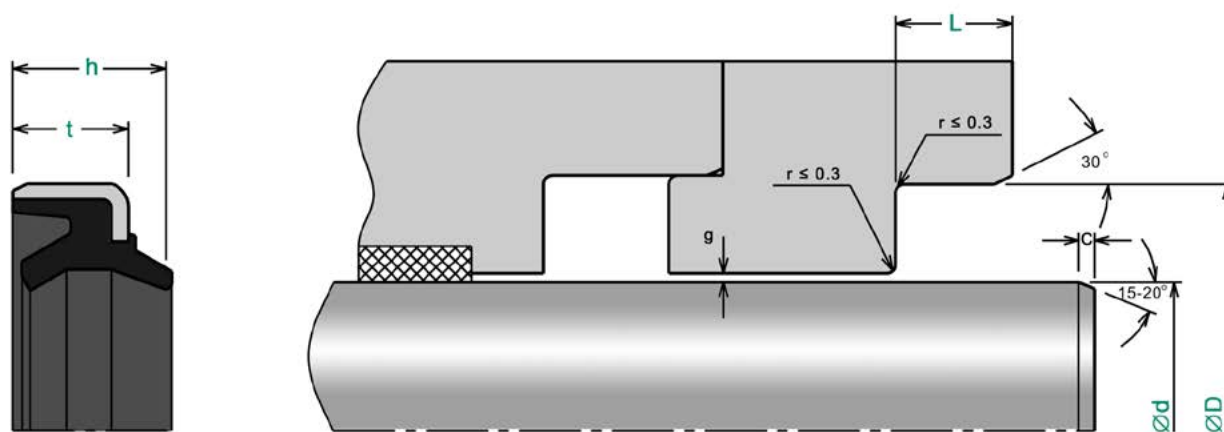
Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

WPK Metric Size


Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U6927	10	16	3	3	4.5	1.5
U6942	16	26	5	5	8	2.5
U5312	16	26	7	7	10	2.5
U8322	17	23	4	4	5	1.5
14049	20	28	3.5	3.5	5	2.0
U3884	20	32	6	6	9	3.0
12589	25	37	6	6	9	3.0
U00219	28	35	4	4	5.5	2.0
U7003	28	38	7	7	10	2.5
U6943	30	40	7	7	10	2.5
12586	30	42	6	6	9	3.0
U4391	32	42	5	5	7	2.5
U00016	32	44	7	7	10	3.0
U7002	32	45	7	7	10	3.5
U4387	35	45	7	7	10	2.5
12536	35	47	7	7	10	3.0
U00405	40	50	7	7	11	2.5
12544	40	52	7	7	10	3.0
U5992	50	58	8	8	10	2.0
U6944	50	60	7	7	10	2.5
U3705	50	62	7	7	10	3.0
U7489	50	64	8	8	11	3.5
U4427	52	62	7	7	10	2.5
U4390	55	65	7	7	10	2.5
U3706	55	69	8	8	11	3.5
U3707	60	74	8	8	11	3.5
U3879	65	79	8	8	11	3.5
U3880	70	84	8	8	11	3.5
U3708	75	89	8	8	11	3.5
U3709	80	94	8	8	11	3.5
U3710	90	104	8	8	11	3.5
U3711	95	109	8	8	11	3.5
U3712	100	114	8	8	11	3.5
U3713	110	126	9	9	12	4.0
U3714	120	136	9	9	12	4.0
U00406	130	140	7	7	11	2.5
U00646	135	155	10	10	14	5.0
U00645	180	200	10	10	14	5.0



WPK1

PRODUCT DESCRIPTION

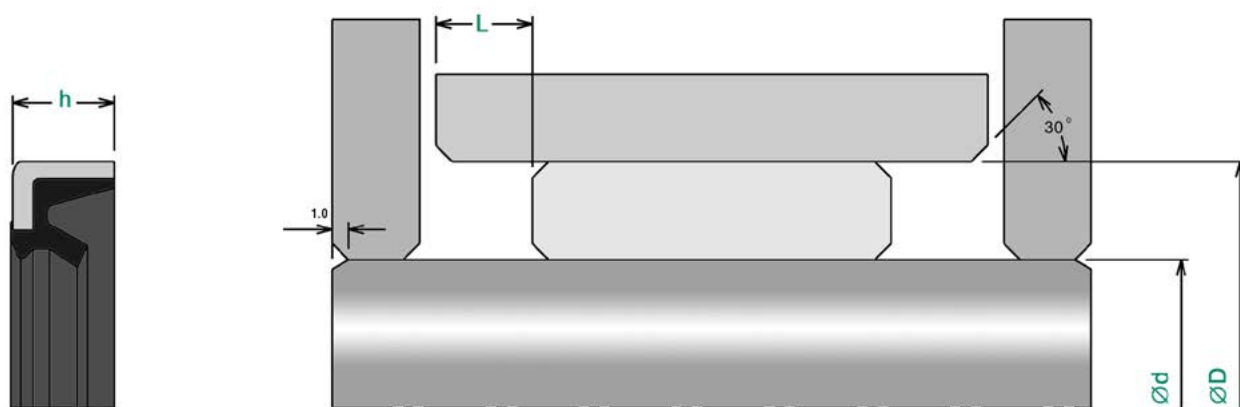
Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

WPK1 Metric Size

Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	h
U6804	30	40	4	4
U6805	35	45	4	4
U00427	38.05	47.63	4.2	4.2
U7016	40	50	5	5
U6293	45	55	5	5
U6294	45	55	6	6
U6295	50	65	6	6
U6296	55	70	6	6
U7338	60	75	6	6
U5052	65	80	6	6
U4743	70	85	6	6
U5051	75	90	6	6
U7203	75	95	7	7
U5050	80	95	6	6
U4944	90	105	6	6
U7269	90	110	9.5	9.5
U5049	95	110	8	8
U7339	95	110	9.5	9.5
U4945	100	115	8	8
U7227	100	120	8	8
U5048	110	125	8	8
U5578	110	125	9.5	9.5
U7202	110	130	8	8
U7340	110	130	9.5	9.5
U5047	120	135	8	8
U5211	120	135	9.5	9.5
U5681	120	140	8	8
U7228	120	140	9.5	9.5
U7226	125	145	9.5	9.5
U5212	130	155	9.5	9.5
U5680	140	160	8	8
U8327	140	170	9.5	9.5
U5213	145	160	8	8
U6806	150	180	8	8
U6535	200	225	8	8
U6536	210	225	8	8
U5127	210	240	9.5	9.5
U5020	260	290	9.5	9.5

4.25



WPK3

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$
Temperature	-35°C , +100°C

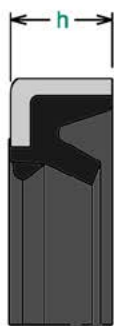
DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

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WPK3

Metric Size



Ordering dimensions in green

4.26



WPM

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

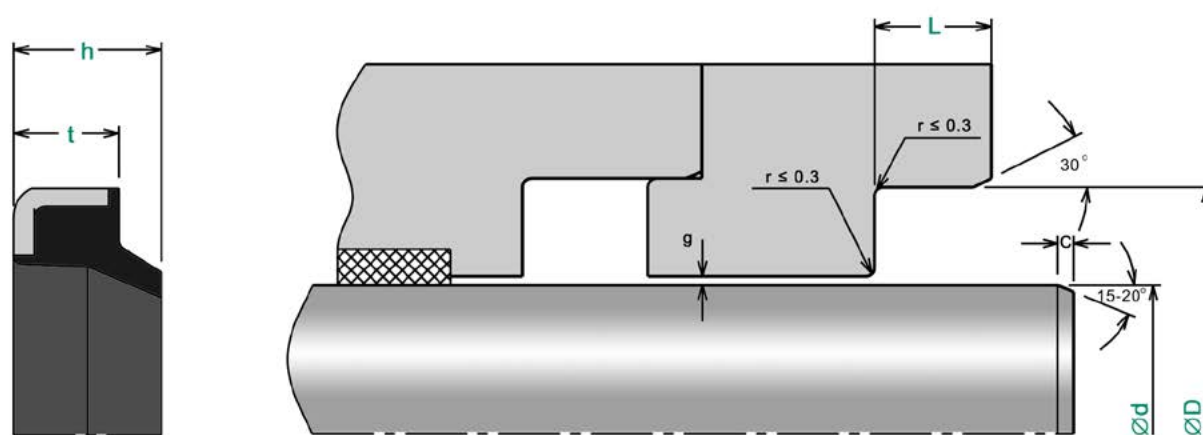
TECHNICAL DATA

MATERIAL	NBR	FKM
MEDIA	Mineral oils	Mineral oils
Running speed v	$\leq 1 \text{ m/s}$	$\leq 1 \text{ m/s}$
Temperature	-35°C , +100°C	-25°C , +250°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	$\leq 0.8 \mu\text{m}$	$\leq 3.2 \mu\text{m}$
Groove flanks	L	$\leq 3.2 \mu\text{m}$	$\leq 12.5 \mu\text{m}$

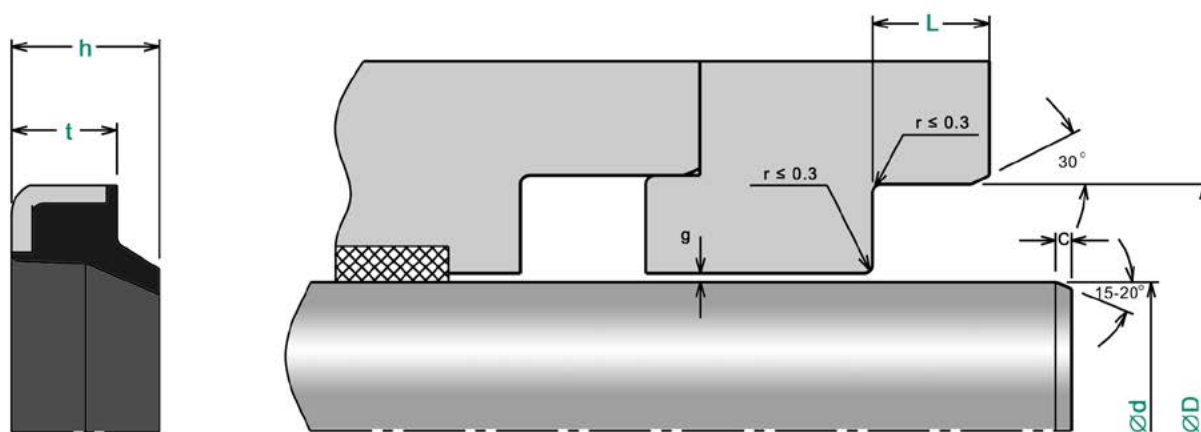
WPM Metric Size



Ordering dimensions in green

NAK NO.	NBR	FKM	Ød f8	ØD H8	L +/- 0.2	t	h	c
09834	•		6	13	3	3	4.5	2.0
U5956	•		8	14	3.3	3.3	5	1.5
U6549	•		8	15	3	3	5	2.0
04732	•		8	22	3	3	4.5	3.5
U9232	•		9.5	14.2	3	3	4	1.5
U1713	•		10	16	3	3	4.5	1.5
U5809	•		10	16	3.5	3.5	5	1.5
05837	•		10	16	5	5	7	1.5
U2986	•		10	18	5	5	8	2.0
U3887	•		10	20	4	4	6	2.5
U4178	•		10	20	5	5	8	2.5
U3885	•		12	18	3.5	3.5	5	1.5
01320	•		12	20	4	4	6	2.0
U1749	•		12	22	5	5	8	2.5
U4517	•		12.7	18	3	3	4.5	1.5
U5957	•		14	19.8	3	3	4.5	1.5
U5919	•		14	20	3.5	3.5	5	1.5
12479	•		14	20	4	4	5	1.5
U1771	•		14	22	3	3	4	2.0
U5920	•		15	21	3.5	3.5	5	1.5
05929	•		15	25	5	5	8	2.5
U1680	•		15	25	7	7	10	2.5
U5145	•		16	20	3	3	4	1.0
01358	•		16	22	3	3	4	1.5
U4180	•		16	22	3.5	3.5	5	1.5
U1258	•		16	26	5	5	8	2.5
01384	•		16	26	7	7	10	2.5
05011	•		17	22	5	5	7	1.5
U5320	•		17	22	5	5	8	1.5
U00076	•		18	26	3	3	4	2.0
U7126	•		18	26	4	4	6.5	2.0
11010	•		18	26	5	5	7	2.0
U9007	•		18	28	5	5	8	2.5
03888	•		18	28	7	7	10	2.5
09886	•		18	28	7	7	12	2.5
U5949	•		18	32	5	5	8	3.5
U2408	•		19	26	4	4	6	2.0
U6728	•		19	26	5	5	8	2.0

4.27

WPM Metric Size

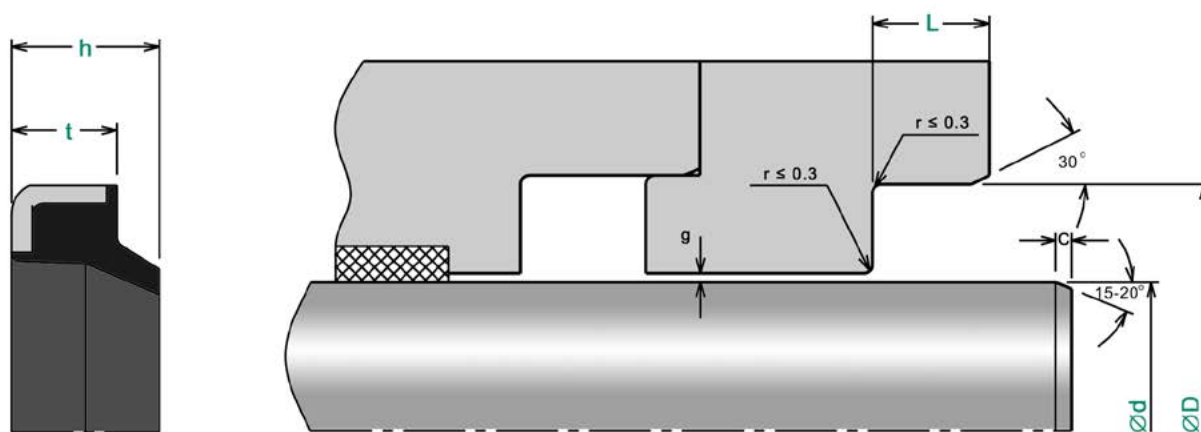
Ordering dimensions in green

NAK NO.	NBR	FKM	Ød f8	ØD H8	L +/- 0.2	t	h	c
U5958	•		20	26	3.5	3.5	5	1.5
12200	•		20	26	4	4	7	1.5
15225	•		20	26	5	5	8	1.5
05079	•		20	28	3.5	3.5	5	2.0
U1554	•		20	28	5	5	8	2.0
U5142	•		20	30	4	4	6	2.5
09396	•		20	30	5	5	8	2.5
U1699	•		20	30	5	5	7	2.5
09396	•		20	30	5	5	8	2.5
05424	•		20	30	7	7	10	2.5
U1678	•		20	35	7	7	10	4.0
U4363	•		21	28	3.5	3.5	5.5	2.0
U5950	•		21	28	3.5	3.5	5	2.0
U7766	•		21	28	4.8	4.8	9	2.0
U4181	•		22	28	5	5	8	1.5
U4005	•		22	28	5	5	9	1.5
10286	•		22	30	5	5	8	2.0
U6690	•		22	30	6	6	11	2.0
09697	•		22	32	5	5	8	2.5
U00432	•		22	32	7	7	10	2.5
09398	•		22	35	5	5	8	3.5
U5948	•		23	33	5	5	8	2.5
13709	•		24	32	5	5	7	2.0
U6587	•		24	32	5	5	8	2.0
12618	•		24	35	7	7	10	3.0
U5767	•		25	33	4	4	7	2.0
11109	•		25	35	3	3	4	2.5
U5257	•		25	35	5	5	7	2.5
U9362	•		25	35	5	5	8	2.5
U1370	•		25	35	7	7	10	2.5
13473	•		25	37	6	6	9	3.0
U3600	•		25	43	5	5	7	4.5
U3025	•		25.4	33.34	4.76	4.76	6.35	2.0
12481	•		26	35	7	7	10	2.5
11108	•		28	34	5	5	7	1.5
U1727	•		28	38	5	5	8	2.5
13474	•		28	38	7	7	10	2.5
06096	•		28	40	7	7	10	3.0

WPM Metric Size

NAK NO.	NBR	FKM	Ød f8	ØD H8	L +/- 0.2	t	h	c
U9028	•		28	42	7	7	10	3.5
11481	•		29.5	40	7	7	10	3.0
11613	•		30	40	5	5	7	2.5
05294	•		30	40	5	5	8	2.5
U7033	•		30	40	7	7	10	2.5
14177	•		30	42	5	5	8	3.0
U1715	•		30	42	7	7	10	3.0
11276	•		30	45	5	5	8	4.0
U6971	•		30	45	7	7	10	4.0
04759	•		32	40	4	4	7	2.0
U1410	•		32	40	7	7	10	2.0
U1731	•		32	42	5	5	7	2.5
U9029	•		32	42	5	5	8	2.5
04992	•		32	42	7	7	10	2.5
U4183	•		32	45	4	4	8	3.5
U1679	•		32	45	5	5	8	3.5
04391	•		32	45	7	7	10	3.5
U1695	•		32	50	7	7	10	4.5
U9008	•		33	43	7	7	10	2.5
U4428	•		35	45	5	5	7	2.5
U1186	•		35	45	7	7	10	2.5
U5888	•		35	47	7	7	10	3.0
14964	•		35	48	5	5	8	3.5
U1576	•		35	48	6.5	6.5	12	3.5
U1676	•		35	50	9	9	12	4.0
U1675	•		35	52	7	7	10	4.5
U1730	•		36	45	7	7	10	2.5
10296	•		36	46	7	7	10	2.5
05807	•		38	48	7	7	10	2.5
U5018	•		38	53	5	5	8	4.0
11482	•		39.5	50	7	7	10	3.0
U1712	•		40	50	3.5	3.5	5	2.5
U2729	•		40	50	5	5	7	2.5
01380	•		40	50	5	5	8	2.5
04565	•		40	50	7	7	10	2.5
U4185	•		40	52	5	5	8	3.0
01390	•		40	52	7	7	10	3.0
U7244	•		40	53	5	5	8	3.5
U1677	•		40	55	9	9	12	4.0
15463	•		40	62	7	7	10	5.5
08993	•		42	52	7	7	10	2.5
U4186	•		45	55	5	5	7	2.5
U2845	•		45	55	7	7	10	2.5
13423	•		47	57	7	7	10	2.5
15260	•		48	60	7	7	10	3.0
U1674	•		50	56	4	4	6	1.5
04896	•		50	56	5	5	7	1.5
U7128	•		50	56	5	5	8	1.5
09856	•		50	60	5	5	7	2.5
U1189	•		50	60	5	5	8	2.5
05278	•		50	60	7	7	10	2.5
U6585	•		50	62	5	5	7	3.0
U6998	•		50	62	7	7	10	3.0
U1702	•		50	65	5	5	8	4.0

4.27

WPM Metric Size

Ordering dimensions in green

NAK NO.	NBR	FKM	Ød f8	ØD H8	L +/- 0.2	t	h	c
U9879	•		50	65	7	7	10	4.0
U00214	•		50.8	63.5	7	7	10	3.5
U2923	•		55	65	7	7	10	2.5
U6597	•		55	70	7	7	10	4.0
U1673	•		56	65	7	7	10	2.5
U6674	•		56	66	5	5	7	2.5
15261	•		56	66	7	7	10	2.5
U5234	•		60	68	5	5	8	2.0
12821	•		60	70	5	5	7	2.5
U1930	•		60	70	7	7	10	2.5
U1732	•		60	74	5	5	8	3.5
U00403	•		60	74	8	8	11	3.5
04902	•		60	75	7	7	10	4.0
U9124	•		60	78	11	11	15	4.5
U1719	•		60	80	7	7	10	5.0
14538	•		62	72	7	7	10	2.5
U5922	•		63	73	5	5	7	2.5
U1716	•		63	73	7	7	10	2.5
12646	•		63	75	7	7	10	3.0
U9532	•		63	78	7	7	10	4.0
U5753	•		63	83	5	5	8	5.0
U4187	•		65	75	5	5	7	2.5
U1419	•		65	75	7	7	10	2.5
01024	•		70	80	5	5	7	2.5
04184	•		70	80	7	7	10	2.5
U7403	•		70	84	7	7	10	3.5
14772	•		70	90	7	7	10	5.0
U6975	•		75	85	7	7	10	2.5
12417	•		78	88	7	7	10	2.5
U5752	•		80	88	7	7	10	2.0
U1742	•		80	90	7	7	10	2.5
U6598	•		80	100	12	12	15	5.0
11794	•		82.55	92	7	7	10	2.5
15393	•		84	94	5	5	8	2.5
U1697	•		85	95	7	7	10	2.5
U7127	•		90	100	5	5	7	2.5
08899	•		90	100	7	7	10	2.5
08994	•		95	105	7	7	10	2.5

WPM Metric Size

NAK NO.	NBR	FKM	Ød f8	ØD H8	L +/- 0.2	t	h	c
15244	•		100	110	5	5	8	2.5
08884	•		100	110	7	7	10	2.5
12401	•		100	115	8	8	10	4.0
U00404	•		100	120	7	7	10	5.0
08944	•		105	115	7	7	10	2.5
08901	•		110	120	7	7	10	2.5
U4188	•		115	125	7	7	10	2.5
11471	•		120	130	7	7	10	2.5
15395	•		125	140	7	7	10	4.0
13240	•		125	140	9	9	12	4.0
08875	•		130	145	9	9	12	4.0
15396	•		135	145	7	7	10	2.5
07392	•		135	150	9	9	12	4.0
U7644	•		140	150	7	7	10	2.5
09209	•		140	155	9	9	12	4.0
06537	•		150	165	9	9	12	4.0
15409	•		150	166	8	8	12	4.0
08902	•		160	175	9	9	12	4.0
15408	•		165	180	12	12	16	4.0
12635	•		170	185	9	9	12	4.0
U8583	•		170	185	10	10	14	4.0
16887	•		175	190	9	9		4.0
09524	•		180	195	10	10	14	4.0
U2838	•		190	210	10	10	14	5.0
U1521	•		200	215	9	9	12	4.0
12514	•		200	220	12	12	16	5.0
U7080	•		210	230	12	12	16	5.0
U9736	•		220	235	10	10	14	4.0
11614	•		220	240	12	12	16	5.0
14867	•		230	255	12	12	16	6.5
14118	•		240	260	12	12	16	5.0
U2622	•		250	270	12	12	16	5.0
U4153	•		270	295	12	12	16	6.5
13411	•		275	300	12	12	16	6.5
15407	•		320	340	12	12	16	5.0
U6555	•		400	420	12	12	16	5.0
U7509		•	5	10	3	3	4	1.5
U4388		•	10	16	3	3	4.5	1.5
U5108		•	12	18	3.5	3.5	5	1.5
01320		•	12	20	4	4	6	2.0
U1830		•	14	25	5	5	8	3.0
U2885		•	15	23	4	4	7	2.0
01358		•	16	22	3	3	4	1.5
16879		•	20	30	7	7	10	2.5
13268		•	25	35	5	5	7	2.5
U00475		•	25	35	7	7	10	2.5
05294		•	30	40	5	5	8	2.5
U1739		•	30	40	7	7	10	2.5
U1730		•	36	45	7	7	10	2.5
U1751		•	36	46	5	5	7	2.5
U00206		•	45	55	7	7	10	2.5
13273		•	50	60	7	7	10	2.5
U3331		•	56	66	7	7	10	2.5
U1823		•	60	70	7	7	10	2.5

4.27

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WPR

PRODUCT DESCRIPTION

Popular dust seals. Used for heavy duty application.

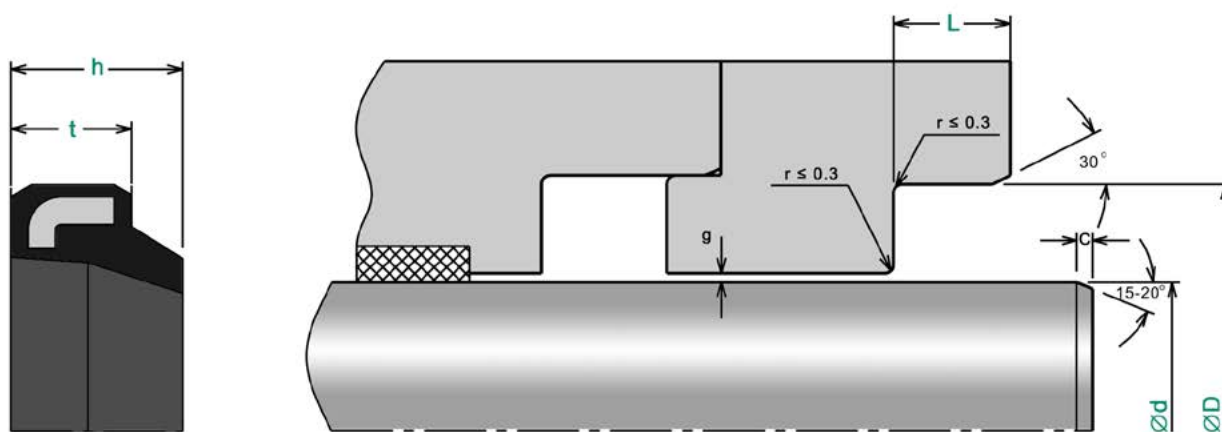
TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	$\varnothing d$	0.2~0.4 μm	0.8~1.6 μm
Groove base	$\varnothing D$	≤ 0.8 μm	≤ 3.2 μm
Groove flanks	L	≤ 3.2 μm	≤ 12.5 μm

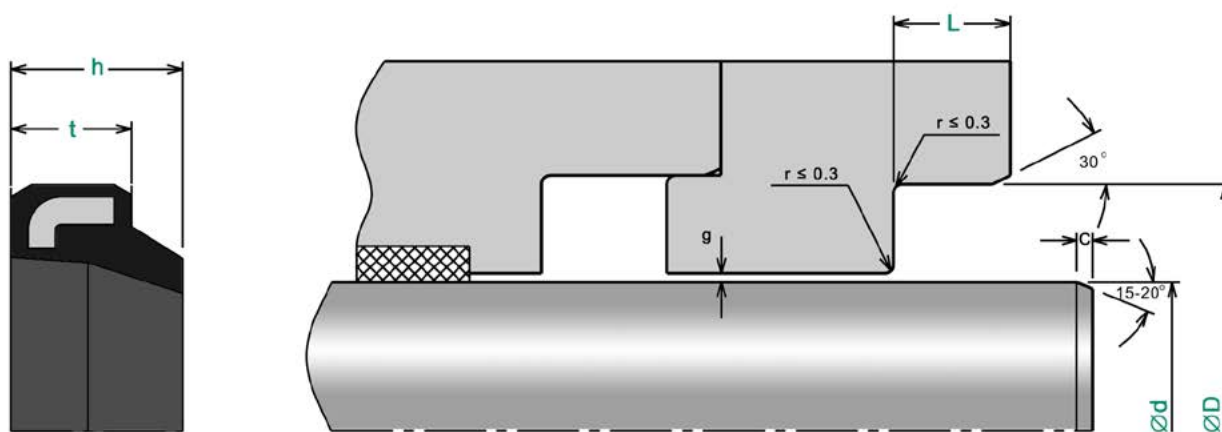
WPR Metric Size



Ordering dimensions in green

NAK NO.	Ød f8	ØD H10	L +/- 0.2	t	h	c
10819	6	13	3	3	4.5	2.0
11628	8	14	3.5	3.5	5	1.5
12862	10	16	3.5	3.5	5	1.5
U3352	10	20	5	5	8	2.5
U3193	12	20	4	4	6	2.0
12109	12	22	5	5	8	2.5
U9233	12.5	17.5	3	3	4	1.5
U5787	12.7	18	3	3	4.5	1.5
12675	14	20	3.5	3.5	5	1.5
U1821	14	22	3	3	4	2.0
10101	15	21	3.5	3.5	5	1.5
12079	15	25	5	5	7	2.5
02122	16	22	3	3	4	1.5
U7734	16	22	3.5	3.5	5	1.5
U1512	16	24	3.5	3.5	5	2.0
U1820	16	26	5	5	8	2.5
07008	18	28	5	5	7	2.5
U9996	18	28	7	7	10	2.5
15406	20	28	3.5	3.5	5	2.0
U2273	20	28	7	7	10	2.0
06006	20	30	5	5	7	2.5
07156	22	32	5	5	7	2.5
10826	22	32	7	7	10	2.5
11466	24	35	5	5	8	3.0
02133	25.4	34	5	5	8	2.5
U7386	28	40	7	7	10	3.0
U2731	30	40	5	5	7	2.5
07369	30	42	6	6	9	3.0
01557	32	42	5	5	7	2.5
U1418	32	42	7	7	10	2.5
04296	32	45	7	7	10	3.5
U2236	35	45	5	5	7	2.5
04582	35	45	7	7	10	2.5
11795	36	45	7	7	10	2.5
12360	38	48	7	7	10	2.5
05147	40	50	5	5	8	2.5
01439	40	50	7	7	10	2.5
11501	42	52	5	5	7	2.5

WPR Metric Size



Ordering dimensions in green

[illegible]



WPV

Wiper Seal 防塵密封件

PRODUCT DESCRIPTION

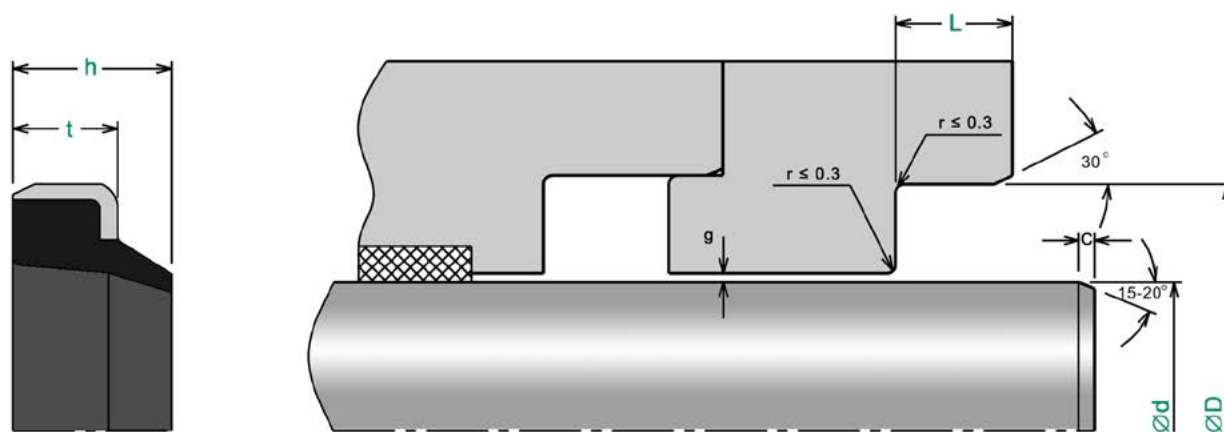
Popular dust seals. Used for heavy duty application.

TECHNICAL DATA

MATERIAL	NBR
MEDIA	Mineral oils
Running speed v	≤ 1 m/s
Temperature	-35°C , +100°C

DESIGN NOTES

Peak-to-valley heights		Ra	Rz
Sliding surface	Ød	0.2~0.4 µm	0.8~1.6 µm
Groove base	ØD	≤ 0.8 µm	≤ 3.2 µm
Groove flanks	L	≤ 3.2 µm	≤ 12.5 µm

WPV Metric Size


Ordering dimensions in green

NAK NO.	Ød f8	ØD H8	L +/- 0.2	t	h	c
U4954	16	22	3	3	4	1.5
U5202	16	26	5	5	8	2.5
U4946	20	30	4	4	6	2.5
U4953	20	30	7	7	10	2.5
U4947	22	28	5	5	8	1.5
U4952	22	32	5	5	7	2.5
U5055	24	32	5	5	7	2.0
U4958	25	35	5	5	8	2.5
U4931	25	35	7	7	10	2.5
U4657	25	37	6	6	9	3.0
U4940	28	38	7	7	10	2.5
U4957	28	40	7	7	10	3.0
U5128	30	40	5	5	8	2.5
U4928	30	40	7	7	10	2.5
U4951	32	42	7	7	10	2.5
U4935	35	45	7	7	10	2.5
U4939	36	46	7	7	10	2.5
U00557	38	48	7	7	10	2.5
U4956	40	50	5	5	8	2.5
U4936	40	50	7	7	10	2.5
U5158	40	52	5	5	8	3.0
U4771	40	52	7	7	10	3.0
U4926	45	55	7	7	10	2.5
U4950	48	60	7	7	10	3.0
U4955	50	56	5	5	7	1.5
U4927	50	60	7	7	10	2.5
U5157	50	62	5	5	8	3.0
U4949	55	65	7	7	10	2.5
U4938	56	66	7	7	10	2.5
U4941	60	70	7	7	10	2.5
U4948	63	73	7	7	10	2.5
U4932	65	75	7	7	10	2.5
U4930	70	80	7	7	10	2.5
U4933	80	90	7	7	10	2.5
U8938	90	100	7	7	10	2.5
U4934	100	110	7	7	10	2.5
U4937	110	120	7	7	10	2.5
U4929	120	130	7	7	10	2.5
U5201	150	165	9	9	12	4.0



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